Cellular Expert Desktop RLP for ArcGIS Pro User Guide

Version 4.9

2026

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About Cellular Expert

Cellular Expert UAB (CE) developed ultra-fast wave propagation, communication systems deployment planning, and radio/optical visibility calculation software for ESRI's ArcGIS mapping environment, which is widely used within Telecom, Defense, IoT, and other companies and organizations.

CE's communication network planning, network asset management, operational support software, and customer-tailored solutions enhance the intelligence and business efficiency of more than 170 communication network companies, regulators, and defense organizations in more than 50 countries.

1. Software Purpose and Functionality

Cellular Expert Desktop for ArcGIS Pro (CE Pro) is a professional radio coverage planning software tool designed on top of the Esri ArcGIS Pro environment. The CE Pro is a highly versatile and functional tool, covering the network design functionality needs of most professional radio communication planners and developers. It could support network planning and optimization for the entire range of wireless technologies in frequencies from 10 kHz to 350 GHz.

The key features and benefits of utilizing Cellular Expert software are as follows:

1. CE Pro is a purpose-built extension of ESRI software, a world-renowned standard in GIS environments. This means that along with wireless network planning and optimization, it gives users various value-added GIS/mapping functionalities to manage, visualize feature classes or roasters, create reports, and graphs, or automate workflows,
2. CE Pro is made of several processing engines and functional modules and therefore may be easily configured for various business functions: network planning, network engineering, technology information support to sales and marketing, network assets management, and network operational maintenance and supervision,
3. CE Pro is maintained and constantly developed by a dedicated agile team of software developers and radio planners, allowing attentive responsiveness to customer needs and developing custom-tailored solutions.

CE Pro performs mobile network coverage analysis based on:

1. Site and cell level details of radio equipment complement, specifications, cell and component carriers configuration,
2. Subscriber characteristics – geographic distribution density, generated traffic to estimate network/cell loading,
3. DTM, buildings, and clutter-based deterministic point-to-area radio wave propagation modeling.

CE Pro allows the user to input and use a broad variety of GIS and network data to support the simulations, as the overall quality of coverage calculations is dependent on the completeness and detail of technical network data and the resolution and quality of the GIS data. CE Pro can efficiently simulate wide area network coverage using the GIS data with a resolution of down to sub-meter.

Calculated results of coverage predictions could be presented as coverage raster maps.

CE Pro is capable of modeling various wireless technologies: cellular (2G/3G/4G/5G), PMR/PAMR (TETRA, APCO, others), FWA/BWA, IoT (LoRa/SigFox, others), as well as fixed microwave links. Therefore, it may be used as a radio planning tool in various industries: Mobile Operators, Integrated Telecom Companies, Wireless Internet Service providers, Regulatory authorities, Utilities, Broadband Infrastructure providers, Defense organizations, as well as any other users of radiocommunication systems.

In the following, we summarize the key functionalities of the tool.

Data Management

The tool allows the importation, storage, and management of detailed technical data on network nodes, such as sites, cells, RF transmitters, and antennas.

Signal Strength Prediction

The tool contains several in-built path loss prediction models that allow the user to easily start simulations based on the evaluation of the most essential pathloss contributing factors. The following two models constitute the starting set:

- Free space – typically used for modeling short-range mobile communications, fixed links, or other radiocommunications applications with prevalent Line-of-Sight conditions on propagation paths,

- UniMacro – a proprietary universal model for wide area mobile communication systems that flexibly accounts for a variety of propagation paths as determined for each specific reception point based on terrain and clutter data vs. configuration of the modeled system.

Further details on path loss models and their configuration options are provided in the relevant section of this manual.

Model Tuning
The tool provides an automated method for fine-tuning propagation model parameters to fit the specific scenario and type of area of network deployment based on analyzing real field strength measurement results.

Radio Coverage Calculation for Different Mobile Technologies
Radio coverage calculation is assisted in the tool by the possibility to use pre-loaded cell configuration templates that are tailored to typical technical parameters of base stations in different Radio Access Technologies. Accordingly, the model settings and outputs will be adjusted to suit the scenario pertinent to that technology, i.e.:

- 2G – radio coverage is calculated in dBm as receive power level of narrow-band (200 kHz) signal,
- 3G – radio coverage is calculated in dBm as the receive power level of a single broadband (3.85 MHz) carrier,
- 4G/5G – radio coverage is calculated in dBm as the equivalent RSRP of a single sub-carrier component in the complex OFDM broadband signal.
- Wi-Fi - wireless communication technology based on the IEEE 802.11 standards, used for setting up local area networks (WLANs) and providing internet access in various settings without requiring cable connections.

Profile Analysis
The tool provides powerful GIS analytical features to analyze the terrain and clutter on the fixed link path, such as allowing to estimate of the Fresnel zone clearance condition, Power Budget, Path loss and Angles between Tx and Rx.

2. System requirements

This chapter will guide you through the minimal hardware and software requirements.

Note: requirements can vary significantly, depending on the acceptable calculation time and task complexity.

2.1 Minimal Requirements for Hardware

Processor (CPU):

- Minimum: 8 cores, hyperthreaded
- Recommended: 16 cores

(Optional) Requirements for GPU-accelerated calculations

- GPU – any NVIDIA GPU with CUDA capabilities (https://developer.nvidia.com/cuda-gpus)
- Driver version: 456.38 or later
- CUDA Toolkit 11.0 or later

Memory/RAM:

- Minimum: 16 GB
- Recommended: 32 GB

Storage:

- Minimum: 500 GB of free space
- Recommended: 2TB or more of free space on a solid-state drive (SSD)

2.2 Minimal Requirements for Software

Cellular Expert runs on Microsoft Windows 10 or higher operating system.

It requires:

- Microsoft .NET 8.0

Download .NET 8.0 Desktop Runtime (v8.0.0) - Windows x64 Installer

- Microsoft Visual C++ Redistributable packages 2015-2022
https://aka.ms/vs/17/release/vc_redist.x64.exe
- .NET support for ArcGIS libraries
- ArcGIS Pro from version 3.3.x

3. License types

Only a Single Use Cellular Expert license is available. The license type is annual and dedicated to one workstation connected with ArcGIS Online, which is used for ArcGIS Pro.

3.1 Single-User Environment

For the Single-User configuration of Cellular Expert, all information about radio network objects is stored in a personal geodatabase (GDB format) or locally on the disc (calculation results, raster data in GeoTIFF format, etc.).

Geographical data can be stored:

- Locally on a disc

An ArcGIS Pro license and an active ArcGIS Named User or ArcGIS Pro Standalone are required to operate in the Single-User environment.

4. Getting Started

Welcome to Cellular Expert.

This chapter will guide you through project creation and analysis. It includes the installation and preparation of the Cellular Expert extension, as well as using it for the first time and creating a project.

Note: the tools are not explained in this chapter. They are referenced in other chapters of this guide.

4.1 Installation

- If applicable, uninstall the previous version of Cellular Expert for ArcGIS Pro
- Make sure that *.NET 8.0* is installed
- Make sure that *ArcGIS Pro 3.3.x* is installed
- Run the *ceProSetup.msi* file to install the new version of Cellular Expert for ArcGIS Pro
- After successful installation, open Task Manager > Services, and run CE_Pro_Prediction_Service

4.2 Activation

This step is required for new users. If the Cellular Expert for ArcGIS Pro license has already been activated, please skip this step.

You can activate the license in two ways:

First way:

1. Open ArcGIS Pro and select **Settings**
2. Navigate to **Licensing** → **External Extensions**
3. Find the **Cellular Expert** entry in the extensions table
4. Click the checkmark in the **Enabled** field. The **License Activation** dialog will appear
5. Copy the **User Key** and send it to support@cellular-expert.com. You will be provided with a **License Activation Key** which must be entered into the same dialogue
6. Press **Activate License** and the license will be activated, enabling the acquired versions of the add-on

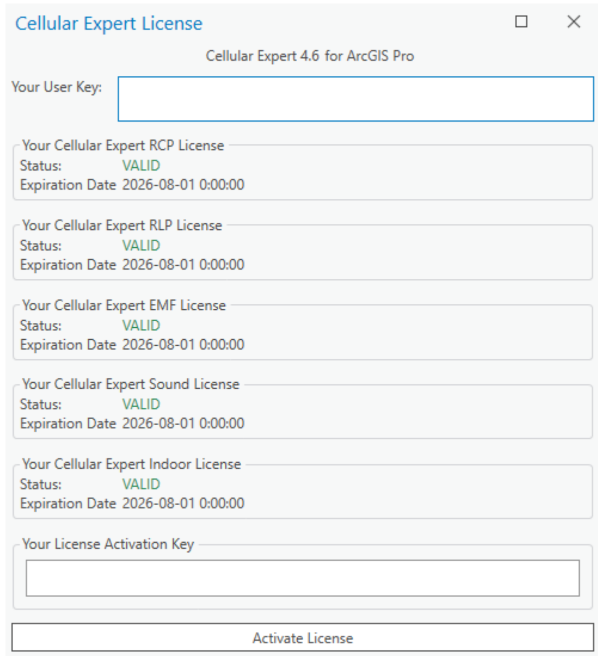
Second way:

1. Create an empty ArcGIS Pro project
2. Navigate to **Insert** and select **New Map**. After the map insertion, the Cellular Expert tabs will be enabled
3. Navigate to any of the Cellular Expert tabs and select **License Information** in the **About** section.
4. Copy the **User Key** and send it to support@cellular-expert.com. You will be provided with a **License Activation Key** which must be entered into the same dialogue
5. Press **Activate License** and the license will be activated, enabling the acquired versions of the add-on

If you want to check the expiration date of the license you can:

1. Open ArcGIS Pro and start/open a project with an active map
2. Navigate to the **Cellular Expert** tab
3. Select **License Information** in the **About** section

If you encounter any problems or want additional details about the license, please contact Cellular Expert at support@cellular-expert.com. For more information, see the chapter **Technical Support**.

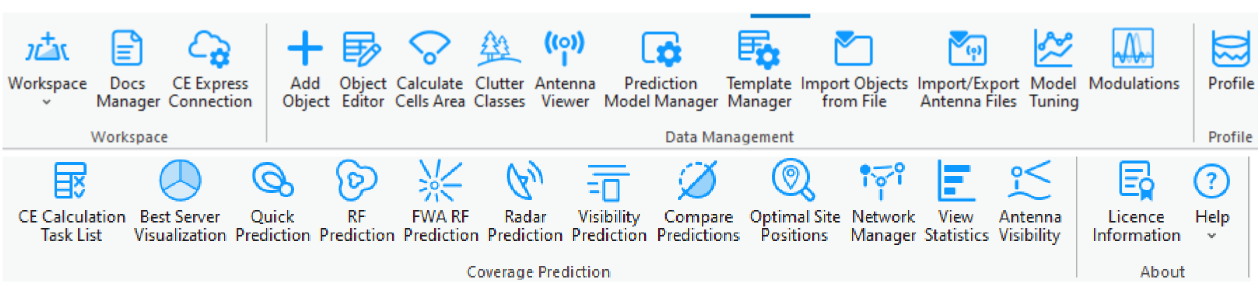


4.3 Tools

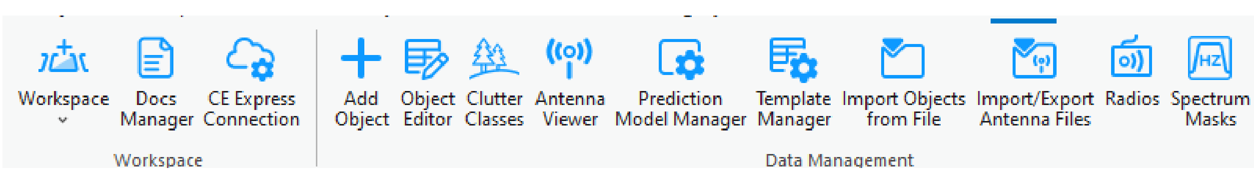
The Cellular Expert tools are in the Cellular Expert add-on. They appear automatically after installation of CE for ArcGIS Pro and will be found in the menu ribbon.

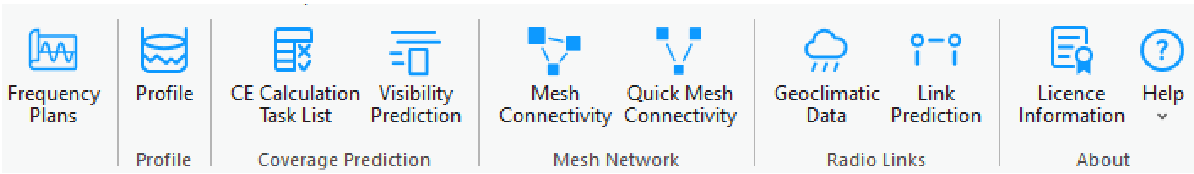
There are 5 types of licenses and therefore 5 different tabs with various tool configurations:

- RCP

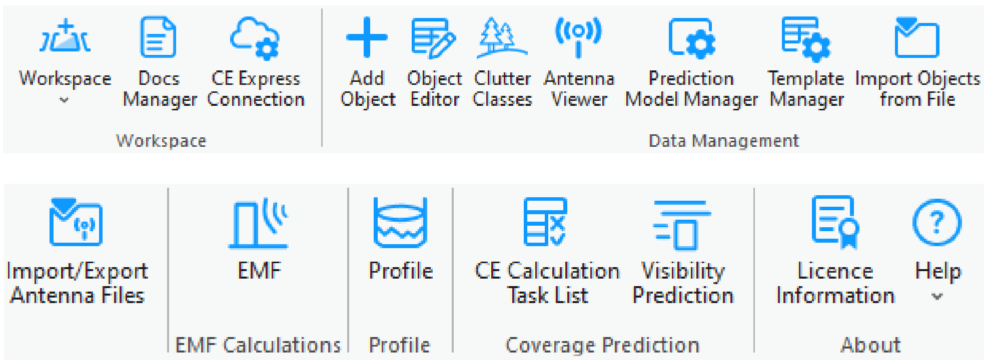


- RLP

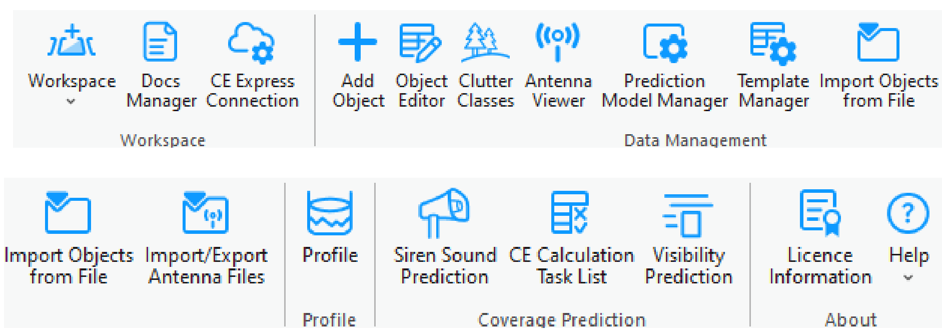




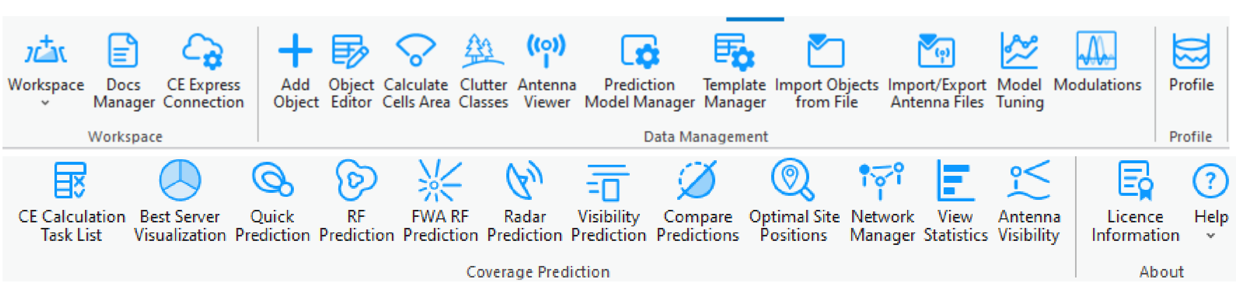
- EMF



- Sound



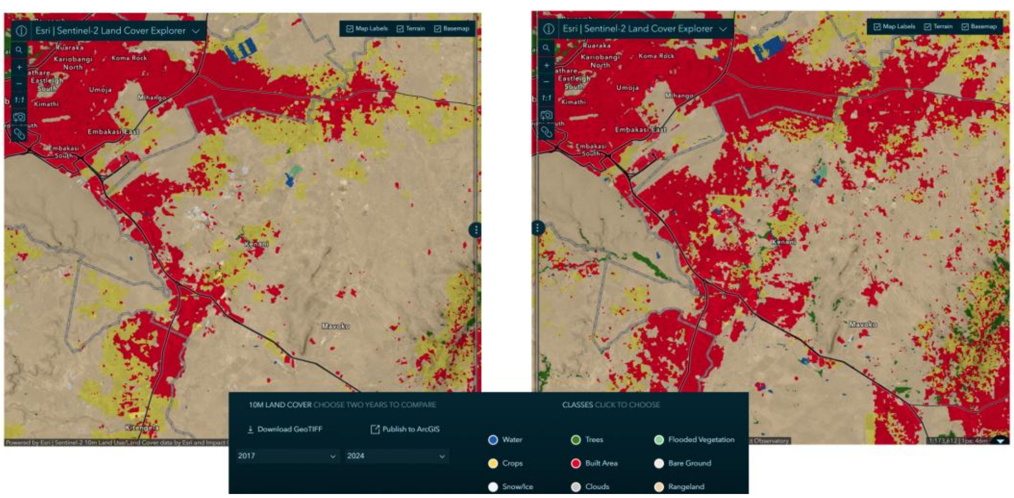
- Indoor



This User Guide describes RLP tools.

5. Geographic data

CE Desktop is designed to work with any geospatial data available to the customer and fully exploit its precision for the most accurate coverage and QoS calculations. The platform supports multi-resolution input datasets — from freely available global sources such as Sentinel-2 10 m land cover and ASTER DEM, to premium high-resolution terrain and 3D building models when provided by the customer or government agencies.



Source: https://livingatlas.arcgis.com/landcoverexplorer/

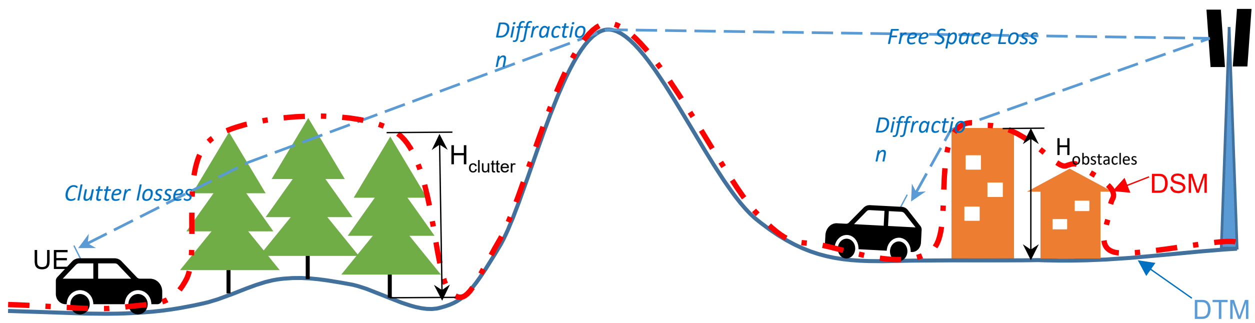
By leveraging whatever data is available locally, CE Express performs nationwide calculations at the maximum feasible resolution, accurately modeling signal propagation even in dense urban environments. Support for 3D multi-height calculations ensures that coverage predictions reflect street-level, indoor, and rooftop conditions, providing regulators with a realistic representation of service availability.

This flexibility ensures that NRAs can use their existing GIS assets, open datasets, or commercial data they already license, turning them into actionable broadband maps without additional data procurement requirements from the solution provider.

By using terrain elevation, obstacles, and clutter classification in every calculation, Cellular Expert accurately models:

- **Line-of-Sight and Non-Line-of-Sight Conditions** – Determining diffraction, reflection, and shadowing effects over hills, valleys, and urban obstacles.

- **Coverage Footprints** – Generating precise signal strength maps at national, regional, and local levels.
- **Capacity and Interference Analysis** – Modeling realistic signal overlaps and interference zones for multi-operator, multi-technology environments.

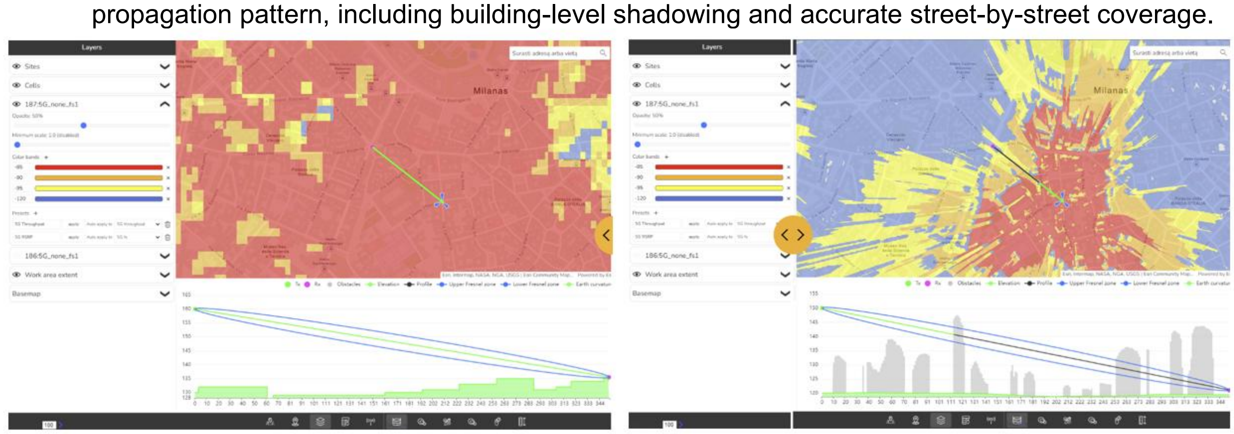


The CE tools make use of three distinct GIS data layers to obtain high precision modelling of radio wave propagation losses:

1. **Digital Terrain Model** (DTM), also known as Digital Elevation Model (DEM), which describes Earth surface, i.e., path terrain profile in terms of ground elevation above uniform sea level.
2. **Obstacles layer**, delineating buildings and other such objects above Earth surface that may be considered to be principal impediments for radio wave propagation.
3. **Clutter layer**, delineating natural occurring or human cultivated ground cover that may be partially penetrable by radio waves, such as natural vegetation (e.g., forests, trees, bushes) or various crops, gardens, parks, etc.

The image above illustrates how Cellular Expert uses different resolutions of topographical data to significantly improve coverage prediction accuracy.

- Left image: Coverage calculated using 25 m resolution ASTER DEM data, showing a general view of signal distribution but with limited detail, especially in dense urban areas.
- Right image: Coverage calculated using 1 m resolution data, revealing a much more precise propagation pattern, including building-level shadowing and accurate street-by-street coverage.



More information: https://blog.maxar.com/earth-intelligence/2022/benefits-of-using-maxars-precision3d-

[telco-suite-for-5g]

Cellular Expert can easily integrate and process 1 m or even sub-meter topographical data, providing highly detailed RF calculations. This level of precision is essential for:

- Modeling 2G/3G/4G/5G, small cells and mmWave networks.
- Identifying exact coverage gaps at the building and street level.
- Supporting regulatory-grade broadband mapping and planning.

By using high-resolution terrain and clutter data, Cellular Expert ensures that its calculations match real-world conditions as closely as possible — resulting in better network design decisions and more reliable broadband planning outcomes.

5.1 Geographic data requirements

The supported geographical data types:

Only GeoTIFF is supported. Topographical data must have specific names:

- The Digital terrain model must be named *elevation.tif*
- The land use (or clutter) grid must be named *clutterClasses.tif*
- The clutter heights (typically building, vegetation height) grid must be named *clutterHeight.tif*

Mandatory geographical data:

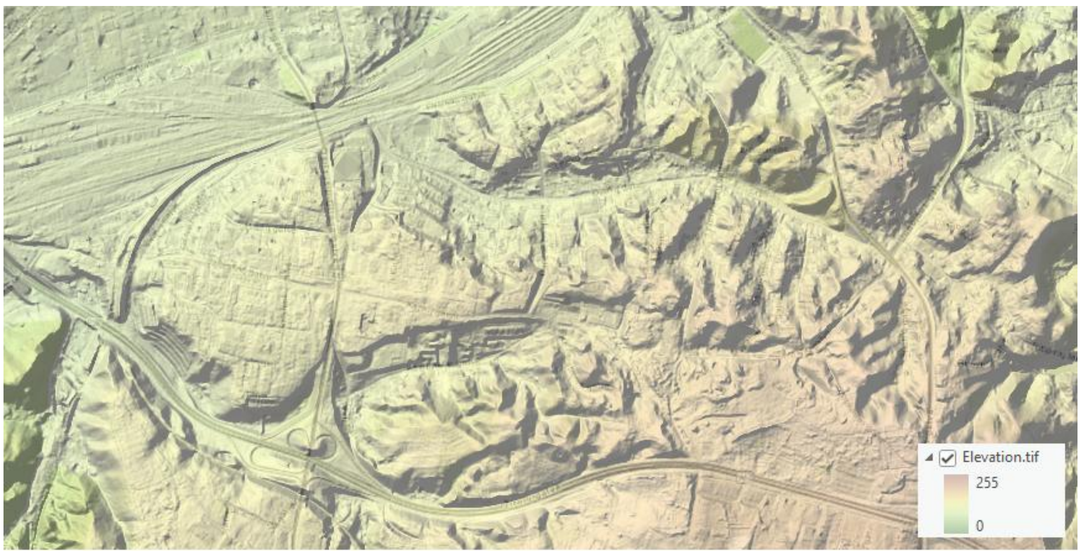
- Digital Terrain Model (DTM) grid

All geodata must be located in one catalog.

5.1.1 Digital Terrain Model (DTM) Grid (Mandatory)

The Digital Terrain Model (DTM), also known as Digital Elevation Model (DEM), represents the Earth’s ground level above sea level. Each raster pixel has its height value.

A sample DTM raster is presented below. Each pixel represents 5 square meters with its height value. In reality, within a one-pixel area, the height is not the same everywhere. Thus, the pixel’s height value is the height in its center or the maximum. The smaller the pixels, the more accurate is the grid - but also more data to calculate.



5.1.1.1 Prepare DTM raster

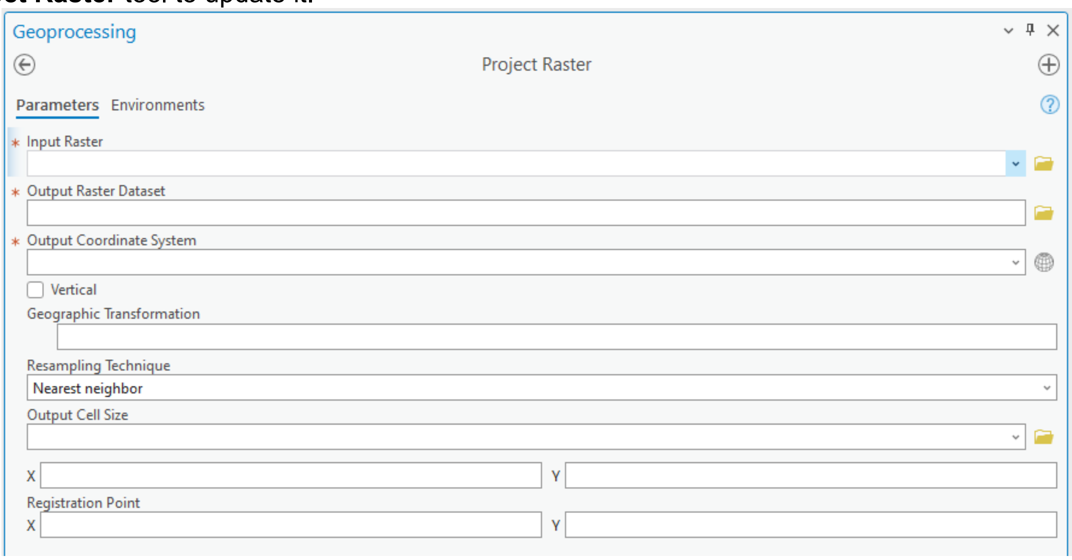
The Digital Terrain Model (DTM) has several requirements, which are listed below.

Projection

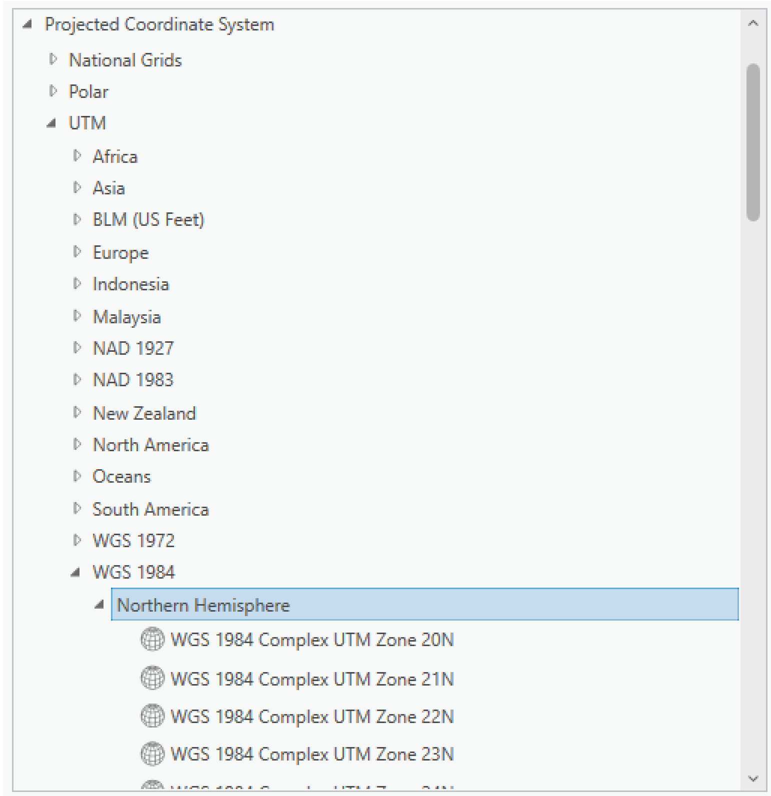
The raster must use a Projected Coordinate System. To check the coordinate system of your raster, use the Properties function in ArcGIS Pro. Add the raster to your project, right-click on it, and select Properties. Then, go to the Source tab > Spatial Reference and check the Coordinate System type parameter to confirm it is in a Projected Coordinate System.

Projected Coordinate System	LKS 1994 Lithuania TM
Projection	Transverse Mercator
WKID	3346
Previous WKID	2600
Authority	EPSG
Linear Unit	Meters (1.0)
False Easting	500000.0
False Northing	0.0
Central Meridian	24.0
Scale Factor	0.9998
Latitude Of Origin	0.0

If your raster is in a Geographic Coordinate System or needs a different projection, use the **Geoprocessing > Project Raster** tool to update it.



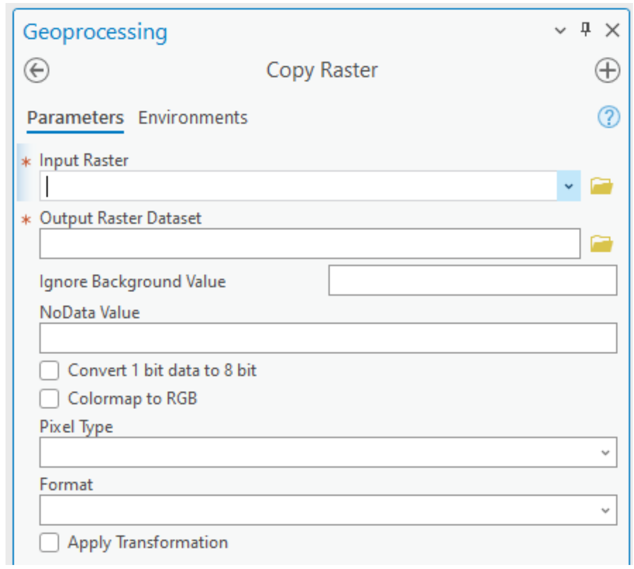
In the Output Coordinate System, specify a new coordinate system. It is recommended to use a UTM coordinate system under the WGS 1984 projection.



You can find the appropriate UTM zone for your area here:
https://www.arcgis.com/apps/mapviewer/index.html?layers=b294795270aa4fb3bd25286bf09edc51

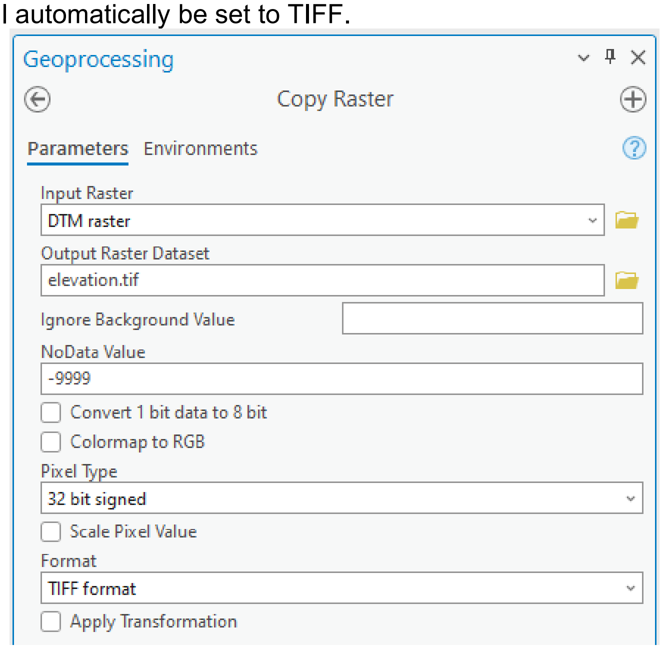
Correct No Data value and raster name

After setting the correct projection, assign the NoData attribute and specify the appropriate name for the DTM raster. To do this, use the **Copy Raster tool in Geoprocessing**.



Configure the following settings:

- **Input Raster:** Select your newly projected DTM raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to elevation.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.



5.1.2 Clutter classes grid

Land use or clutter refers to the classification of the earth's surface into categories such as urban, suburban, rural, forest, water, and open land, each of which affects radio propagation differently. Clutter data is crucial because it determines how signals are absorbed, reflected, or diffracted by the environment, directly influencing coverage, interference, and quality of service. The naming and classification of land use types may vary. An example is the Sentinel-2 Land Cover dataset from the Living Atlas: [Living Atlas Sentinel-2 Land Cover]

This data is freely available worldwide and is detailed in Cellular Expert databases. Once the workspace is created, a default clutter class table is automatically applied for each land use class.

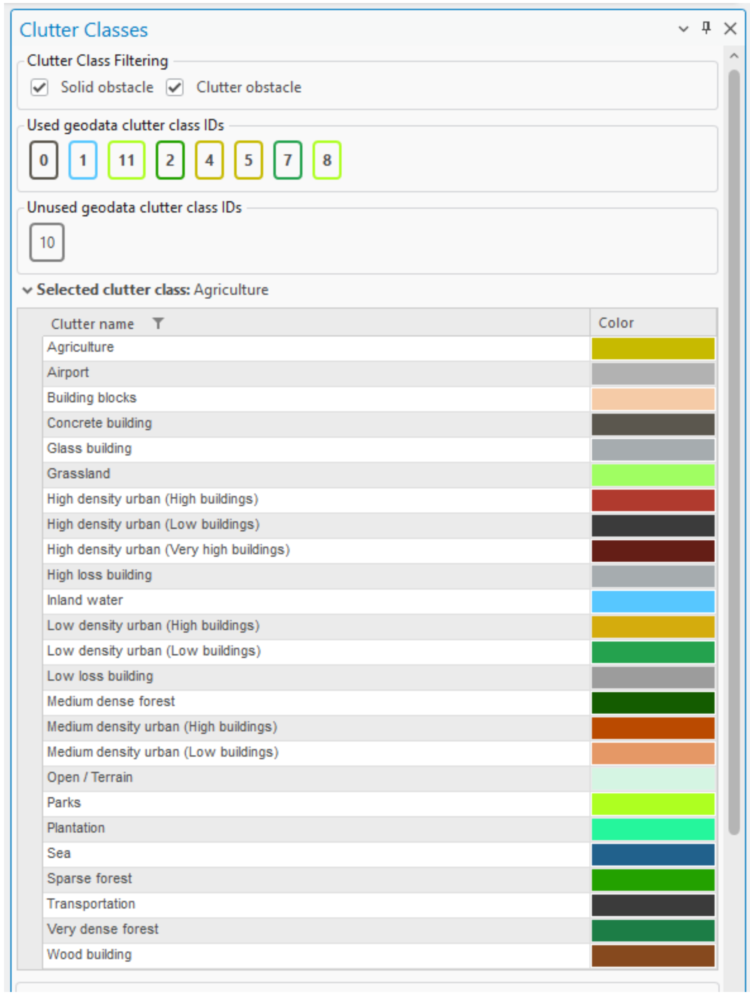
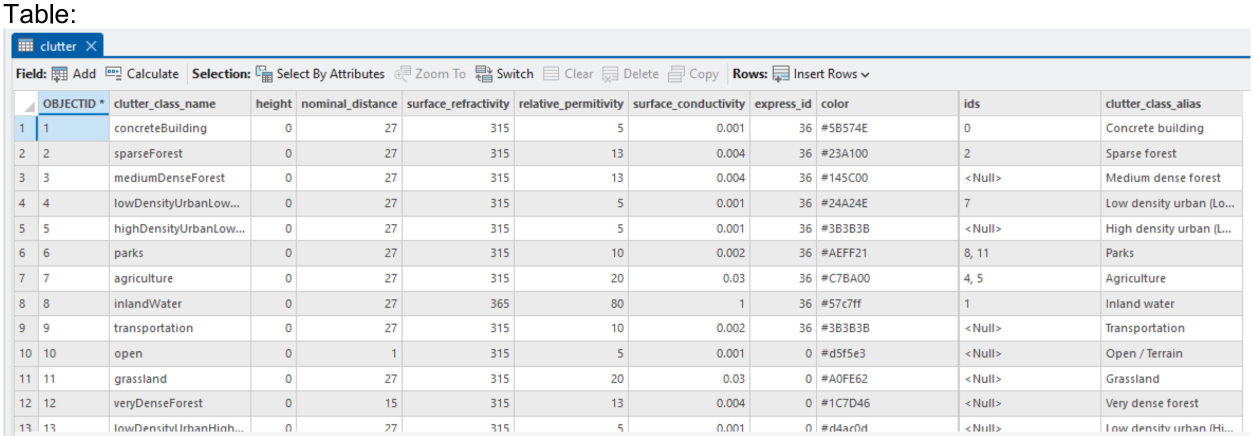
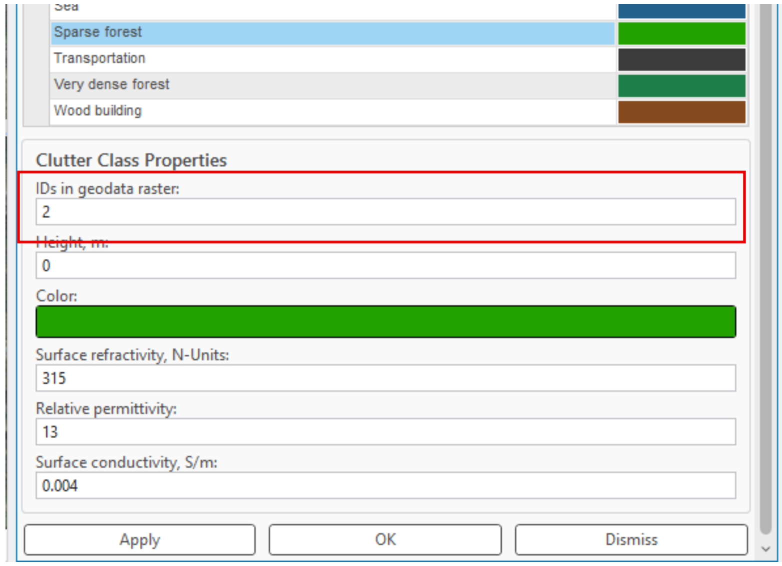


Table:

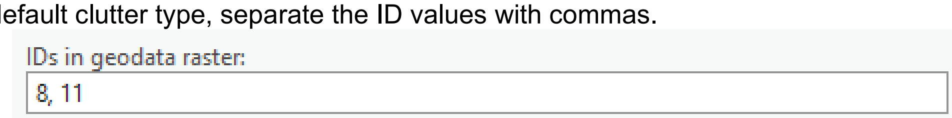


	OBJECTID *	clutter_class_name	height	nominal_distance	surface_refractivity	relative_permitivity	surface_conductivity	express_id	color	ids	clutter_class_alias
1	1	concreteBuilding	0	27	315	5	0.001	36	#5B574E	0	Concrete building
2	2	sparseForest	0	27	315	13	0.004	36	#23A100	2	Sparse forest
3	3	mediumDenseForest	0	27	315	13	0.004	36	#145C00	<Null>	Medium dense forest
4	4	lowDensityUrbanLow...	0	27	315	5	0.001	36	#24A24E	7	Low density urban (Lo...
5	5	highDensityUrbanLow...	0	27	315	5	0.001	36	#3B3B3B	<Null>	High density urban (L...
6	6	parks	0	27	315	10	0.002	36	#AEFF21	8, 11	Parks
7	7	agriculture	0	27	315	20	0.03	36	#C7BA00	4, 5	Agriculture
8	8	inlandWater	0	27	365	80	1	36	#57c7ff	1	Inland water
9	9	transportation	0	27	315	10	0.002	36	#3B3B3B	<Null>	Transportation
10	10	open	0	1	315	5	0.001	0	#d5f5e3	<Null>	Open / Terrain
11	11	grassland	0	27	315	20	0.03	0	#A0FE62	<Null>	Grassland
12	12	veryDenseForest	0	15	315	13	0.004	0	#1C7D46	<Null>	Very dense forest
13	13	lowDensityUrbanHigh...	0	27	315	5	0.001	0	#d4ac0d	<Null>	Low density urban (Hi...

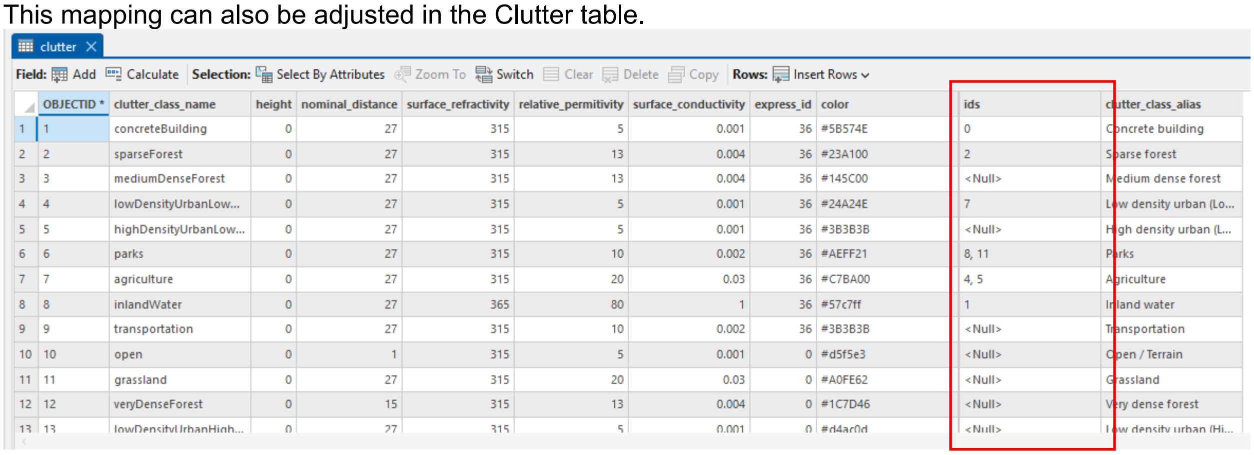
These are standard clutter types in the default workspace database, which cannot be edited. You must map your clutter raster to these predefined clutter types. Standard mapping has already been configured for the Sentinel-2 Land Cover dataset from the Living Atlas: [Living Atlas Sentinel-2 Land Cover]



If you have a different clutter class layer, it can be used for predictions by remapping it using the Clutter Classes tool and specifying the IDs in the geodata raster parameter. If multiple clutter classes correspond to a single default clutter type, separate the ID values with commas.

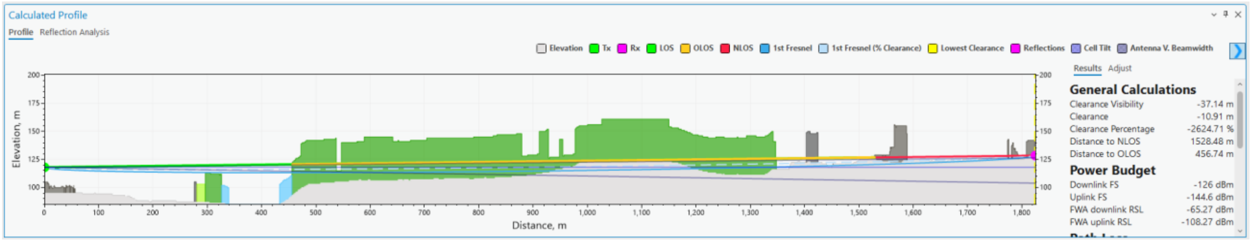


This mapping can also be adjusted in the Clutter table.



	OBJECTID *	clutter_class_name	height	nominal_distance	surface_refractivity	relative_permitivity	surface_conductivity	express_id	color	ids	clutter_class_alias
1	1	concreteBuilding	0	27	315	5	0.001	36	#5B574E	0	Concrete building
2	2	sparseForest	0	27	315	13	0.004	36	#23A100	2	Sparse forest
3	3	mediumDenseForest	0	27	315	13	0.004	36	#145C00	<Null>	Medium dense forest
4	4	lowDensityUrbanLow...	0	27	315	5	0.001	36	#24A24E	7	Low density urban (Lo...
5	5	highDensityUrbanLow...	0	27	315	5	0.001	36	#3B3B3B	<Null>	High density urban (L...
6	6	parks	0	27	315	10	0.002	36	#AEFF21	8, 11	Parks
7	7	agriculture	0	27	315	20	0.03	36	#C7BA00	4, 5	Agriculture
8	8	inlandWater	0	27	365	80	1	36	#57c7ff	1	Inland water
9	9	transportation	0	27	315	10	0.002	36	#3B3B3B	<Null>	Transportation
10	10	open	0	1	315	5	0.001	0	#d5f5e3	<Null>	Open / Terrain
11	11	grassland	0	27	315	20	0.03	0	#A0FE62	<Null>	Grassland
12	12	veryDenseForest	0	15	315	13	0.004	0	#1C7D46	<Null>	Very dense forest
13	13	lowDensityUrbanHigh...	0	27	315	5	0.001	0	#d4ac0d	<Null>	Low density urban (Hi...

Once clutter classes are successfully mapped, the prediction algorithms will recognize the clutter types, apply distinct symbols, and adjust path loss calculations accordingly, based on the parameters set in the prediction model.

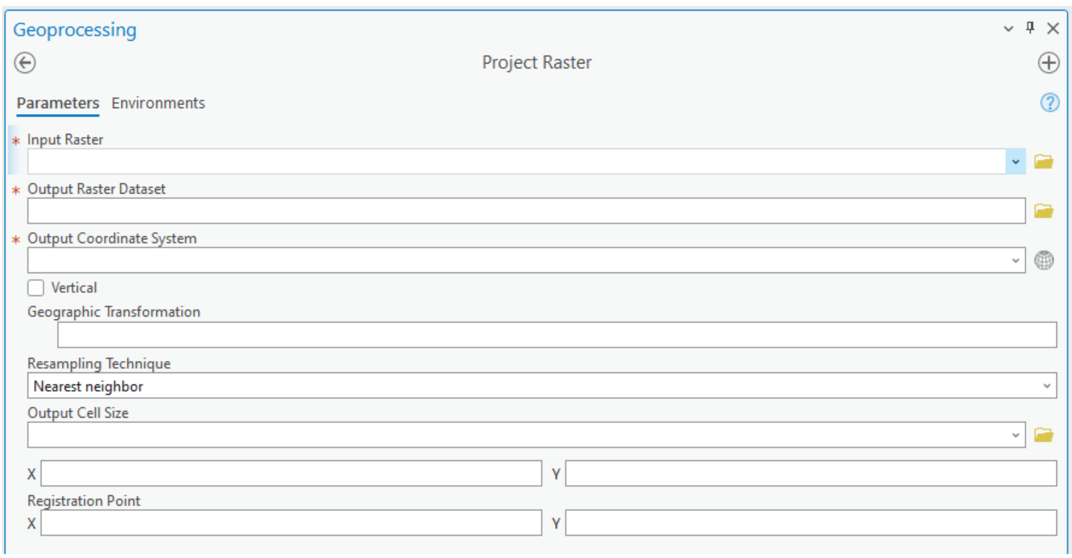


5.1.2.1 Prepare Clutter Classes raster

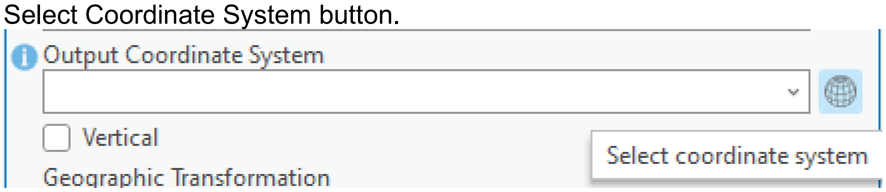
The Clutter Raster has several requirements, which are the same as for DTM raster listed above.

Projection

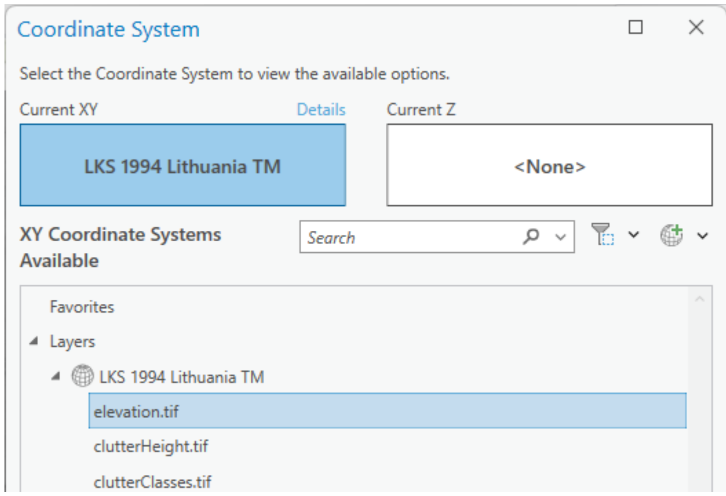
It must have the same coordinate system as your elevation.tif raster. If your raster has different coordinate system, then use the **Geoprocessing tool** → **Project Raster** to fix it.



In the Output Coordinate System you would need to define the same coordinate system as your elevation.tif raster. Click on Select Coordinate System button.

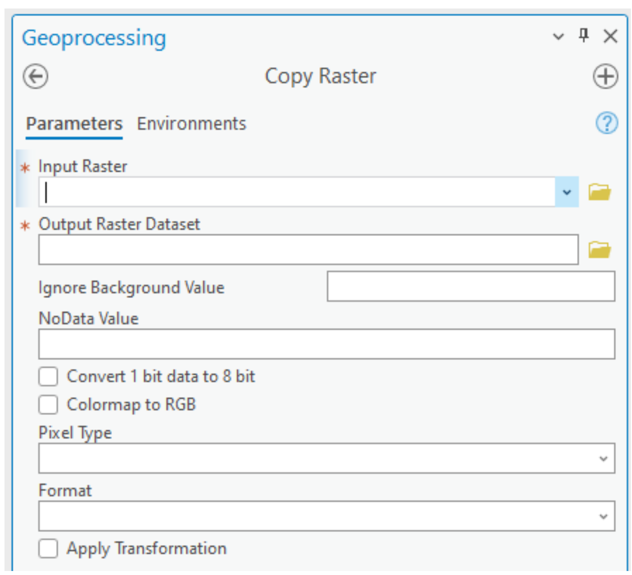


And choose the same coordinate system as your elevation.tif.



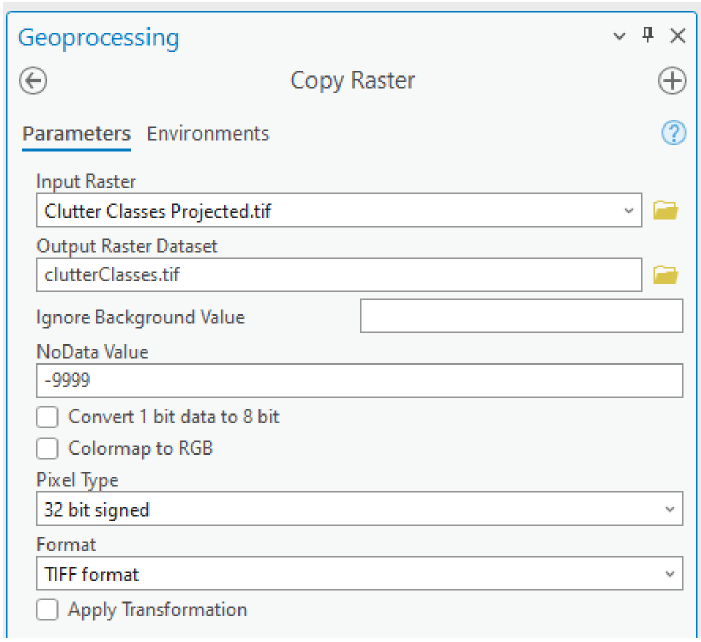
Correct No Data value and raster name

After setting the correct projection, assign the NoData attribute and specify the appropriate name for the Clutter Class raster. To do this, use the **Copy Raster tool in Geoprocessing**.



Configure the following settings:

- **Input Raster:** Select your newly projected Clutter Class raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to clutterClasses.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.

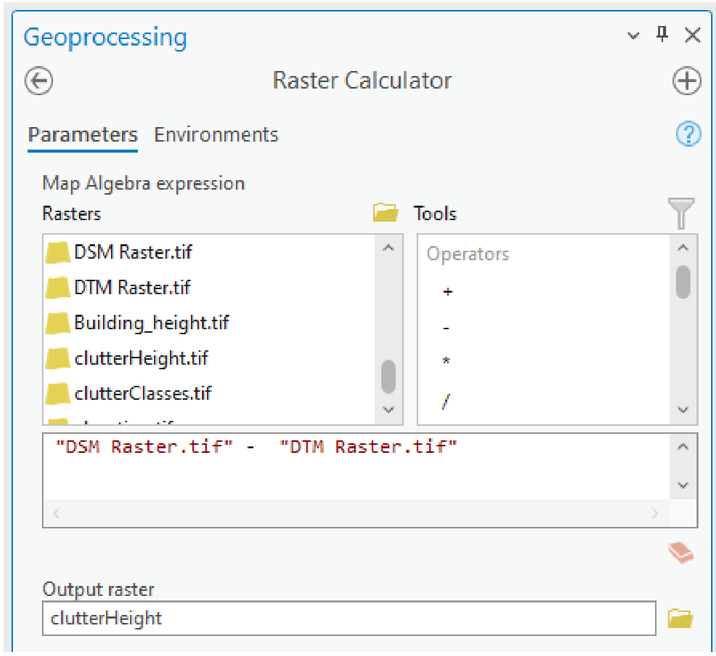


5.1.3 *Clutter heights*

Represents actual clutter heights, which override the default heights specified in the Clutter table. The clutter heights raster requires the accompanying clutterClasses.tif raster and cannot be used independently.

A clutter height raster can be derived from a Digital Surface Model (DSM) raster and a Digital Terrain Model (DTM) raster using the ArcGIS Raster Calculator tool. To access this tool, open Geoprocessing tools and navigate to Spatial Analyst > Map Algebra > Raster Calculator. Use the following formula:

DSM – DTM



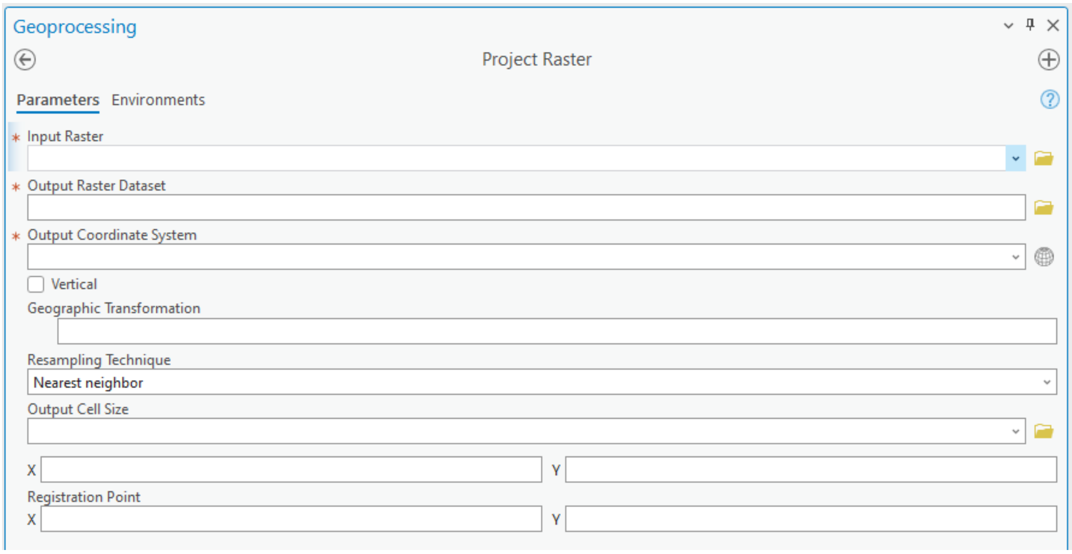
The calculation output will be the difference between the DSM and DTM grids, representing the clutter heights.

5.1.3.1 Prepare Clutter Height raster

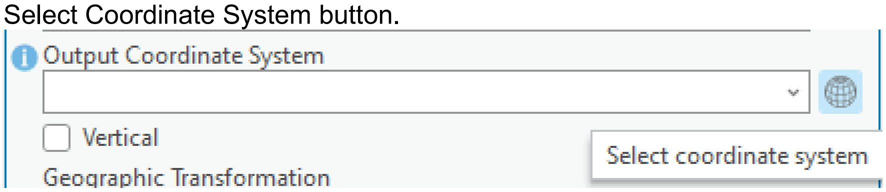
The Clutter Height has several requirements, which are the same as for DTM raster listed above.

Projection

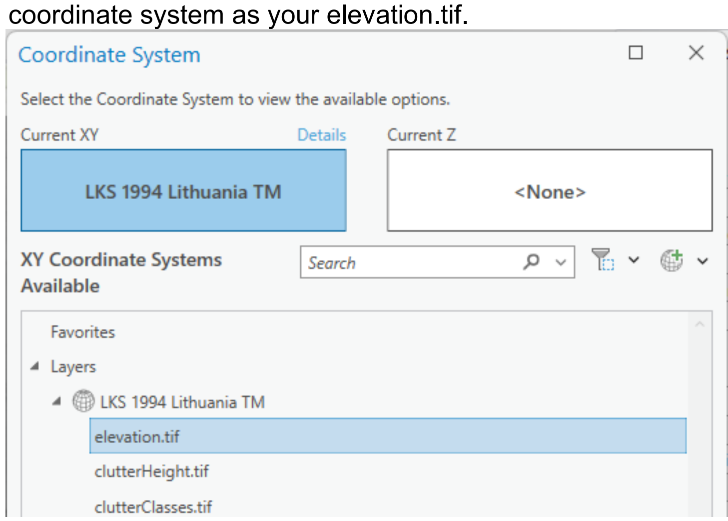
It must have the same coordinate system as your elevation.tif raster. If your raster has different coordinate system, then use the **Geoprocessing tool** → **Project Raster** to fix it.



In the Output Coordinate System you would need to define the same coordinate system as your elevation.tif raster. Click on Select Coordinate System button.

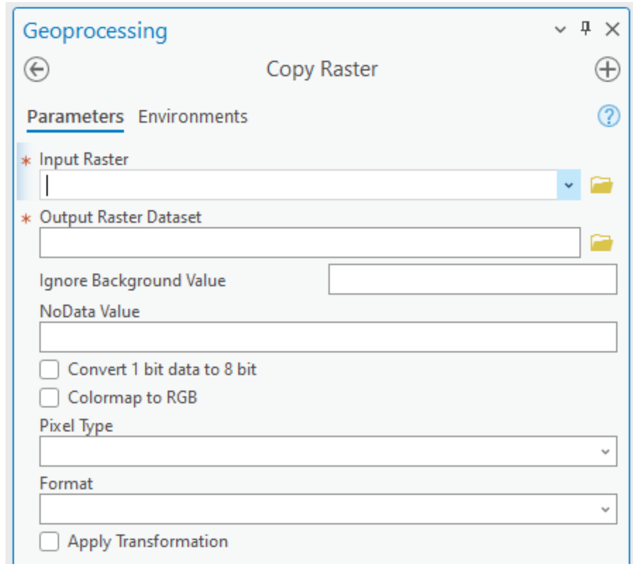


And choose the same coordinate system as your elevation.tif.



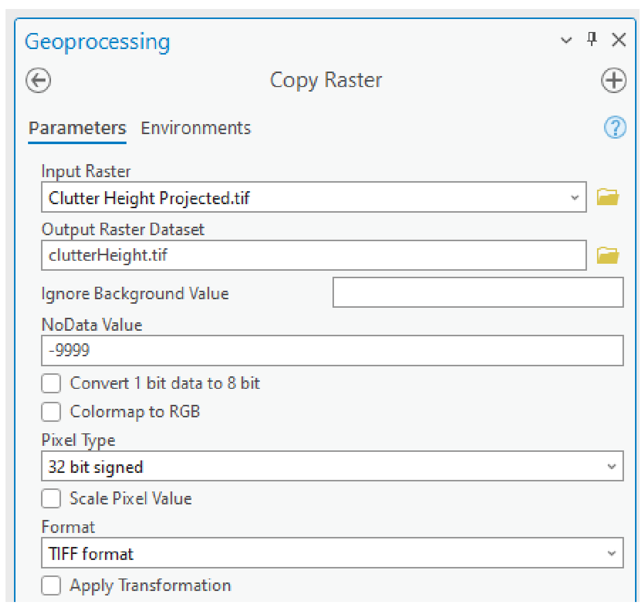
Correct No Data value and raster name

After setting the correct projection, assign the NoData attribute and specify the appropriate name for the Clutter Height raster. To do this, use the **Copy Raster tool in Geoprocessing**.



Configure the following settings:

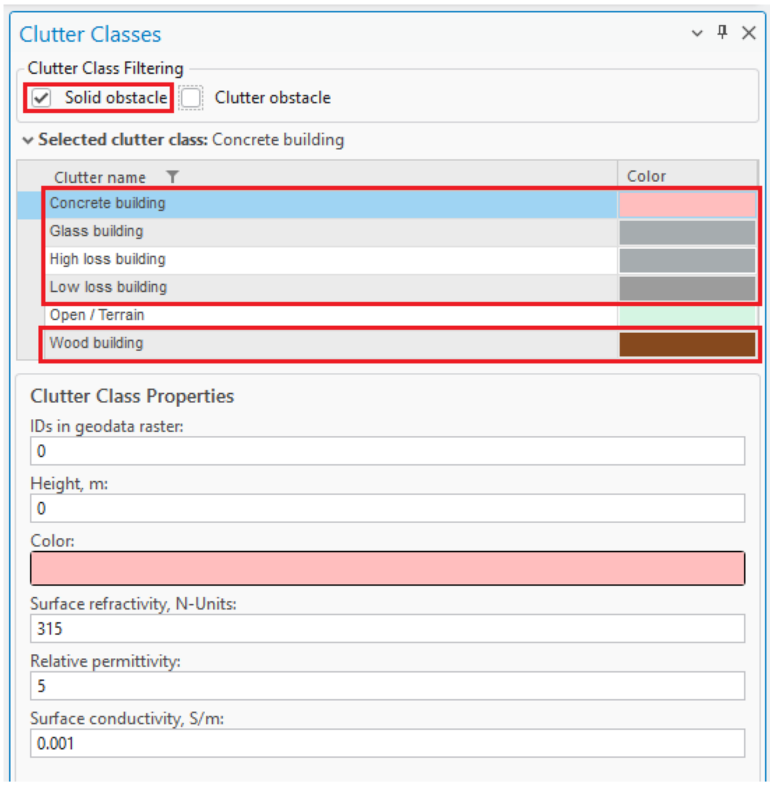
- **Input Raster:** Select your newly projected Clutter Height raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to clutterHeight.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.



5.1.4 Buildings

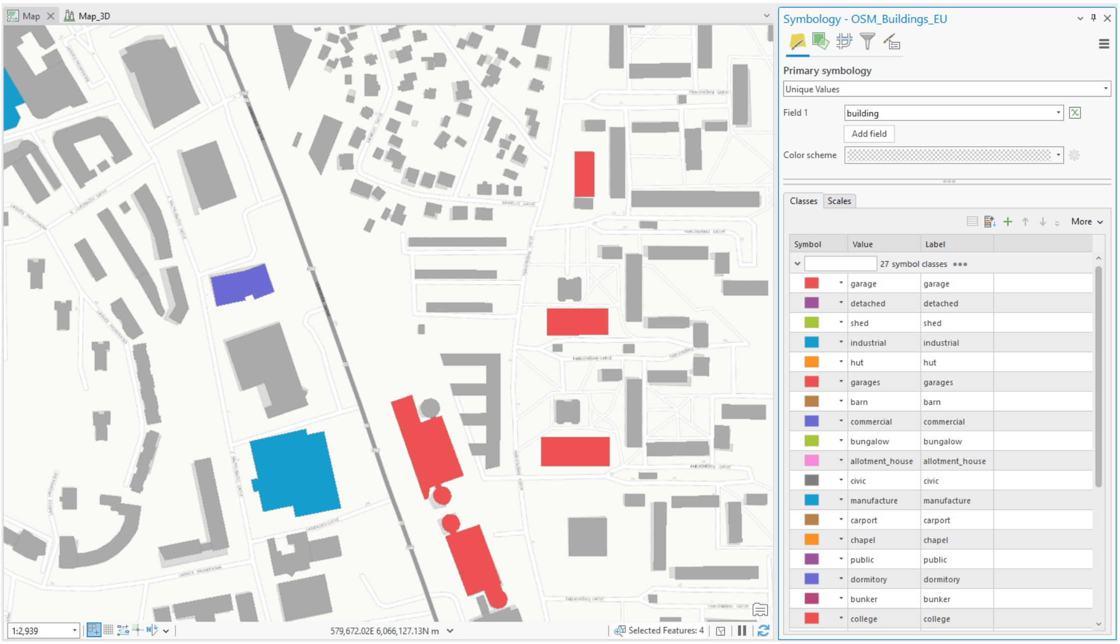
Building features within the Clutter Classes raster are automatically identified and categorized using a range of dedicated **building-specific clutter types**. These clutter types are available within the **Clutter Classes tool** and are specifically designed to represent different architectural materials and structural characteristics.

Importantly, all building-related clutter classes are treated as **Solid Obstacles** – ensuring accurate modeling of signal attenuation and reflection in propagation calculations. This classification enhances the realism and precision of both indoor and outdoor network predictions by accounting for the physical impact of built environments.



You can seamlessly **edit your Clutter Classes raster** using standard **ArcGIS Geoprocessing tools** to integrate detailed building data into your existing clutter layer.

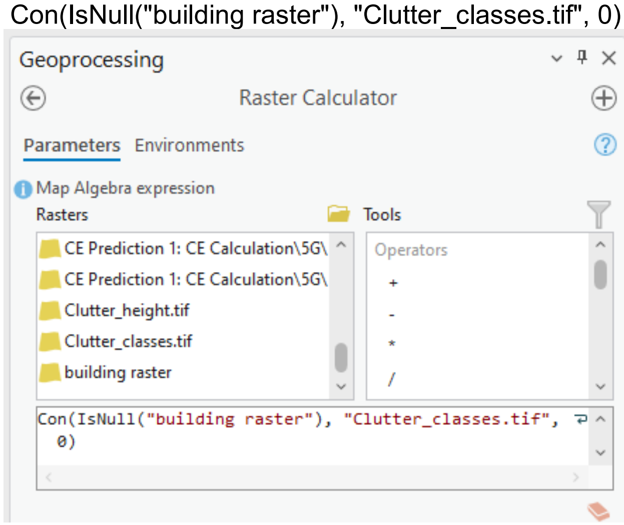
The process supports both **vector and raster input formats**, giving you flexibility depending on your data source. Whether you're working with CAD-derived vectors or pre-classified raster layers, these tools allow you to enrich the clutter map with accurate building representations – essential for high-fidelity propagation modeling.



If your building data is in **vector format** (e.g., polygons), you'll first need to **convert it to raster** before incorporating it into the Clutter Classes layer.

1. **Convert Vector to Raster**
   Use the **Polygon to Raster** tool available in the **Geoprocessing** pane. This will transform your vector-based building footprints into a raster format suitable for clutter classification.
2. **Update Clutter Classes Using Raster Calculator**
   After conversion, open the **Raster Calculator** to integrate the new raster into your existing clutter layer:
   - Navigate to **Geoprocessing**
   - Go to **Spatial Analyst > Map Algebra > Raster Calculator**
   - Use map algebra expressions to merge or replace values as needed, assigning appropriate clutter class codes to building areas.

Con(IsNull("building raster"), "Clutter_classes.tif", 0)

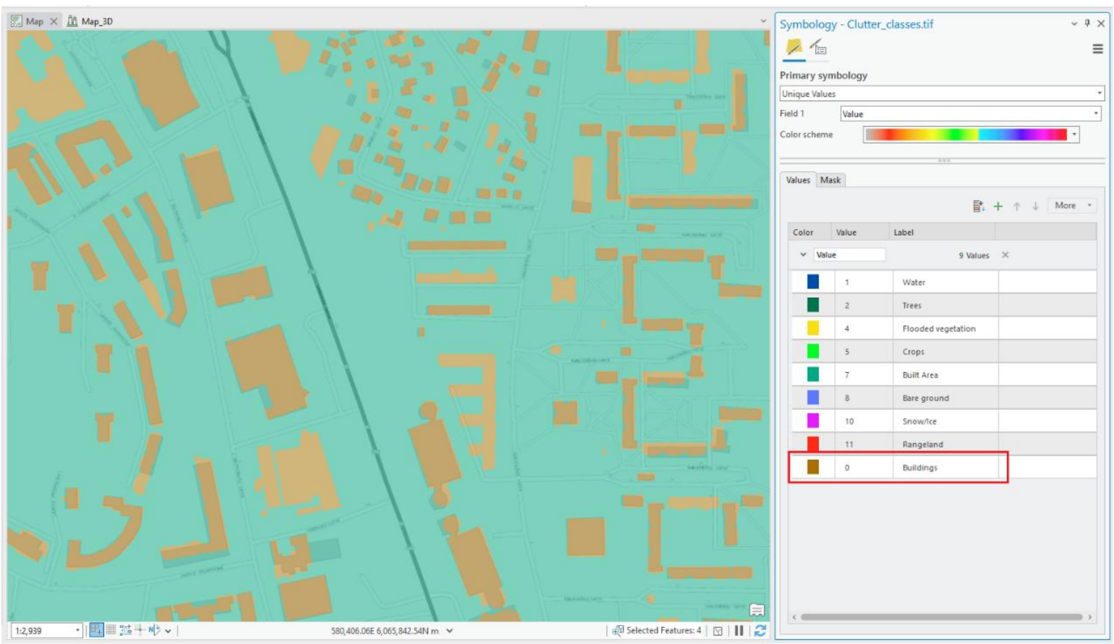


This workflow allows you to accurately reflect building data in your clutter map, improving prediction accuracy in urban and mixed-use environments.

When incorporating building data into the Clutter Classes raster, you have the flexibility to **assign any numeric value** to represent the **"Buildings"** clutter class. In the example provided, a value of **0** was used, but you may choose any value – **as long as it does not conflict with existing clutter class values** already present in your raster.

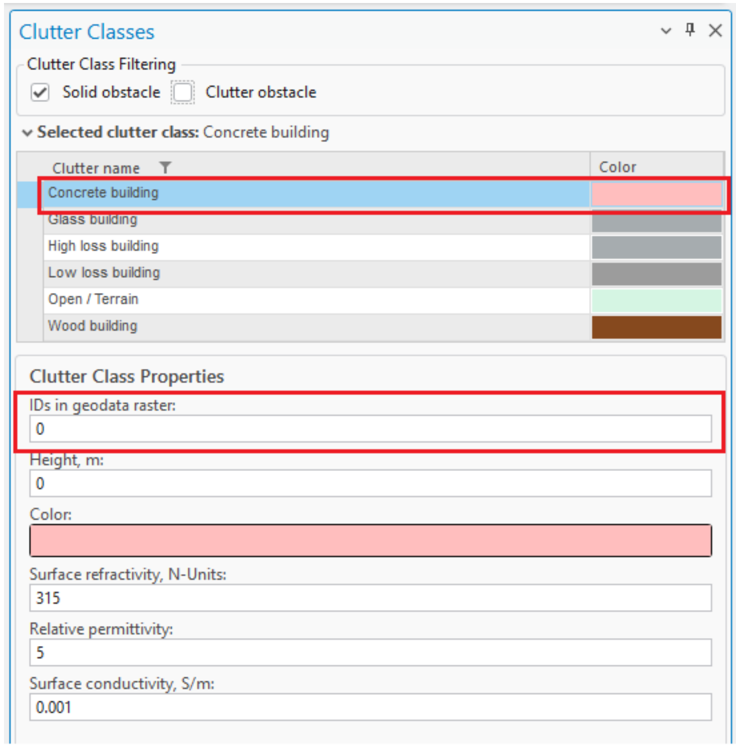
Once applied, the Clutter Classes raster will be updated to include a new entry for **Buildings**, enabling more precise signal propagation modeling that accounts for structural obstructions.

Be sure to document your chosen value and update your Clutter Class definitions accordingly within the Clutter Classes tool to maintain consistency across your modeling environment.



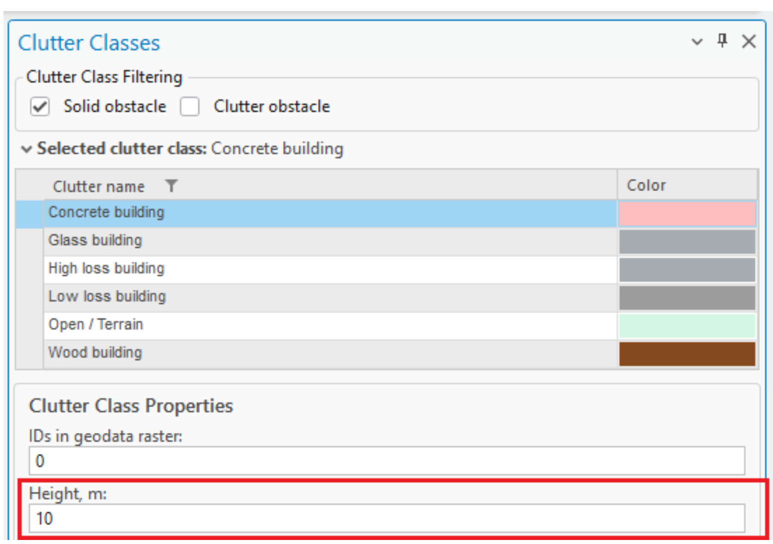
All propagation and prediction calculations reference **clutter types defined in the Clutter Classes table**. Each **building-related clutter class** is assigned a **unique ID value**, which is used during modeling to apply the appropriate path loss parameters for solid structures.

This ensures that **Buildings** are accurately represented in simulations, contributing to more realistic signal behavior in both indoor and outdoor environments.

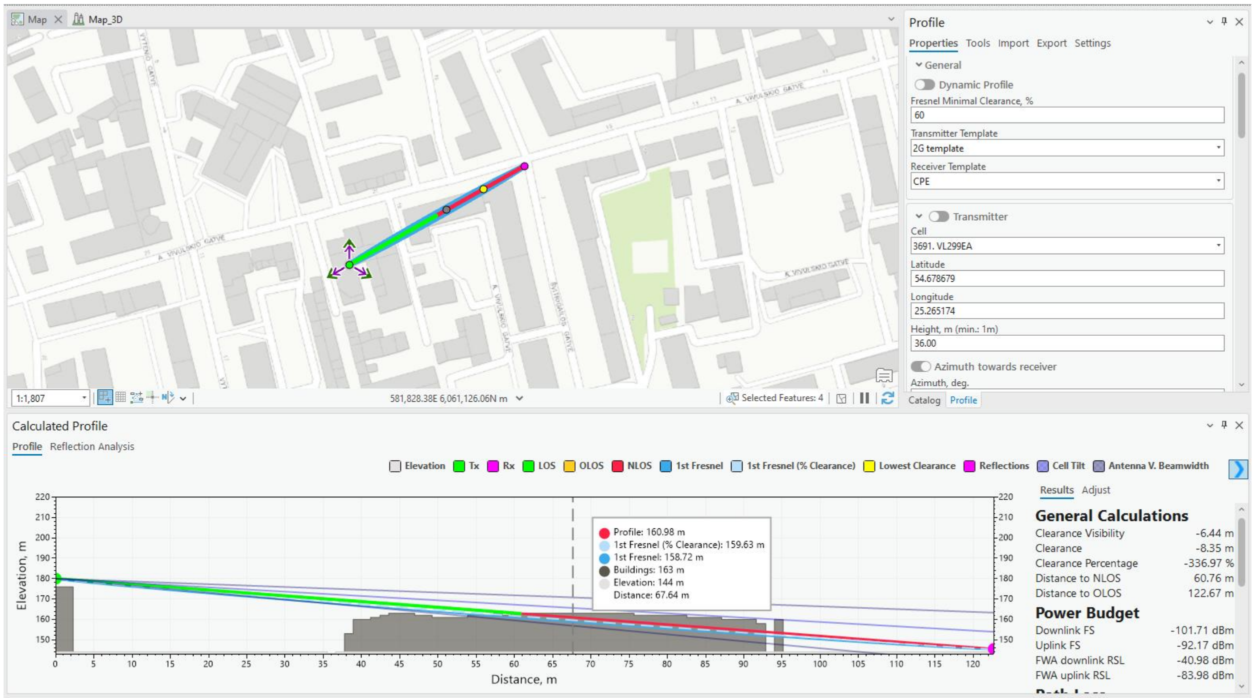


Building Height Determination in Clutter-Based Modeling
Pixels assigned to a **building clutter class ID** will be automatically recognized as **solid obstacle** during prediction calculations. Their heights are determined using the following priority:

- **Option 1**: From the associated **Clutter Height raster**, if available. This provides the most accurate, location-specific height information.
- **Option 2**: If no Clutter Height raster is present, the system will default to the **height value defined in the Clutter Classes table** for the corresponding clutter ID.

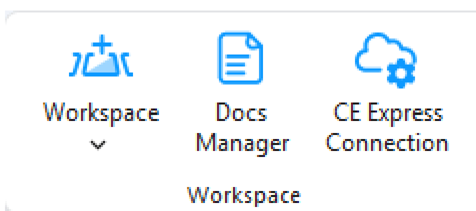


The result:



6. Workspace

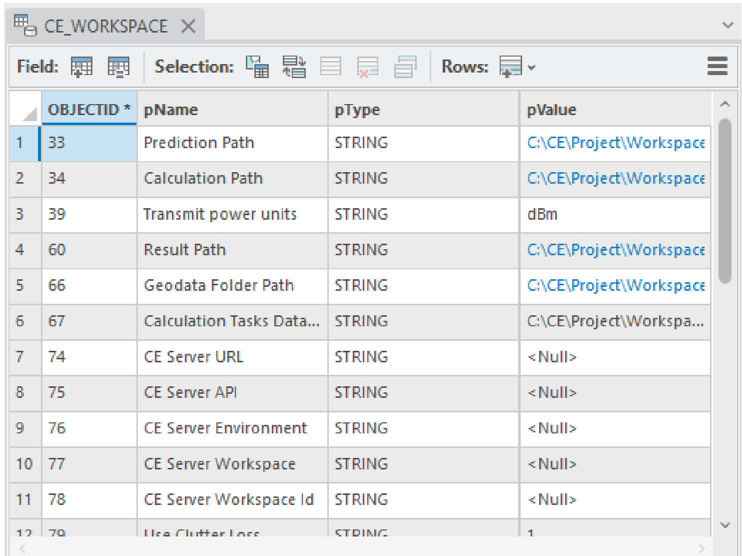
This chapter describes the Cellular Expert workspace functionality.



6.1 Workspace Tool

6.1.1 Workspace Table

Cellular Expert workspace is a geodatabase containing data tables, feature datasets, and the workspace definition table. After creating a new workspace database, the workspace definition table will be named *CE_WORKSPACE* and contain the information about the dataset.



	OBJECTID *	pName	pType	pValue
1	33	Prediction Path	STRING	C:\CE\Project\Workspace
2	34	Calculation Path	STRING	C:\CE\Project\Workspace
3	39	Transmit power units	STRING	dBm
4	60	Result Path	STRING	C:\CE\Project\Workspace
5	66	Geodata Folder Path	STRING	C:\CE\Project\Workspace
6	67	Calculation Tasks Data...	STRING	C:\CE\Project\Workspa...
7	74	CE Server URL	STRING	<Null>
8	75	CE Server API	STRING	<Null>
9	76	CE Server Environment	STRING	<Null>
10	77	CE Server Workspace	STRING	<Null>
11	78	CE Server Workspace Id	STRING	<Null>
12	79	Use Clutter Loss	STRING	1

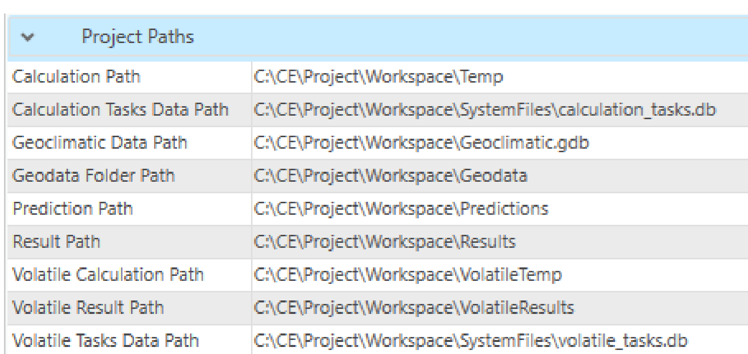
Data field types and values:

- OBJECTID – long integer type
- pName – text type
- pType – text type
- pValue – text type

When moving your project's directory to another location (to another computer), it is necessary to update the workspace parameters by referencing the new paths to properly load the project:

- Prediction Path – path for storing prediction grids
- Calculation Path – path for temporary calculations
- Calculation Tasks Data Path – path for saving calculation tasks
- Geodata Folder Path – path for geodata (prediction models do not have geodata options, topographical data are taken from the Geodata Folder Path)
- Result Path – path for final results

Workspace calculation paths and settings can be previewed in the dedicated tool, navigate to Workspace > Properties. It would show all parameters listed in CE_WORKSPACE table.

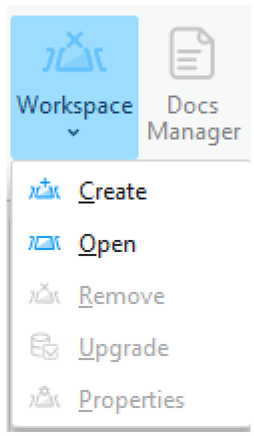


Project Paths	
Calculation Path	C:\CE\Project\Workspace\Temp
Calculation Tasks Data Path	C:\CE\Project\Workspace\SystemFiles\calculation_tasks.db
Geoclimatic Data Path	C:\CE\Project\Workspace\Geoclimatic.gdb
Geodata Folder Path	C:\CE\Project\Workspace\Geodata
Prediction Path	C:\CE\Project\Workspace\Predictions
Result Path	C:\CE\Project\Workspace\Results
Volatile Calculation Path	C:\CE\Project\Workspace\VolatileTemp
Volatile Result Path	C:\CE\Project\Workspace\VolatileResults
Volatile Tasks Data Path	C:\CE\Project\Workspace\SystemFiles\volatile_tasks.db

6.1.2 Create Workspace

Steps to create a new workspace:

1. Click the **Create** button in the **Cellular Expert Workspace** menu.

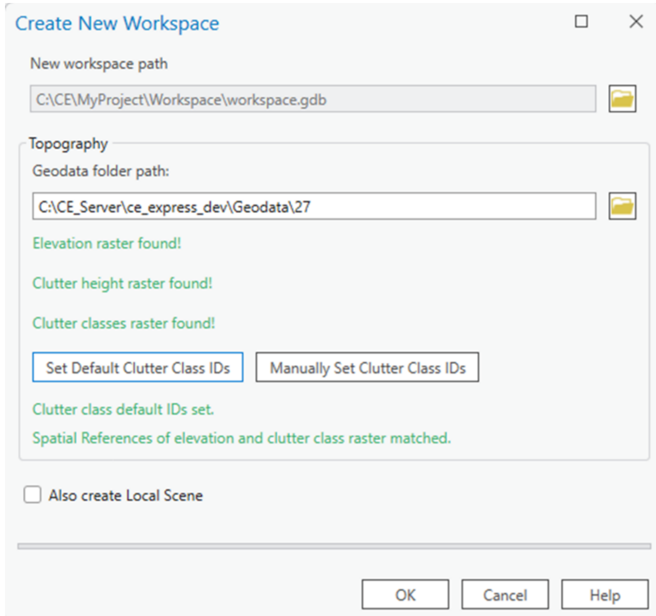


2. The **Create** dialogue will appear. Fill in the minimum required data:

New workspace path
It is automatically filled based on the ArcGIS Pro project location. The recommendation is to first save your ArcGIS Pro project, and then open the Workspace > Create tool. The project's workspace folder will be automatically created in the ArcGIS Pro project location catalog. This way, you will have both the ArcGIS Pro and CE workspace in the same location.

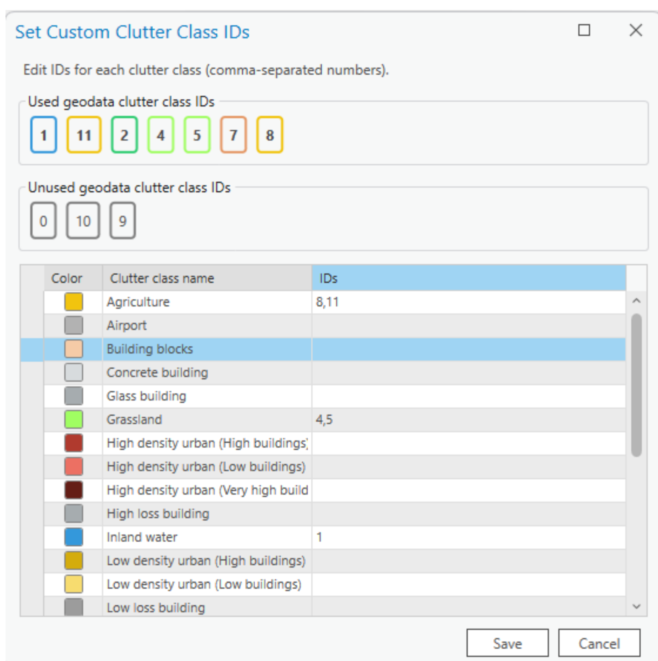
Geodata folder path
Catalog where geodata files are stored. More about Geodata requirements are listed in **5. Geographic data topic**. Click on  *Browse* button to define Geodata catalog. Geodata must have specific names:

- The Digital terrain model must be named *elevation.tif*
- The land use (or clutter) grid must be named *clutterClasses.tif*
- The clutter heights (typically building and vegetation height) grid must be named *clutterHeight.tif*



Note: *The projected Coordinate System* has been filled automatically and taken from the defined *Elevation grid*. This means that this coordinate system will be assigned to your project feature layers.

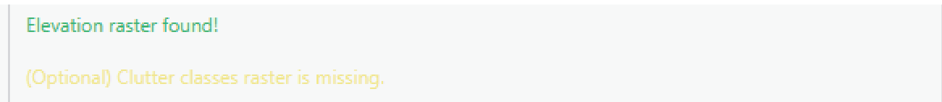
When creating a new workspace using geodata containing clutter classes raster, default class IDs can now be set by clicking *Set Default Clutter Class IDs* button. The default ID values are based on Living Atlas clutter. Clutter class IDs can also be set manually, either before or after the default values are set. To initiate the manual editing of clutter class IDs, click the *Manually Set Clutter Class IDs* button. Each clutter class can be edited to designate its used and unused types, also indicated by their distinct colors.



Note: *The projected Coordinate System* has been filled automatically and taken from the defined *Elevation grid*. This means that this coordinate system will be assigned to your project feature layers.

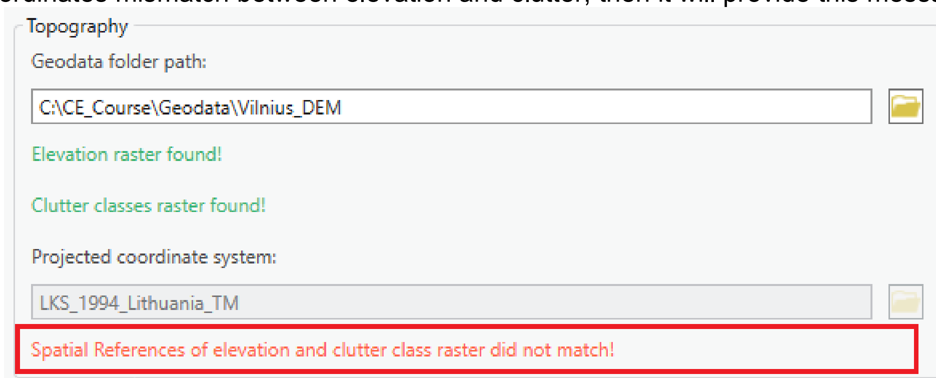
Several messages related to Geodata:

- If only Elevation exists in Geodata catalog, then tool will be filled with such information:

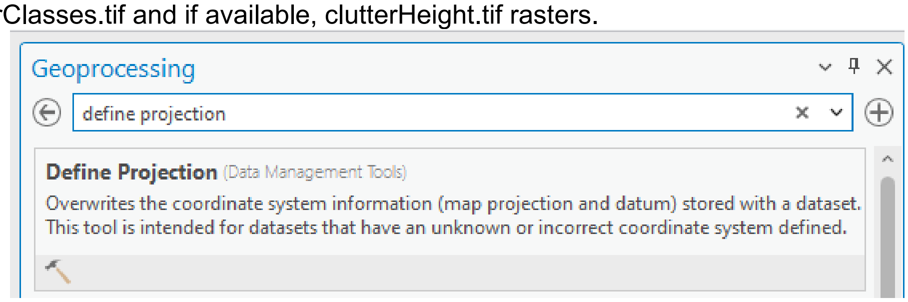


  Clutter Classes and Clutter Height are optional.

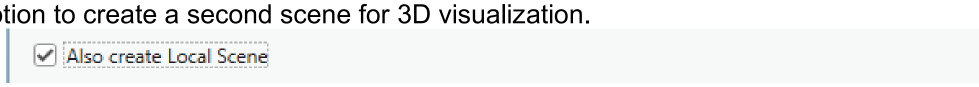
- If coordinates mismatch between elevation and clutter, then it will provide this message:



  Please close Workspace Create tool and fix your geodata. You can use Geoprocessing tools, Define Projection tool to fix this issue. Define the same coordinate as elevation.tif for your clutterClasses.tif and if available, clutterHeight.tif rasters.

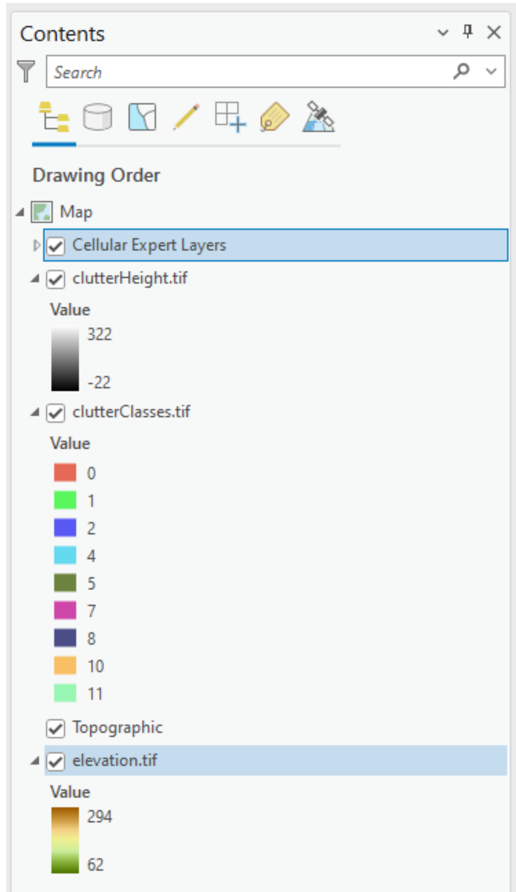


Also Create Local Scene
Use the option to create a second scene for 3D visualization.

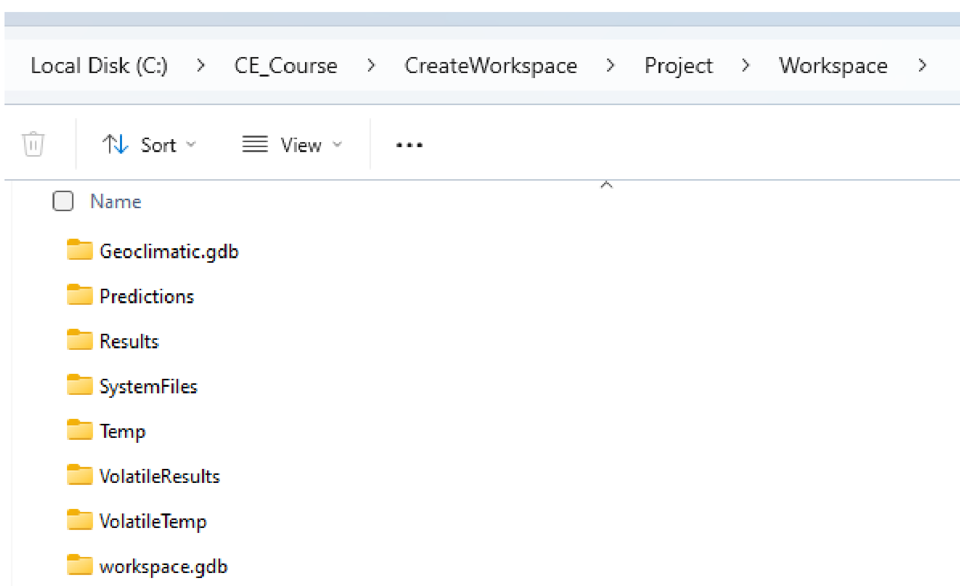


3. Press OK button to start workspace creation procedure.

Cellular Expert layer and geodata will be added to the project.



Workspace geodatabase and required folders will be created within successful workspace creation procedure.



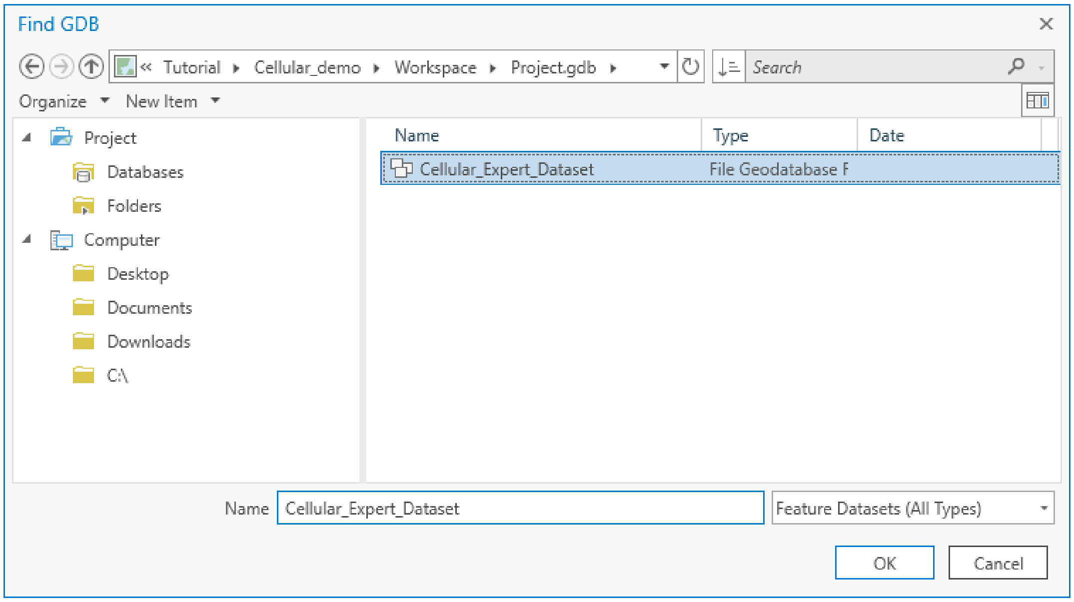
The Project Paths will be filled in the ***Workspace Properties*** → ***Properties*** tab.

Project Paths	
Calculation Path	C:\CE_Course\CreateWorkspace\Project\Workspace\Temp
Calculation Tasks Data Path	C:\CE_Course\CreateWorkspace\Project\Workspace\SystemFiles\calculation_tasks.db
Geoclimatic Data Path	C:\CE_Course\CreateWorkspace\Project\Workspace\Geoclimatic.gdb
Geodata Folder Path	C:\CE_Course\Geodata\Vilnius
Prediction Path	C:\CE_Course\CreateWorkspace\Project\Workspace\Predictions
Result Path	C:\CE_Course\CreateWorkspace\Project\Workspace\Results
Volatile Calculation Path	C:\CE_Course\CreateWorkspace\Project\Workspace\VolatileTemp
Volatile Result Path	C:\CE_Course\CreateWorkspace\Project\Workspace\VolatileResults
Volatile Tasks Data Path	C:\CE_Course\CreateWorkspace\Project\Workspace\SystemFiles\volatile_tasks.db

6.1.3 Open Workspace

Steps to open a workspace:

1. Click the **Open** option in the **Workspace** menu of the ***Cellular Expert*** toolbar.
2. Specify the workspace name and click the OK button.



The specified workspace will be activated automatically.

6.1.4 Remove Workspace

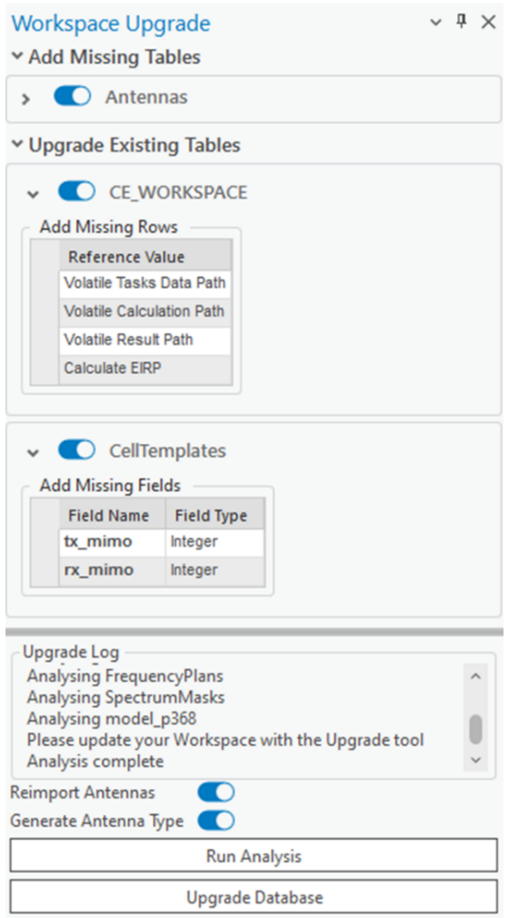
Upon selecting **Remove** in the Workspace dropdown menu, the workspace will be removed from the currently open ArcGIS Pro project.

6.1.5 Workspace Upgrade

Workspace Upgrade is a tool that enables the user to update missing tables from the default database (*default.gdb*). If the table exists, Workspace Upgrade also checks if all default fields are present in that table. If the tables are toggled, they will be updated (upgraded) by the tool.

Usually, with new versions of *CE for ArcGIS Pro* come changes to default data tables. This tool is especially useful when checking if the newest version tables correspond to the current project tables.

Workspace Upgrade automatically checks all these parameters and notifies the user upon the start-up of the project.



Reimport Antennas
Imports the deleted default antennas back to the Antennas table.

Generate Antenna Type
Based on the Antenna data assign antennas a type if they are missing it.

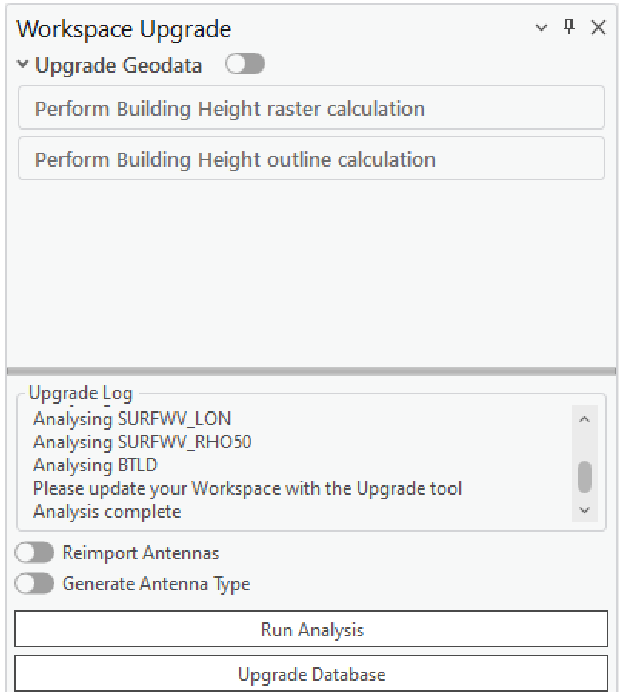
Run Analysis
Manually checks if the default tables and their fields exist in the project.

Upgrade Database
Creates the missing tables and/or creates/adds back the missing fields from default tables.

If a project with incompatible geodata is loaded, the Workspace Upgrade tool analyzes the current geodata, as well as the owned user Esri Extension licenses to offer the optimal geodata upgrade path. This process is one-time only, and can be done by checking the Upgrade Geodata toggle.

- If only elevation and buildingHeight rasters exist, the buildingHeight can automatically be renamed to clutterHeight.
- (**Image Analyst and Spatial Analyst licenses only**) If elevation, buildingHeight and clutterHeight rasters exist, raster calculations are performed to merge the buildingHeight and clutterHeight rasters.
- (**Image Analyst and Spatial Analyst licenses only**) If elevation, buildingHeight, clutterHeight and

clutterClasses rasters exist, raster calculations are performed to merge the buildingHeight and clutterHeight rasters, and modify the clutterClasses raster to add the building outlines with ID of 0.



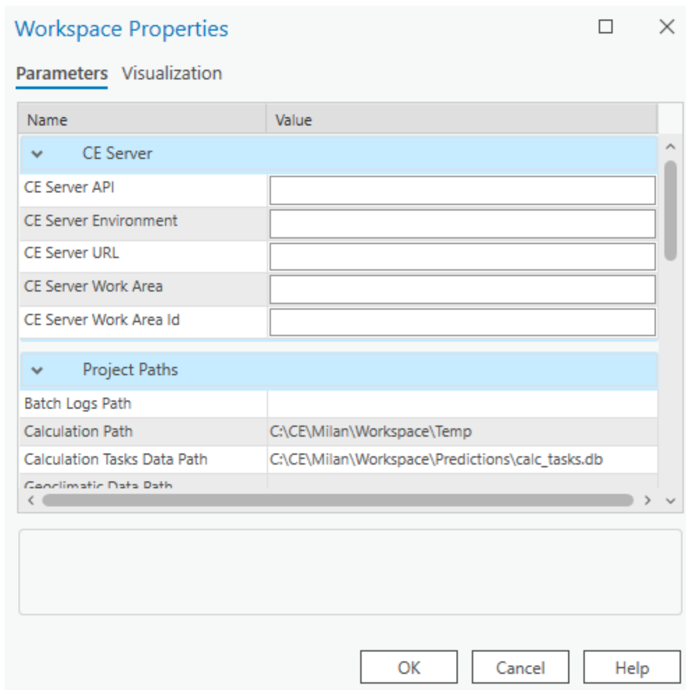
6.1.6 Workspace Properties

The Workspace properties dialogue shows all the workspace information from the “CE_WORKSPACE” data table. It also enables the user to customize the symbol visualization.

To open the **Workspace Properties** dialogue, click on the **Workspace** menu icon and choose **Properties.**

6.1.6.1 Parameters

All information from the “CE_WORKSPACE” table is represented in the **Parameters** tab of the Workspace Properties.



OK

Click this button to save any changes and close the dialogue.

Cancel

Click this button to cancel any changes and close the dialogue.

Help

Get helpful information about the dialogue.

<u>CE Server Parameters</u>

ArcGIS Portal URL
Connection URL to the ArcGIS Portal website.

CE Server API
Connection URL to Cellular Expert Server API.

CE Server Environment
Cellular Expert Server environment in which it is run.

CE Server URL
Connection URL to Cellular Expert Server.

CE Server Workspace
The workspace used in the Cellular Expert Server.

CE Server Workspace ID
The ID of the workspace which is used in the Cellular Expert Server.

<u>Project Paths Parameters</u>

Calculation Path
Path for temporary calculation results.

Calculation Tasks Data Path
Path for the calculation results that will be displayed in the Calculation Task List.

Geodata Folder Path
Path to a folder in which the geodata from the Cellular Expert Workspace is stored.

Prediction Path
Path for storing prediction grids.

Result Path
Path for storing the final calculation results

Volatile Calculation Path
Path for temporary Quick Prediction calculation results.

Volatile Result Path
Path for storing the final Quick Prediction calculation results

Volatile Tasks Data Path
Path for the Quick Prediction calculation results that will be displayed in the Calculation Task List.

<u>Project Settings Parameters</u>

Calculate EIRP
Determines whether calculate EIRP or no in the prediction calculations.

- Value YES. EIRP will be calculated based on Power, Antenna Gain and Misc. Loss values.
- Value NO. EIRP will be taken as single value defined in Power field.

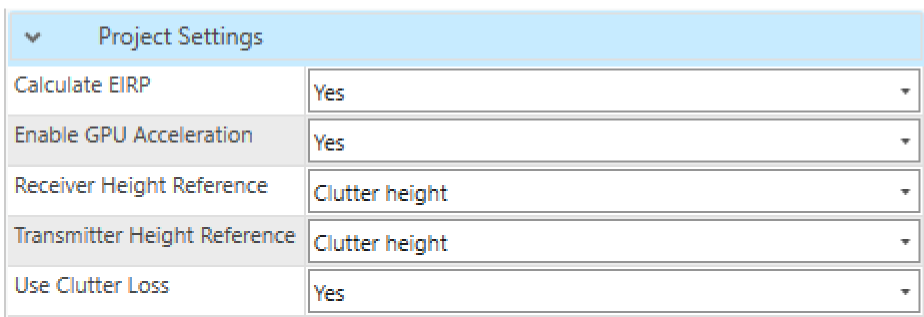
Enable GPU Acceleration
Enables GPU Accelerations that optimizes the prediction calculations and makes them run faster. Possible values Yes/No.

Transmit Power Units
Represents power value in dBm or Watts.

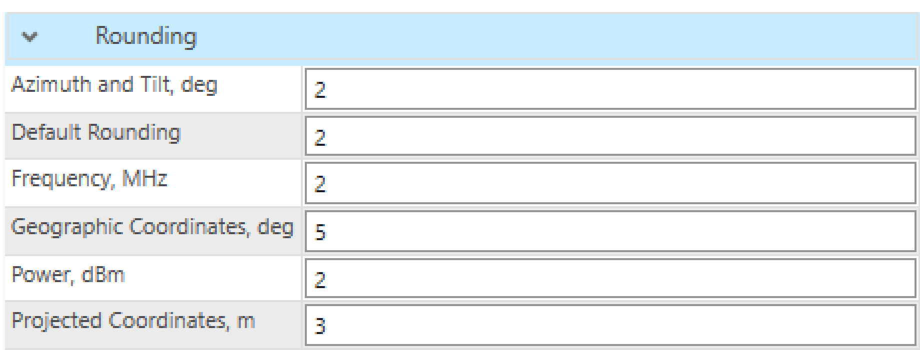
Receiver/Transmitter Height Reference
The reference raster for calculating the receiver's and transmitter's height for these calculations: Profile, RF Predictions, Quick Predictions, Visibility and other prediction tools. Possible values:

- Elevation – reference layer will be used elevation.tif raster. This is a default parameter.
- Clutter height – reference layer will be used clutterHeight.tif raster. Receiver/transmitter height will be calculated over Clutter Height raster. clutterHeight.tif raster is a must to enable this parameter.
- Clutter height (buildings only) – Height reference is calculated using the Building class from the clutterClasses.tif raster. The Building class is defined in the Clutter Class dialog and linked to a specific clutter class ID in the clutterClasses.tif raster.
- Absolute – receiver and/or transmitter's absolute height (relative to sea level) will be used in relevant calculations.

Use Clutter Loss
Determines whether clutterClasses.tif and clutterHeight.tif rasters are used in prediction calculations. Possible values Yes/No.

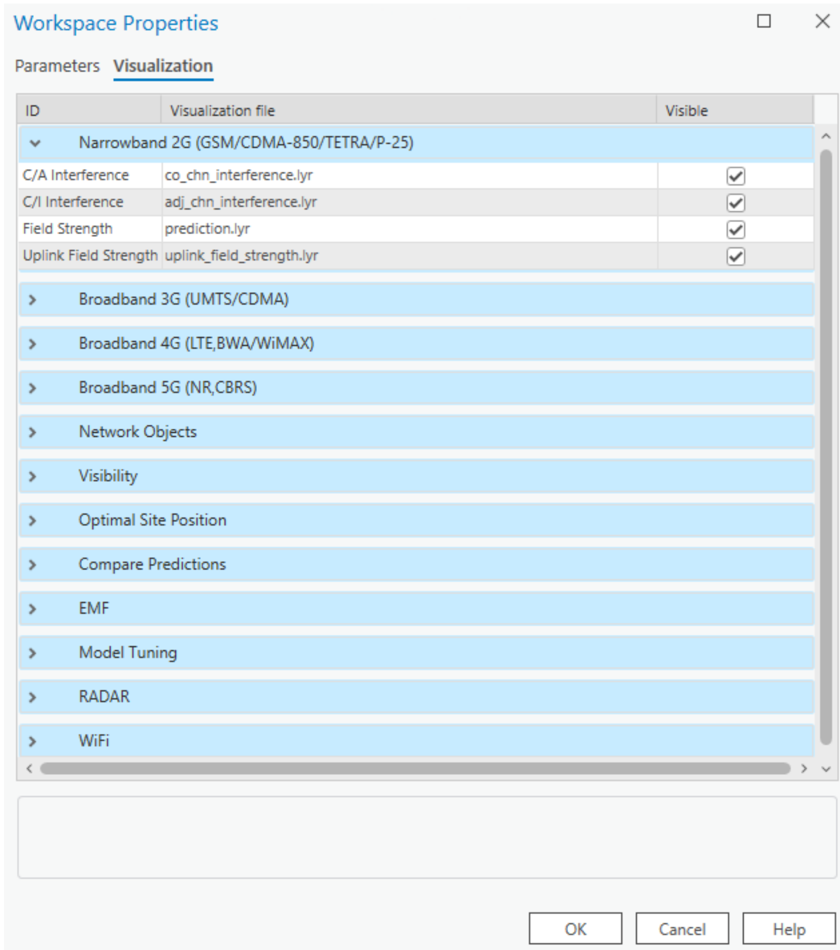


Rounding
The rounding value for different parameters.



6.1.6.2 Visualization

The Cellular Expert network objects (Sites, Cells, OMEN) and calculation result rasters are represented in ArcGIS with the symbology as defined in the *.lyr* files, located in the **Visualization** tab ("CE_LAYERS" table).



OK

Click this button to save any changes and close the dialogue.

Cancel

Click this button to cancel any changes and close the dialogue.

Help

Get helpful information about the dialogue.

All visualization settings have the “Visible” option set to On by default. Suppose the Visible option is set to off, the next time a relevant calculation is performed. In that case, the rasters associated with the calculation are added to the map with the visibility disabled. This is done only for the calculations performed after the setting is changed, and does not impact already added rasters on the map.

The symbology is defined as a list of Layers and *.lyr* files:

- Narrowband 2G (GSM/CDMA-850/TETRA/P-25) – second generation network (like GSM) calculations or technology-independent calculations (for example, the symbology for antenna loss by tilt is defined by the same file for WiMAX, LTE, and other technology)

- Broadband 3G (UMTS/HSDPA) – results for UMTS, HSDPA and other 3 - 3.5 generation technologies
- Broadband 4G (LTE, BWA/WiMAX) – results of LTE technology calculations
- Broadband 5G (NR, CBRS) – results of 5G-NR technology calculations
- Network objects – Sites, Cells, OMEN
- Visibility – results of Visibility Calculations
- Optimal Site Position – results of optimal site positioning.
- Compare predictions – the results of comparing several predictions.
- Model Tuning – results of model calibration
- Radar – results of radar coverage calculations
- Wi-Fi – results of Wi-Fi coverage calculations

For example, the layer *4G Downlink Throughput* with defined path *dl_ul_throughput.lyr* means that the 4G downlink bitrate prediction raster will be represented in ArcGIS using the symbology file *…/Cellular Expert/Layers/dl_ul_throughput.lyr*

<u>*Note*</u>: if you change the symbology with ArcGIS tools, it will be saved only in the current ArcGIS project. So, when you re-open the same Cellular Expert workspace in another ArcGIS project, the symbology will be defined in the **Visualization** tab of the **Workspace Properties** dialogue.

To create a new visualization and locate it in the **Workspace Properties**:

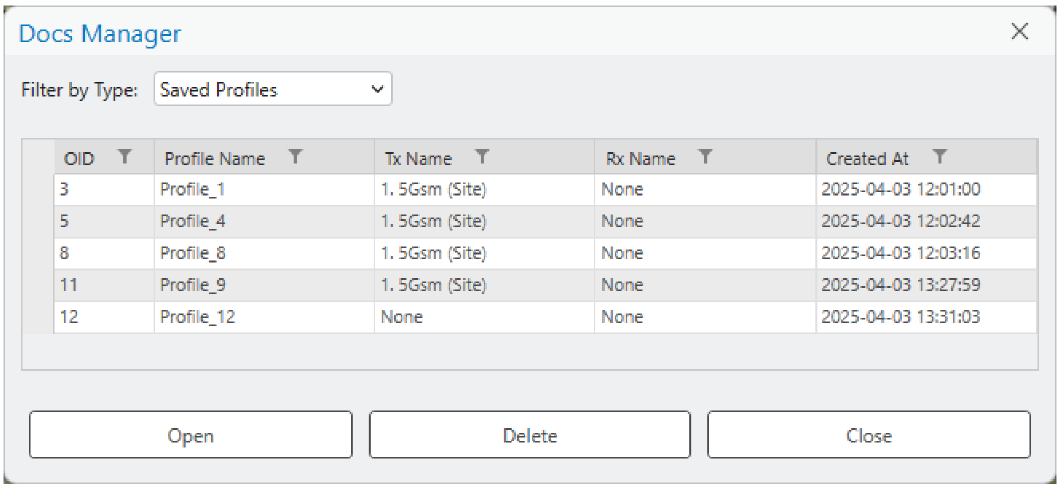
- Right-click on the layer with the modified symbology (for example *Sites*) in the ArcGIS **Table Of Contents**
- Choose **Sharing > Save As Layer File…** from the opened menu
- Save the file with a given name (for example *Sites_my_symbol.lyrx*)
- Copy/Paste it to C:\Program Files\Cellular Expert\Layers
- Open the Cellular Expert workspace in which you want to use your symbology
- Open the **Workspace Properties** dialogue and select the **Visualization** tab
- Select the row with the corresponding layer name (for example *Sites*)
- Click the browse button and locate your layer file (for example "*C:\Program Files\Cellular Expert\Layers\Sites_my_symbol.lyrx*") which contains the modified symbols. Press the **OK** button.
- The symbology for the Cellular Expert network objects (Sites, Cells, OMEN) will be applied to the layers when you open the workspace next time. To see the changes immediately re-open the workspace (close it using the **Remove** option and open it again with the **Open** option from the workspace menu)

The new symbology will be uploaded and remain assigned to the workspace database independently of the ArcGIS project file.

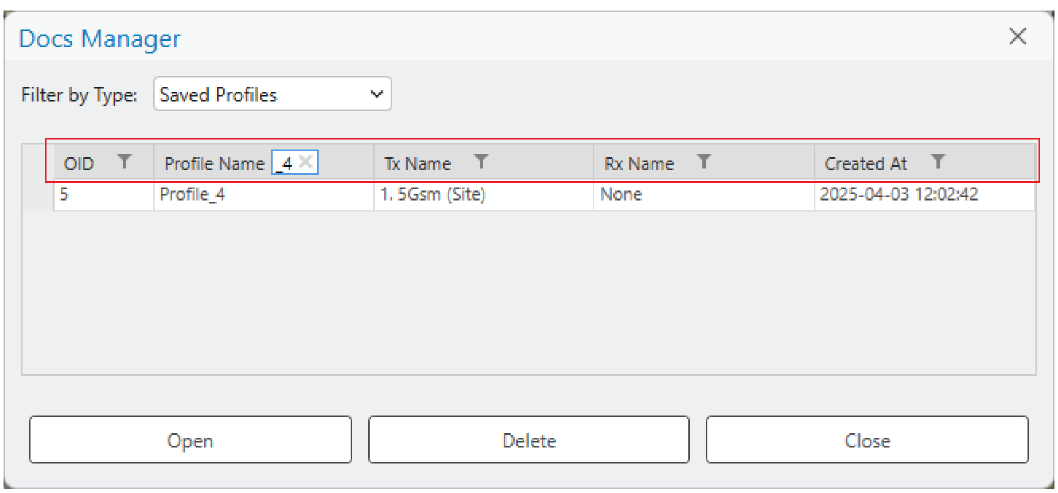
When you use layer files for visualization, do not forget about them. Remember that when you move your workspace to another location, the location path settings can become incorrect. If Cellular Expert is not able to find your defined symbology file, it will use the default file from the location *../Cellular Expert/Layers*.

6.2 Docs Manager

Docs Manager is a tool for managing Saved Profiles between the transmitter (Tx) and receiver (Rx), which are generated in the Profile tool, as well as saved Link Prediction results, Profile Reports, and Link Prediction Reports. When a profile is saved in the Profile tool, it is automatically stored in Docs Manager, allowing users to reopen it at any time. This ensures that all parameters and calculations related to Tx and Rx are preserved for future reference, eliminating the need to reconfigure settings repeatedly. The same applies for Link Prediction results, and the Reports can be opened if the original exported document is, for example, deleted from Desktop or other saved location.

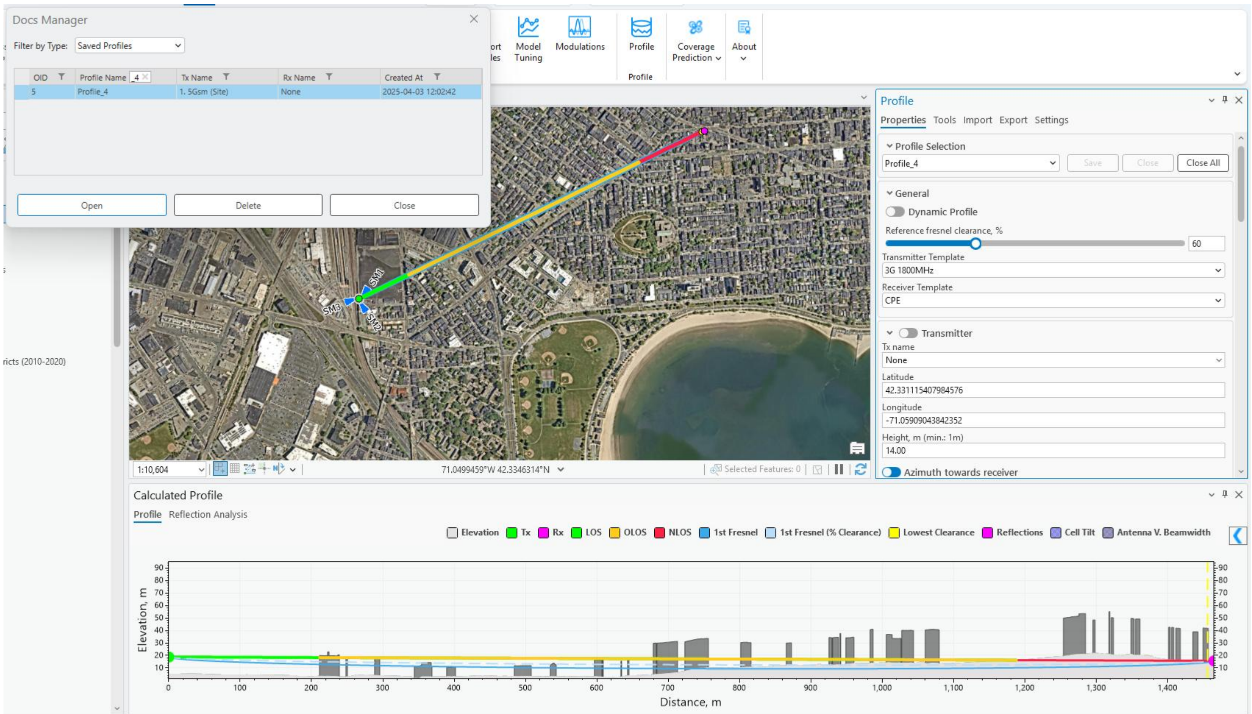


How to Find a Saved Profile
Use the filter option for each field to quickly locate the required profile from the list.



How to Open a Profile
A saved profile can be accessed in two ways:

1. **Double-click** on the desired profile.
2. **Select the profile** and click **Open**.



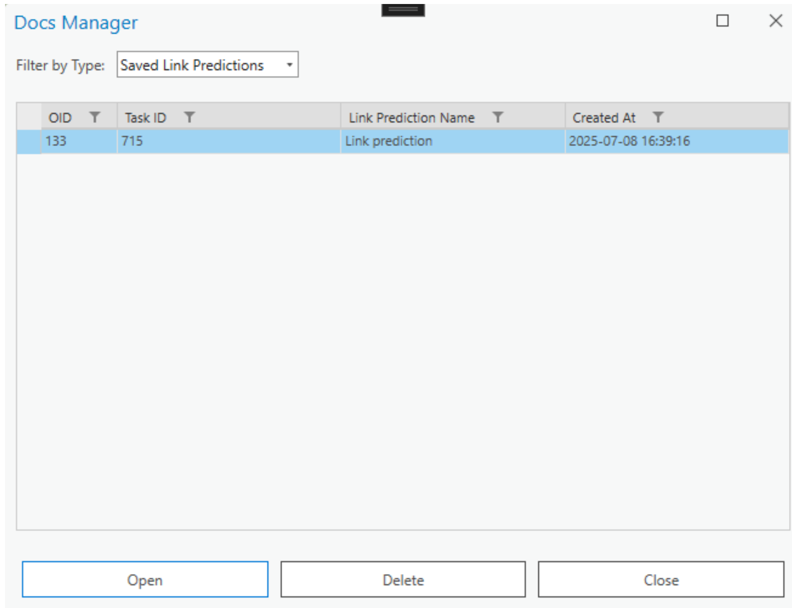
Additional Functions

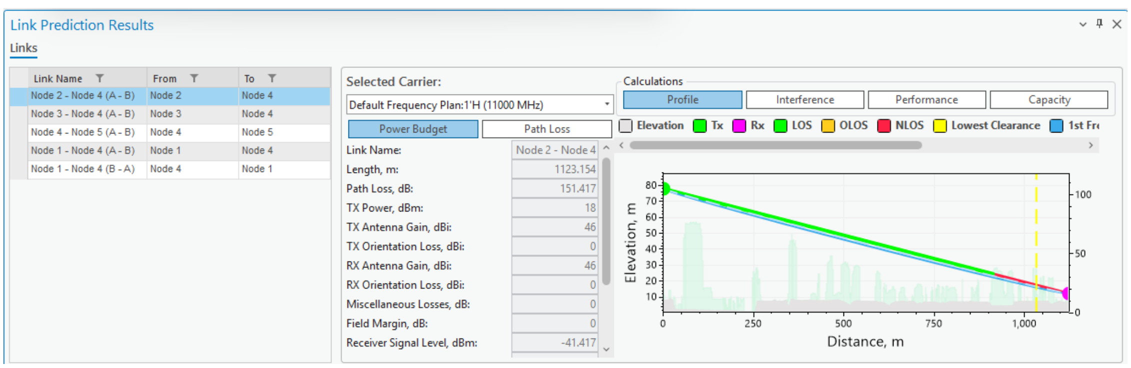
- **Delete**: Removes the selected profile.
- **Close**: Closes the Docs Manager dialog.

By using Docs Manager, users can efficiently store and retrieve profiles, ensuring consistency and accuracy in their Tx and Rx calculations.

How to Open a Link Prediction

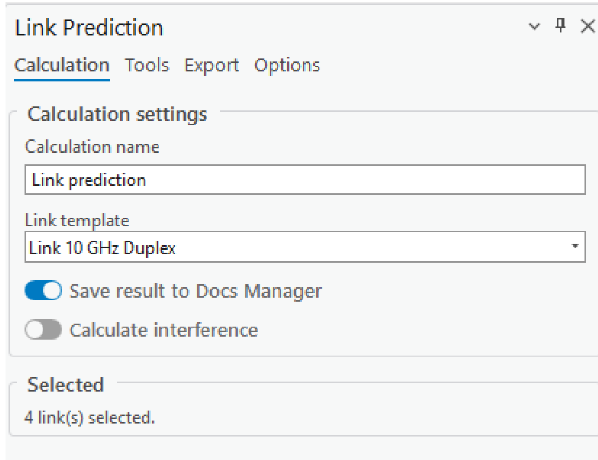
A saved Link Prediction, just like Profile, can be accessed either by double-clicking the desired Link Prediction result, or by select it and clicking Open.





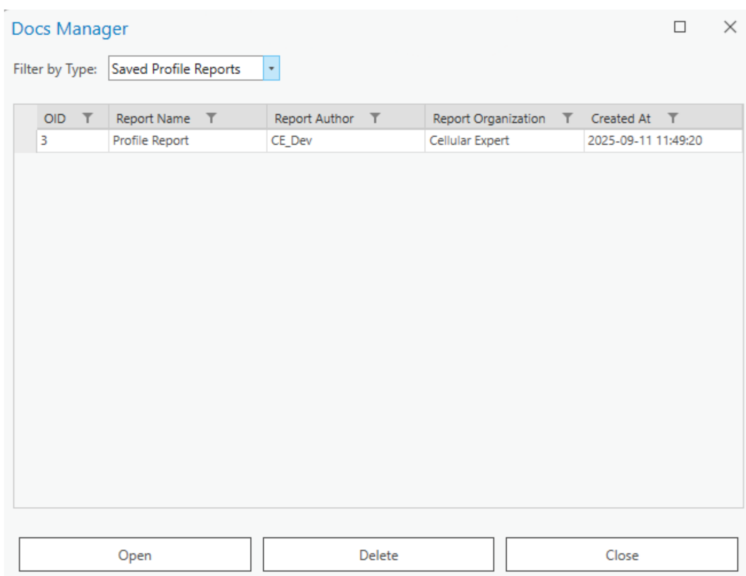
How to Save a Link Prediction

Link Prediction result can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Link Prediction tool.



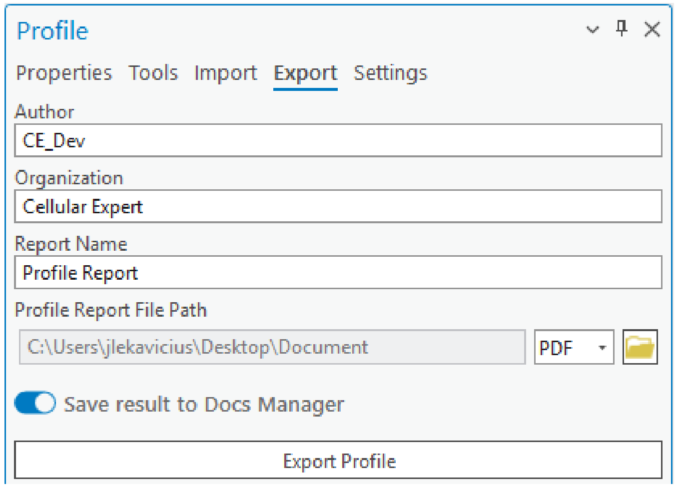
How to Open a Profile Report

A Profile Report document can be accessed either by double-clicking the desired Profile Report, or by select it and clicking Open. The document will be opened using your default PDF document reader.



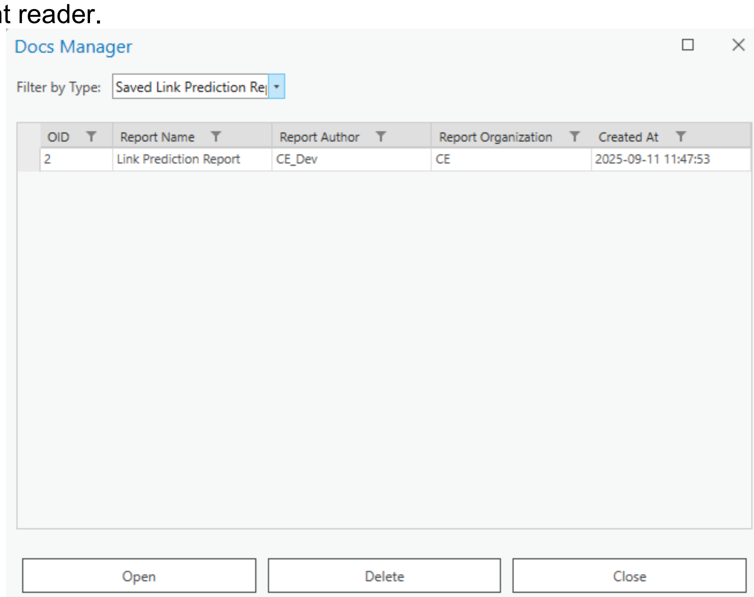
How to Save a Profile Report

Profile Report of the current drawn profile can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Export tab of the Profile tool.



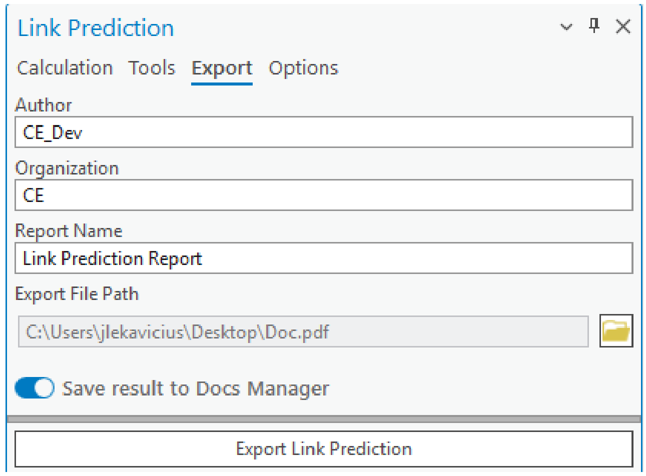
How to Open a Link Prediction Report

A Link Prediction Report document can be accessed either by double-clicking the desired Link Prediction Report, or by select it and clicking Open. The document will be opened using your default PDF document reader.



How to Save a Link Prediction Report

Link Prediction Report can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Export tab of the Link Prediction tool.



6.3 CE Express Connection

CE Express Connection is a tool that lets you establish a connection between the *CE Express* database and *CE for ArcGIS Pro*. When the connection is established, data can be retrieved from *CE Express* and uploaded to *CE for ArcGIS Pro* workspace.

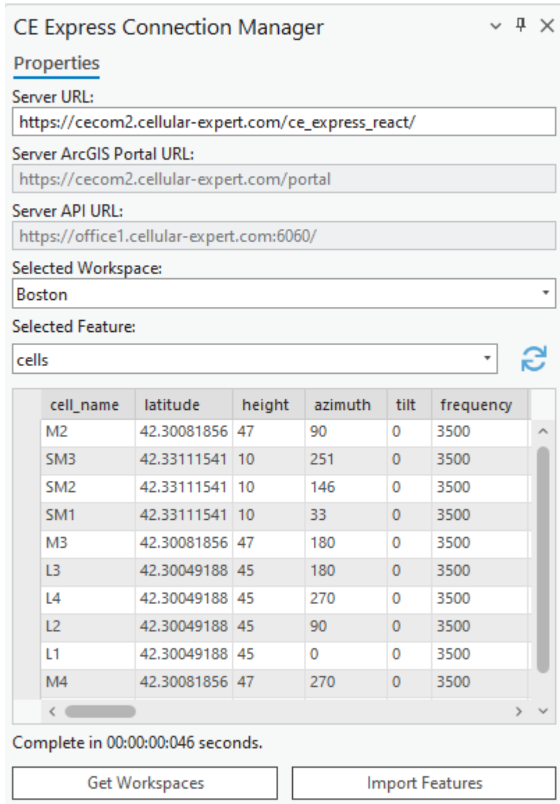
For the connection to be established you must have access to **ArcGIS Portal** and have a valid **CE Express URL**.

6.3.1 Properties

Click the 

button and select the **Properties** tab to open the **CE Express Connection** dialogue.

To connect to *CE Server Express* and get the list of workspaces, insert the **Server URL** in the corresponding field. Then press the **Get Workspaces** button.
If you select one of the appearing workspaces, the properties of that workspace will be saved to your current ArcGIS Pro project automatically.



Server URL
The URL to the *CE Express* database where the workspaces will be located. This parameter must be defined to connect to the database.

Server ArcGIS Portal URL
The URL to your organization's ArcGIS Portal (filled in automatically).

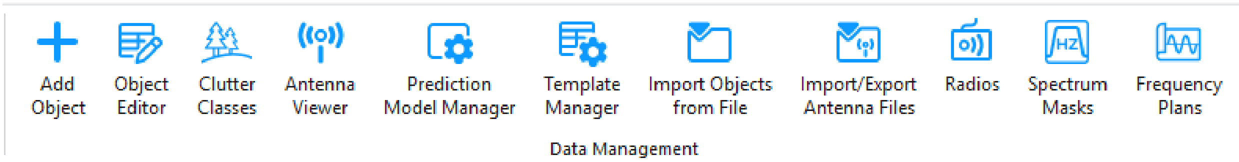
Server API URL
The API is used in the connection process (filled in automatically).

Selected Workspace
The list of all workspaces was retrieved from the *CE Express Database.*

Get Workspaces
Establishes the connection between the provided *Server URL* and *CE for ArcGIS Pro*. Upon clicking the button, the user may be redirected to a browser window in which he will have to log in to the **ArcGIS Portal**. After doing so, the workspaces will be retrieved.

Import Features
Imports the retrieved objects to the currently opened CE workspace.

7. Data Management



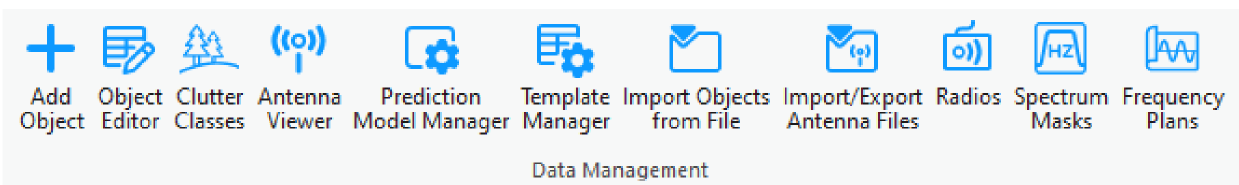
7.1 Network Objects

Cellular Expert network objects are:

Sites – represent the geographical location of a radio station. They are identified by the unique site ID and contain information about the geographical coordinates, ground altitude, and base height.

Links – represents a radio connection between a transmitter and a receiver. It can be established between the selected transmitter and receiver. The link operates at a fixed frequency from the transmitter's carriers list.

With the data management tools located in the **Data Management** section, you can view and edit the Cellular Expert network objects.



Note: most of these tools work only in editing sessions on the currently active workspace.

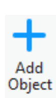
7.2 Add Object

New network objects can be created in several ways. They can be:

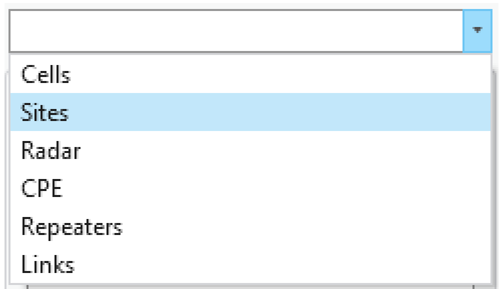
- Imported using the CE for ArcGIS Pro functionality
- Created with Cellular Expert tools from zero (define all parameters in the process)
- Created from templates

7.2.1 Add Site

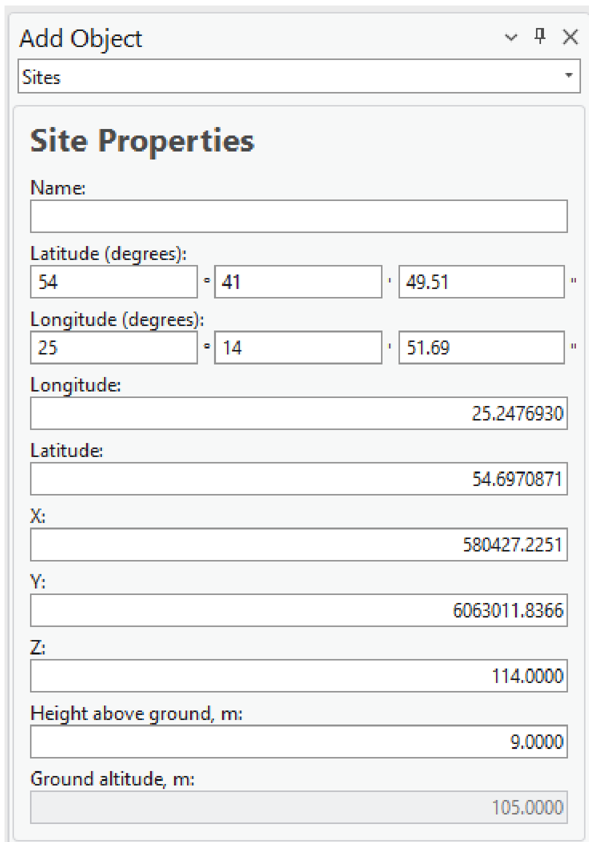
The object represents Tower or Site location. It has several parameters, such as Site Name, coordinate or height. It is used only in 4G or 5G carrier aggregation calculation, when Total Downlink Throughput is calculated.

1. Choose the 

button from the toolbar and select **Sites** from the dropdown list.

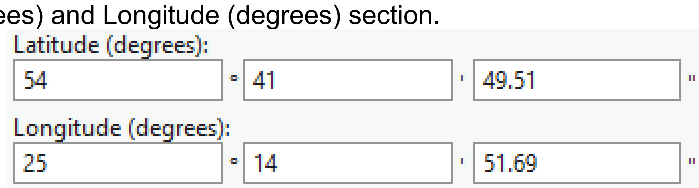


2. Define the location of the new Site by pressing the mouse left button on the map. The new Site will be placed right in that location.

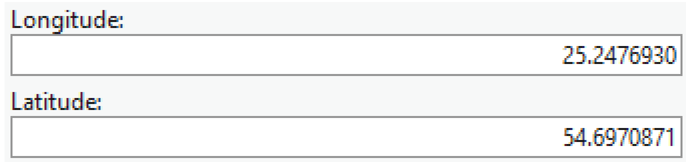


The Site object can be created by entering exact coordinates in:

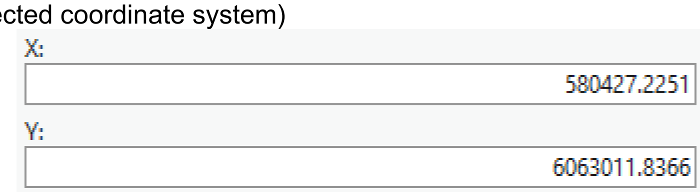
- Latitude (degrees) and Longitude (degrees) section.



- Latitude and Longitude



- X and Y (projected coordinate system)



3. Define Site name and press Save Changes to save object to the database.

Save Changes
Creates the object with the given parameters.

Dismiss
Cancels object creation and closes the dialogue.

<u>Site Properties</u>

Name
Site identification.

Latitude (degrees)
Latitude degrees, minutes, and seconds Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude (degrees)
Longitude degrees, minutes, and seconds X type coordinate in the WGS 1984 geographical coordinate system.

Latitude
Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude
Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

X
Coordinate in the projected coordinate system.

Y
Coordinate in the projected coordinate system.

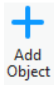
Z
3D dimensions representing an object's height above sea level, used for visualizing objects in a 3D scene.

Height Above Ground
Object’s height above the terrain.

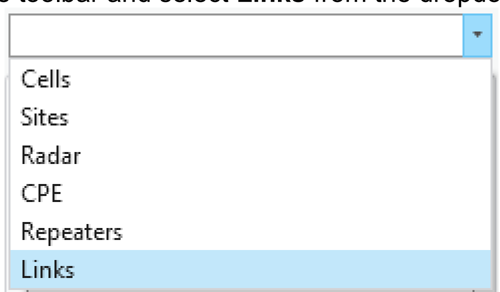
Ground Altitude
Ground elevation above sea level at the network object's location.

7.2.2 Add Link

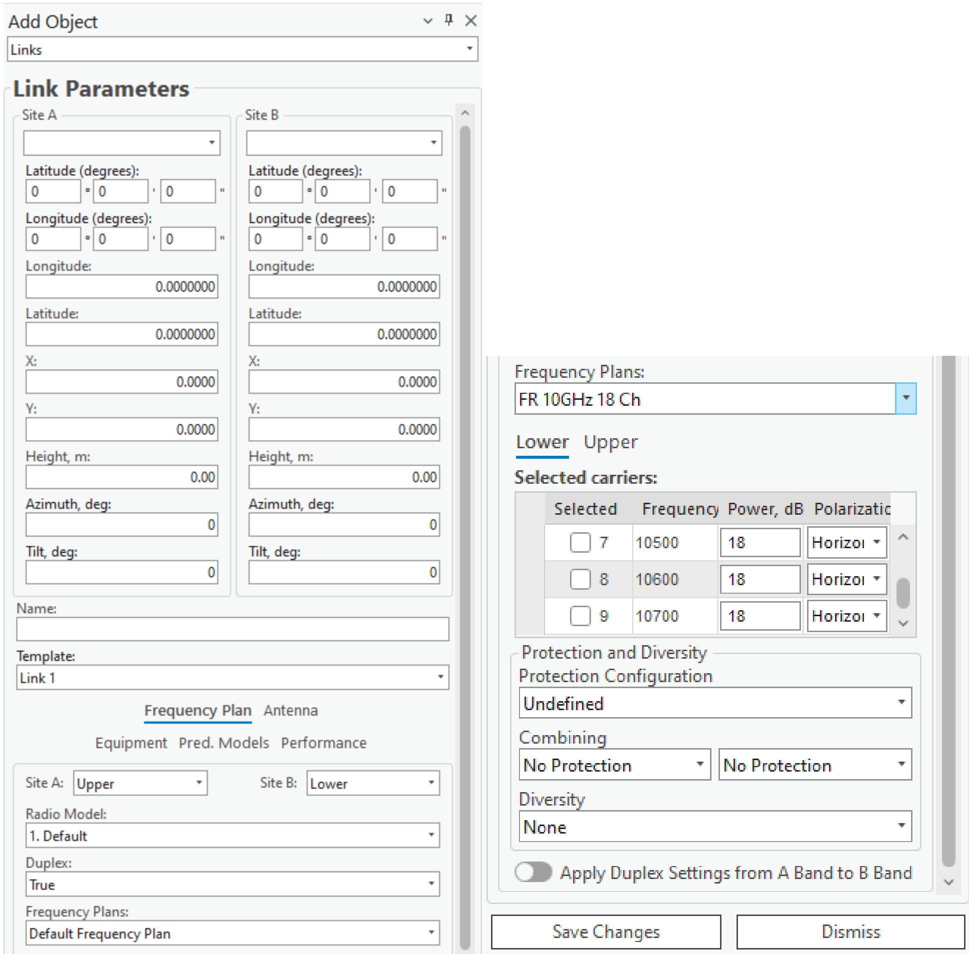
The object represents a Microwave link that holds all the information about frequencies, antennas, radio equipment, etc. Object type is a line feature class.

Choose the 

button from the toolbar and select **Links** from the dropdown list.



Define the location for the new Link by left-clicking on the map and selecting two distinct points, or two existing Sites. The new Link will be created as a geometry between those two points.



Save Changes
Creates the object with the given parameters.

Dismiss
Cancels object creation and closes the dialogue.

Link Parameters

Site (A, B)
Sites that have been assigned for the transmitter and receiver points of the link. If upon link creation no sites were assigned, new sites will be created and assigned to the link.

Latitude (degrees A, B)
Latitude degrees, minutes, and seconds Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude (degrees A, B)
Longitude degrees, minutes, and seconds X type coordinate in the WGS 1984 geographical coordinate system.

Latitude (A, B)
Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude (A, B)
Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

X (A, B)
Coordinate in the projected coordinate system.

Y (A, B)
Coordinate in the projected coordinate system.

Z (A, B)
3D dimensions representing an object's height above sea level, used for visualizing objects in a 3D scene.

Height (A, B)
Object's height above the terrain.

Azimuth (A, B)
Direction from the North in degrees.

Tilt (A, B)
Mechanical tilt in telecommunications repeaters is the physical angling of the antenna to optimize signal coverage.

Link Name
Link identification.

Template
The template will fill all empty or not specified fields with default values that are not necessary for predictions.

7.2.2.1 Frequency Plan

Band (A, B)
Specifies whether the transmitter operates in Upper or Lower frequencies. Additionally, if the link type is duplex, we also determine the receiver band.

Radio Model
The radio equipment parameters which are assigned to the Link.

Duplex
Specifies whether the signal can be transmitted and received simultaneously. Changing the duplex mode will affect the carrier selection.

Frequency Plan
Frequency plan that is applied to the link. Changing the frequency plan will affect the carrier selection

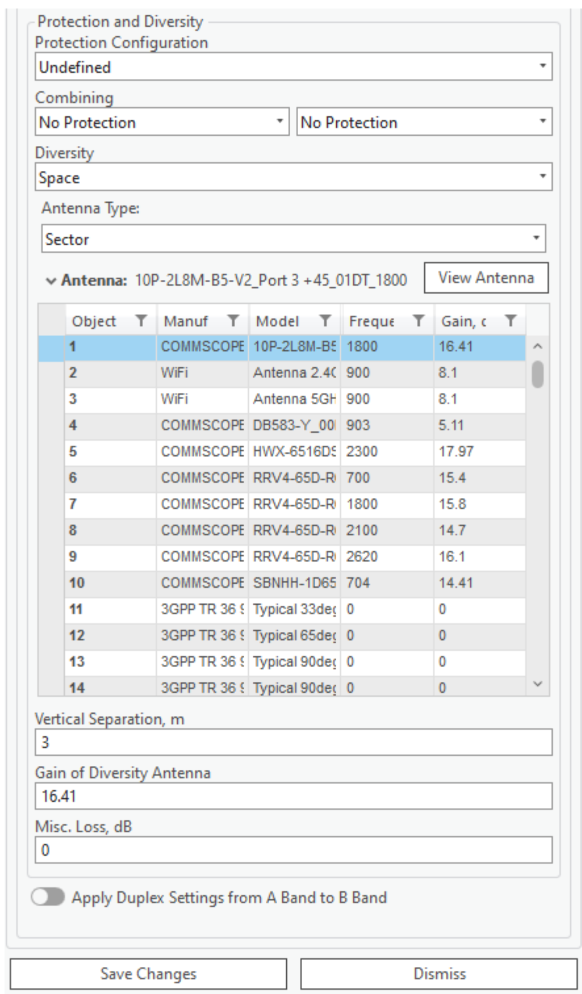
Carrier Selection
Carriers that will be assigned to the created Link. Carriers must be selected for the link to be created. If duplex mode is enabled, the user will be able to edit both Upper and Lower band carriers. Also, selecting one carrier will result in the selection of both Upper and Lower carriers based on their ID.

Protection configuration
1+1 or M:N protection configuration.
Based on the selected protection configuration the following transmitter and receiver switching schemes

are displayed:

Link configuration	Tx combining	Rx combining
1+1	Hot standby	Combining/Switching
1:1	Switching	Combining/Switching
M:N	Switching	Switching

Diversity
Can be the following: space, frequency, space and frequency with two or four receivers, and angle diversity. To define diversity options (antenna or frequency separation values) use the Diversity properties command. For space diversity options, the antenna can be selected, as well as the parameters for vertical separation (in meters), gain of diversity antenna, and misc. loss (in decibels).



Protection and diversity improvement calculation algorithms are provided in Cellular Expert Technical Reference.

Apply parameter settings to duplex link
If toggled, saves the receiver band carrier parameters to the transmitter band carrier parameters, overwriting any changes that were made to the selected carriers.

Read more about Frequency Plans in *Frequency Plans*.

7.2.2.2 Antenna

Antenna (A, B)
Antenna name, parameters, and patterns for predictions.

Misc Loss (Optional)
Miscellaneous loss value in dB.

View Antenna
Opens the Antenna Viewer with the corresponding antenna patterns.

Read more about antennae in *Antenna Viewer*.

7.2.2.3 Equipment

Spectrum Mask
Mask form for selected carriers.

Radio Model
The radio equipment parameters which is assigned to the Link.

View Mask
Opens the Spectrum Masks with the selected spectrum mask patterns.

Read more about equipment in *Radios*, *Spectrum Masks*.

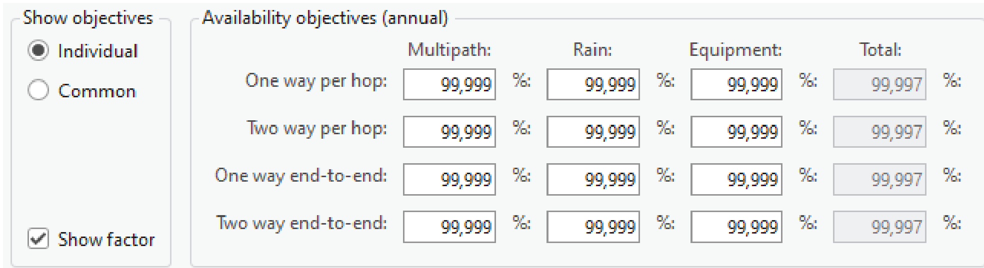
7.2.2.4 Prediction Models

Read more about prediction models in *Prediction Model Manager*.

This tab lets the user select which prediction model and configuration should be used for calculations.

7.2.2.5 Performance Objectives

Performance Objectives Opens a dialog for the assignment of individual performance objectives for the current link. It allows for defining availability and performance objectives. Predefined performance objectives are related to the radio links under analysis.



The first group of objectives is **annual availability** objectives for multipath, rain, and equipment. Total availability is recalculated automatically as a sum after entering objectives for multipath, rain, and equipment. The objectives are expressed as a percentage over the average year, separately for one-way and two-way paths, for single hop, and end-to-end radio paths.

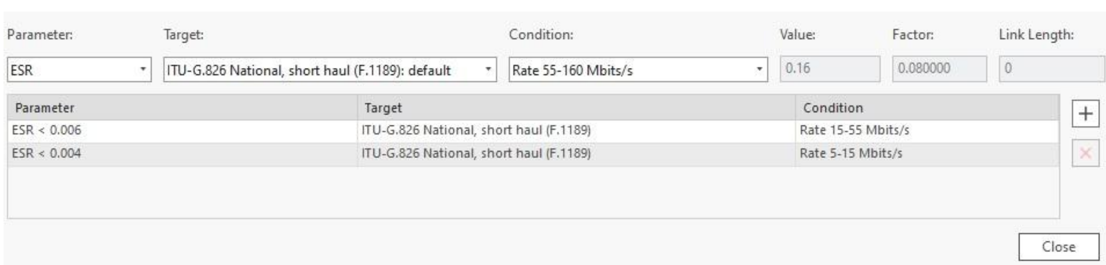
The other group of objectives is performance objectives. There are two kinds of performance objectives

- **Individual** - prescribed for a given link
- **Common** - valid for all links subjected to the prediction procedure

Such a framework enables the evaluation of one group of performance objectives for a whole radio link network, together with specialized objectives for separate parts of a network.

Individual performance objectives are accessible only when the performance objectives dialog is launched from the selected link's properties dialog.
To create a new performance objective, first select the required performance parameter, the target defining an objective for this parameter, and finally, the condition which applies to a given link, or a group of links.

Add the performance objective by pressing the [+] button.



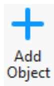
To delete the objective, select it in the table and press the [×] button respectively.

Value
If the value of a given performance objective depends on the path length of a link, it can be represented by the value

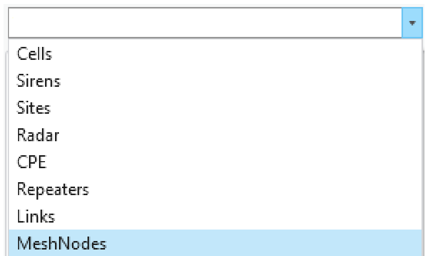
Show Factor
If the value of a given performance objective depends on the path length of a link, it can be represented by a multiplication factor as a formatted string.

7.2.3 Add Mesh Node

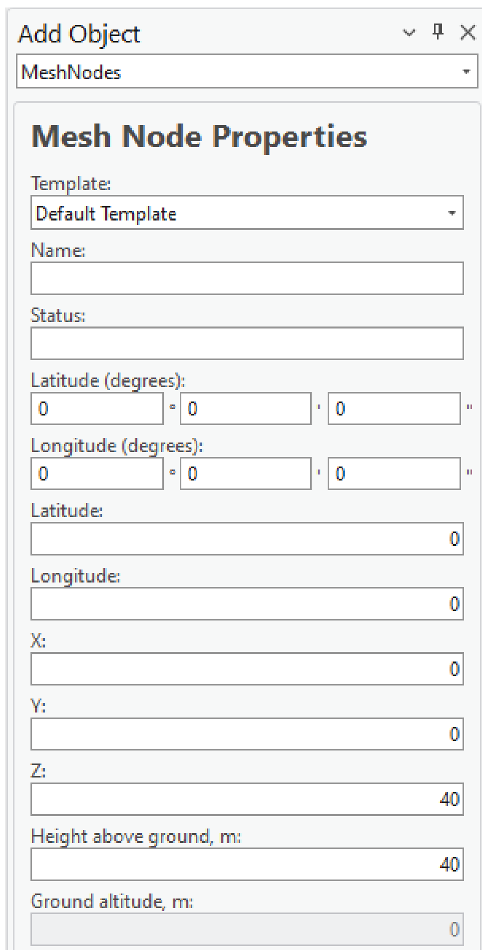
The object represents a node used in mesh network calculations and serves as an endpoint for Link objects. It includes several configurable parameters such as name, geographic coordinates, height, maximum number of connections, group affiliation, and additional technical attributes.

Choose the 

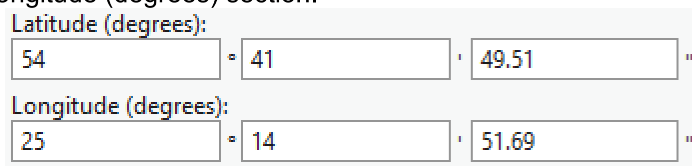
button from the toolbar and select **MeshNodes** from the dropdown list.



Define the location of the new Mesh Node by pressing the mouse left button on the map. The new Mesh Node will be placed right in that location.

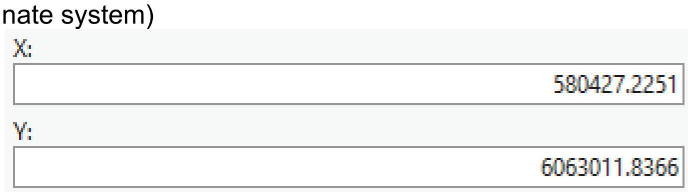


The Mesh Node object can be created by entering exact coordinates in:
Latitude (degrees) and Longitude (degrees) section.



Latitude and Longitude

X and Y (projected coordinate system)



Define Mesh Node name and press Save Changes to save object to the database.

Save Changes
Creates the object with the given parameters.

Dismiss
Cancels object creation and closes the dialogue.

Mesh Node Properties

Name
Mesh node identification.

Status
Mesh node status.

Latitude (degrees)
Latitude degrees, minutes, and seconds Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude (degrees)
Longitude degrees, minutes, and seconds X type coordinate in the WGS 1984 geographical coordinate system.

Latitude
Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude
Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

X
Coordinate in the projected coordinate system.

Y
Coordinate in the projected coordinate system.

Z
3D dimensions representing an object's height above sea level, used for visualizing objects in a 3D scene.

Height Above Ground
Object’s height above the terrain.

Ground Altitude
Ground elevation above sea level at the network object's location.

Frequency, MHz
Frequency of the mesh node.

Antenna
Define antenna patterns for the mesh node object.

Power, dBm
Power value of the mesh node.

Misc Loss, dB
Miscellaneous loss value of the mesh node.

Misc Loss, dBm
Sensitivity value of the mesh node.

Type
Type of the mesh node (text description).

Max connections
The maximum number of connections the mesh node can have (used for mesh connectivity calculations).

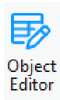
Layer
The layer of the mesh node (number value). Used for priority calculations in Automatic Frequency Planning.

Group name
The group name of the mesh node (text description). Several mesh nodes may belong to the same group.

Prediction model
The prediction model that will be used for the mesh node in Mesh Connectivity and Quick Mesh Connectivity calculations. The default selection is LOS ITU-R P.525 (6GHz – 100GHz) – Default.

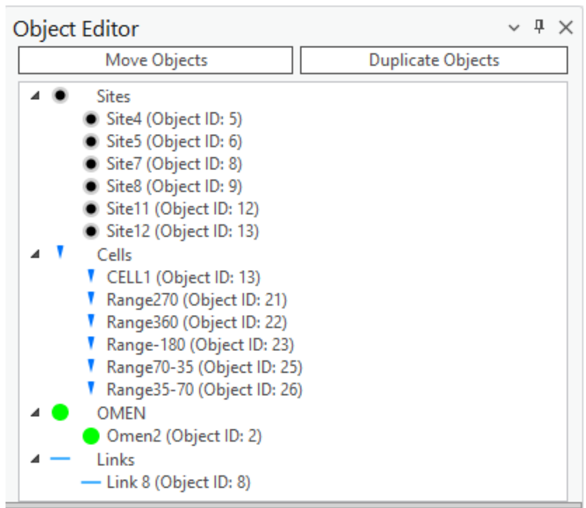
7.3 Object Editor

The Object Editor enables the user to make changes to a network object after it is created and placed on the map.

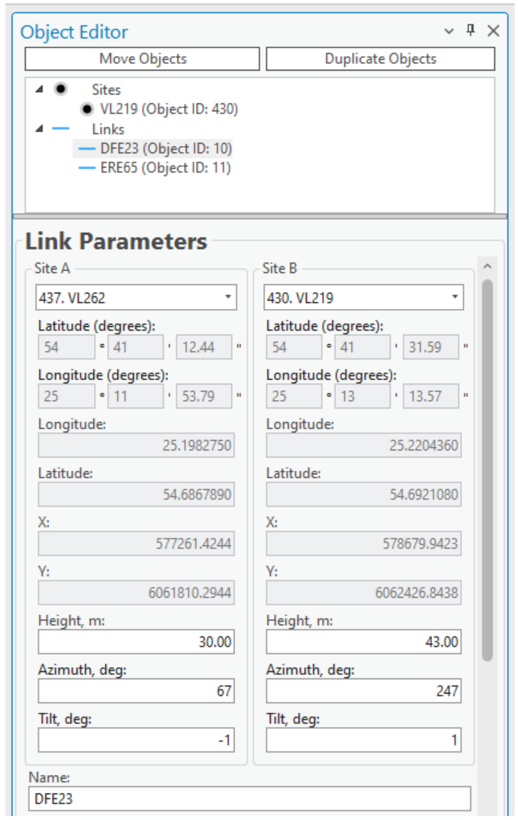
Choose the 

button to open the **Object Editor** dialogue.

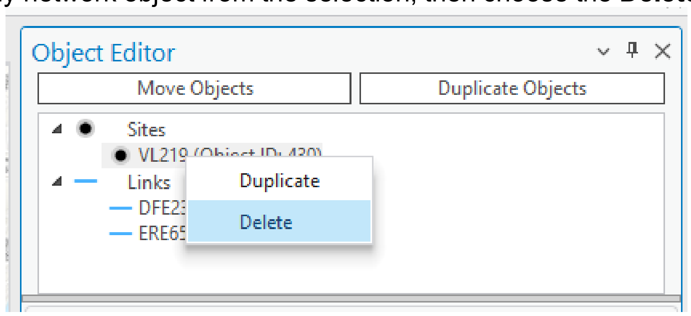
Select objects by navigating to the ArcGIS Pro **Edit** → **Selection** section and choosing the **Select** tool. The selected objects will appear in the Object Editor in a tree hierarchy.



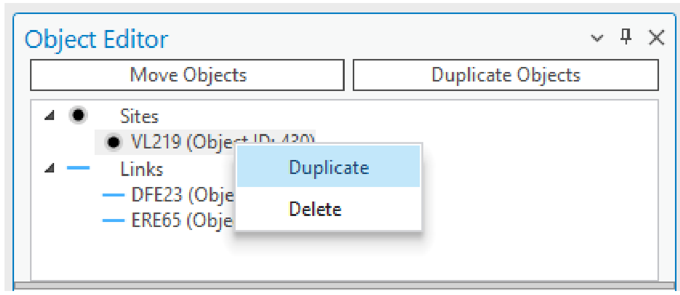
To edit one of the selected objects, left-click on that object and the corresponding editing menu will open below the list.



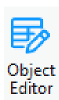
Delete Object
Select and right-click any network object from the selection, then choose the **Delete** option from the popup.



Duplicate Object
Select and right-click any object from the selection, then choose the **Duplicate** option from the popup. The duplicated object will retain all information from the original object including coordinates/meridians. If you want to duplicate objects and change their coordinates/meridians at the same time, use the separate **Duplicate Objects** button from above the selection tree.

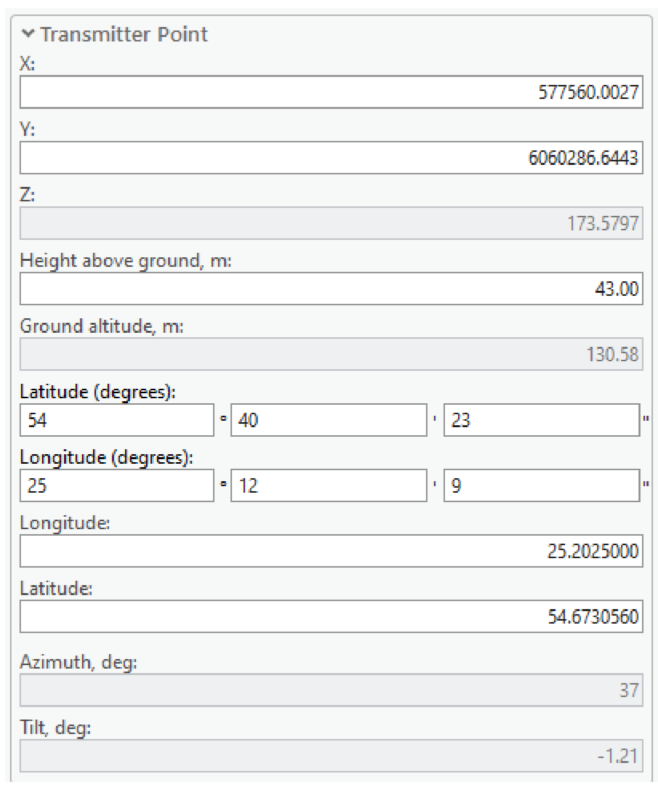


7.3.1 Move Link Objects

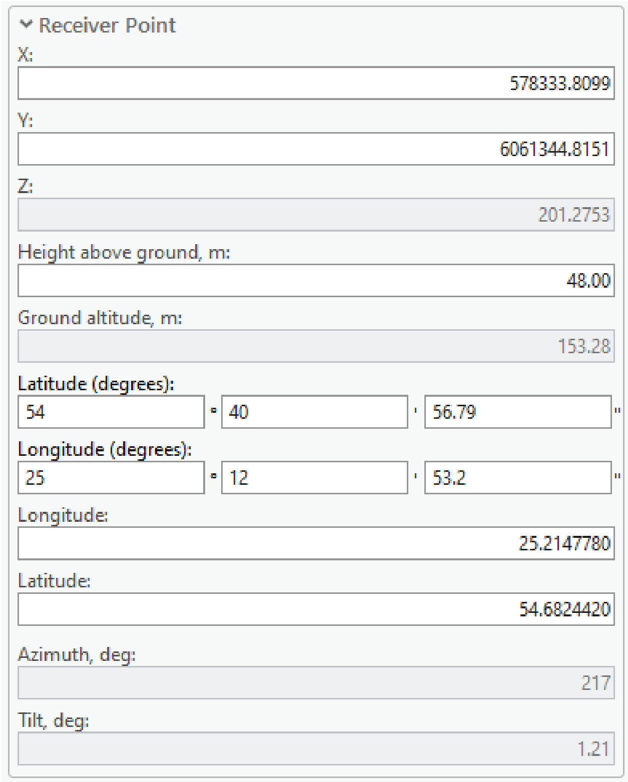
Choose the 

button to open the **Object Editor** dialogue. Select the desirable Link objects with the **Select** tool and press the **Move Objects** button in the Object Editor.

The Links object has Transmitter parameters.

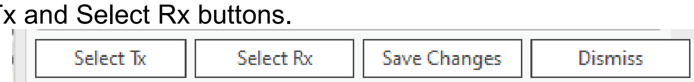


And Receiver parameters.



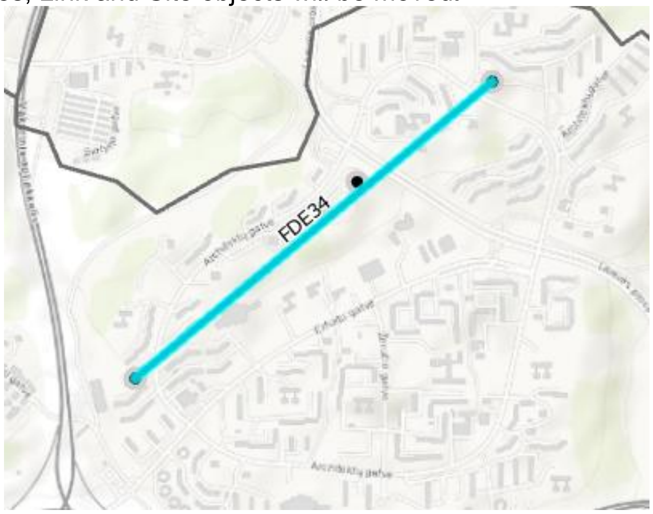
The location of Transmitter and Receiver can be changed:

- By defining exact coordinates in Transmitter and Receiver section.
- Using Select Tx and Select Rx buttons.



If new location is defined, then potential link will be visible on the map.

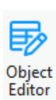
By pressing Save Changes, Link and Site objects will be moved.



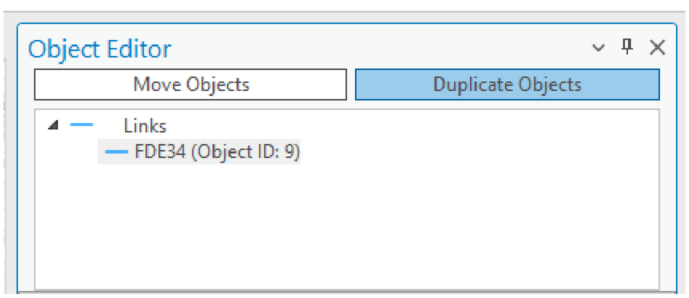
Save Changes
Saves the changes to the objects.

Dismiss
Cancels the changes to the objects and closes the dialogue.

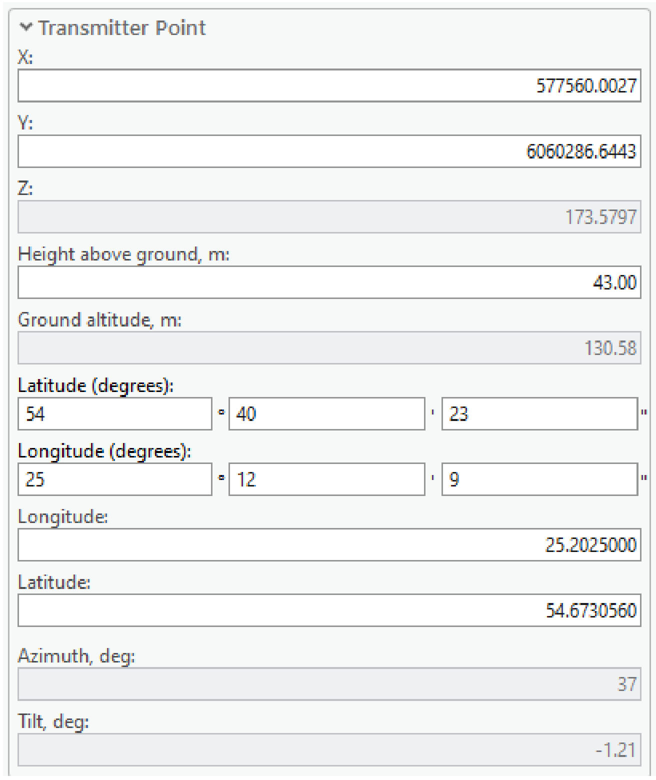
7.3.2 Duplicate Link Objects

Choose the 

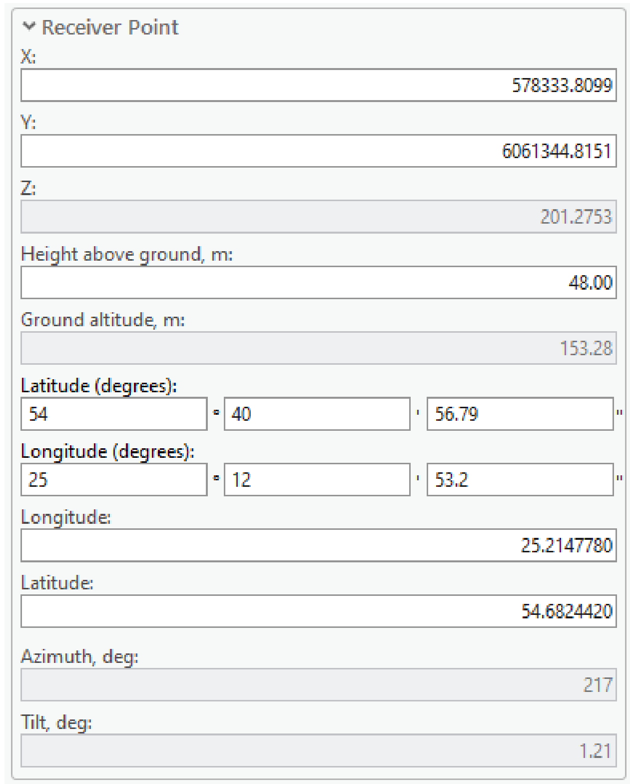
button to open the **Object Editor** dialogue. Select the Link object with the **Select** tool and press the **Duplicate Objects** button in the Object Editor dialogue.



The Links object has Transmitter parameters.

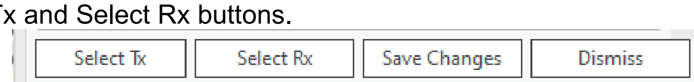


And Receiver parameters.



The location of Transmitter and Receiver can be changed:

- By defining exact coordinates in Transmitter and Receiver section.
- Using Select Tx and Select Rx buttons.



If new location is defined, then potential link will be visible on the map.

By pressing Save Changes, Link and Site objects will be duplicated.

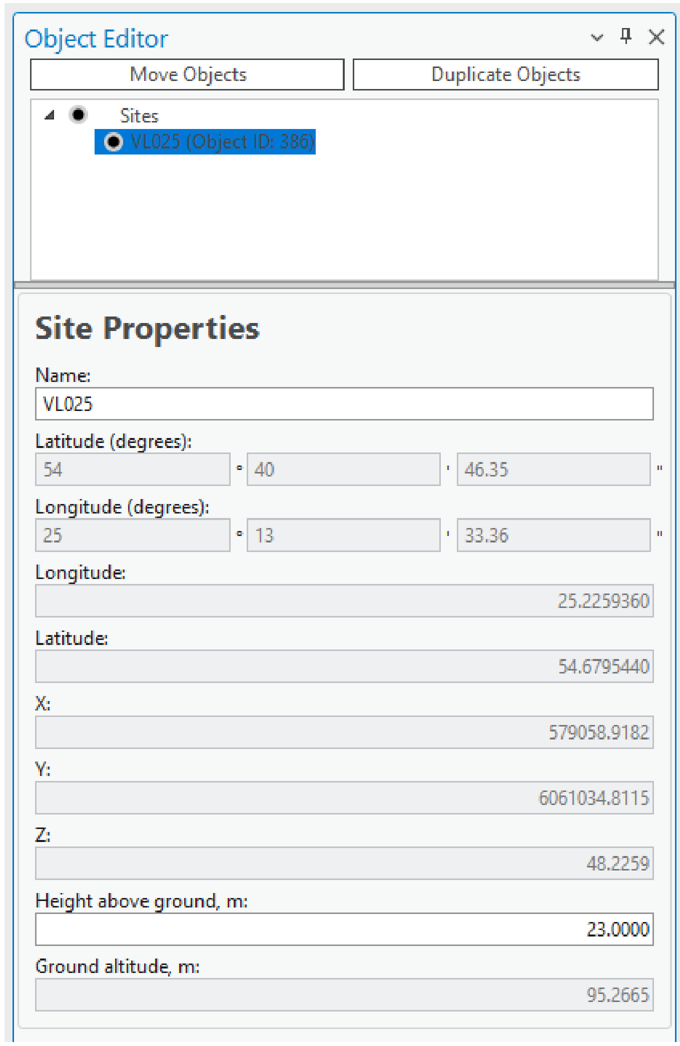
Save Changes
Saves the changes to the objects.

Dismiss
Cancels the changes to the objects and closes the dialogue.

7.3.3 Move Site (Point) Objects

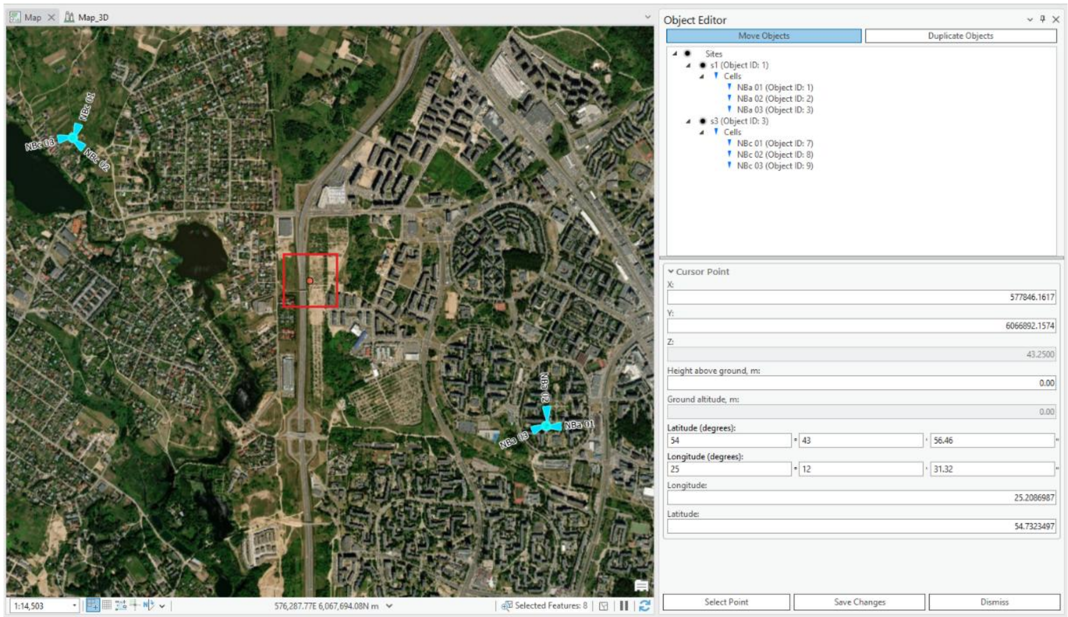
Choose the 

button to open the **Object Editor** dialogue. Select the desirable objects with the **Select** tool and press the **Move Objects** button in the Object Editor.

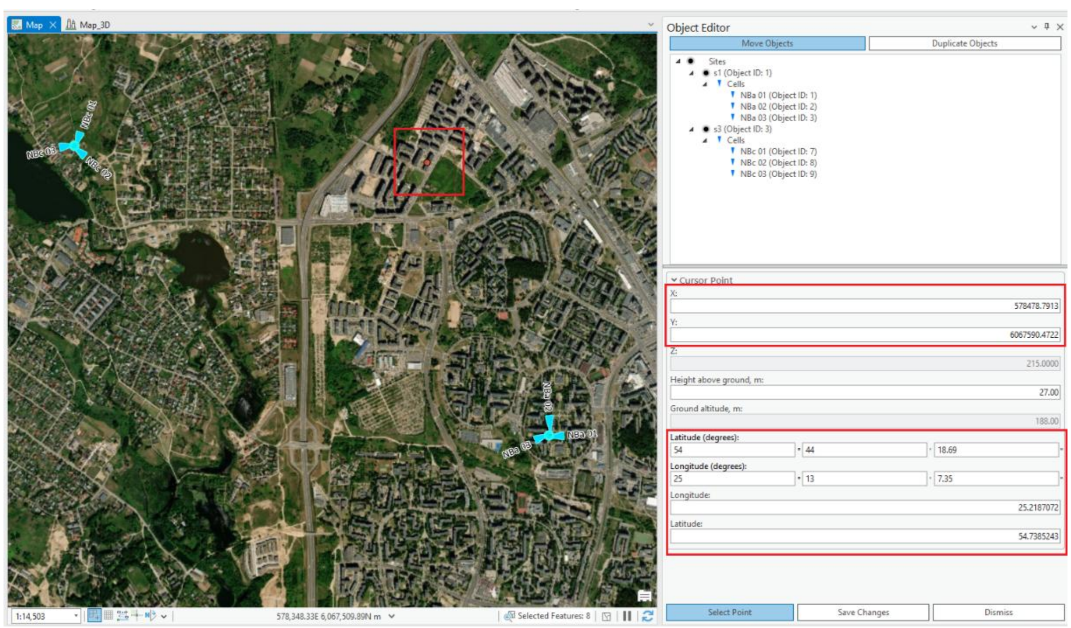


There are multiple ways to move objects:

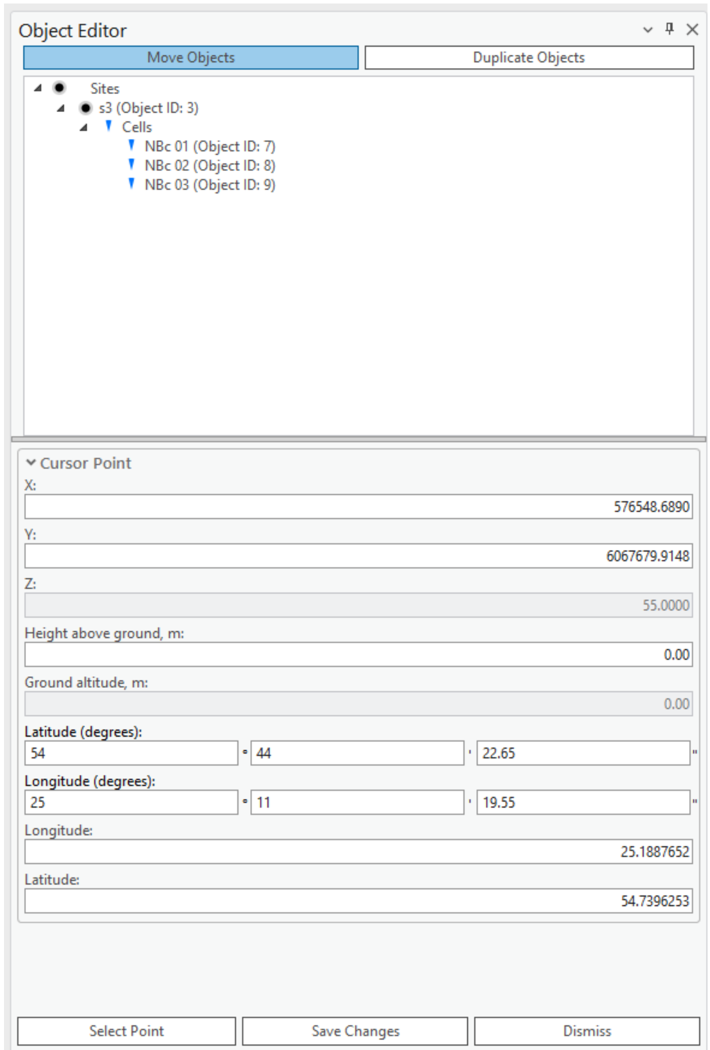
- If multiple objects are selected, the move objects display will show the geospatial properties of the center point between the objects denoted as the “Cursor point”.



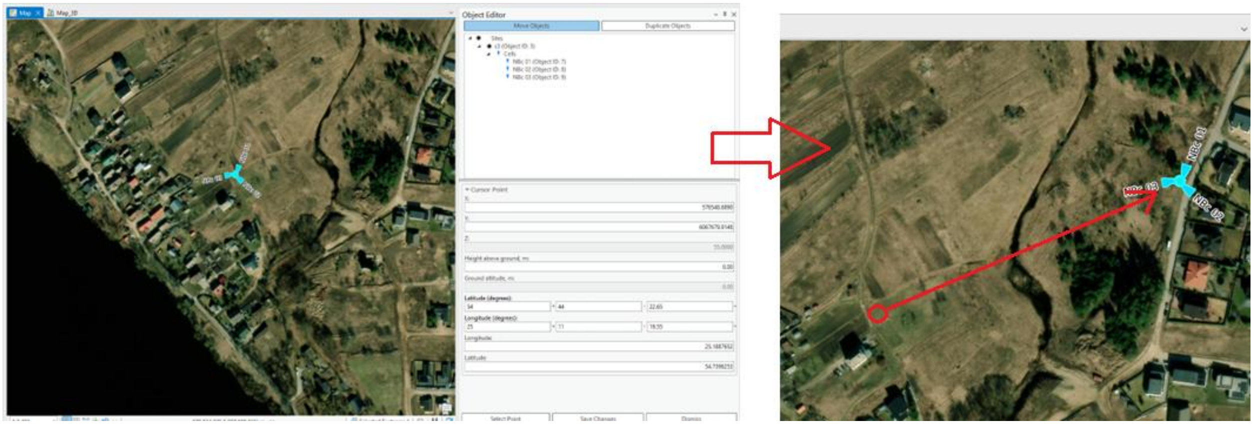
The "Cursor Point" functions as a reference marker depicted by a red dot on the map. This marker is centrally located among the objects on the map and shows where the cursor was placed. You can adjust the position of the Cursor Point either by entering new coordinates directly or by clicking “Select Point” and choosing a different location on the map. When you move the Cursor Point, all the objects on the map will shift their position to maintain their relative distances from this central point.



- If a single point object is selected or several objects at the same location, the move objects display will show the geospatial properties of these objects denoted as “Cursor Point”.



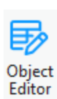
If a single point object is selected on the map, the "Cursor Point" serves as its positional anchor. Positioned as a red dot, the Cursor Point represents the current coordinates of the selected object and shows where the cursor was last placed. You can alter the location of the selected object by manually updating the Cursor Point's coordinates or by clicking “Select Point” and choosing a new location on the map. Any movement of the Cursor Point will directly translate the selected object to the corresponding new location while keeping its spatial relationship with the Cursor Point consistent.



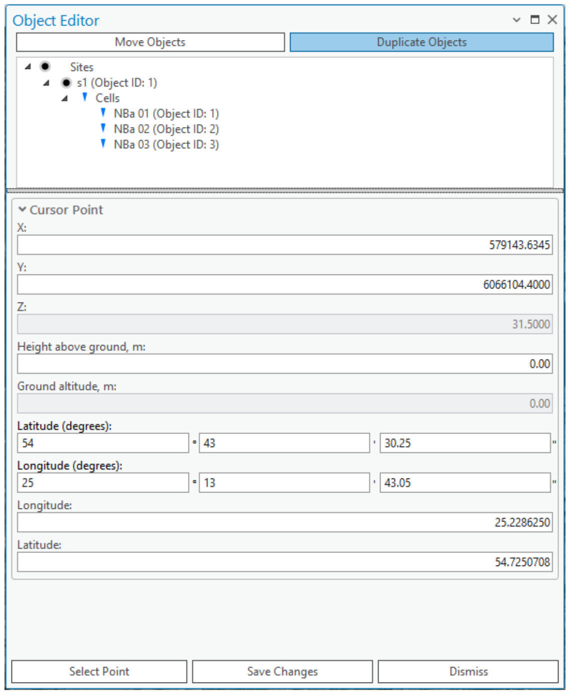
Save Changes
Saves the changes to the objects.

Dismiss
Cancels the changes to the objects and closes the dialogue.

7.3.4 Duplicate Site (Point) Objects

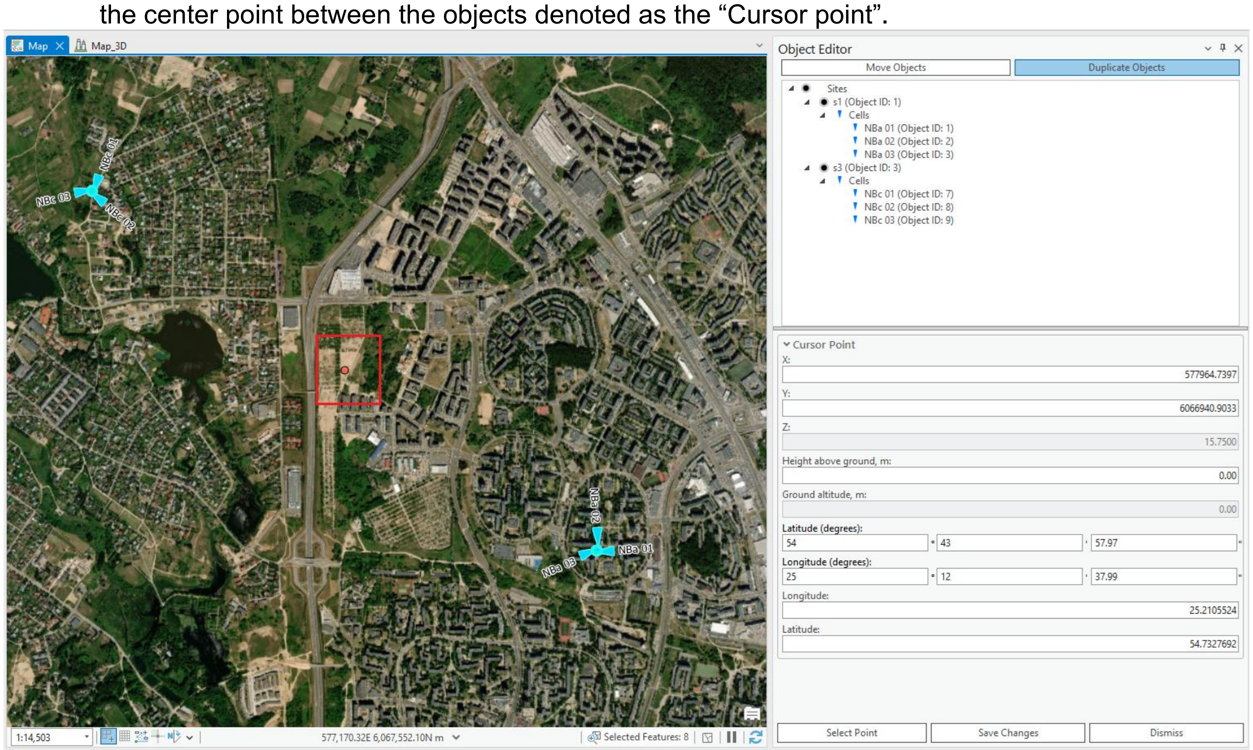
Choose the 

button to open the **Object Editor** dialogue. Select the object with the **Select** tool and press the **Duplicate Objects** button in the Object Editor dialogue.

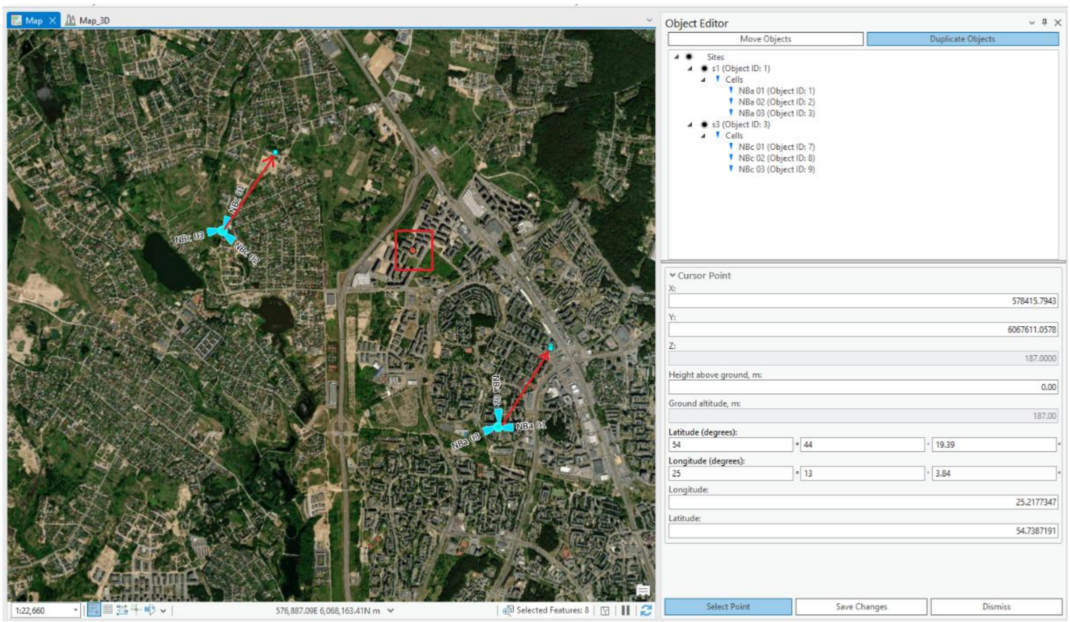


There are multiple ways to duplicate objects:

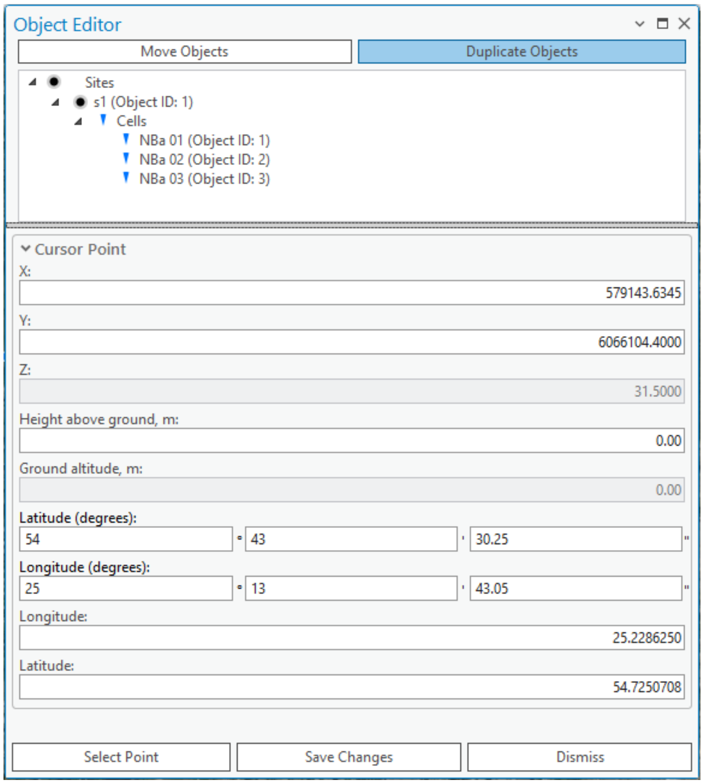
- If multiple objects are selected, the duplicate objects display will show the geospatial properties of the center point between the objects denoted as the “Cursor point”.



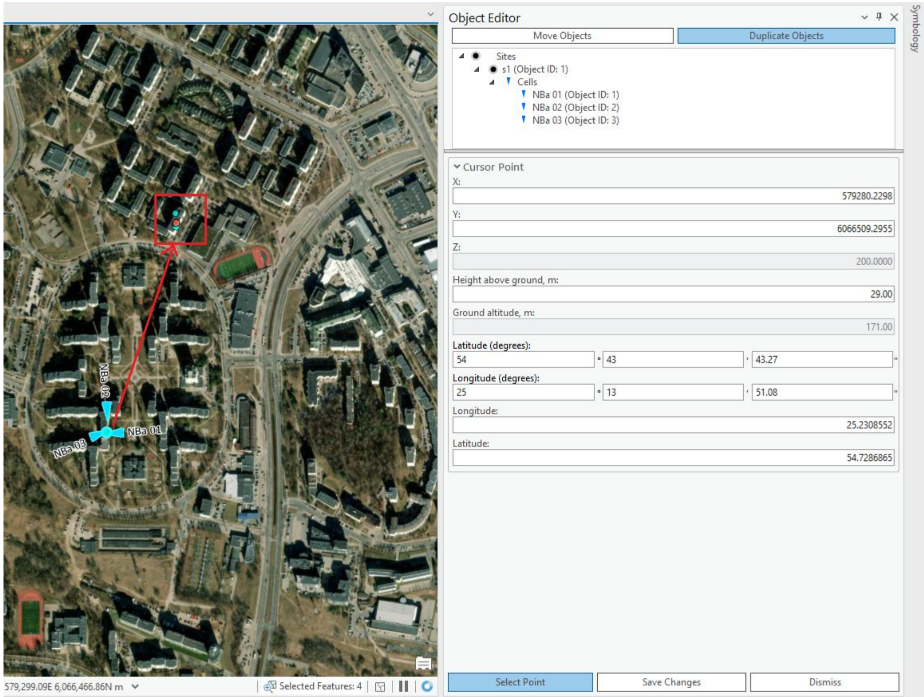
The "Cursor Point" functions as a reference marker depicted by a red dot on the map. This marker is centrally located among the objects on the map and shows where the cursor was placed. You can adjust the position of the Cursor Point either by entering new coordinates directly or by clicking “Select Point” and choosing a different location on the map. When you move the Cursor Point, all the objects on the map will shift their position to maintain their relative distances from this central point.



- If a single point object or several objects on the same location are selected, the duplicate objects display will show the geospatial properties of that object denoted as “Cursor Point”.



If a single point object is selected on the map, the "Cursor Point" serves as its positional anchor. Positioned as a red dot, the Cursor Point represents the current coordinates of the selected object and also shows where the cursor was last placed. You can alter the location of the selected object by manually updating the Cursor Point's coordinates or by clicking “Select Point” and choosing a new location on the map. Any movement of the Cursor Point will directly translate the selected object to the corresponding new location while keeping its spatial relationship with the Cursor Point consistent.



Save Changes
Duplicates the objects.

Dismiss
Cancels the changes to the objects and closes the dialogue.

7.4 Clutter Classes

The **Clutter Classes** tool is designed to manage categories describing different types of environments in telecommunication networks. It relies on the Clutter Classes raster used in the project. The raster values must match the ID values in the Clutter Classes dialog. For example, if "Trees" has a value of 2 in the Clutter Classes raster, this ID must be defined in the Clutter Classes tool. By default, ESRI's Sentinel-2 Land Use values are used after workspace creation: [ESRI Sentinel-2 Land Cover]

Here is the table view of Clutter table, which comes with default database.

clutter

Field: Add Calculate Selection: Select By Attributes Switch Clear Delete Copy Rows: Insert

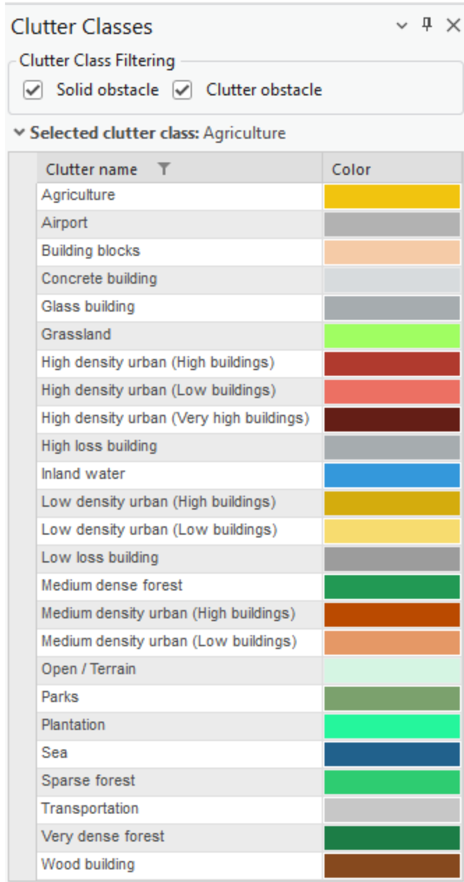
	OBJECTID *	clutter_class_name	height	nominal_distance	surface_refractivity	relative_permitivity	surface_conductivity	express_id	color	ids	clutter_class_alias
1	1	open	0	1	315	5	0.001	<Null>	#d5f5e3	<Null>	Open / Terrain
2	2	grassland	0	27	315	20	0.03	<Null>	#A0FE62	<Null>	Grassland
3	3	sparseForest	0	27	315	13	0.004	<Null>	#2ecc71	<Null>	Sparse forest
4	4	mediumDenseForest	0	27	315	13	0.004	<Null>	#229954	<Null>	Medium dense forest
5	5	veryDenseForest	0	15	315	13	0.004	<Null>	#1C7D46	<Null>	Very dense forest
6	6	lowDensityUrbanLowB...	0	27	315	5	0.001	<Null>	#f7dc6f	<Null>	Low density urban (Low buildings)
7	7	lowDensityUrbanHigh...	0	27	315	5	0.001	<Null>	#d4ac0d	<Null>	Low density urban (High buildings)
8	8	mediumDensityUrbanL...	0	15	365	5	0.001	<Null>	#e59866	<Null>	Medium density urban (Low buildings)
9	9	mediumDensityUrban...	0	15	315	5	0.001	<Null>	#ba4a00	<Null>	Medium density urban (High buildings)
10	10	highDensityUrbanLow...	0	10	315	5	0.001	<Null>	#ec7063	<Null>	High density urban (Low buildings)
11	11	highDensityUrbanHig...	0	10	315	5	0.001	<Null>	#b03a2e	<Null>	High density urban (High buildings)
12	12	highDensityUrbanVery...	0	10	315	5	0.001	<Null>	#641e16	<Null>	High density urban (Very high buildings)
13	13	buildingBlocks	0	15	315	5	0.001	<Null>	#f5cba7	<Null>	Building blocks
14	14	transportation	0	27	315	10	0.002	<Null>	#c7c7c7	<Null>	Transportation
15	15	agriculture	0	27	315	20	0.03	<Null>	#f1c40f	<Null>	Agriculture
16	16	plantation	0	27	315	20	0.03	<Null>	#25f69c	<Null>	Plantation
17	17	parks	0	27	315	20	0.03	<Null>	#7ba16d	<Null>	Parks
18	18	airport	0	27	315	5	0.001	<Null>	#b2b2b2	<Null>	Airport
19	19	sea	0	27	315	80	1	<Null>	#21618c	<Null>	Sea
20	20	inlandWater	0	27	315	80	1	<Null>	#3498db	<Null>	Inland water
21	21	concreteBuilding	0	1	315	5	0.001	<Null>	#d7dbdd	<Null>	Concrete building
22	22	glassBuilding	0	1	315	5	0.001	<Null>	#a6acaf	<Null>	Glass building
23	23	woodBuilding	0	1	315	5	0.001	<Null>	#86491E	<Null>	Wood building
24	24	lowLossBuilding	0	1	315	5	0.001	<Null>	#9C9C9C	<Null>	Low loss building
25	25	highLossBuilding	0	1	315	5	0.001	<Null>	#a6acaf	<Null>	High loss building
	Click to add new row.										

Sentinel-2 clutter classes raster provides information about these classes:

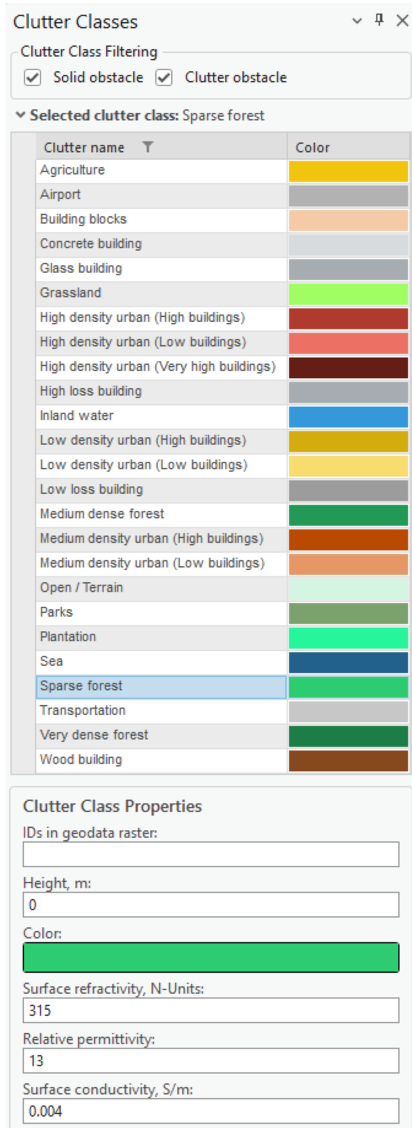
1	Water
2	Trees
4	Flooded...
5	Crops
7	Built Area
8	Bare...
10	Snow/Ice
11	Rangeland

Choose the 

button to open the **Clutter Classes** dialogue.



Select one of the clutter classes to open their properties. Clutter class list can also be filtered by solid obstacle and/or clutter obstacle categories.



New mapping between land use raster and clutter class data should be done in Clutter Classes dialog.

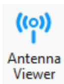
Apply
Saves changes made to clutter classes.

OK
Saves changes made to clutter classes and cloes the dialogue.

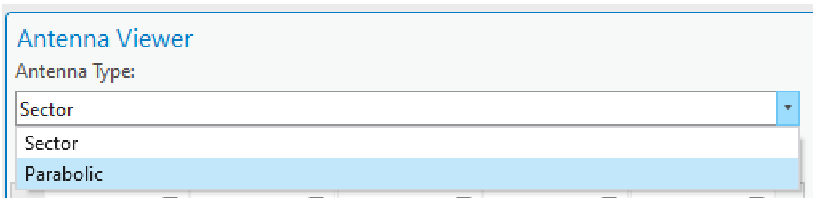
Dismiss
Cancels clutter class changes and closes the dialogue.

7.5 Antenna Viewer

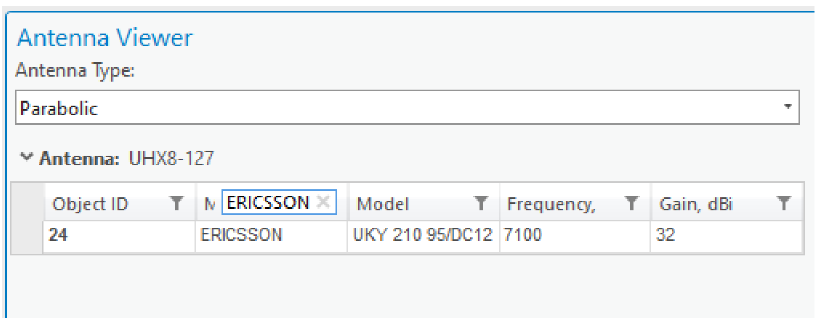
The Antenna Viewer enables the user to preview the default antennae, compare their vertical and horizontal patterns as well as view the values of these patterns in a table.

Click the 

button to open the **Antenna Viewer**. Change Antenna Type to Parabolic.

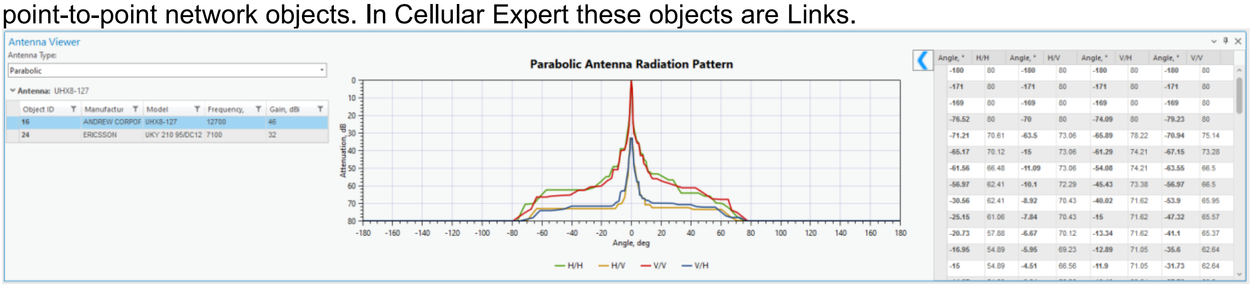


You can filter the entries of the Antenna Table by pressing the **Filter** symbol near one of the field names and entering the desired filter value.



7.5.1 Preview Antenna Patterns

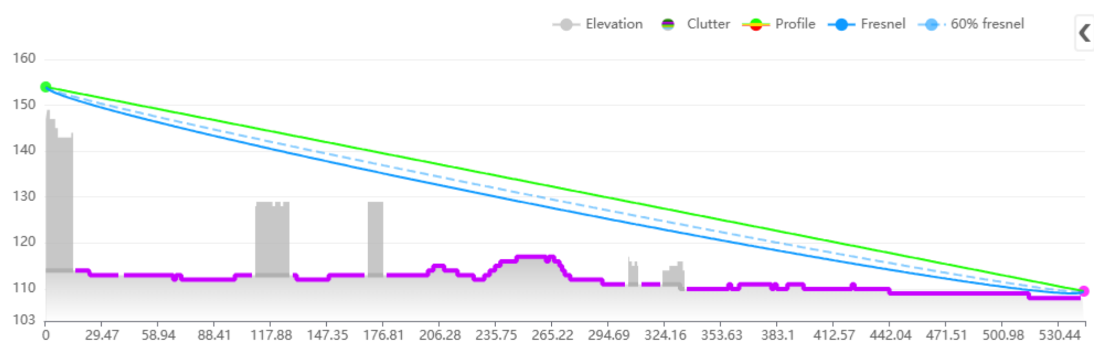
The antenna Type dropdown list contains two antenna types: **Sector** and **Parabolic**. Sector antennas have 2 patterns and pertain to a single network object. Parabolic antennas have 4 patterns and pertain to point-to-point network objects. In Cellular Expert these objects are Links.



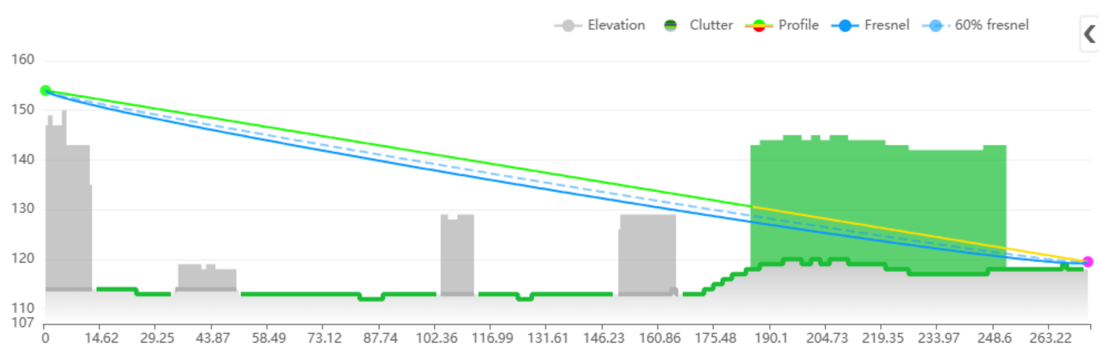
7.6 Prediction Model Manager

The CE Path Loss Modelling aims to perform near-deterministic calculation of received signal levels at each specific point (pixel) in the network's target coverage area by applying selective path loss model depending on the radio visibility condition between the transmitter antenna vis-à-vis a receiver antenna located at a given point in coverage area. The radio visibility is evaluated based on the DTM, Obstacles and Clutter path profile information, as described in previous section. This verification of radio visibility will result in the receiver antenna point assigned into one of three possible radio visibility conditions:

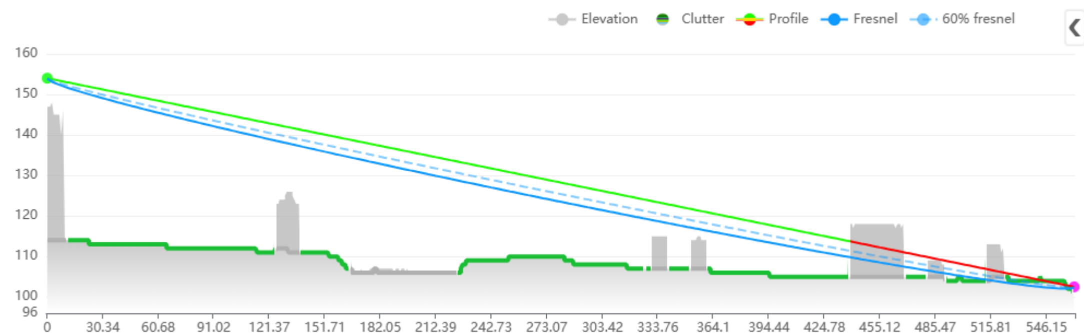
- **Line-of-Sight (LOS)** – occurs when there are neither terrain irregularities, obstacles or clutter interposing the direct radio path between the transmitter and receiver antennas. The radio path is understood to include the 1st Fresnel zone around the direct line and account for Spherical Earth effect. The LOS condition is illustrated by the path profile depicted in Fig. 3(a).
- **Obstructed LOS (OLOS)** – occurs when the direct radio propagation line is interposed by clutter, see illustration in Fig. 3(b).
- **Non-LOS (NLOS)** – occurs when the direct radio propagation line is interposed by terrain bulges or obstacles, see illustration in Fig. 3(c).



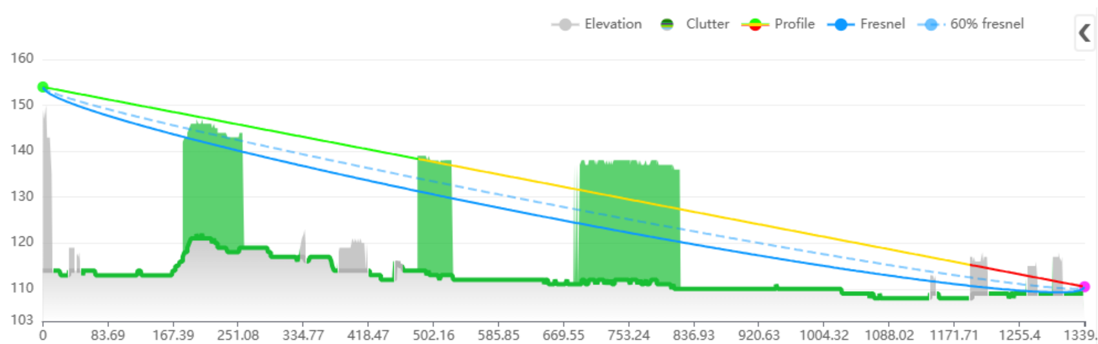
(a) Example of path profile with LOS condition (green line of direct radio link)



(b) Example of path profile with OLOS condition (yellow segment of radio link path)



(c) Example of path profile with NLOS condition (red segment of radio link path)



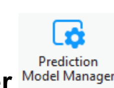
(d) Example of path profile with OLOS+NLOS condition (yellow+red segment of radio link path)

Fig. 3. Illustration of different LOS conditions

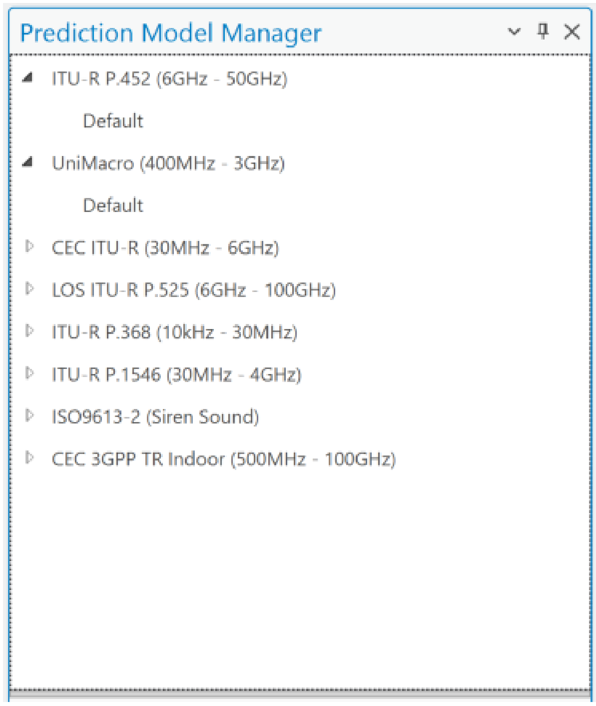
Depending on the LOS condition for the receive antenna at specific location (area map pixel), the CE tools will apply the specific sub-set of path loss prediction model, as explained in the following section.
Note that when the receiver is located indoors, the special Outdoor-to-Indoor propagation function will be applied in addition to basic path loss, as explained in the separate section at the end of this chapter.

7.6.1 Models

Prediction models available in Cellular Expert support frequencies from 10kHz to 350 GHz.

To open the **Prediction Model Manager** dialogue, click on the **Prediction Model Manager** 

tool in the **Data Management** section.



CEC ITU-R 3GPP Model (100MHz – 6GHz) is a combination model intended for use in a variety of different radiocommunication systems which is derived explicitly from ITU-R path loss modelling methods as follows:

a. Receive antenna in LOS condition – path loss calculated as FSL based on Recommendation ITU-R P.525 (ref URL).
b. Receive antenna in OLOS condition – total path loss modelled as a combination of basic FSL calculated based on Recommendation ITU-R P.525 (ref URL) with included dual slope option and clutter loss modelling based on Recommendation ITU-R P.2108 (ref URL).
c. Receive antenna in NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 (ref URL) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 (ref URL).
d. Receive antenna in OLOS+NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 (ref URL) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 (ref URL), and clutter loss modelling based on Recommendation ITU-R P.2108 (ref URL).
e. Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation (if receiver is in vegetation type clutter) or based on 3GPP TR 38.901 (ref URL) (if receiver is in a building).

ITU-R P.452 Model (6GHz – 50GHz) is provided as a universally applicable model with a very wide frequency range from 0.1-50 GHz. Its implementation is based on the methodology described in the Recommendation ITU-R P.452 (ref URL). This model does not provide for definition of OLOS visibility condition; instead, it considers clutter as part of the general obstacles category and accordingly distinguishes only two radio visibility cases:

a. Receive antenna in LOS condition – path loss model based on FSL principle.
b. Receive antenna in NLOS condition – total path loss modelled using a combination of basic transmission losses and losses due to diffraction.

ITU-R P.1546 Model (30MHz – 4GHz) (ref URL) is a widely recognized radio propagation prediction method developed by the **International Telecommunication Union (ITU)**. It is primarily used for estimating **point-to-area radio signal coverage** in the frequency range from **30 MHz to 4000 MHz** over **terrestrial paths**. This model is especially suitable for **broadcasting, land mobile, and fixed services**.

Key Features

- **Versatile Application**: Supports predictions over **land, sea, and mixed paths**, making it adaptable to various geographic conditions.
- **Input Parameters**: Takes into account factors such as **transmitter and receiver heights**, **terrain profile**, **clutter (buildings, vegetation)**, **climate**, and **time/location variability**.
- **Time and Location Variability**: Predictions can be tailored for different statistical reliability levels (e.g., 50% or 10% time availability).
- **Clutter and Terrain Handling**: The model can incorporate detailed **digital elevation models (DEM)** and **clutter data** for more accurate predictions, reflecting the influence of buildings, forests, and other surface features.

Use in Cellular Expert

In the **Cellular Expert** software, the ITU-R P.1546 model is implemented to support **real-world coverage planning** and **regulatory studies**. Users can configure environmental parameters and resolution settings to match local conditions and improve prediction accuracy.

LOS ITU-R P.525 Model (6GHz – 100GHz) is the FSL path loss calculated based on the method in Recommendation ITU-R P.525 (ref URL). As such it could be used for modelling radio links where LOS is considered a necessary condition, e.g., for Fixed (Point-to-Point) Links or Mobile Systems in mmWave bands.

UniMacro Model (400MHz – 3GHz) is the CE's proprietary combination model developed over the years of practical experience with the operational planning of cellular mobile networks in the frequency ranges from 400-3000 MHz. It had been fine-tuned to produce coverage predictions that are most closely aligned with what could be expected to be experienced by the actual mobile network users in the field. The model will model different path losses depending on radio visibility conditions as follows:

a. Receive antenna in LOS condition – path loss model based on FSL principle and dual slope based on breakpoint distance.
b. Receive antenna in OLOS, or NLOS condition – path loss modelled using Extended Hata (Open Area) model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 (ref URL).
c. Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation (if receiver is in vegetation type clutter) or based on 3GPP TR 38.901 (ref URL) (if receiver is in a building).

ITU-R P.368 (10kHz – 30MHz) provides a standardized prediction method for assessing the ground-wave field strength of radio waves in the 10 kHz to 30 MHz frequency range. This frequency band is primarily associated with long-range communication systems using amplitude modulation (AM) and shortwave bands, often for maritime, aeronautical, military, and broadcasting services.

This model offers guidance for engineers, planners, and researchers working on system design and analysis in the MF (Medium Frequency) and HF (High Frequency) bands.

The ITU-R P.368 model calculates signal strength based on several key environmental and system parameters:

Frequency (f)
Higher frequencies tend to attenuate more rapidly over ground. The attenuation rate increases significantly above 3 MHz.

Distance (d)
Field strength diminishes with increasing distance due to geometrical spreading and absorption by the ground and atmosphere.

Surface refractivity, Surface Conductivity (σ) and Relative Permittivity (ε_r)
The surface over which the wave propagates critically affects signal strength:

- Sea water: High conductivity, minimal loss
- Dry land or desert: Low conductivity, high loss
- Typical values range from:
    - Conductivity: 10^{-4} to 5 S/m
    - Relative permittivity: 4 to 81

ISO 9613 standard provides a validated, practical method for predicting the outdoor propagation of sound, and is increasingly applied in siren sound modeling within public security, emergency warning systems, and defense operations. This modeling ensures that acoustic alert systems (e.g. civil defense sirens, disaster warnings, military alert signals) achieve their intended coverage, intelligibility, and effectiveness across various terrain and urban environments.

Purpose in the Public Security Context
In emergency and defense scenarios, reliable audibility of sirens is critical for:

- Civil alert and evacuation systems
- Military base perimeter alarms
- Air raid or missile defense warning networks
- Disaster alert systems (e.g. earthquakes, tsunamis, nuclear incidents)

By applying ISO 9613-2, engineers can model how far a siren can be heard under specific environmental conditions, optimizing:

- Placement and spacing of sirens
- Sound power selection
- Minimization of acoustic shadow zones
- Compliance with national safety and civil defense regulations

The model assumes standard favorable propagation:

- Downwind or moderate inversion conditions
- Ambient temperature ~10 °C
- Relative humidity ~70%

These are conservative conditions ensuring that siren reach is never overestimated, supporting public safety margin planning.
CEC 3GPP TR Indoor (500MHz – 100GHz) Propagation Model is a high-frequency path loss model designed for indoor radiocommunication systems operating within the 500 MHz to 100 GHz range. It builds upon the CEC ITU-R 3GPP model (100 MHz – 6 GHz) by adapting it to complex indoor environments, such

as:

- Office buildings
- Residential units
- Shopping centers
- Industrial halls

This model integrates core **ITU-R recommendations** for free-space loss and penetration effects, while leveraging 3GPP-specific methods for accurate simulation of indoor multipath, wall attenuation, and frequency-dependent fading.

Purpose and Use Cases
The model is intended for:

- Indoor wireless access network design (e.g., Wi-Fi, 5G NR, mmWave)
- System-level simulations for indoor coverage planning
- Performance evaluation of in-building penetration for outdoor base stations
- Integration with dual-slope and multi-scenario path loss modeling frameworks

7.6.1.1 CEC ITU-R 3GPP Model

Model application

This deterministic model is designed for precise tracking of the main, strongest radio ray, while also empirically modeling the scattering of other rays around the receiver. It applies to all ranges of cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within the 30 MHz to 6 GHz frequency range.

The model is recommended for accurate wide-area propagation and coverage modeling, especially when precise and up-to-date topographic data are available. This includes Digital Terrain Models (DTM), building data (including height information), and vegetation data (with height details) derived from a Digital Surface Model (DSM). Ideally, these data should be created with LiDAR or similar methods, at a resolution of at least 10 meters, though 5, 2, or even 1 meter or higher is preferable for optimal accuracy.

When building data and their heights are not available, and only DTM and clutter data at a resolution of 10 meters or lower are accessible, the UniMacro Model should be considered for wide-area propagation and coverage modeling in a slightly narrower frequency range of 400 MHz to 3 GHz.

Default settings

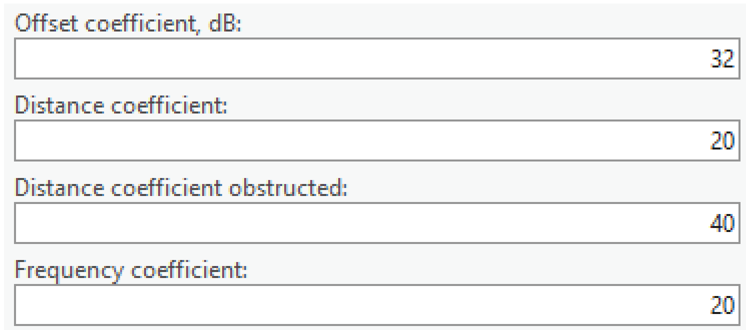
General settings to calculate Model loss

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.

- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

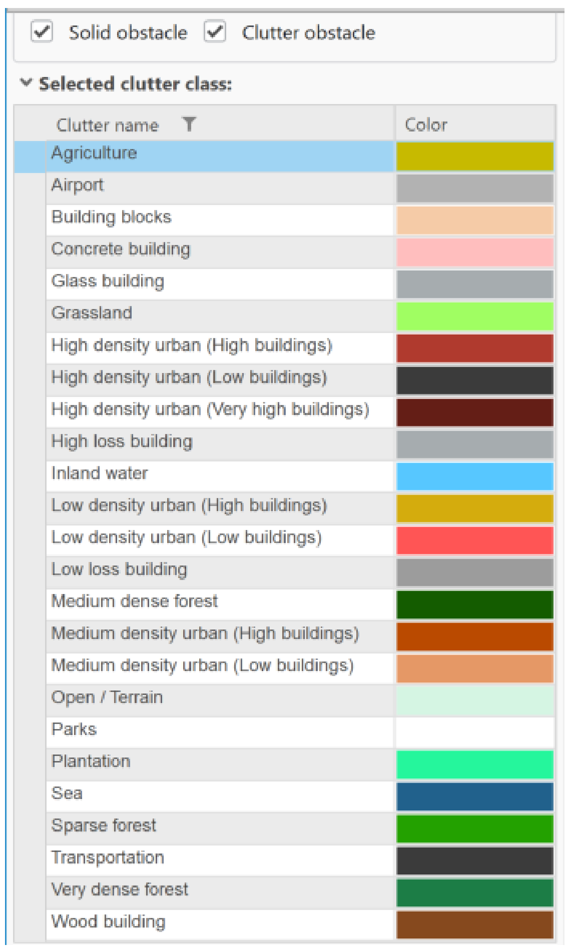
- Distance coefficient obstructed – represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 40.

- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

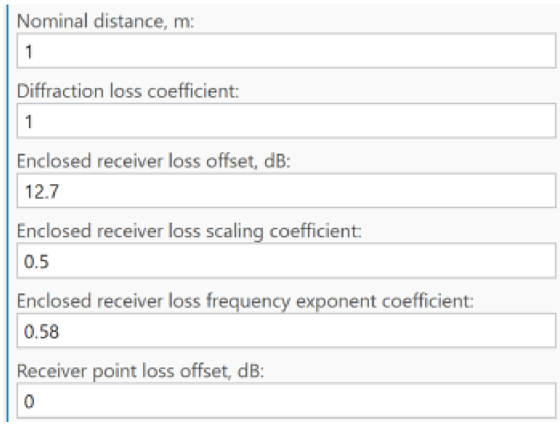


Clutter class to calculate diffraction, clutter loss, penetration loss and receiver loss

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients.



These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.



Key Parameters:

- Nominal distance, m – the average distance between objects within the clutter class, ranging from 1 to 100 meters.
- Diffraction loss coefficient – a multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter types.ltiplier for diffraction calculations. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher then forest or other clutter classes.
- Enclosed receiver loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.
- Enclosed receiver loss scaling coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Enclosed receiver loss frequency exponent coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.
- Receiver point loss offset, dB – an additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

Clutter Classes default values

	Penetration receiver loss offset	**Penetration receiver loss scaling coefficient**	**Penetration receiver loss frequency exponent coefficient**
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65
Medium density urban (Low buildings)	0	0.89	0.65

Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1
Wood building	2	0.25	1
Low loss building	8.5	0.25	1
High loss building	17	0.25	1

7.6.1.2 ITU-R P.452 Model

Model application

This model is designed to estimate radio signal propagation over long distances, including terrestrial paths. It is specifically designed to predict signal attenuation caused by various mechanisms, such as diffraction, tropospheric scatter, ducting, and reflections from the Earth's surface, across frequencies ranging from 0.1 GHz to 50 GHz. While other prediction model covers frequencies from 100MHz to 6GHz, we recommend to use it from 6GHz to 50GHz frequencies.

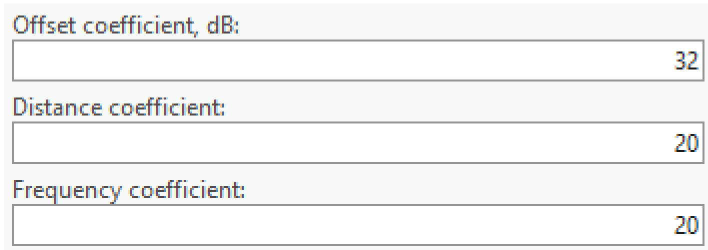
This model is particularly well-suited for microwave links and is widely used for planning and interference analysis in fixed and mobile radio communication systems. By accounting for the effects of terrain, atmospheric conditions, and other factors, it helps engineers assess link reliability and optimize network performance in a variety of environmental scenarios.

Default settings

General settings to calculate Model loss

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.

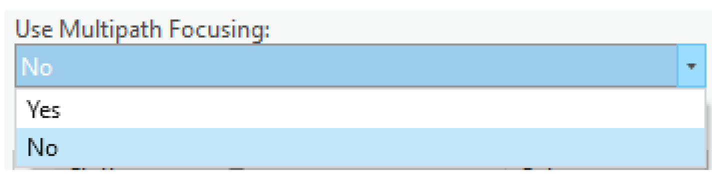
- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.



Multipath and focusing

In ITU-R P.452, the correction for multipath and focusing effects accounts for signal enhancements caused by constructive interference and atmospheric focusing. This adjustment reduces the total path loss under favorable conditions, such as over-water paths or specific atmospheric gradients, ensuring more accurate signal predictions.

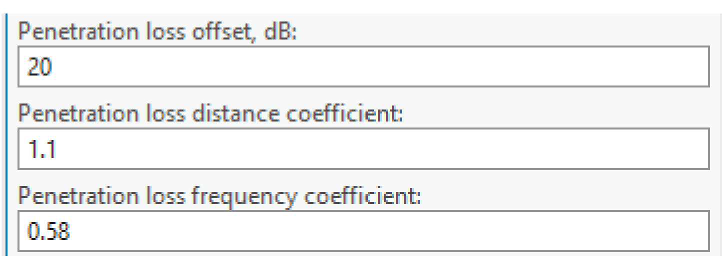
Possible values Yes or No.



Clutter class to calculate penetration loss

Clutter class option describes several fixed Clutter names with penetration loss coefficients.

These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.



Key Parameters:

- Penetration loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.

- Penetration loss distance coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Penetration loss frequency coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.

Clutter Classes default values

	Penetration receiver loss offset	Penetration receiver loss scaling coefficient	Penetration receiver loss frequency exponent coefficient
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65
Medium density urban (Low buildings)	0	0.89	0.65
Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1
Wood building	2	0.25	1
Low loss building	8.5	0.25	1
High loss building	17	0.25	1

7.6.1.3 UniMacro Model

Model application

This model is designed for deterministic tracking of the main, strongest radio ray in Line of Sight (LOS) areas, while propagation modeling in Obstructed Line of Sight (OLOS) and Non-Line of Sight (NLOS) areas uses empirically determined parameters defined in ITU-R and 3GPP recommendations. It also models the scattering of other rays around the receiver. The model applies empirically validated values for the 400 MHz to 3 GHz frequency range and is suitable for modeling all cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within that frequency range.

This model is recommended for wide-area propagation and coverage modeling when building data and their heights are unavailable, and only DTM and clutter data at a resolution of 10 meters or lower are accessible.

For accurate wide-area propagation and coverage modeling where building data and their heights are available, the CEC ITU-R 3GPP Model is recommended, offering a broader application frequency range of 100 MHz to 6 GHz.

Default settings

General settings to calculate Model loss

If Tx and Rx is in LOS condition:

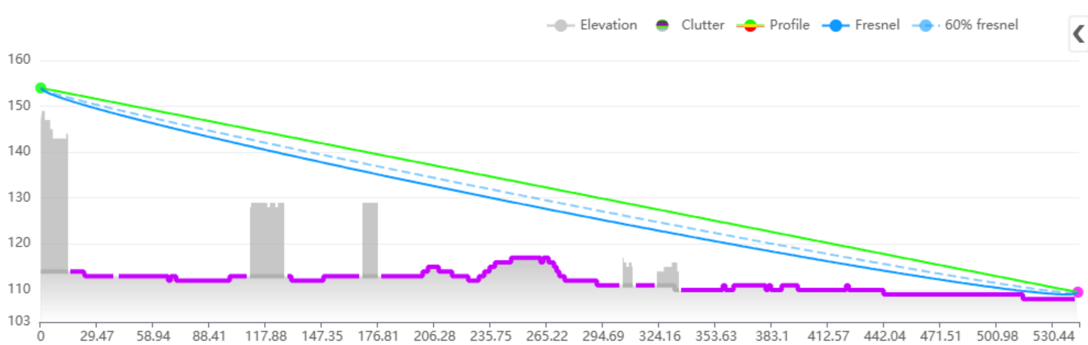


Fig. 4. Illustration of LOS conditions

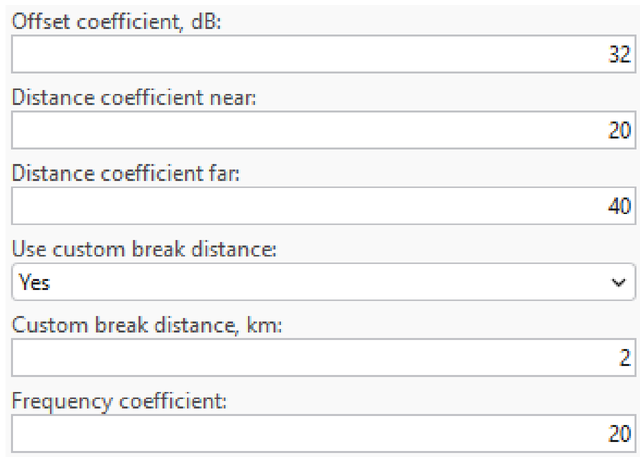
Line of Sight coefficients are used to calculate general model loss.

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient near – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Distance coefficient far – represents the slope based on breakpoint distance between the cell and the receiver location. The default value is 40.
- Use custom break distance – if value Yes, enables custom Fresnel breakpoint distance value. The path loss dependence on the distance is split into near and far zones by the breakpoint distance

and the effect is only applied for Line of Sight condition.

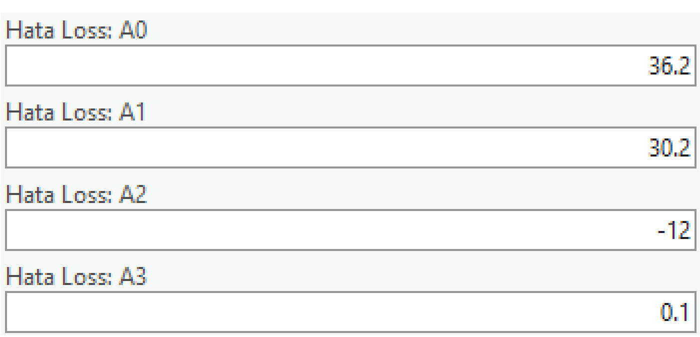
- Custom break distance, km – Fresnel breakpoint distance then path loss is calculated using Distance coefficient far parameter.

- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.



If Tx and Rx is in OLOS or NLOS condition, then Hata 9999 equation is used. 9999 Model is the Ericsson's implementation of Hata Model. Ericsson provides the steering parameters of 9999 Model for different environments; therefore it's very convenient just to apply in this form as the default parameters.

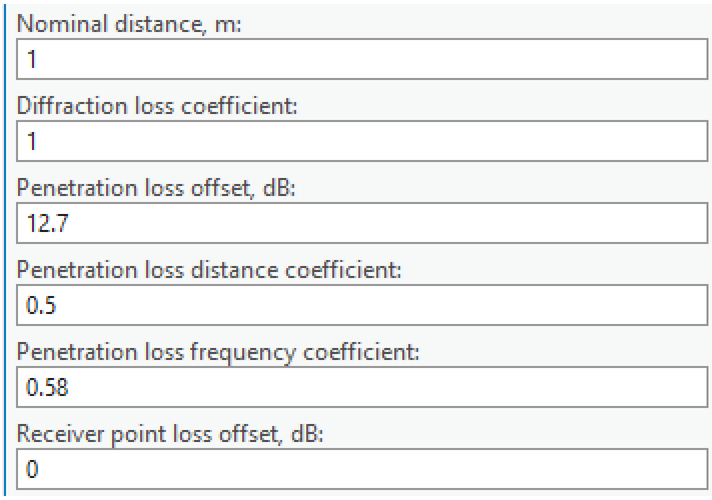
- Hata Loss: A0 - constant offset in dB this value simply added to loss grid. Adjusting this value, you can minimize mean error. It regulates the absolute level of the loss curve. Default value 36.2.
- Hata Loss: A1 - distance influence coefficient. Physically it represents loss dependant on distance such as atmospheric (dust, hydrometeors, etc...) losses. It regulates slope of the curve. Default value 30.2.
- Hata Loss: A2 - transmitter height influence coefficient. It is related to errors in DTM, real Earth curvature, etc. It regulates loss curve vertical position like the A0, but with respect to antenna height. Default value -12.
- Hata Loss: A3 - Okumura-Hata type of multiplying factor for log(h_M)log(d). Default value 0.1.



Clutter class to calculate diffraction, clutter loss, penetration loss and receiver loss

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients.

These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.



Key Parameters:

- Nominal distance, m – the average distance between objects within the clutter class, ranging from 1 to 100 meters.
- Diffraction loss coefficient – a multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter types.ltiplier for diffraction calculations. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher then forest or other clutter classes.
- Penetration loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.
- Penetration loss distance coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Penetration loss frequency coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.
- Receiver point loss offset, dB – an additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

Clutter Classes default values

	Penetration receiver loss offset	**Penetration receiver loss scaling coefficient**	**Penetration receiver loss frequency exponent coefficient**
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65

Medium density urban (Low buildings)	0	0.89	0.65
Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1
Wood building	2	0.25	1
Low loss building	8.5	0.25	1
High loss building	17	0.25	1

7.6.1.4 LOS ITU-R Model

Model application

Line of Sight model is typically used for mmWave band frequencies within the 6 GHz – 100 GHz frequency range and provides results only for line-of-sight areas.

Default settings

General settings to calculate Model loss

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

7.6.1.5 ITU-R P.368 Model

Model application

ITU-R P.368 is ground-wave propagation of radio signals model. It is specifically designed to estimate the field strength and attenuation of radio waves over the Earth's surface, particularly for frequencies below 30 MHz.

This model is widely used in planning and designing long-distance communication systems, such as maritime, broadcasting, and low-frequency navigation systems, where ground-wave propagation plays a critical role. It accounts for factors such as terrain conductivity, dielectric properties, and surface roughness to deliver accurate predictions of signal behavior.

Default settings

The general model parameters include the radius and receiver height. Additional parameters used for path loss calculations are derived from the clutter classes. Each clutter class has its own unique set of parameters:

- Surface refractivity - a measure of the refractive index's influence on electromagnetic wave propagation, particularly in the lower atmosphere close to the Earth's surface. It is expressed as a dimensionless value, typically dependent on atmospheric pressure, temperature, and humidity. High surface refractivity can significantly affect radio wave bending and propagation, such as ducting or anomalous refraction.
- Relative permittivity - quantifies a material's ability to permit electric field propagation relative to vacuum. It is a complex quantity with the real part representing energy storage capability and the imaginary part representing energy dissipation within the material.
  ITU-R P.368 uses relative permittivity to model how radio waves interact with various surface materials, such as soil, water, or vegetation. These interactions influence reflection, refraction, and absorption phenomena at the surface.
- Surface conductivity - refers to a material's ability to conduct electrical currents across its surface. It is measured in siemens per meter (S/m). Higher conductivity indicates that a surface can easily allow current flow, affecting the reflection and absorption of radio waves.
  According to ITU-R P.368, surface conductivity is a critical factor in determining the reflective properties of surfaces, such as dry versus wet soil, or metallic versus dielectric surfaces.

	Buildings	**Forest**	**Dense forest**	**Urban**	**Dense urban**	**Bare ground**	**Crops**	**Water**	**Road**
Surface refractivity, N-	315	315	315	315	315	315	315	365	315

Units									
Relative permittivity	5	13	13	5	5	10	20	80	10
Surface conductivity, S/m	0.001	0.004	0.004	0.001	0.001	0.002	0.03	1	0.002

7.6.1.6 ITU-R P.1546 Model

Model application

This model is designed for wide-area radio propagation prediction based on empirical data and statistical analysis of measured field strengths. It provides path loss estimations over land, sea, and mixed terrain for frequencies ranging from 30 MHz to 3 GHz, using parameters defined in ITU-R Recommendation P.1546. The model is applicable to terrestrial broadcasting, mobile, and public safety networks, including technologies such as 2G, 3G, and 4G, within its frequency range.

The ITU-R P.1546 model accounts for antenna heights, terrain elevation (DTM), land cover types (clutter), and environmental conditions, incorporating corrections for time variability and location-specific effects. It is particularly suitable for modeling line-of-sight (LOS) and non-line-of-sight (NLOS) propagation over long distances where detailed building data is not available.

This model is recommended for national or regional coverage planning, especially in cases where high-resolution DTM (e.g., 30m) and clutter data (e.g., 10m resolution) are available, but building heights and detailed 3D structures are not. Its statistical approach allows for reliable estimations of field strength in both urban and rural environments, without requiring ray-tracing or detailed geometry-based modeling.

For scenarios where accurate building geometry and heights are available and a higher modeling frequency range (up to 6 GHz) is required, the CEC ITU-R 3GPP Model is recommended instead, offering greater precision in dense urban and high-frequency applications.

Default settings

The following key parameters are used to configure the ITU-R P.1546 radio propagation predictions within the software. These settings directly influence the coverage calculation and modeling accuracy:

1. **Radius**

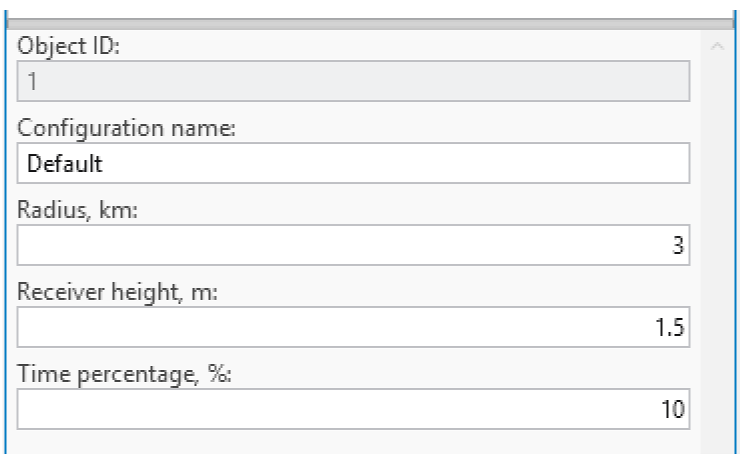
- Description: This setting defines the maximum distance from the transmitter (or prediction center) over which the propagation prediction will be performed.
- Purpose: It limits the spatial extent of the coverage area to optimize calculation time and resource usage.
- Typical Values: Often set between 10 km to 100 km, depending on the transmitter's power, terrain, and target coverage region.

2. **Receiver Height (m)**

- Description: Specifies the height of the receiving antenna above ground level, in meters.
- Purpose: Receiver height impacts the predicted signal strength, especially in terrain with elevation changes or obstacles.
- Guidelines:
  - 1.5 to 2 m for handheld/mobile users (e.g., mobile phones, public safety devices).
  - 10 m or higher for fixed installations (e.g., rooftop or vehicular antennas).
- Note: Accurate setting of receiver height is essential for meaningful signal level predictions.

3. **Time Percentage (%)**

- Description: Indicates the percentage of time during which the predicted field strength is expected to be met or exceeded.
- Purpose: Reflects the statistical variability of signal propagation due to atmospheric and environmental effects.
- Common Use Cases:
  - 50% time: Typical for general service coverage maps (median conditions).
  - 10% time: Used for high-reliability or interference studies, ensuring signal presence under less favorable conditions.
- Interpretation: A 10% time prediction means the signal level is met or exceeded during 10% of the time, capturing worst-case propagation conditions (i.e., stronger signal occurrence during favorable propagation).



7.6.1.7 ISO9613 Model

Model application

ISO 9613 is an international standard used for predicting outdoor sound propagation, including the assessment of siren noise levels. It provides a structured method for calculating sound attenuation over distance, considering factors such as geometric spreading, atmospheric absorption, ground effects, reflections, and obstacles.

In the context of siren sound prediction, ISO 9613 helps determine the effective coverage area, ensuring that warning signals reach the intended audience with sufficient audibility. This standard is essential for optimizing siren placement, regulatory compliance, and designing effective emergency alert systems.

The primary factors included in the standard are:

1. Geometric Spreading

- This refers to how sound spreads out as it moves away from its source. Sound intensity decreases as the distance from the source increases, following the inverse square law (with spherical spreading) or other forms depending on terrain.
- As the distance from the sound source increases, the intensity of sound diminishes, which is considered in the calculation of sound levels at various receiver points.

$$A_{div} = 20 * LOG10(d) + 11$$

2. Atmospheric Absorption

- Sound is absorbed by the atmosphere as it travels, particularly at higher frequencies. The absorption depends on factors like temperature, humidity, and air pressure, and is often more significant over longer distances.
- Atmospheric absorption reduces the intensity of sound as it propagates, especially at higher frequencies. The standard takes into account these effects to calculate the reduction in sound level.

3. Obstacles

- Physical barriers such as walls, hills, and buildings block or scatter sound waves, reducing the sound level that reaches certain areas.
- Obstacles can cause significant sound reduction, especially if they are large or located between the sound source and the receiver. The standard accounts for the shadow zones created by these barriers.

4. Directivity of the Source

- This parameter accounts for the directionality of the sound source, which may not emit sound equally in all directions. Sirens, for example, may have directional characteristics that focus their sound output in certain directions.
- The directivity of the sound source influences how the sound energy is distributed, and therefore, how far and in what pattern the sound propagates.

5. Meteorological Conditions

- Weather conditions such as wind speed, temperature gradients, and humidity can significantly affect sound propagation. For example, sound may travel farther downwind or be absorbed more by humid air.
- These conditions are integrated into the model to adjust the calculations of sound attenuation, ensuring more accurate predictions for different weather scenarios.

Default settings

- **Distance coefficient: 20.**

  Describes slope coefficient based on distance. Value is included in Geometric spreading calculations.

- **Temperature: 20 Cº**
- **Humidity: 50%**
- **Meteorological conditions: 3 dB.** It is a factor, in decibels, which depends on local meteorological statistics for wind speed and direction, and temperature gradients. Experience indicates that values of Meteorological conditions (C0) in practice are limited to the range from zero to approximately + 5 dB.

Ground factor parameter for the ISO9613-2 model's clutter classes allows users to define the acoustic reflectivity (from 0 to 1, indicating hard-soft respectively) of the ground surface for more accurate sound level predictions.

Clutter Classes default values

	Ground factor
Open / Terrain	1
Grassland	1
Sparse forest	1
Medium dense forest	1
Very dense forest	1
Low density urban (Low buildings)	0.5
Low density urban (High buildings)	0.3
Medium density urban (Low buildings)	0.3
Medium density urban (High buildings)	0.1
High density urban (Low buildings)	0.1
High density urban (High buildings)	0
High density urban (Very high buildings)	0
Building blocks	0
Transportation	0
Agriculture	1
Plantation	1
Parks	0.7
Airport	0

Sea	0
Inland water	0
Concrete building	0
Glass building	0
Wood building	0
Low loss building	0
High loss building	0

7.6.1.8 CEC 3GPP TR Indoor (500MHz – 100GHz)

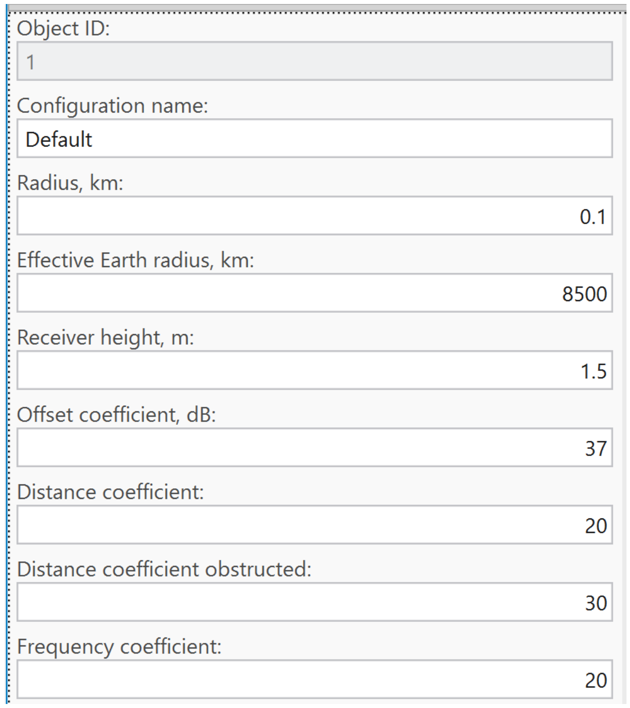
Model application

The **CEC 3GPP TR Indoor Model** (500 MHz – 100 GHz) is a robust, scalable prediction method based on ITU-R principles and 3GPP extensions, specifically engineered for indoor radiowave propagation. By supporting a wide range of scenarios — from clean LOS corridors to deeply obstructed NLOS paths — it enables highly accurate modeling of next-generation wireless systems, ensuring reliable, secure, and efficient communication in the most demanding indoor environments.

Key Enhancements for Indoor Use

- **Frequency scaling up to 100 GHz** supports mmWave and terahertz
- **Wall material database** from 3GPP TR 38.901:
    - Standard drywall: ~2–8 dB per wall
    - Concrete: 5–15 dB
    - Glass: 2–10 dB
- **Multi-floor attenuation** (floor penetration factor)
- **Path loss floors**: ensures minimum attenuation beyond near field
- **LOS probability models**: stochastic treatment of LOS in large buildings

Default settings



Configuration name
Name of the prediction configuration name.

Radius
Maximum prediction radius in kilometers to calculate path loss.

Receiver height
Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius
Earth radius in kilometers, used for the calculations.

Offset coefficient
Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient
Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Distance coefficient obstructed
Represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 30.

Frequency coefficient
Indicates the slope determined by the frequency value, with a default value of 20.

Penetration loss offset
The initial entry loss applied when crossing an obstacle within the clutter class, expressed as an offset in

dB, which is added to the path loss grid.

Penetration loss scaling coefficient
Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss.

Penetration loss frequency exponent coefficient
Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.

Note: penetration losses stack when multiple obstacles are crossed.

Clutter Classes default values

	Penetration receiver loss offset	Penetration receiver loss scaling coefficient	Penetration receiver loss frequency exponent coefficient
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65
Medium density urban (Low buildings)	0	0.89	0.65
Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1

Wood building	2	0.25	1
Low loss building	8.5	0.25	1
High loss building	17	0.25	1

7.6.1.9 CNOSSOS-EU Model

Model application

Developed by the European Commission's Joint Research Centre as the Common Noise Assessment Methods in Europe, CNOSSOS-EU is the reference methodology adopted under the Environmental Noise Directive (2002/49/EC) for strategic noise mapping across EU Member States, making it the preferred choice for projects that must align with European regulatory requirements.

Unlike ISO 9613-2, which assumes downwind or moderate temperature inversion conditions throughout, CNOSSOS-EU evaluates sound propagation under two distinct meteorological scenarios — favourable conditions (downwind or temperature inversion, which enhance propagation toward the receiver) and homogeneous conditions (neutral atmosphere) — and combines the two results using the long-term occurrence probability of favourable conditions for the given direction. This produces long-term average sound pressure levels that more realistically reflect the variability of outdoor propagation over extended periods.

The model also features more refined calculations of geometrical divergence, atmospheric absorption, ground effect, diffraction over obstacles and around vertical edges, and reflections from building façades and other vertical surfaces. Ground effect is handled separately for the source-side, middle, and receiver-side regions of the propagation path, with acoustic reflectivity defined per clutter class via the Ground factor parameter shared with the ISO 9613-2 model. Diffraction is computed using path-difference geometry consistent with the CNOSSOS-EU specification, and terrain profiles, clutter classification, and building data from the workspace are used directly in the calculation. The primary factors included in the standard are:

Default settings

- **Distance coefficient: 20.**

Describes slope coefficient based on distance. Value is included in Geometric spreading calculations.

- **Temperature: 15 C°**
- **Humidity: 70%**
- **Favourable conditions occurrence: 0.25:** Probability or fraction of time during which meteorological conditions are "favourable" for sound propagation from a source to a receiver.

Ground factor parameter for the CNOSSOS-EU model’s clutter classes allows users to define the acoustic reflectivity (from 0 to 1, indicating hard-soft respectively) of the ground surface for more accurate sound level predictions. When the clutter classes raster is absent from the geodata, a ground factor value of 0.5 is used.

Clutter Classes default values

	Ground factor
Open / Terrain	1

Grassland	1
Sparse forest	1
Medium dense forest	1
Very dense forest	1
Low density urban (Low buildings)	0.5
Low density urban (High buildings)	0.3
Medium density urban (Low buildings)	0.3
Medium density urban (High buildings)	0.1
High density urban (Low buildings)	0.1
High density urban (High buildings)	0
High density urban (Very high buildings)	0
Building blocks	0
Transportation	0
Agriculture	1
Plantation	1
Parks	0.7
Airport	0
Sea	0
Inland water	0
Concrete building	0
Glass building	0
Wood building	0
Low loss building	0
High loss building	0

7.7 Template Manager

Template Manager allows the user to edit current project templates residing in the various Template tables of the *default.gdb*. The user may change the field values of the templates, create new or detele existing templates.

Templates are essential tools designed to streamline and simplify the configuration process by predefining parameters automatically. Their primary purpose is to save time and reduce the risk of human error during setup, particularly in complex scenarios like wireless network configuration.
Here's why templates are beneficial:

1. **Automatic Parameter Filling**: Templates automatically populate required parameters, eliminating

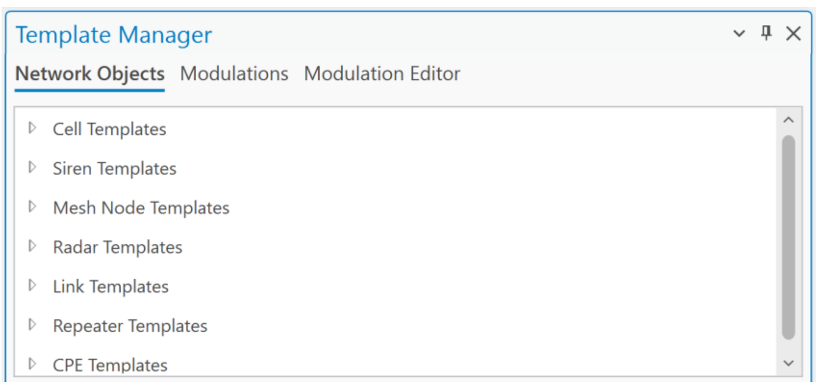
the need for users to input them one by one. This ensures consistency and speeds up the setup process, especially when working with large datasets or multiple network layers, such as cells, sites, or CPEs.

2. **Error Prevention**: In cases where a parameter might be inadvertently missed during manual configuration, templates act as a safeguard. They ensure all necessary parameters are accounted for, reducing the likelihood of incomplete or incorrect setups.
3. **Coverage Prediction Assurance**: Templates are particularly valuable when performing tasks like coverage predictions. If a parameter in the wireless network layer is forgotten or overlooked (e.g., for a cell, site, or CPE), the template provides a fallback, ensuring that the prediction can still be performed accurately and without interruption.

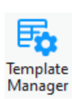
By using templates, users can maintain efficiency, accuracy, and reliability in network planning and analysis, even in "just-in-case" scenarios where manual inputs might fall short.

The templates are divided into categories:

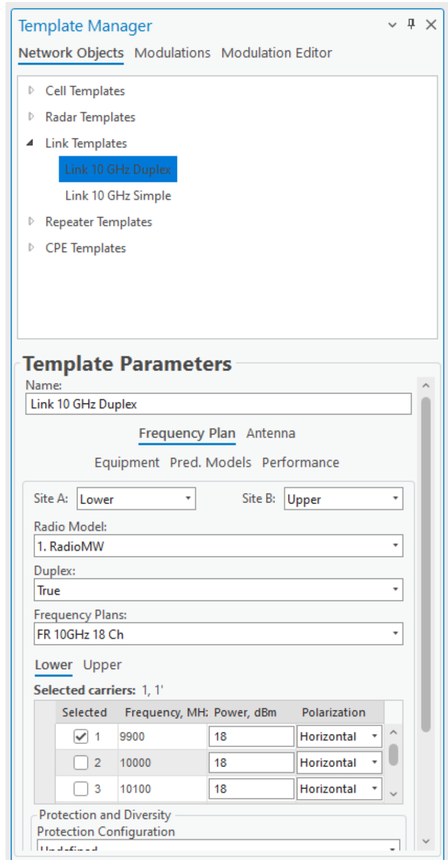
- Network Objects – edit and manage network objects (cells, sites, links, etc) templates. Templates are divided into different network layers, each network have unique template structure based on available parameters in network layer.
- Modulations – create modulation configurations that can be used for MW links > Radios.
- Modulation Editor – create single modulations, which can be used in the Modulations tab.



7.7.1 Edit Network Objects template

Click the 

button to open the **Template Manager** dialogue. Select one of the opened templates to edit them.

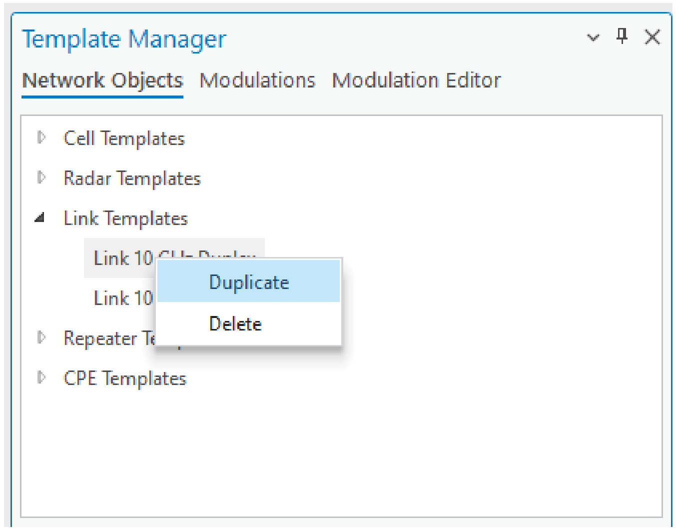


Save Changes
Saves the changes to the objects.

Dismiss
Cancels the changes to the objects and closes the dialogue.

7.7.2 Manage Network Object Template

Right-click on a selected template to open the context menu with Duplicate and Delete options.

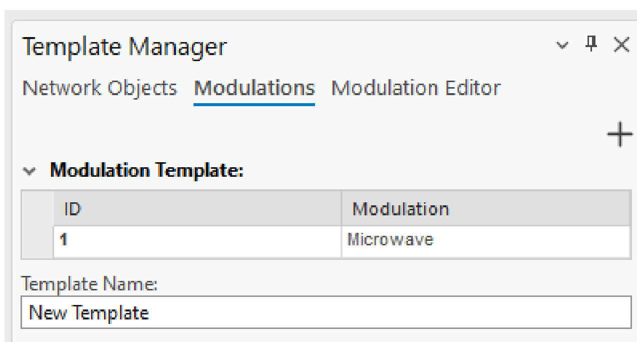


Duplicate will create a new template with the same parameters.

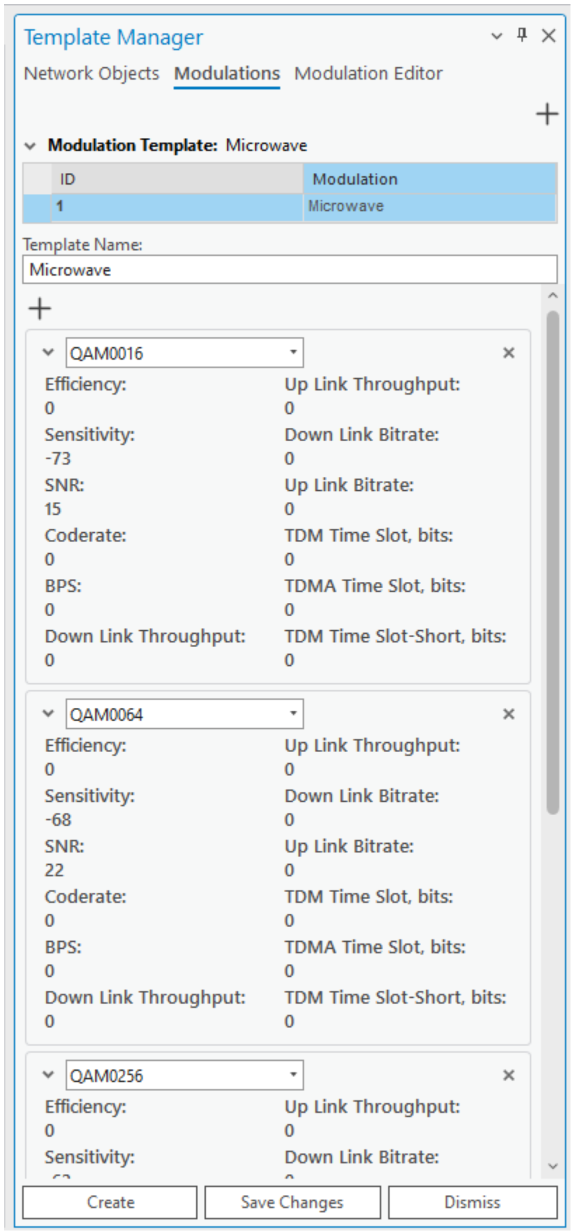
Delete will remove this selected template.

7.7.3 Modulations

The modulations are used in MW link calculations and specifically can be defined for Radios. Instead of specifying Modulations one by one, the customer can create a set of them, and it is available in this tab.



To preview and edit the Modulations template, click on it in the table.



 (top right)
A new modulation with default values can be initialized.

X
Modulation can be removed.

Save changes
All changes will be saved.

Dismiss

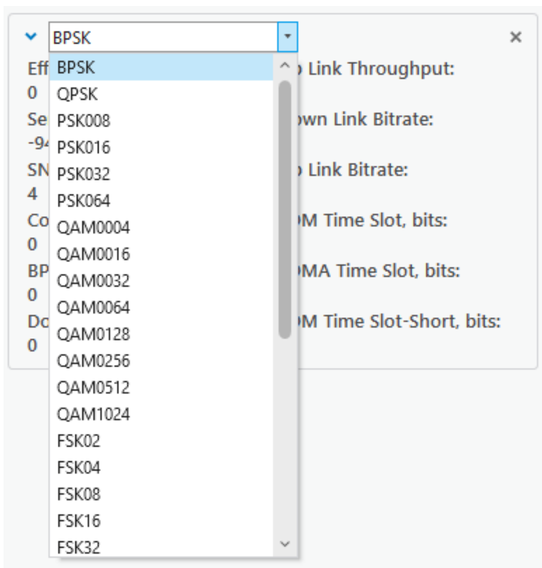
Remove changes.

To create a new Modulation template, press + button in the top right corner of the dialog to initialize a new template.

Template name
The name of the modulation template.

\+ (under the template name field)
Assign modulation to the template.

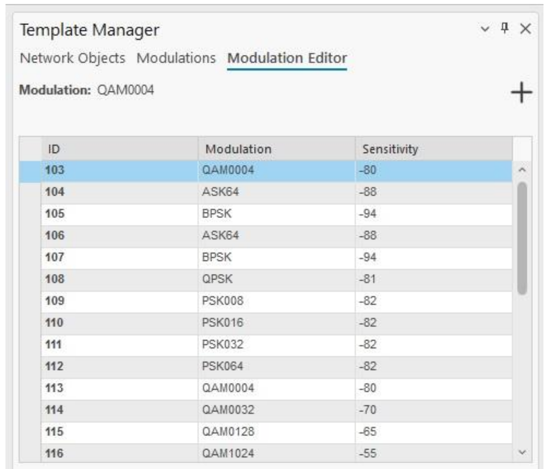
Use the modulation drop-down list menu to apply and include the modulations.



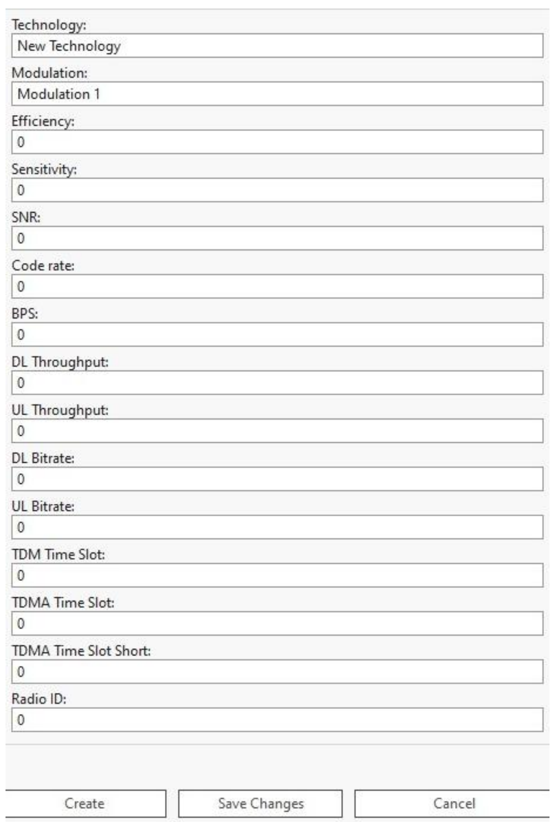
Once the changes are made, click the Create button at the bottom of the window to create a new modulation template.

7.7.4 Modulation Editor

The list of available modulations is taken from the Modulation Editor tab.



To add a new modulation, click the  at the top right of the Modulation Editor window to initialize a new modulation with default values, and press the Create button at the bottom of the window. Alternatively, click on an existing one to edit its parameters.



Create
Create a new modulation with the specified parameters.

Save Changes
Apply changes to a currently selected modulation. The button is disabled if the modulation is not created.

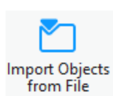
Cancel
Dismiss changes made to the modulation.

7.8 Import Objects

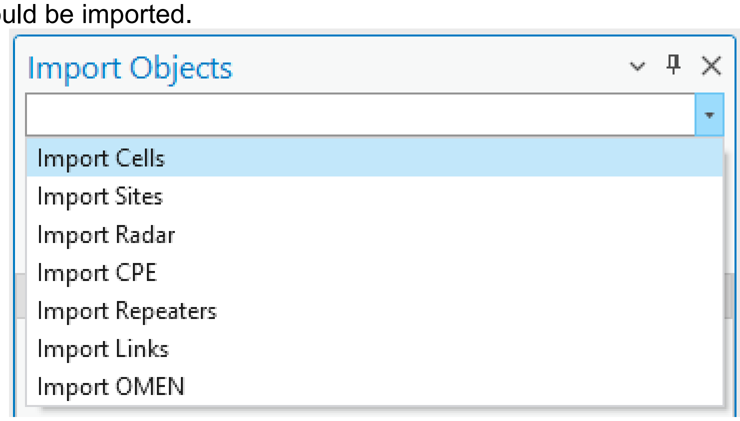
The Import Objects feature further enhances efficiency by allowing users to bring in network objects from multiple external documents without manually creating them. This capability is invaluable when dealing with extensive or complex datasets. Cellular Expert for ArcGIS Pro supports the import of three widely used file formats: .xls, .xlsx, and .csv. Additionally, for mapping files, it supports the .json format.

Key benefits of the Import Objects feature include:

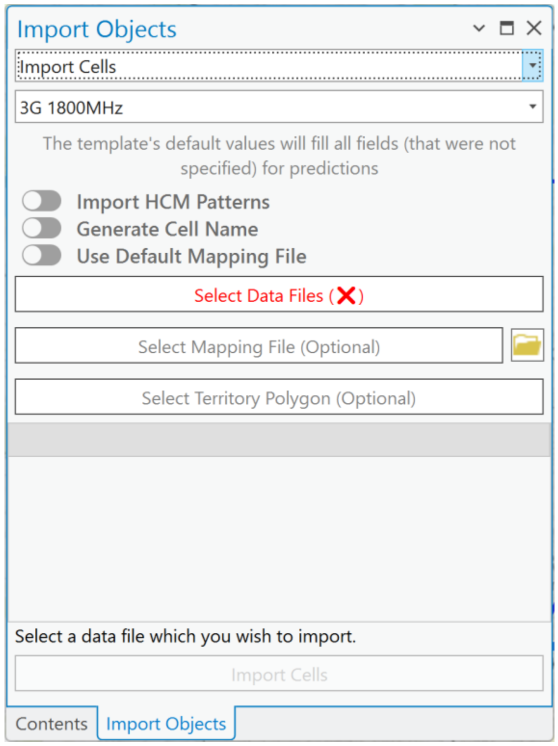
- Time-Saving: Instead of manually entering or recreating network objects, users can quickly import data from existing documents, significantly reducing setup time.
- Data Consistency: Importing ensures that all network objects maintain consistency with the original data source, minimizing discrepancies and errors.
- Support for Multiple Formats: With support for commonly used file formats like .xls, .xlsx, and .csv, the tool is versatile and can integrate seamlessly with various data workflows.

Click the 

button to open the **Import Objects** dialogue. Expand Import drop-down menu and select the object, which should be imported.



The dialog will be filled with the options to define data and mapping files.



7.8.1 Import Cells

The option enables possibility to import Cells in the Cellular Expert workspace. It has additional parameters compared to other import options.

Template
Take necessary parameters from the template during the import. Template values are taken if some parameters are missing in text file and mapping file.

Import HCM Patterns
Antenna patterns would be created and imported based on specific antenna name values provided in HCM agreement. HCM antenna pattern import requires the data files to have an *antenna_type* field.

Generate Cell Name
Generate a cell name for all selected cells based on these parameters in this exact order: "longitude", "latitude", "height", "azimuth", "power", "antenna_gain", and "frequency". If any of the fields are missing, that field will be skipped.

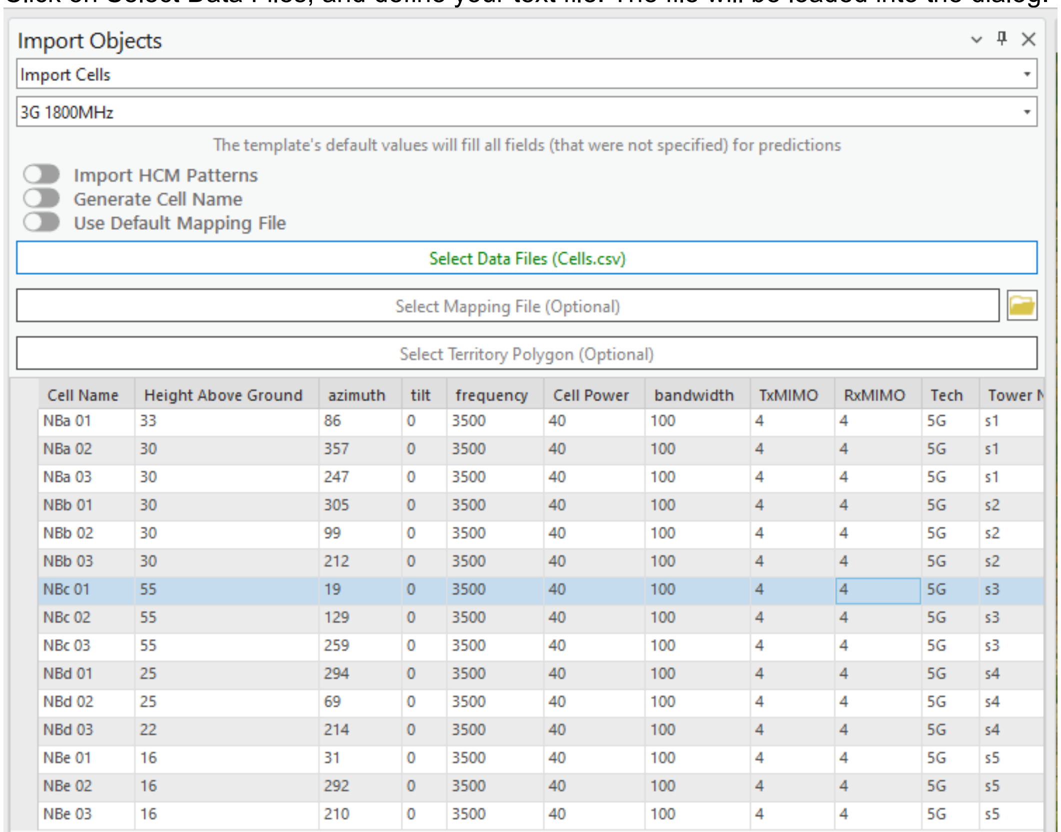
Use Default Mapping File
A default mapping file will be applied to the imported data.

Select Data Files
Opens a dialogue window where the user can select the files that define the network objects to be imported. The supported formats: *.xls*, *.xlsx*,*.csv,* and tables from *.sde* connection. Upon successful selection, the button will light up green.

Select Mapping File (Optional)
Opens a dialogue window where the user can select a file that defines the data to be imported and the conditions under which said data is processed. The supported format is *.json*. Upon successful selection, the button will light up green. More on mapping files are below.

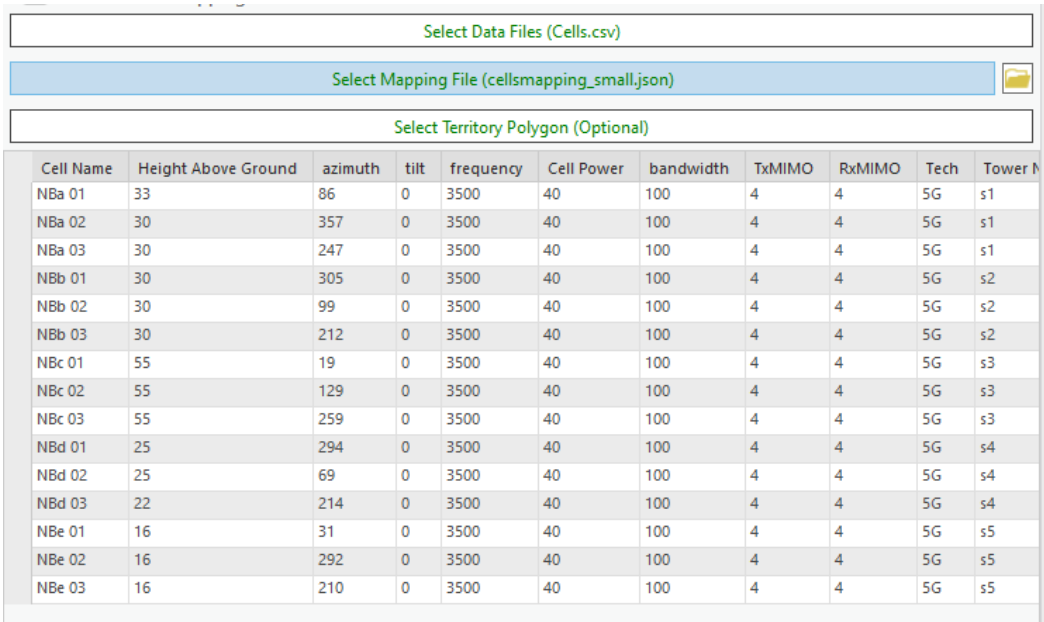
Steps.

1. Click on Select Data Files, and define your text file. The file will be loaded into the dialog.



Cell Name	Height Above Ground	azimuth	tilt	frequency	Cell Power	bandwidth	TxMIMO	RxMIMO	Tech	Tower N
NBa 01	33	86	0	3500	40	100	4	4	5G	s1
NBa 02	30	357	0	3500	40	100	4	4	5G	s1
NBa 03	30	247	0	3500	40	100	4	4	5G	s1
NBb 01	30	305	0	3500	40	100	4	4	5G	s2
NBb 02	30	99	0	3500	40	100	4	4	5G	s2
NBb 03	30	212	0	3500	40	100	4	4	5G	s2
NBc 01	55	19	0	3500	40	100	4	4	5G	s3
NBc 02	55	129	0	3500	40	100	4	4	5G	s3
NBc 03	55	259	0	3500	40	100	4	4	5G	s3
NBd 01	25	294	0	3500	40	100	4	4	5G	s4
NBd 02	25	69	0	3500	40	100	4	4	5G	s4
NBd 03	22	214	0	3500	40	100	4	4	5G	s4
NBe 01	16	31	0	3500	40	100	4	4	5G	s5
NBe 02	16	292	0	3500	40	100	4	4	5G	s5
NBe 03	16	210	0	3500	40	100	4	4	5G	s5

2. If your data structure is different compared to Cellular Expert database, then Mapping file should be used to map your data structure and Cellular Expert workspace, Cells layer structure. More information about mapping file is available below.



Cell Name	Height Above Ground	azimuth	tilt	frequency	Cell Power	bandwidth	TxMIMO	RxMIMO	Tech	Tower N
NBa 01	33	86	0	3500	40	100	4	4	5G	s1
NBa 02	30	357	0	3500	40	100	4	4	5G	s1
NBa 03	30	247	0	3500	40	100	4	4	5G	s1
NBb 01	30	305	0	3500	40	100	4	4	5G	s2
NBb 02	30	99	0	3500	40	100	4	4	5G	s2
NBb 03	30	212	0	3500	40	100	4	4	5G	s2
NBc 01	55	19	0	3500	40	100	4	4	5G	s3
NBc 02	55	129	0	3500	40	100	4	4	5G	s3
NBc 03	55	259	0	3500	40	100	4	4	5G	s3
NBd 01	25	294	0	3500	40	100	4	4	5G	s4
NBd 02	25	69	0	3500	40	100	4	4	5G	s4
NBd 03	22	214	0	3500	40	100	4	4	5G	s4
NBe 01	16	31	0	3500	40	100	4	4	5G	s5
NBe 02	16	292	0	3500	40	100	4	4	5G	s5
NBe 03	16	210	0	3500	40	100	4	4	5G	s5

3. Click on Import Cells button to start importing procedure.

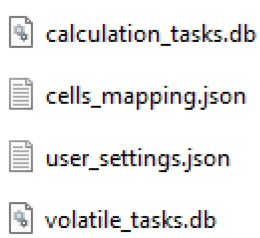
Sites can be imported together with Cells, if:

- The mapping file is not used, and text file has site_id field, which contains information about Site name. It must be text data.
- The mapping file is used, then the data should be mapped with site_id field. It must be text data.

7.8.1.1 Mapping file

The data in the import files may have names, values, and units that do not match the data in the Cellular Expert database. To resolve such issues an additional Mapping file should be imported in which these data conflicts ought to be addressed.

The empty mapping file can be found in the project's workspace catalog, SystemFiles folder.



The mapping files are not necessary if the import file data corresponds to the Cellular Expert database data, otherwise mapping files are a must for a successful import.

Values that are not mentioned in the mapping file will not be affected.

The structure of a mapping file:

```
"table_name": "Cells",

"fields": [
        {
                "current_name": "lat",
                "destination_name": "latitude",
                "default_value": "41.268"
        },
```

"table_name" - defines which network object's values are being mapped.
"current_name" - the name of the value that is written in the data file. In this example "lat" is a column name in the data file and will be changed to "latitude" when the mapping file is applied, and objects are imported.
"destination_name" - the proper name of the property (table column name) in the Cellular Expert database. This property will be checked when the mapping file is imported.
"default_value" – value that will be used when an object in the data file has no value for a particular property. In this example, if "latitude" is not set, then it will by default be assigned the value of 41.258. Leaving the default to empty quotation marks ("") means that no default value will be applied.

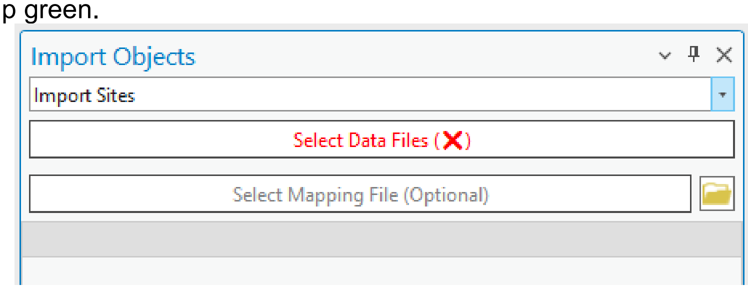
7.8.2 Import Sites, Radar, CPE, Repeaters, Links, or Mesh Nodes

Other network objects can be imported to Cellular Expert workspace. The dialog contains information about.

Template
Take necessary parameters from the template during the import. Template values are taken if some parameters are missing in text file and mapping file.

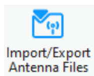
Select Data Files
Opens a dialogue window where the user can select the files that define the network objects to be imported. The supported formats: *.xls*, *.xlsx*,*.csv,* and tables from *.sde* connection. Upon successful selection, the button will light up green.

Select Mapping File (Optional)
Opens a dialogue window where the user can select a file that defines the data to be imported and the conditions under which said data is processed. The supported format is *.json*. Upon successful selection, the button will light up green.

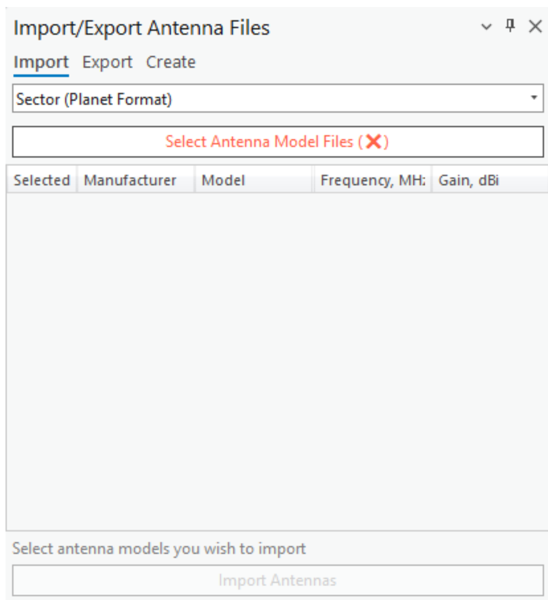


7.9 Import/Export Antenna Files

7.9.1 Import Antennas

Click the toolbar button 

and select **Import** to import antenna patterns. The command opens a dialogue window where the user can select the antenna pattern files to be imported into the Cellular Expert database. Select the antenna type in the dropdown list and proceed.



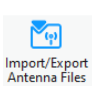
Select Files
This button opens a dialogue in which you can select one or more antenna pattern files to be imported. The supported type formats are *Planet*, *Andrew,* and *NSMA*.

Import Antennas
Imports the selected Antenna pattern files to the Cellular Expert database.

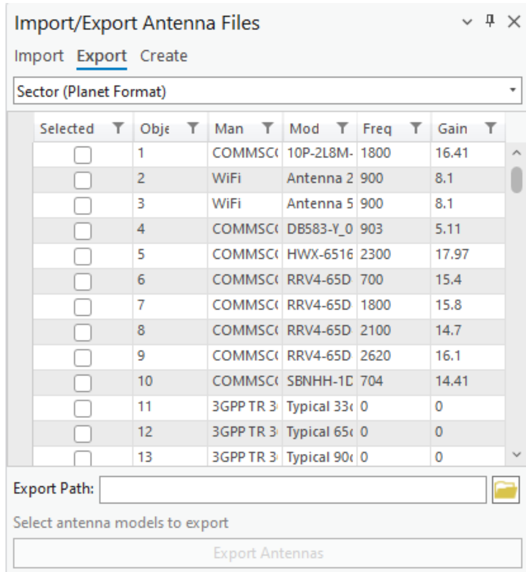
Import Antennas
Refreshes the data table if changes have been made to it.

7.9.2 Export Antennas

Exports selected antennas to a desired format.

Click on the 

button and select **Export** to export antenna patterns. Select the **Export path** on your local hard drive, check the desired antennas, and click the *Export* button.



Export Path

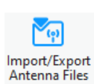
The destination folder on your local hard drive to which the selected antennas will be exported. Can be entered manually or by clicking the **Open** button and selecting it in the dialog. After defining the antenna format, all available antennas of that format will be displayed in the Antennas table. The supported type formats are *Planet, Andrew, and NSMA*.

Export Antennas

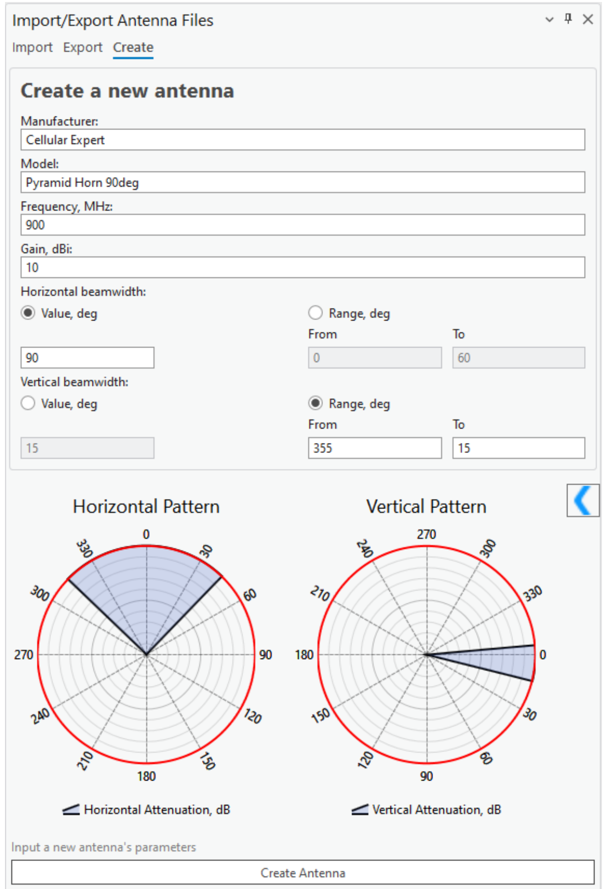
Exports the chosen antennas to the selected destination folder.

7.9.3 Create Antennas

Create new antennas based on input parameters.

Click on the 

button and select **Create** to create antenna patterns. Various parameters of the antenna can be entered, as well as horizontal and vertical beamwidth values or ranges. For the selected horizontal and vertical values or ranges, the horizontal and vertical attenuations are set to 0, while all other attenuations are set to 1000. Based on the horizontal and vertical beamwidths, the horizontal and vertical antenna patterns are displayed.



Manufacturer
Antenna manufacturer (company or entity).

Model
Antenna identification or name.

Frequency
Antenna frequency value in MHz.

Gain
Antenna gain value in dBi.

Horizontal beamwidth
Antenna's horizontal beamwidth value or range in degrees. By default, the value is 60.

Vertical beamwidth
Antenna's vertical beamwidth value or range in degrees. By default, the value is 15.

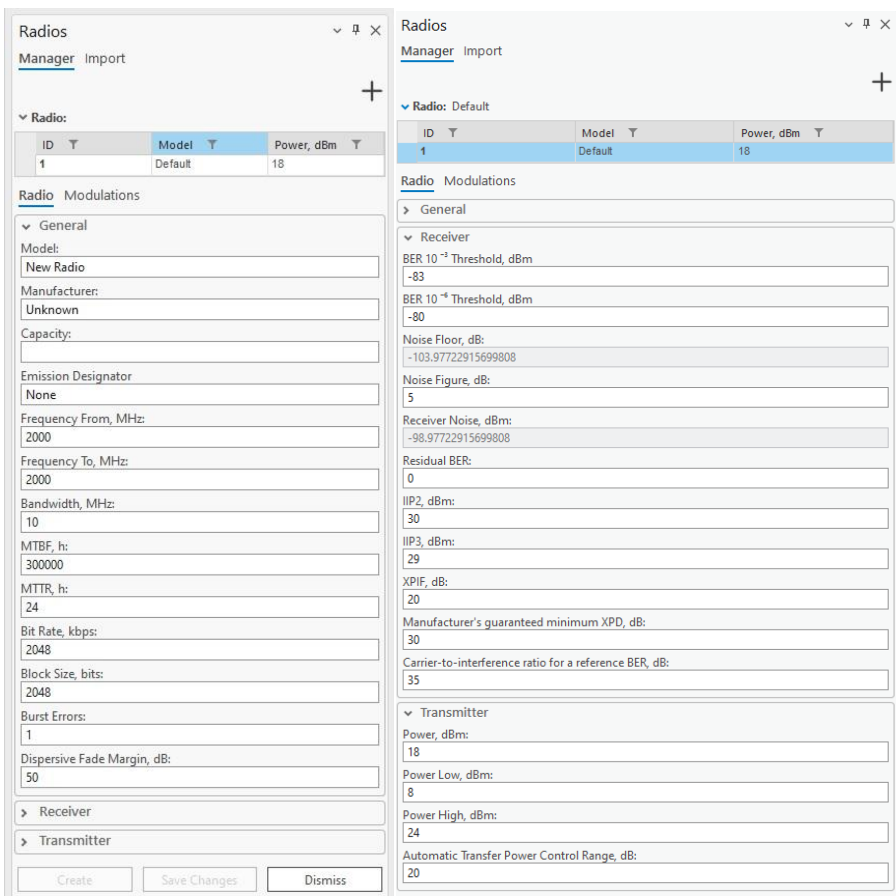
Create Antenna
Creates the antenna pattern in the database.

7.10 Radios

Click the  button to open the **Radios** Dockpane. The dock pane's Manager tab includes the functionality for creating new radios and deleting, editing, or duplicating existing ones. The Import tab allows for importing new antennas from supported format (.raf) files.

Radios are specific to the *CE for the ArcGIS Pro RLP* license. The tool enables you to create radios that will be necessary to create a link and later be used in calculations.

7.10.1 Manager Tab



By clicking an already existing radio from the table displayed at the top of the dockpane, you can edit its properties and save them. Right-clicking an already existing radio displays a context menu with options **Duplicate** and **Delete**, for respectively either creating a copy of the selected antenna or deleting it.

Initialize a new radio with default parameters.

Save Changes
Save changes to the radio that is currently being edited.

Create
Create a new radio with the specified parameters.

Dismiss
Cancels all changes made to the radio and closes the dock pane.

Radio Properties: General

Model
Radio identification or name.

Manufacturer
Radio manufacturer (company or entity).

Capacity
Maximum data throughput of the radio.

Emission Designator
Code describing the characteristics of the radio emission, including modulation type, signal nature, and bandwidth.

Frequency From, MHz
The lower limit of the frequency range for the radio communication channel, measured in MHz.

Frequency To, MHz
The upper limit of the frequency range for the radio communication channel, measured in MHz.

Bandwidth, MHz
Range of frequencies within which the radio operates, value in MHz. Required for 4G and 5G technologies.

MTBF, h
Mean time between failures: average time (in hours) between successive failures of the radio system during normal operation, indicating reliability.

MTTR, h
Mean time to repair: average time (in hours) required to repair the radio system after a failure occurs, indicating maintainability.

Bit Rate, kbps
Number of bits transmitted per second over the communication channel, measured in kbps.

Block Size, bits Number of bits grouped as a single unit of data for transmission or processing.

Burst Errors
Sequences of errors occurring in clusters or bursts, typically caused by noise or interference affecting multiple consecutive bits.

Dispersive Fade Margin, dB
Additional signal strength required to overcome dispersion and fading effects in the radio communication system, measured in dB.

Radio Properties: Receiver

BER 10^{-3} Threshold, dBm

Receiver sensitivity threshold for BER = 10^{-6}.

BER 10^{-6} Threshold, dBm

Receiver sensitivity threshold for BER = 10^{-3}.

Noise Floor, dB
Refers to the minimum power level of unwanted noise or interference at the receiver at the base station.

Noise Figure, dB
Measure of the degradation of the signal-to-noise ratio (SNR) as a signal passes through a radio component or system. Value in dB. Required for 4G and 5G technologies.

Receiver Noise, dBm
The sum of Noise Floor and Noise Figure.

Residual BER
The remaining bit error rate after all error correction methods have been applied.

IIP2, dBm
Second-order intercept point: indicates the linearity of a radio receiver's resistance to second-order intermodulation distortion, measured in decibels relative to 1 milliwatt (dBm).

IIP3, dBm
Third-order intercept point: indicates the linearity of a radio receiver's resistance to third-order intermodulation distortion, measured in decibels relative to 1 milliwatt (dBm).

XPIF, dB
Cross-polarization interference factor: interference caused by signal components with orthogonal polarization states, indicating the impact of cross-polarization on system performance, expressed in decibels (dB). The default value is 20. The value should be higher than 0 for rain ITU fading calculations to work.

Manufacturer’s guaranteed minimum XPD, dB
The manufacturer’s guaranteed minimum cross-polar discrimination, expressed in decibels (dB). The default value is 30.

Carrier-to-interference ratio for a reference BER, dB
Carrier-to-interference ratio for a reference bit error rate, expressed in decibels (dB). The default value is 35.

Radio Properties: Transmitter

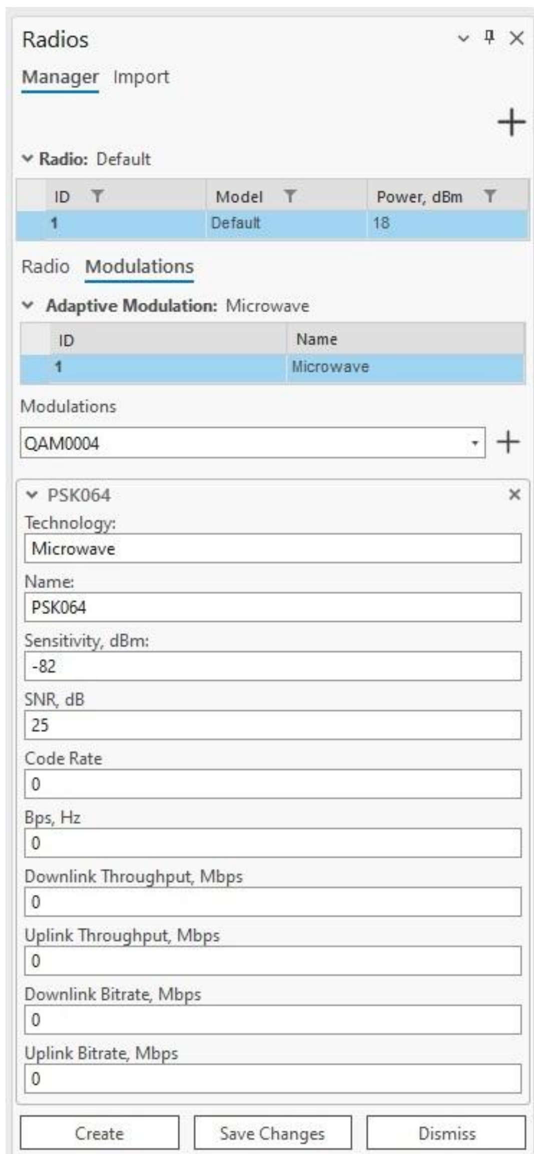
Power, dBm
Power value in dBm.

Power Low, dBm
The minimum power level at which the radio transmits, measured in decibels relative to 1 milliwatt (dBm).

Power High, dBm
The maximum power level at which the radio transmits, measured in decibels relative to 1 milliwatt (dBm).

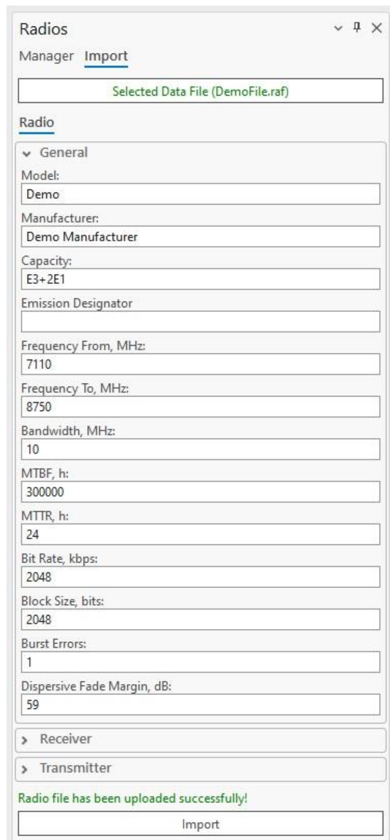
Automatic Transfer Power Control Range, dB

The range over which the radio can automatically adjust its transmission power, measured in dB.



Selecting the Modulations tab allows you to assign a modulation to a radio. The adaptive modulations are displayed in the table by their ID and technology name. The modulations available for assigning to a radio are listed in a dropdown below the table. The selected modulation can be assigned to a radio by pressing the  button. Several modulations can be assigned to a radio, and the parameters of the assigned modulations can be edited.

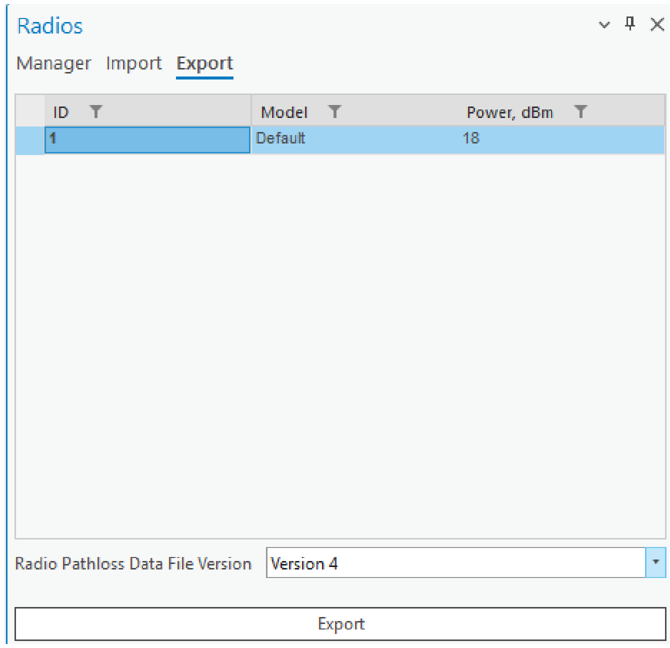
7.10.2 Import Tab



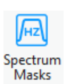
Selecting the Import tab displays a new window with the Select Data File button for importing radio data files. RAF files are supported. Once a supported radio file is selected, its parameters can be edited (General, Receiver, and Transmitter categories). The text “Radio file has been uploaded successfully” appears at the bottom of the dock pane, and the Import button becomes enabled. Clicking the Import button adds the radio to the database, and it also appears in the Manager tab.

7.10.3 Export Tab

Radios can be exported to the RAF data files version 4 or 5. Depending on the selected file version, the structure of the data will be different.



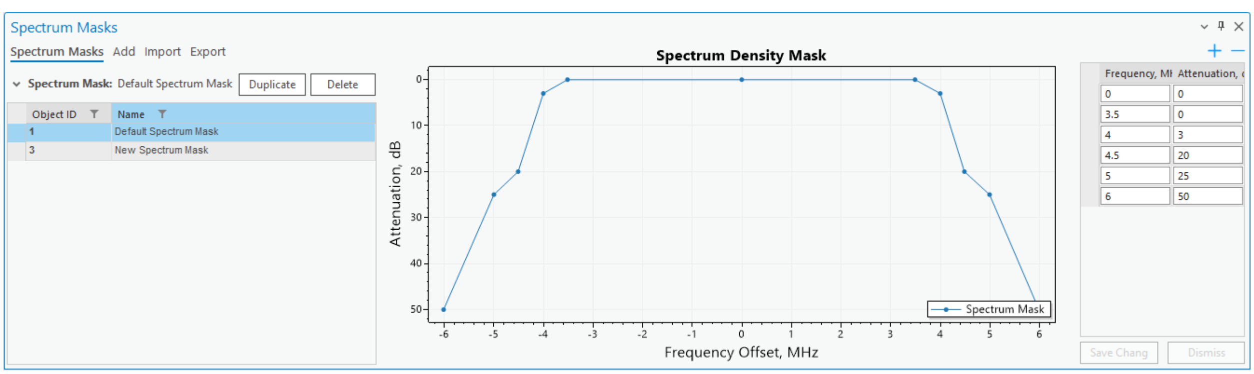
7.11 Spectrum Masks

Click the  button to open the **Spectrum Masks** dialogue.

Spectrum Masks are specific to the *CE for ArcGIS Pro RLP* license. The tool enables you to create spectrum masks that will be necessary to be used in calculations.

7.11.1 View Spectrum Masks

You can view, edit, and delete all spectrum masks by navigating to the **Spectrum Masks** tab on the Spectrum Masks dockpane. By selecting a spectrum mask you can view its properties, change them, and delete the spectrum mask altogether. Changes in the mask point values will be reflected in the graph.

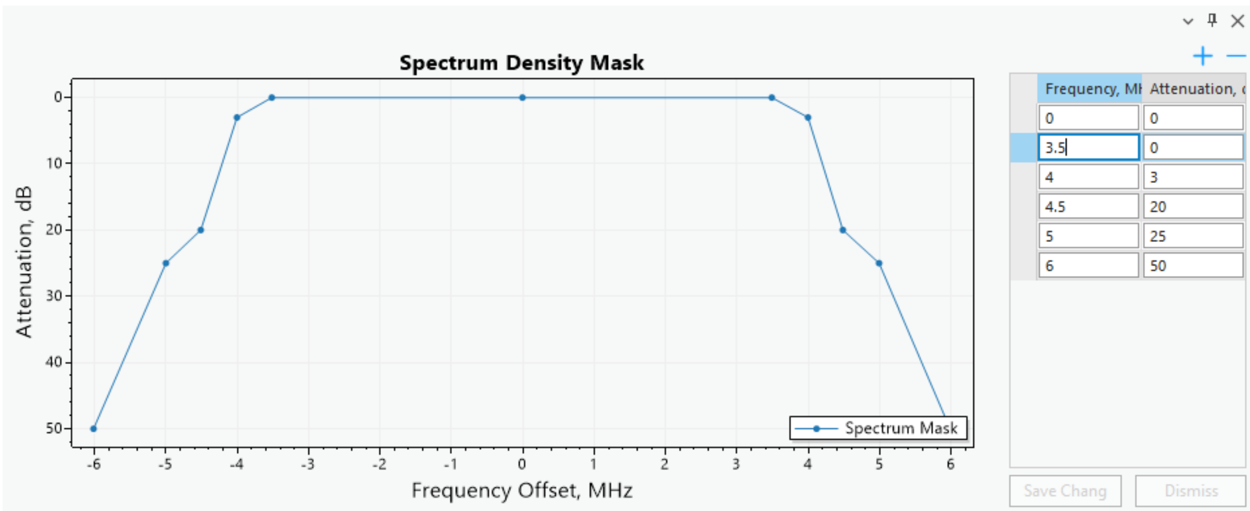


Delete

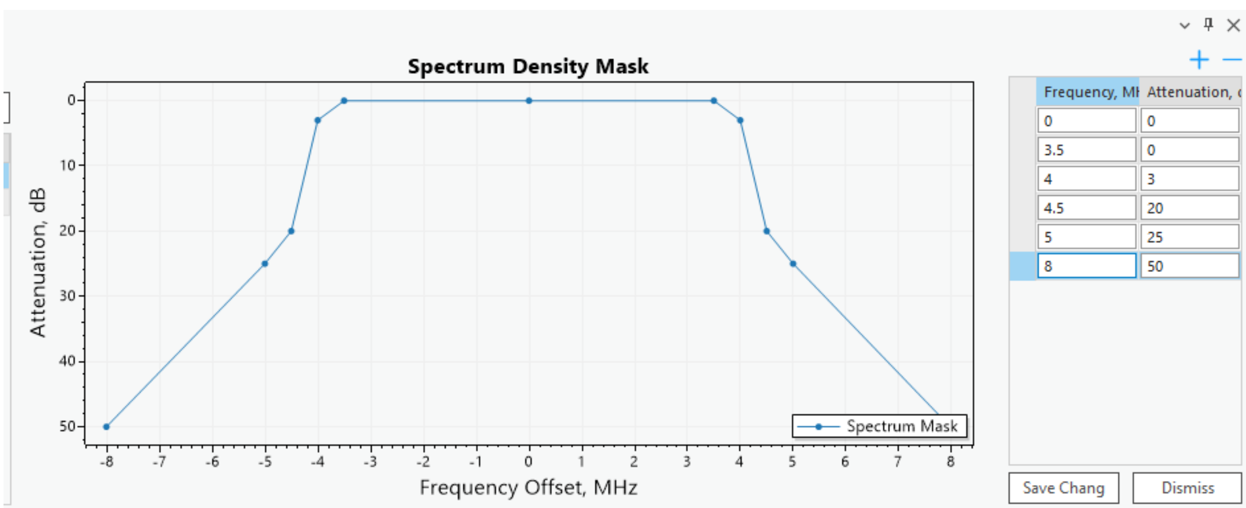
Delete the selected spectrum mask.

Duplicate
Create a copy of the selected spectrum mask.

When you create a spectrum mask, you can see its visual representation as well as the values of each spectrum mask point (frequency and attenuation).



You can change any values of these points and those changes will be reflected in the graph.



—
Deletes the currently selected mask point. If none are selected, does nothing.

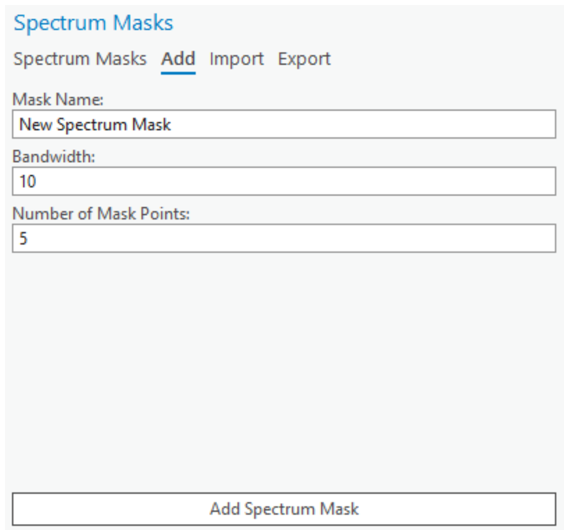
+
Adds a new mask point to the spectrum mask.

Save Changes
Save changes made to the current spectrum mask.

Dismiss
Cancel any changes made to the current spectrum mask.

7.11.2 Add Spectrum Mask

You can add a spectrum mask by selecting the **Add** tab located on the Spectrum Masks dockpane.



Add Spectrum Mask
Creates a new spectrum mask.

<u>Spectrum Mask Parameters</u>

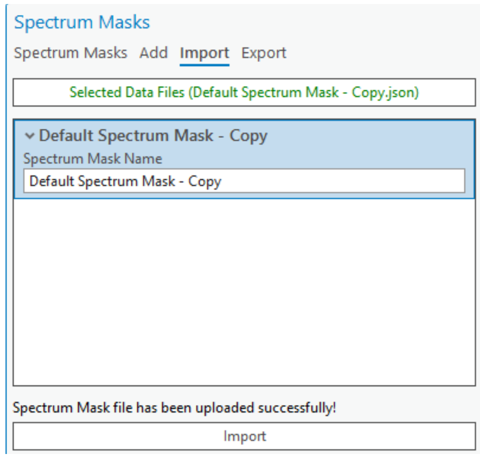
Mask Name
Spectrum mask identification.

Bandwidth
Value in MHz. Required for 4G and 5G technologies.

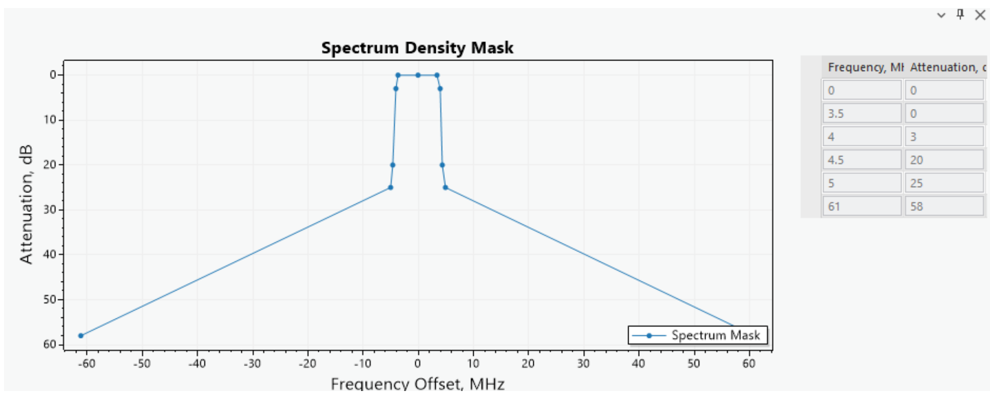
Number of Mask Points
The number of points the resulting spectrum mask will have.

7.11.3 Import Spectrum Mask

You can add a spectrum mask by importing it from a JSON format file.

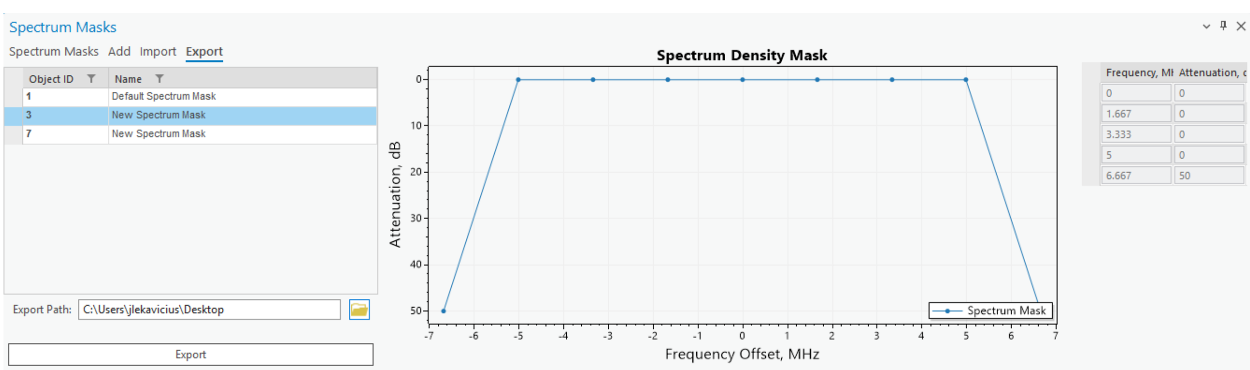


The selected spectrum mask can be visually inspected before importing, by clicking on the selection of the spectrum mask in the container.

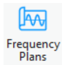


7.11.4 *Export Spectrum Mask*

Spectrum Masks can be exported to JSON file format.



7.12 Frequency Plans

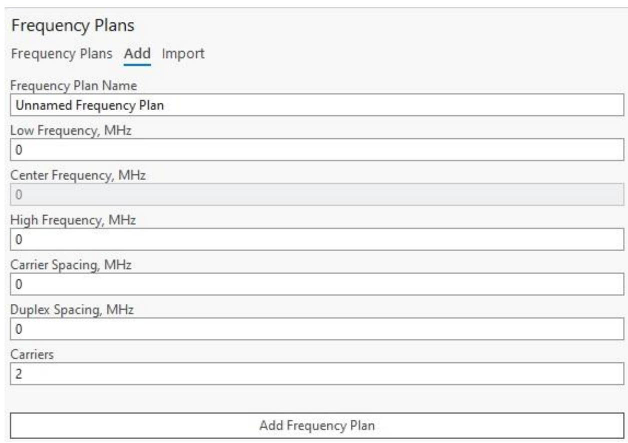
Click the 

button to open the **Frequency Plans** dialogue.

Frequency Plans are specific to the *CE for ArcGIS Pro RLP* license. The tool enables you to create frequency plans that will be necessary to create a link and later be used in calculations.

7.12.1 Add Frequency Plan

You can add a frequency plan by selecting the **Add** tab located on the Frequency Plans dockpane.



Add Frequency Plan
Adds a new frequency plan.

<u>Frequency Plan Parameters</u>

Frequency Plan Name
Frequency Plan identification

Low, Center, High Frequency
Frequencies in MHz.

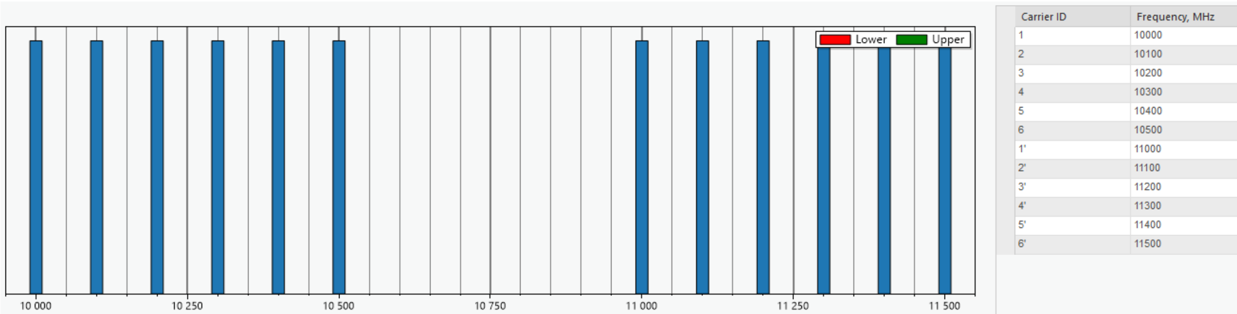
Carrier Spacing, MHz
The frequency separation between adjacent carrier frequencies in a communication system, ensuring non-interference between carriers.

Duplex Spacing, MHz
Frequency separation between the transmit and receive bands in a two-way communication system.

Carriers
Specific frequencies or waves used to modulate and transmit information over a communication medium.

Once you create a frequency plan, you will be able to see its graphical representation as well as the frequencies of each frequency plan's carrier. Select entries in the carrier table to highlight a specific carrier on the plot.

Carriers with the prime symbol (‘) are for Upper carriers, and others are for Lower carriers.

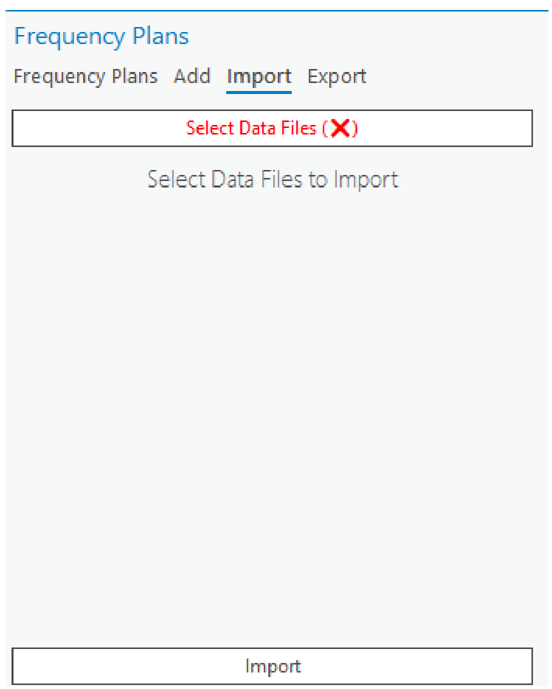


Carrier ID
Carrier identifier.

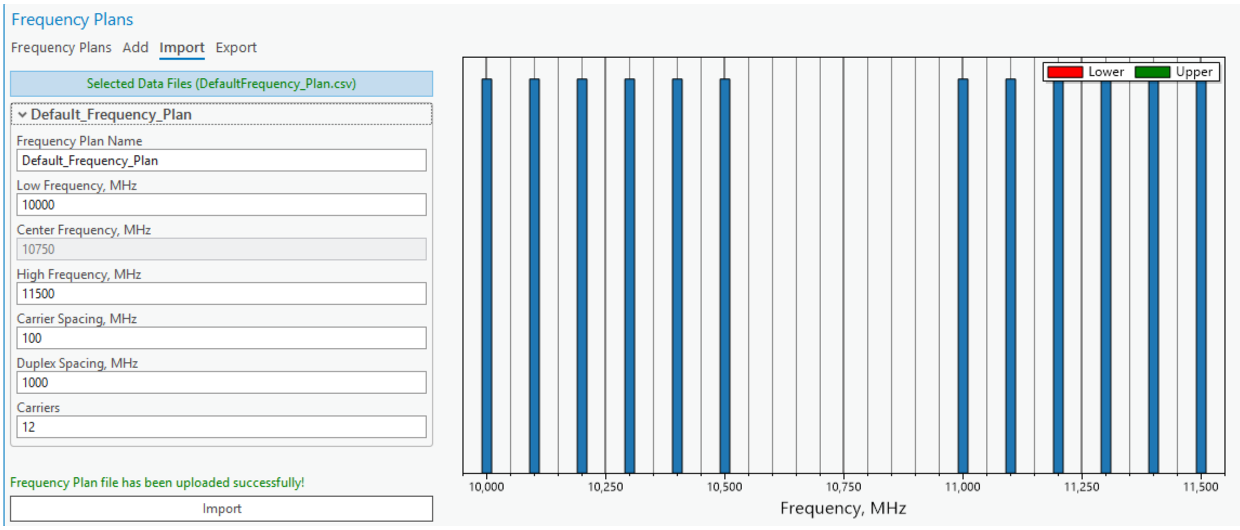
Frequency
Carrier’s frequency value in MHz.

7.12.2 Import Frequency Plan

Frequency plans can be imported into CE workspace as CSV format files. Press **Select Data Files,** then navigate to the frequency plan data CSV format file.

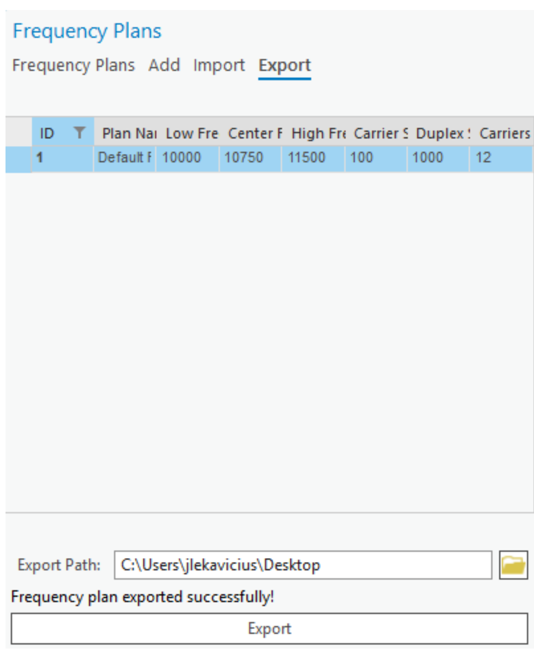


The frequency plan can also be edited before importing. Click Import to finalize the frequency plan import procedure.



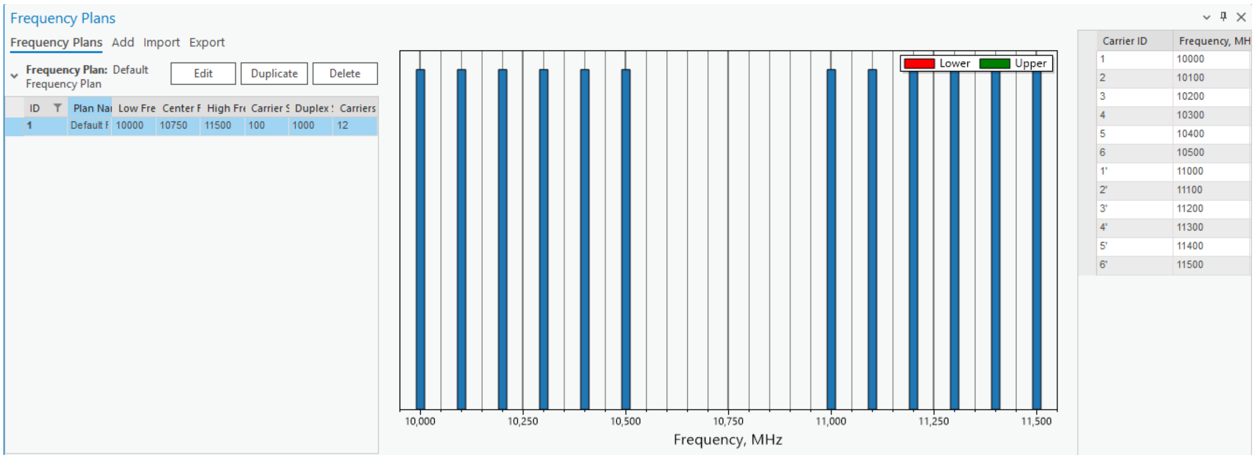
7.12.3 *Export Frequency Plan*

Selected frequency plan can be exported to CSV format file. Select the frequency plan in the table, then define the export path by clicking the **Folder** icon to open the path selection dialog, and click Export to save the frequency plan to CSV format file.



7.12.4 *View Frequency Plans*

You can view, edit, and delete all frequency plans by navigating to the **Frequency Plans** tab on the Frequency Plans dockpane.

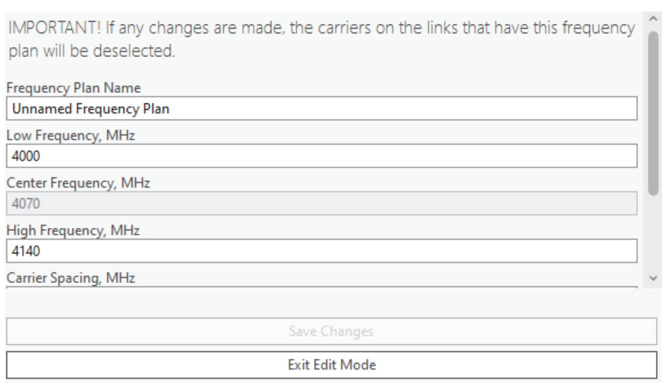


By selecting a frequency plan you can view its properties, change them, and delete the frequency plan.

Delete
Delete the selected frequency plan.

Duplicate
Create a copy of the selected frequency plan.

Edit
Edit the currently selected frequency plan. When this button is pressed you will be redirected to a separate tab to make the edits. Keep in mind that any links that have this changed frequency plan will lose their carrier selection.



Exit Edit Mode
Discard all changes made to the currently editable plan and leave edit mode. On exit, you will be returned to the Frequency Plans tab.

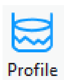
8. Profile

8.1 Profile Tool

A Profile in wireless communication represents the geographical and environmental characteristics of the path between a transmitter and a receiver. It includes detailed information such as elevation data, terrain heights, and any obstacles (e.g., buildings, trees, or mountains) that might impact signal propagation.

All relevant calculations are already performed within the system. These include the power budget, which factors in the total power available for transmission and reception while accounting for system gains and losses, and the path loss, which measures the reduction in signal strength due to distance, frequency, and environmental obstructions. The profile also provides the calculated angles, including the elevation angle (vertical angle between the transmitter and receiver) and the azimuth angle (horizontal direction). Additionally, it determines whether a direct line-of-sight exists between the two points.

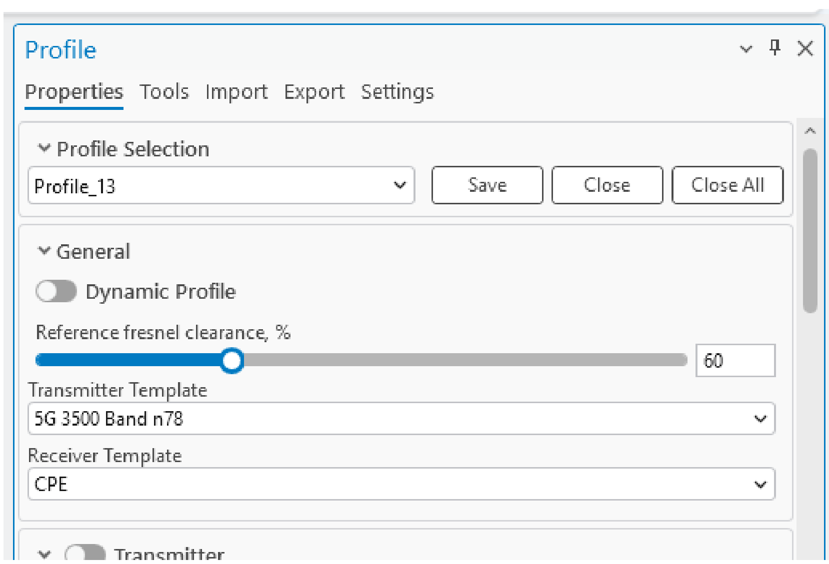
This comprehensive information is ready for use, enabling users to assess the feasibility and performance of a communication link for network planning, optimization, and troubleshooting.

Click the 

button to open the **Profile** dialogue.

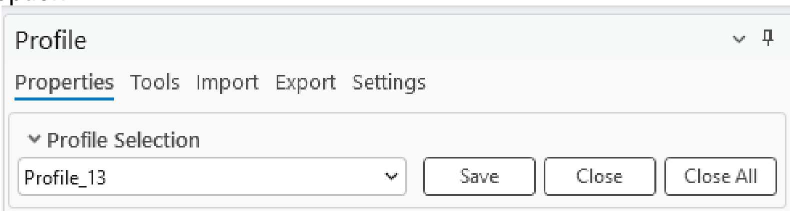
Profile tool enables you to determine the obstructions, elevation, and Fresnel zones between two points on a map.

8.1.1 Properties



Profile: Profile Selection

Profile selection allows for quick switching between profiles while retaining the most recently defined parameters. This feature is especially useful for comparing and verifying calculations across differentTx and Rx configurations. When a new profile is created, it is automatically assigned a name and appears in the Profile Selection option.



Profile can be saved or removed from the list.

Save
The option saves the profile to Docs Manager tool with all defined parameters.

Close
Removes profile from the list in Profile Selection.

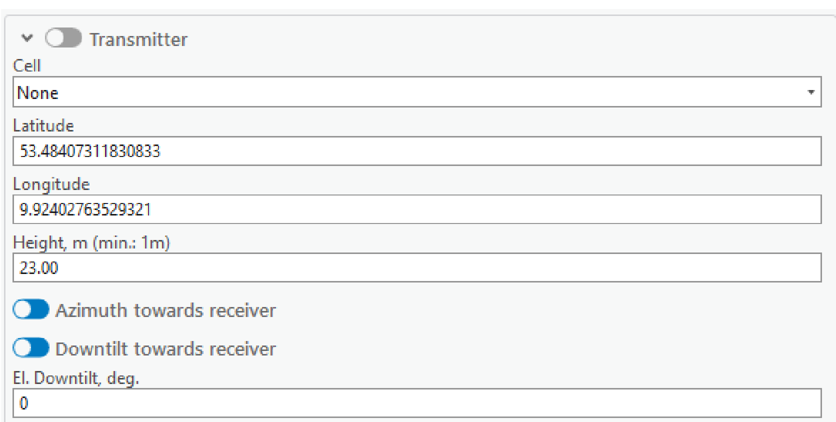
Close All
Removes all profiles from Profile Selection list.

Profile: General

Fresnel Minimal Clearance, %
Percentage by which the primary Fresnel zones will be scaled up or down thus creating a secondary Fresnel zone. The percentage must be in the range of 1 to 200. The value can be changed either by inputting it manually or by using the slider.

Transmitter Template
The template that is used for transmitter's default values.

Receiver Template
The template that is used for receiver's default values.



Profile: Transmitter

Toggling the switch to the left of **Transmitter** will enable the **Fixed Transmitter** functionality, which will modify only the receiver's positioning when drawing the profile on the map.

Cell
A cell from which the profile will be drawn. The parameters of the cell will be taken into calculation if the cell is snapped to by the profile tool.

Latitude
Decimal degrees Y type coordinate.

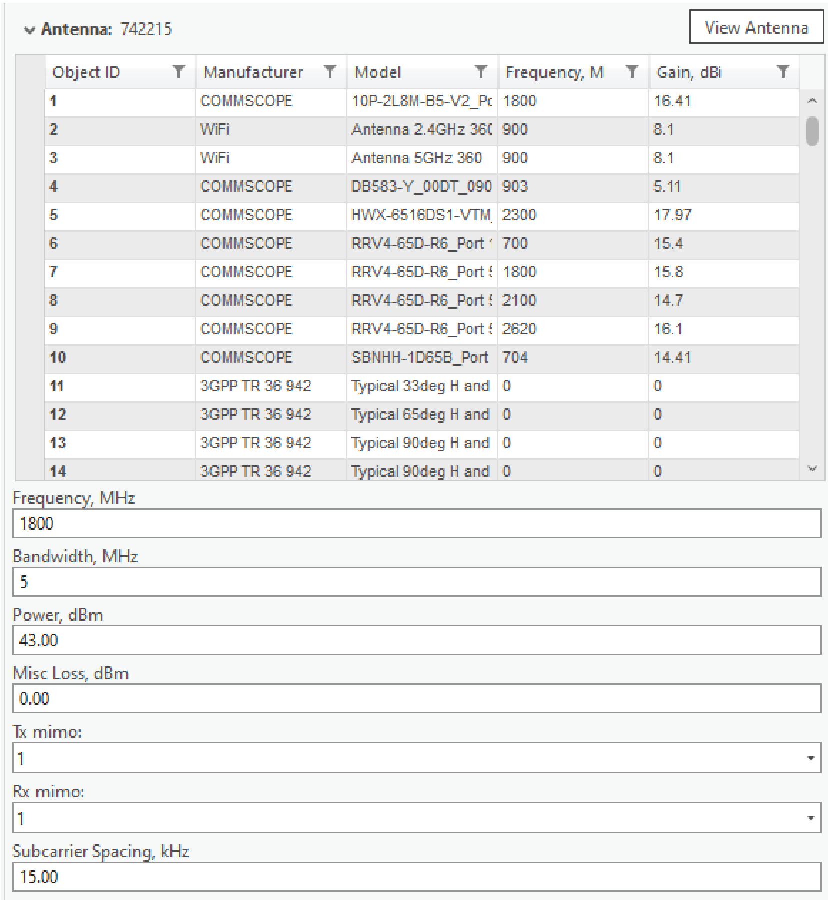
Longitude
Decimal degrees X type coordinate.

Height, m
Height above the ground in meters. The minimum value must be 1m.

Azimuth towards receiver
Enabled by default. When enabled, the transmitter's azimuth is towards the receiver. Disabling this option it would take azimuth value from the Cell object, and use it for FWA Power Budget calculations, it also allows the user to enter a custom azimuth value for the transmitter.

Downtilt towards receiver
Enabled by default. When enabled, the transmitter's tilt is towards the receiver. Disabling this option it would take tilt value from the Cell object, and use it for FWA Power Budget calculations, it allows the user to enter a custom tilt value for the transmitter.

El. Downtilt, deg.
Electrical downtilt value for the transmitter, in degrees.



The antenna for the transmitter can be selected from the table below the **El. Downtilt, deg** input. Its pattern can be viewed by clicking the **View Antenna** button on the right side.

Frequency, MHz
Frequency of the transmitter.

Bandwidth, MHz
Bandwidth of the transmitter.

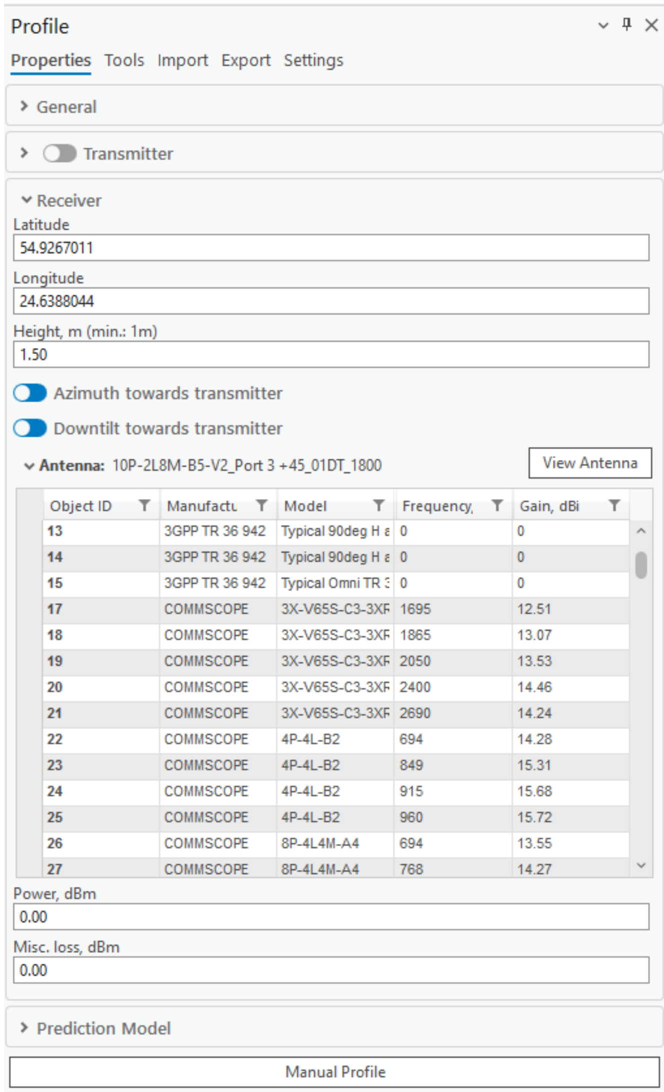
Power, dBm
A power that the transmitter possesses.

Misc Loss, dB
Miscellaneous loss value in dB. Value is not required.

Tx mimo
Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Rx mimo
Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Subcarrier spacing
Value in kHz.



Profile: Receiver

Latitude
Decimal degrees Y type coordinate.

Longitude
Decimal degrees X type coordinate.

Height, m
Height above the ground in meters. The minimum value must be 1m.

Azimuth towards transmitter

Enabled by default. When enabled, the receiver's azimuth is towards the transmitter. Disabling this option it would take azimuth value from the receiver (Cell) object, and use it for FWA Power Budget calculations, it also allows the user to enter a custom azimuth value for the receiver.

Downtilt towards transmitter
Enabled by default. When enabled, the receiver's tilt is towards the transmitter. Disabling this option it would take tilt value from the receiver (Cell) object, and use it for FWA Power Budget calculations, it allows the user to enter a custom tilt value for the receiver.

Power, dBm
A power that the receiver possesses.

Misc Loss, dB
Miscellaneous loss value in dB. Value is not required.

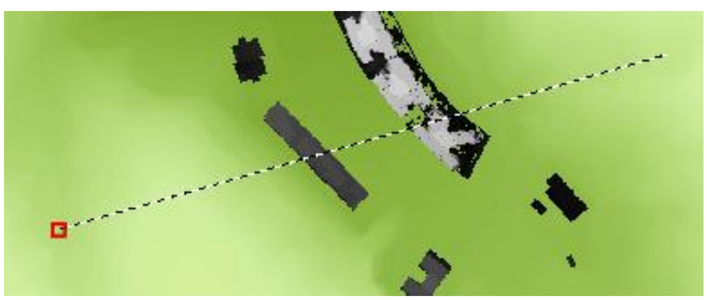
Profile: Prediction Model
Double-click the prediction model from the list to select it for the profile.

8.1.2 Draw Profile

When the Profile tool is selected and enabled, you will be able to select two points on the map in turn creating a Profile line.

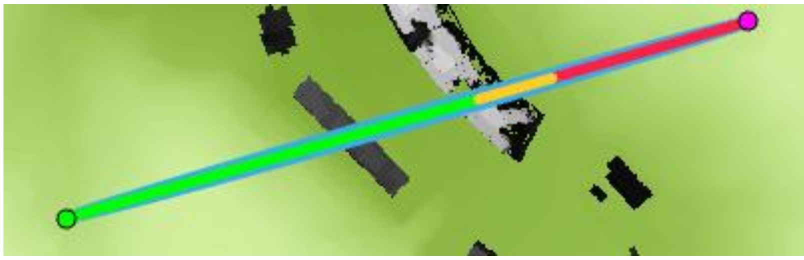
The profile also lets you **snap to different Cellular Expert network objects**. If the object to which the user snaps is a cell, all parameters of that cell that are necessary to draw a profile will be read and included in the calculations. If the object is not a cell, then only the name and the object's coordinates will be taken.

Most of these values will be displayed in the Profile pane shown above. If you change the values in the pane, these changes will be reflected in the Profile plot by redrawing it.

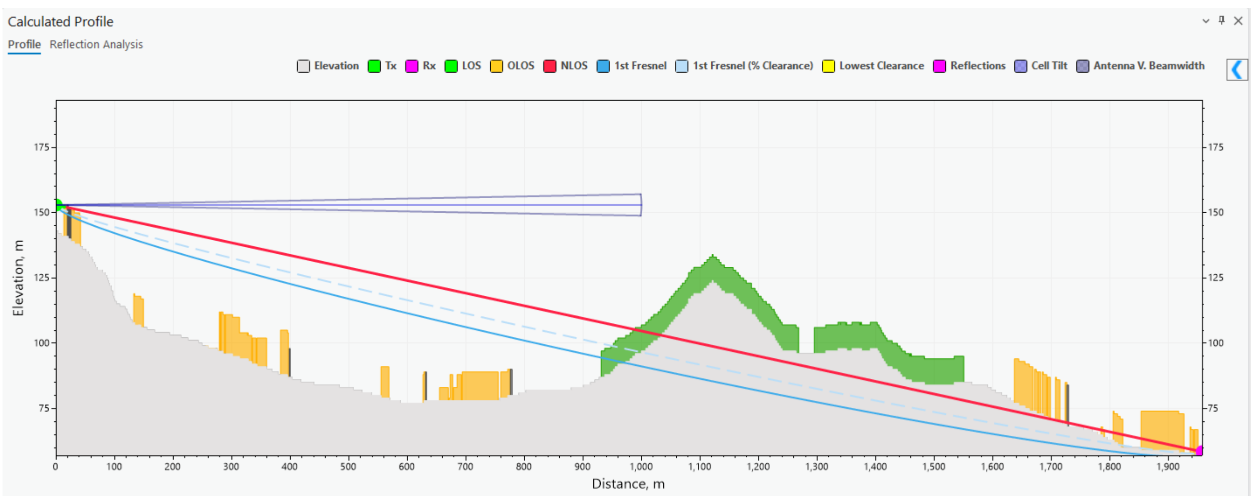


Upon the selection of a second point, these geometries will be created between the points:

- LOS (green) - the distance until the first obstruction in the profile's way
- OLOS (orange) – the distance until the first clutter LOS obstruction
- NLOS (red) – the obstructed path between the sender and receiver
- Lowest Clearance (yellow) – the point at which NLOS has the lowest clearance to the obstacle
- Rx (purple) – the receiver point
- Tx (green) – the transmitter point
- Fresnel (blue) – the Fresnel lines



The Profile plot illustrating these geometries, obstacles (buildings), and the Fresnel zone will appear in a dockpane below. You can inspect the values at particular points by moving the cursor around the plot. The cursor movement on the plot will be projected as a moving point on the map. If a cell is selected for the transmitter, the cell’s tilt (the sum of mechanical and electrical tilts) and vertical beamwidth are displayed as additional symbols: a light blue line and a darker blue cone respectively. The length of the cell tilt and antenna vertical beamwidth symbols is the radius of the prediction model selected for the cell.



The  button allows you to see the Prediction Calculation results.

General Calculations

Path type	LOS
Visibility clearance	4.16 m
Fresnel clearance	-1.01 m
Clearance percentage	80.51 %
Distance to NLOS	1118.87 m
Distance to OLOS	1118.87 m

Power Budget

Downlink FS	-63.12 dBm
Uplink FS	-89.7 dBm
FWA downlink RSL	-4.61 dBm
FWA uplink RSL	-48.4 dBm

Path Loss

Total path loss	98.08 dB
Basic loss	98.08 dB
Multipath focusing loss	0 dB
Terrain / building diffraction loss	0 dB
Enclosed receiver attenuation	0 dB
Enclosed receiver diffraction loss	0 dB

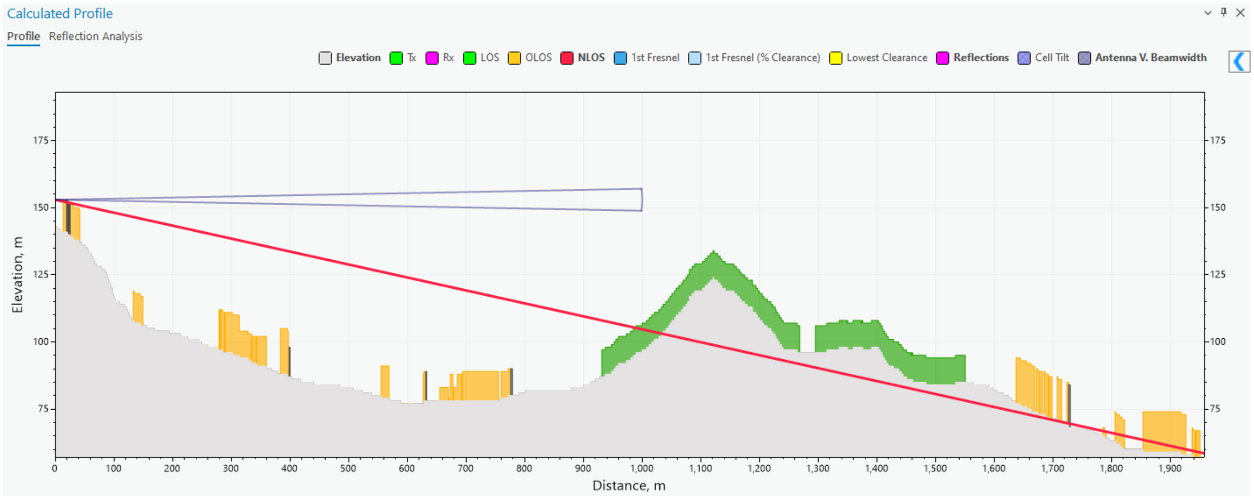
Angles

Transmitter azimuth	34 °
Transmitter tilt	1.01 °
TX-RX azimuth	34 °
TX-RX tilt	1.01 °
Receiver azimuth	214 °
Receiver tilt	-1.01 °
RX-TX azimuth	214 °
RX-TX tilt	-1.01 °

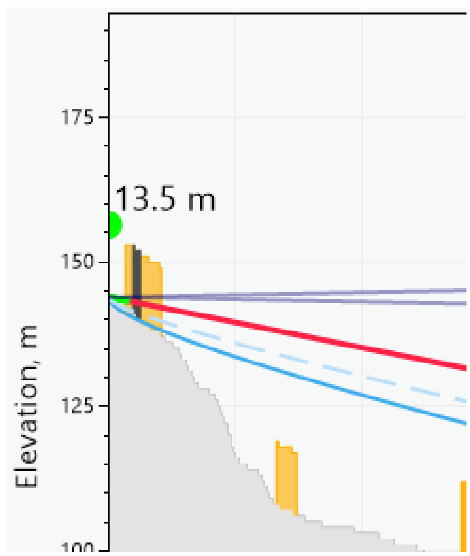
You can change the colors of the profile by clicking the colored squares near the names of the parameters.

Elevation **Tx** **Rx** **LOS** **OLOS** **NLOS** **1st Fresnel** **1st Fresnel (% Clearance)** **Lowest Clearance** **Reflections** **Cell Tilt** **Antenna V. Beamwidth**

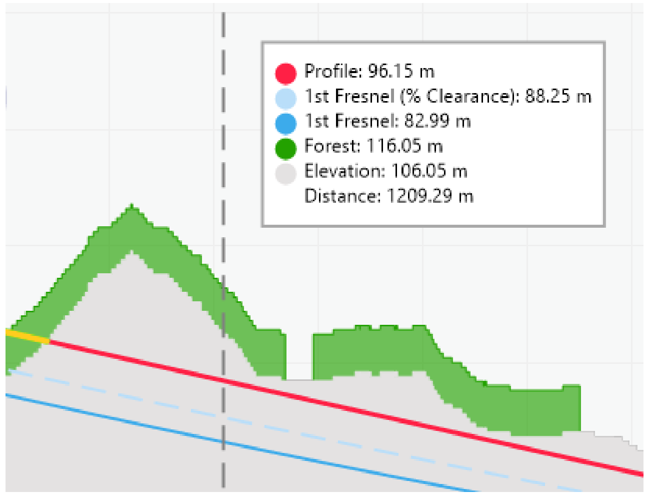
The colors will be updated automatically. You can toggle the visibility of each separate parameter of the profile by clicking on the name of the element. Enabled elements are indicated by **bold** text, and disabled elements are indicated by regular text.



You can change the height of the transmitter/receiver points by dragging their ends on the plot.

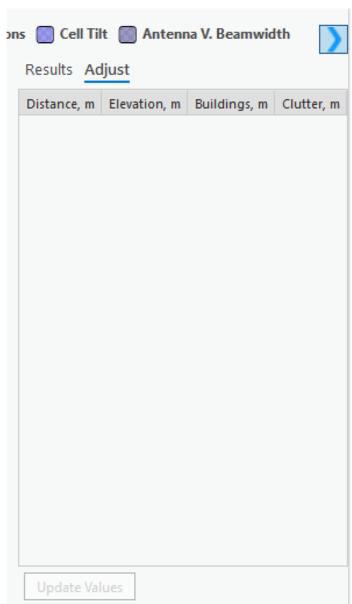


Hovering the cursor over the plot displays a tooltip with the meter values for profile, building, clutter, elevation, distance, etc., as well as their representations in colors.

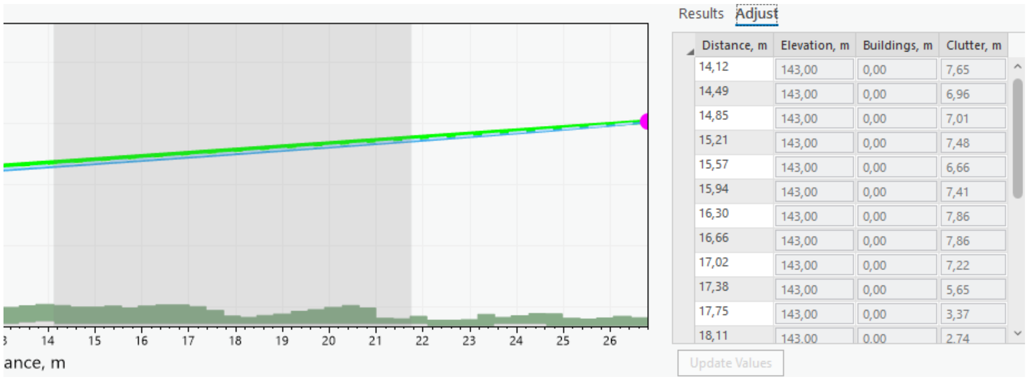


8.1.2.1 Adjust Data

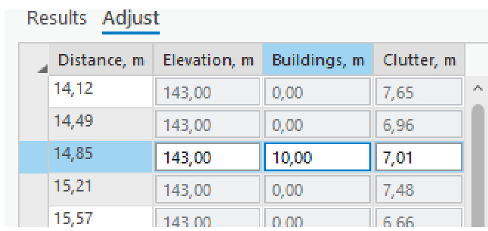
Adjust data is found on the Profile Plot dockpane near the Results. The tool lets you change the elevation, building, and clutter data of the area visible in the profile plot.



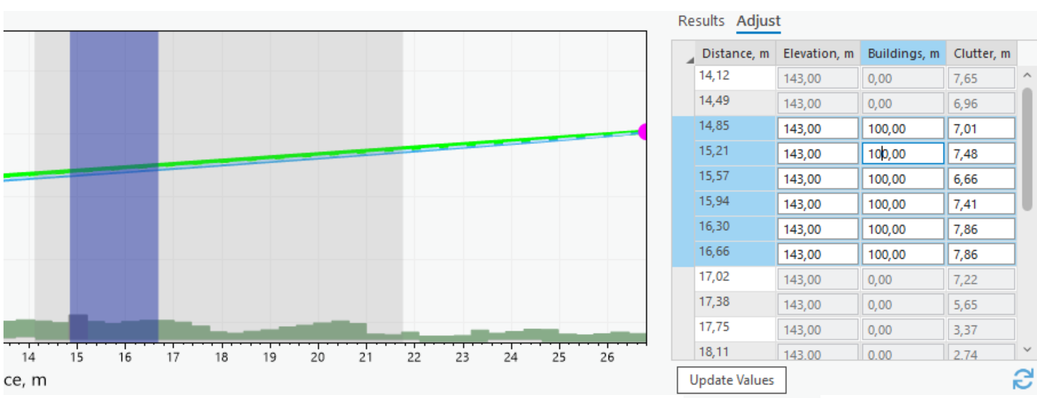
When the **Adjust** tab is opened, select a desirable range for the data adjustment on the plot.



You can make slight changes to the range of the selection area by hovering over the area's edges and dragging them. To adjust the values, click on one of the text boxes and insert the value.

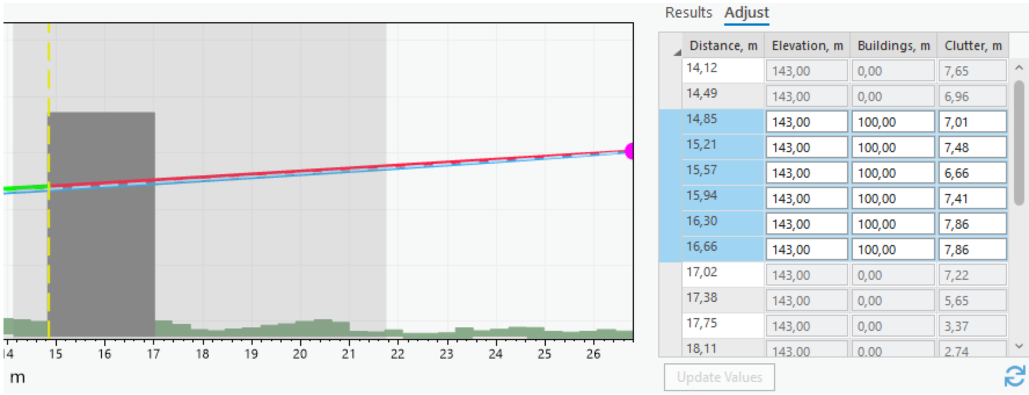


To change multiple values simultaneously, drag across the adjustment table and select multiple rows. Changing the value of a single text box will also change all the other chosen rows' values in that text box. The selected area will be highlighted on the profile plot.

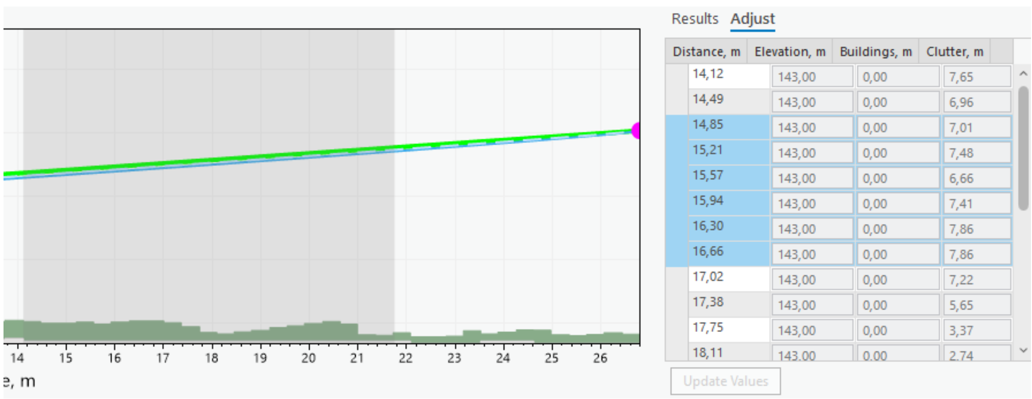


To update the values, either select an unselected row or press the 

button.



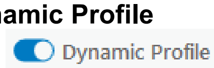
To reset the adjusted values to defaults, click the **Refresh button** .



Manual Profile

If you want to insert specific coordinates and draw a profile that way, you can insert these values in the Profile pane and click the **Manual Profile** button.

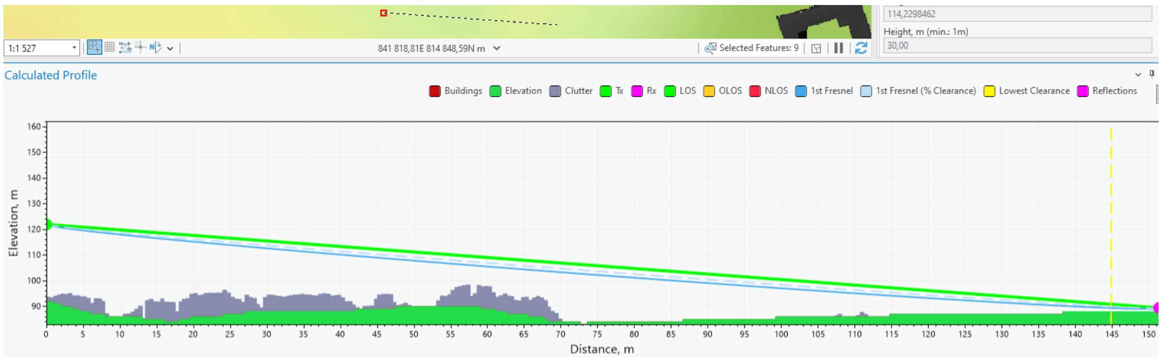
Dynamic Profile

The 

button toggles the Dynamic Profile.

Dynamic Profile lets you see Profile calculations and the plot be drawn in real-time as the cursor moves. Whenever the cursor stops momentarily – the profile gets drawn. The transmitter point needs to be entered for the dynamic profile to be activated. The previous transmitter point will be chosen if the point is not currently entered.

If a second point is selected while the profile is being drawn, the dynamic profile will be disabled, and the LOS lines will appear on the map.

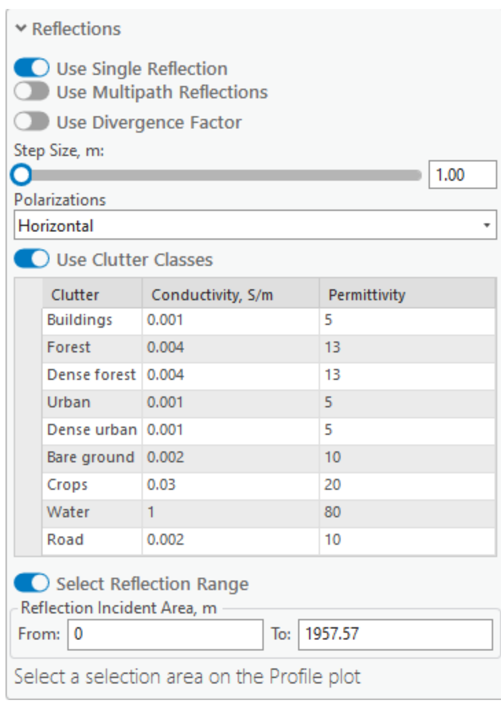


8.1.3 Tools

Reflections are considered in profile and RF calculations to assess how radio waves bounce off surfaces, affecting signal path and strength.

To enable reflections select either Single or Multipath Reflection. Be aware that if reflections are not visible you may need to adjust the height of the receiver/transmitter accordingly.

It is recommended to disable the Step Plot of the profile before using Reflections (see *Settings)*.



Use Single Reflection
Enable a reflection that will reflect straight from the transmitter to the receiver point with the smallest angle.

Use Multipath Reflections

Enable all reflections that happen along the profile line.

Use Divergence Factor
Quantifies the extent to which a reflected signal spreads out, affecting its strength and coverage.

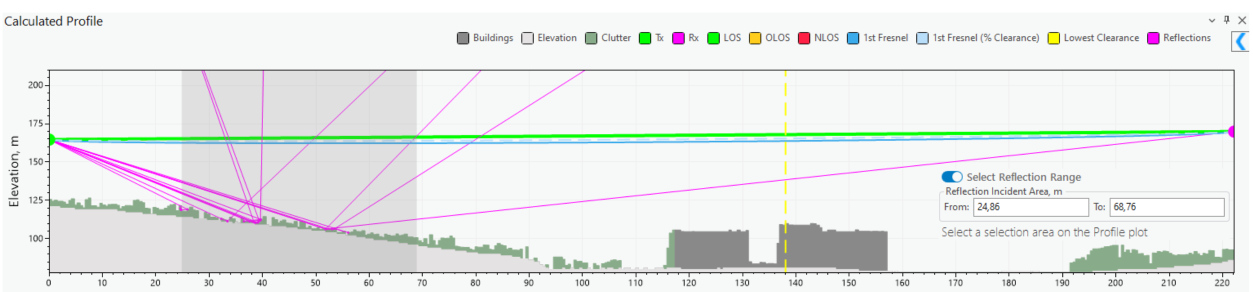
Step Size
The length period by which the reflection will be calculated. The step size is determined by taking the average cell size of the elevation raster.

Polarizations
The polarization of reflections (vertical or horizontal).

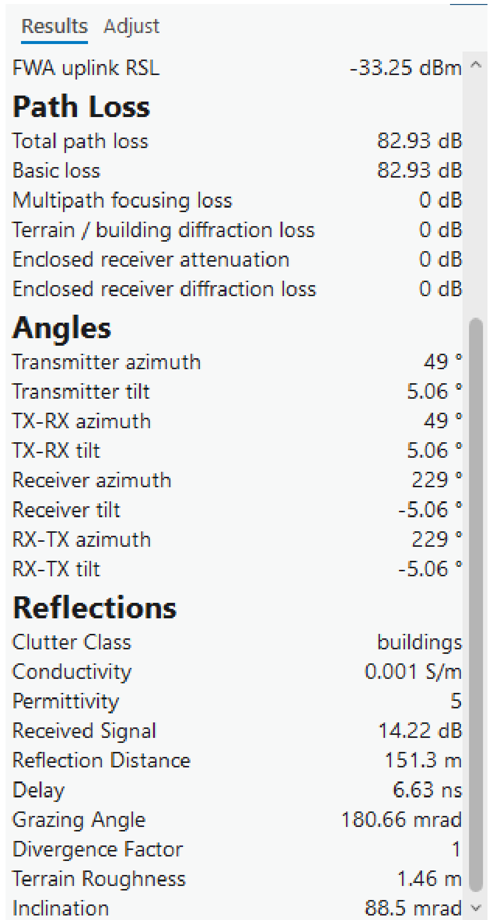
Use Clutter Classes
Enables the use of default clutter classes for the reflection calculations (see *Clutter Classes*). If disabled, custom values may be used for conductivity and permittivity.

Select Reflection Range
Lets you set a range in which the reflection calculations should happen. This will affect both Single and Multipath reflections and help highlight specific important areas along the profile as well as speed up the calculation process.
The range will be enabled as soon as a profile is drawn.

You can select this range on the plot, by holding down the left mouse button and dragging across the screen.

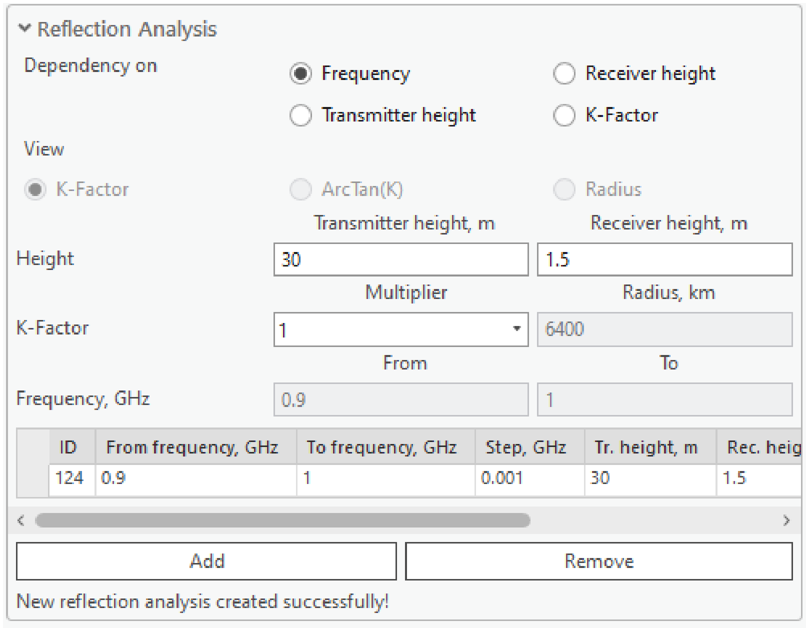


Reflection results will appear in the Profile Results table.



8.1.3.1 Reflection Analysis

The **Reflection Analysis** tool for Profile is designed to help analyze and visualize signal reflections based on the changes in various profile parameters like frequency, transmitter height, receiver height, and K-factor. Reflections must be enabled to perform analysis, and the Single Reflection option is automatically enabled when the tool is selected.

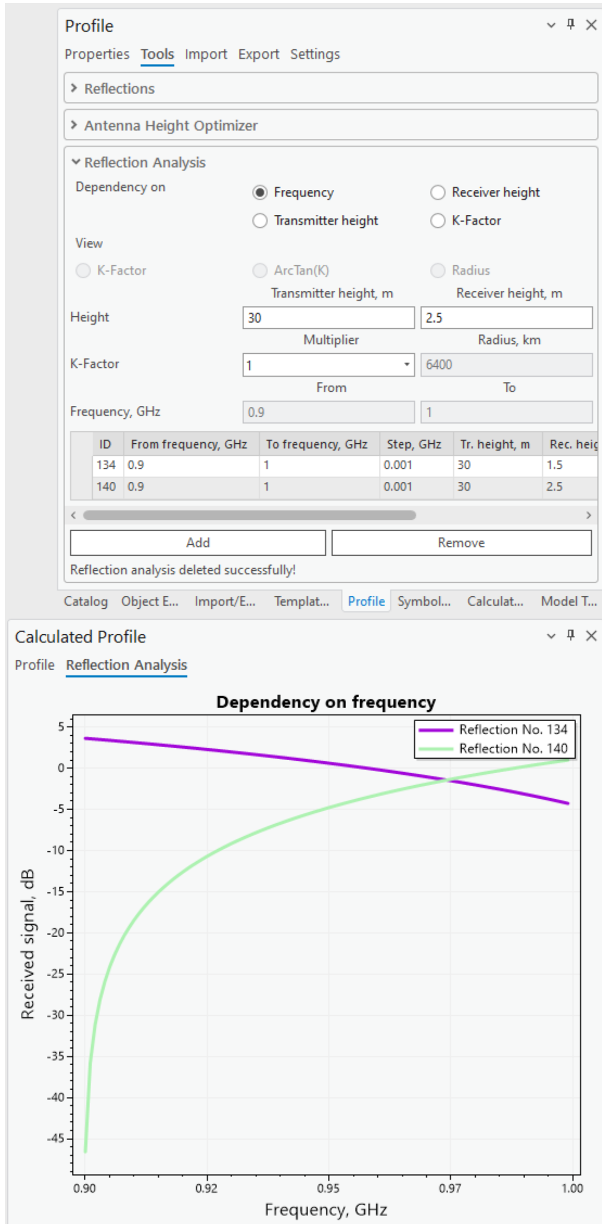


Dependency on Frequency
Calculate reflection analysis based on frequency range (from GHz to GHz)

Dependency on Receiver height
Calculate reflection analysis based on receiver height range (from m to m)

Dependency on Transmitter height
Calculate reflection analysis based on transmitter height range (from m to m)

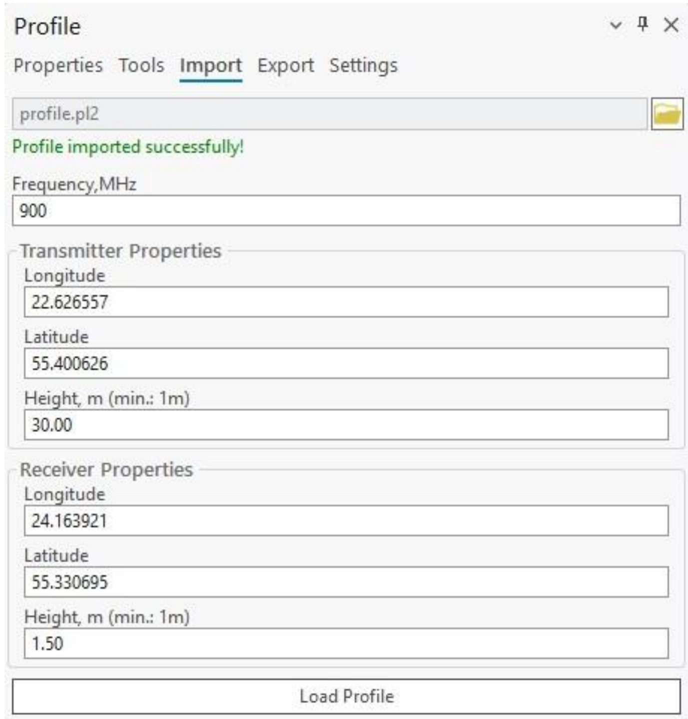
Dependency on K-Factor
Calculate reflection analysis based on K-factor range (from radius, km to radius, km)



The reflection analysis results for each type of dependency are displayed in the **Reflection Analysis** tab of the **Calculated Profile** window.

8.1.4 Import

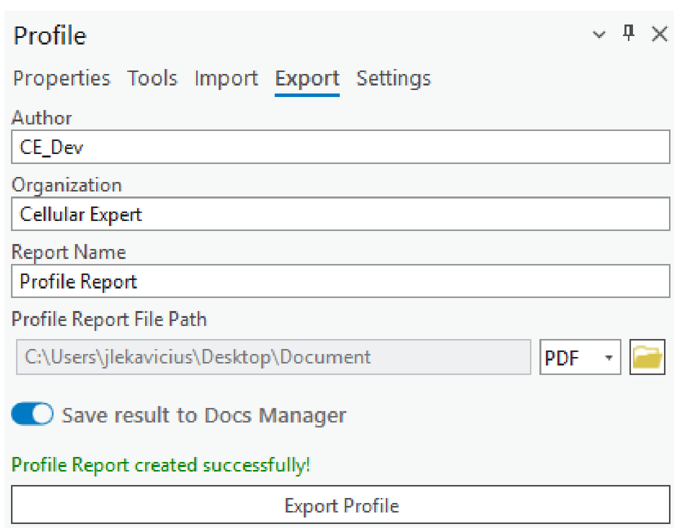
Import a profile by selecting a profile file in the Import section. Supported formats: *.pl2 (path loss file)*. Once imported successfully, the profile data may then be customized.



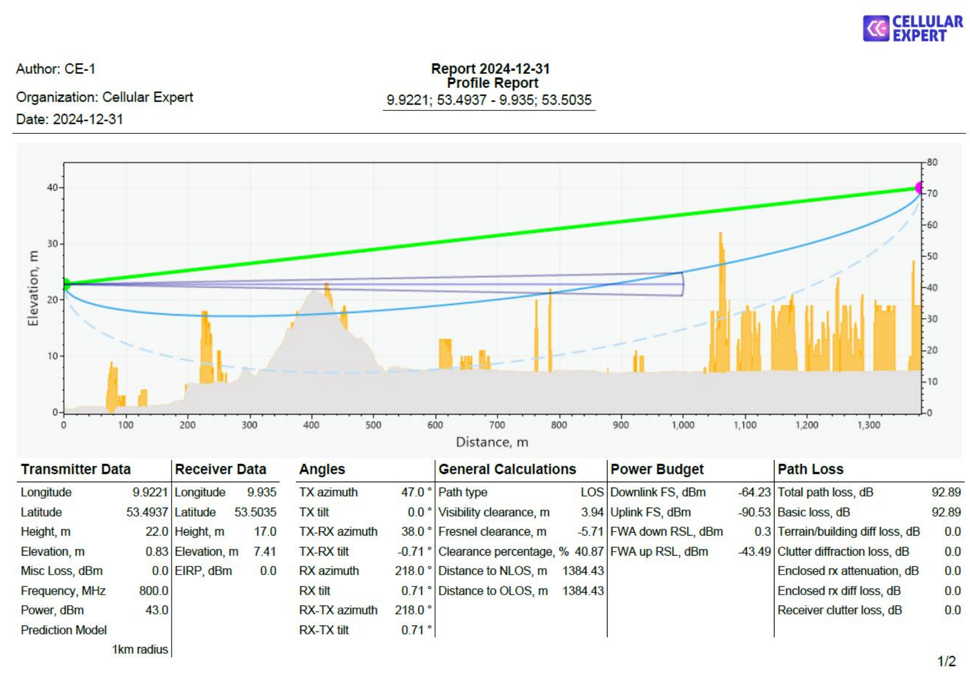
Load Profile
Creates the profile with the provided data.

8.1.5 Export (Profile Report)

The input data and calculation results can be automatically transferred into a **Profile Report**. This report will show transmitter/receiver input data, calculation results as well as the Profile plot and map view in which the profile was drawn. The report can be exported in PDF and PL2 formats. The Profile Report can also be saved to Docs Manager by selecting **Save result to Docs Manager**.



The resulting Profile report will look similar to this example:



Author: CE-1

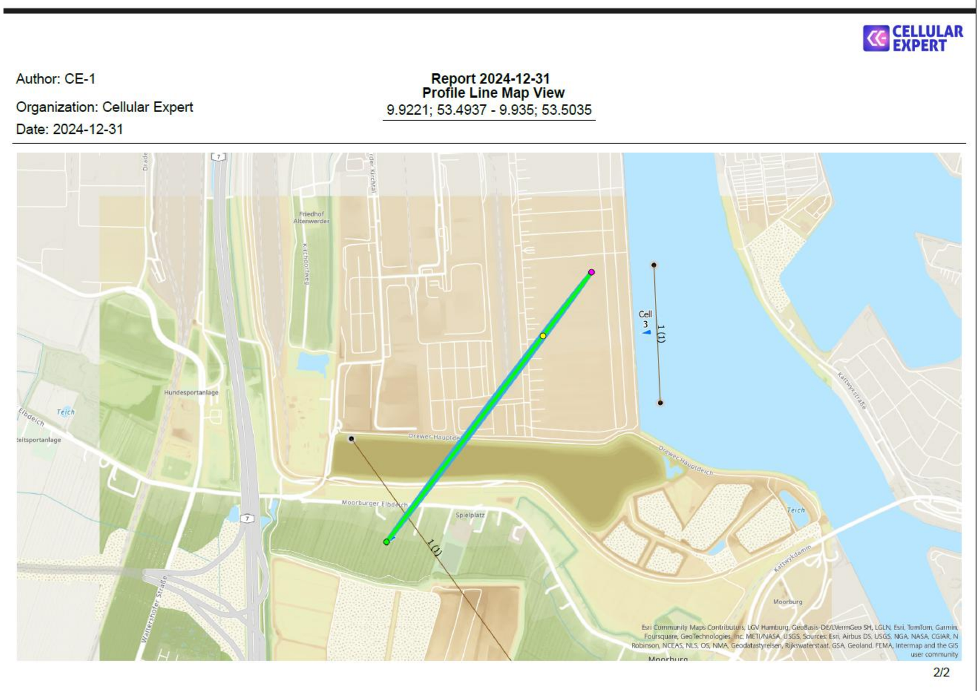
Organization: Cellular Expert

Date: 2024-12-31

Report 2024-12-31
Profile Report
9.9221; 53.4937 - 9.935; 53.5035

Transmitter Data		Receiver Data		Angles		General Calculations		Power Budget		Path Loss	
Longitude	9.9221	Longitude	9.935	TX azimuth	47.0 °	Path type	LOS	Downlink FS, dBm	-64.23	Total path loss, dB	92.89
Latitude	53.4937	Latitude	53.5035	TX tilt	0.0 °	Visibility clearance, m	3.94	Uplink FS, dBm	-90.53	Basic loss, dB	92.89
Height, m	22.0	Height, m	17.0	TX-RX azimuth	38.0 °	Fresnel clearance, m	-5.71	FWA down RSL, dBm	0.3	Terrain/building diff loss, dB	0.0
Elevation, m	0.83	Elevation, m	7.41	TX-RX tilt	-0.71 °	Clearance percentage, %	40.87	FWA up RSL, dBm	-43.49	Clutter diffraction loss, dB	0.0
Misc Loss, dBm	0.0	EIRP, dBm	0.0	RX azimuth	218.0 °	Distance to NLOS, m	1384.43			Enclosed rx attenuation, dB	0.0
Frequency, MHz	800.0			RX tilt	0.71 °	Distance to OLOS, m	1384.43			Enclosed rx diff loss, dB	0.0
Power, dBm	43.0			RX-TX azimuth	218.0 °					Receiver clutter loss, dB	0.0
Prediction Model	1km radius			RX-TX tilt	0.71 °						

1/2



Author: CE-1

Organization: Cellular Expert

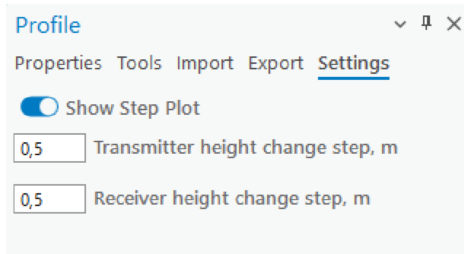
Date: 2024-12-31

Report 2024-12-31
Profile Line Map View
9.9221; 53.4937 - 9.935; 53.5035

2/2

8.1.6 Settings

Currently, you can configure the profile's visual properties and controls in profile settings. The settings can be changed once a profile is drawn.

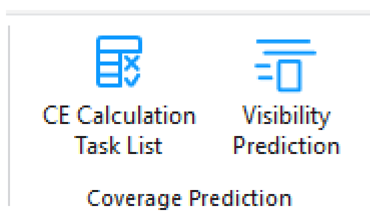


Show Step Plot
Display data points as a series of steps rather than a smooth line or individual points.

Transmitter height change stepnter
The step (in meters) for changing the transmitter height when dragging it in the Profile graph. The default value is 0.5 meters.

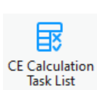
Receiver height change step
The step (in meters) for changing the receiver height when dragging it in the Profile graph. The default value is 0.5 meters.

9. Coverage Prediction

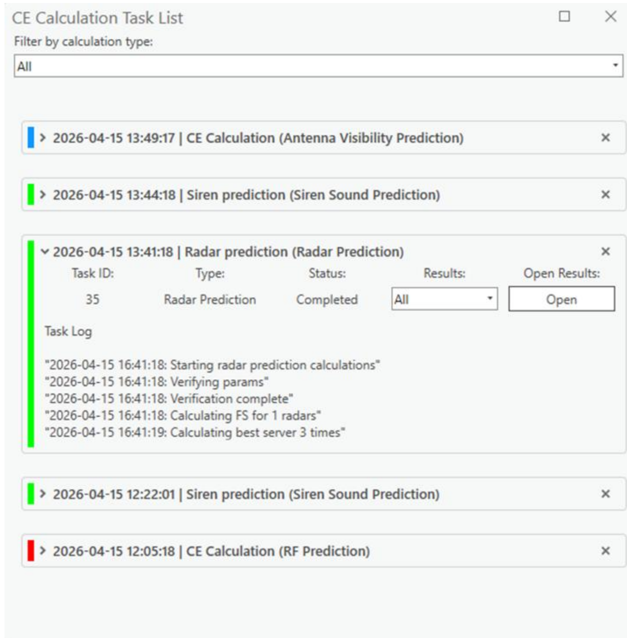


9.1 CE Calculation Task List

All results of the prediction calculations can be found in the **CE Calculation Task List** tab. This includes failed and successful calculations.

Click on the 

button to open the **CE Calculation Task List** dialogue.



The task list refreshes automatically once calculation tasks are run. The task status is indicated by three main colors: blue (in progress), green (completed), and red (failed). Calculation tasks can be deleted from the task list by clicking × on the right side of the task. To open a result raster, select it from the results dropdown and click Open Results. Filtering by calculation spans these types: Antenna Visibility Prediction, EMF Calculation, Link Prediction, Model Tuning, Optimal Site Positions Calculation, RF Prediction, Siren Sound Prediction, and Visibility Prediction.

9.2 Visibility Prediction

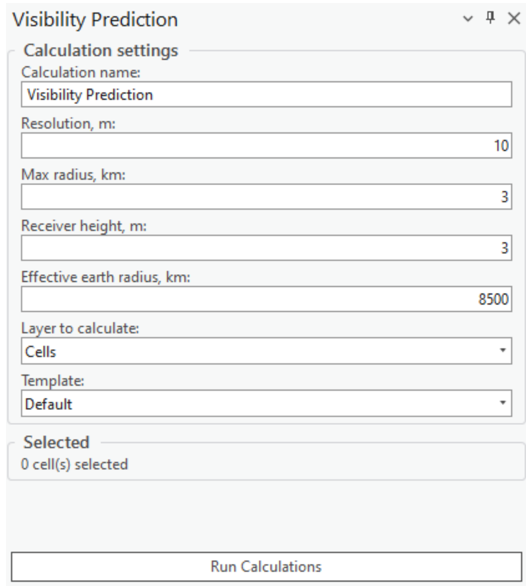
Visibility calculations refer to the determination of line-of-sight between transmitting and receiving antennas, assessing whether any obstructions might impede direct signal transmission.

Visibility Prediction is a tool that calculates 4 different results:

- Minimum Receiver Height – the minimum height of a receiver that could be visible to the transmitter
- Line of Sight – confirms whether visibility exists between the receiver and transmitter with the provided receiver height
- Clearance Height – the distance from the profile that is covered or is not covered.
- Best Server – the same calculation as for **RF Prediction**

Click the 

button to open the **Visibility Prediction** dialogue.



Calculation Name
Name of the calculation that will be displayed in the CE Calculation Task List.

Resolution
Resolution for raster calculations. The output rasters will be produced with the indicated cell size.

Max Radius
The maximum radius of the prediction

Receiver height
Receiver height above the ground.

Effective Earth Radius
Earth radius in kilometers, used for the calculations.

Layers to calculate
All present CE layers on which predictions can be performed

Template
A template that corresponds to the selected layers. When the layer changes the templates change as well.

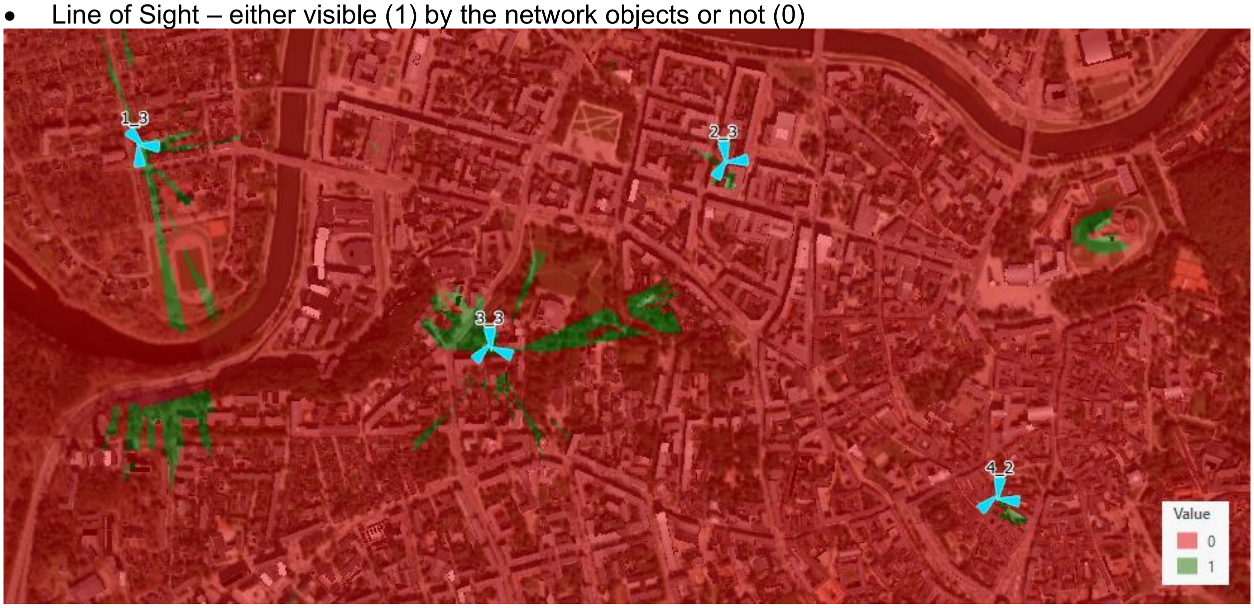
Selected
Network objects that are present on the selected network layer. The visibility prediction will be performed on all of them.

Run Calculations
Starts the prediction calculation.

Results:

- Minimum Receiver Height in meters

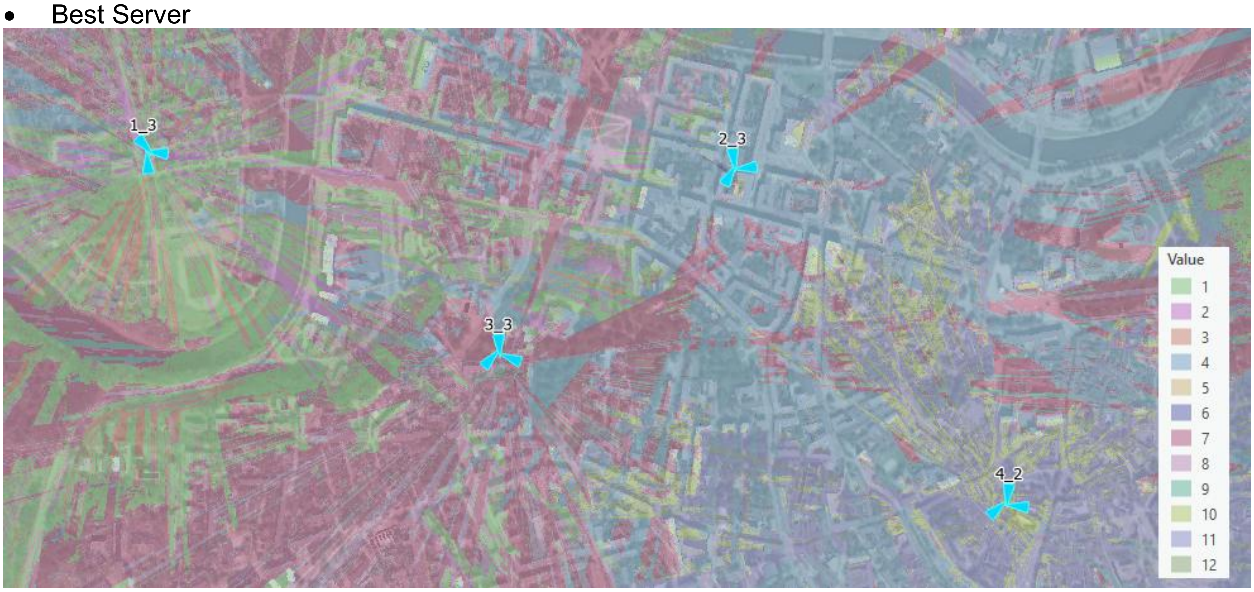
- Line of Sight – either visible (1) by the network objects or not (0)



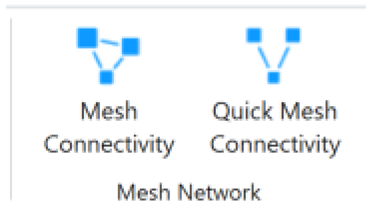
- Clearance in meters


1_3
2_3
3_3
4_2
Value
< -10m
-10 - 0m
0 - 5m
> 5m

- Best Server


1_3
2_3
3_3
4_2
Value
1
2
3
4
5
6
7
8
9
10
11
12

10. Mesh Networks

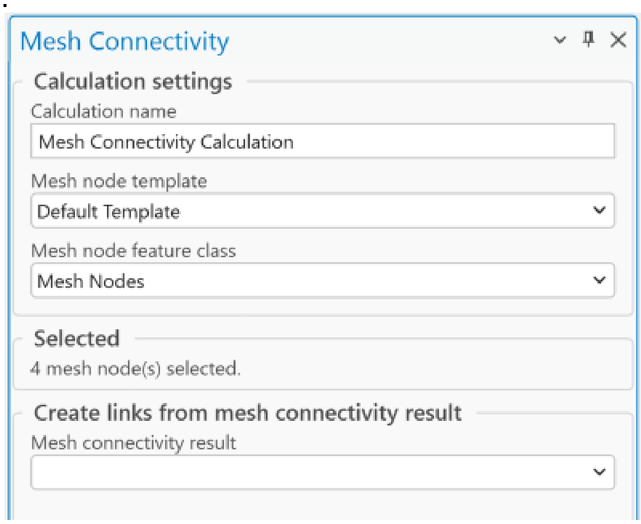


10.1 Mesh Connectivity

Click the 

button to open Mesh Connectivity dialogue.

This tool enables rapid and accurate assessment of wireless peer-to-peer link quality during mesh network planning and optimization.



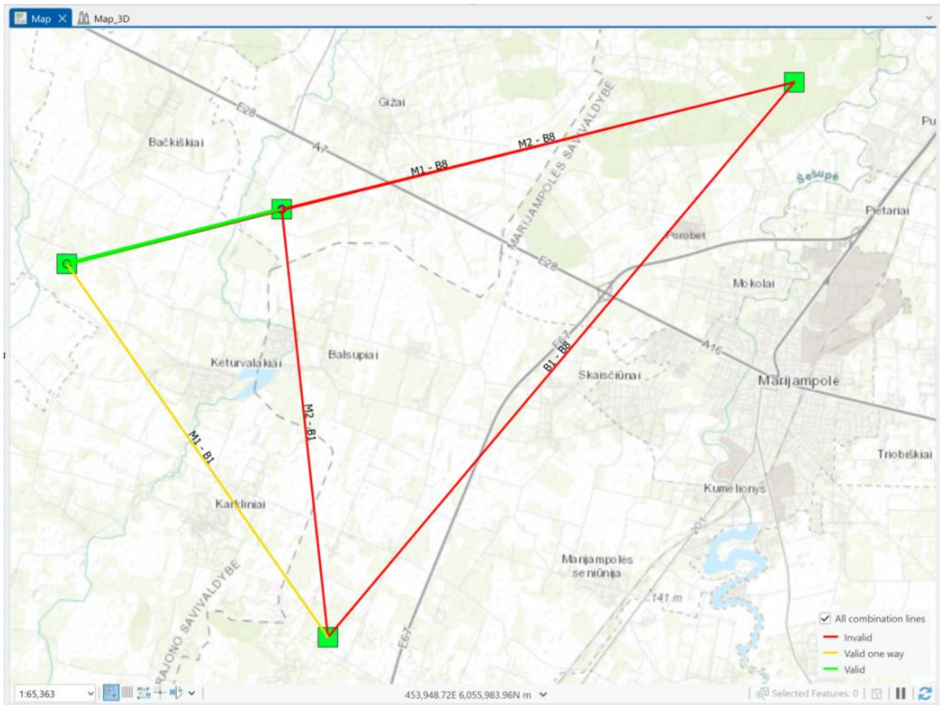
How It Works:

- Select one or more Mesh Nodes in the workspace.
- Open Mesh Connectivity tool.
- Define:
  - Calculation name – the calculation name which will be visible in Contents.
  - Mesh node template – mesh node template if selected features won't have all necessary parameters.
  - Mesh node feature class – by default Mesh Nodes are using for the calculations, but external point type feature class can be used to estimate connectivity with each other.
  - Press Run button to start simulation.
- The tool calculates the RSL between each node pair and compares it against the configured **RSL sensitivity threshold** defined for the Mesh Node object.
- Connectivity status is then **visually rendered** using a clear, color-coded scheme:
  - 🟢 **Green**: Bi-directional connectivity – both nodes meet or exceed the RSL sensitivity for each other.
  - 🟡 **Yellow**: Uni-directional connectivity – only one node meets the RSL threshold for the

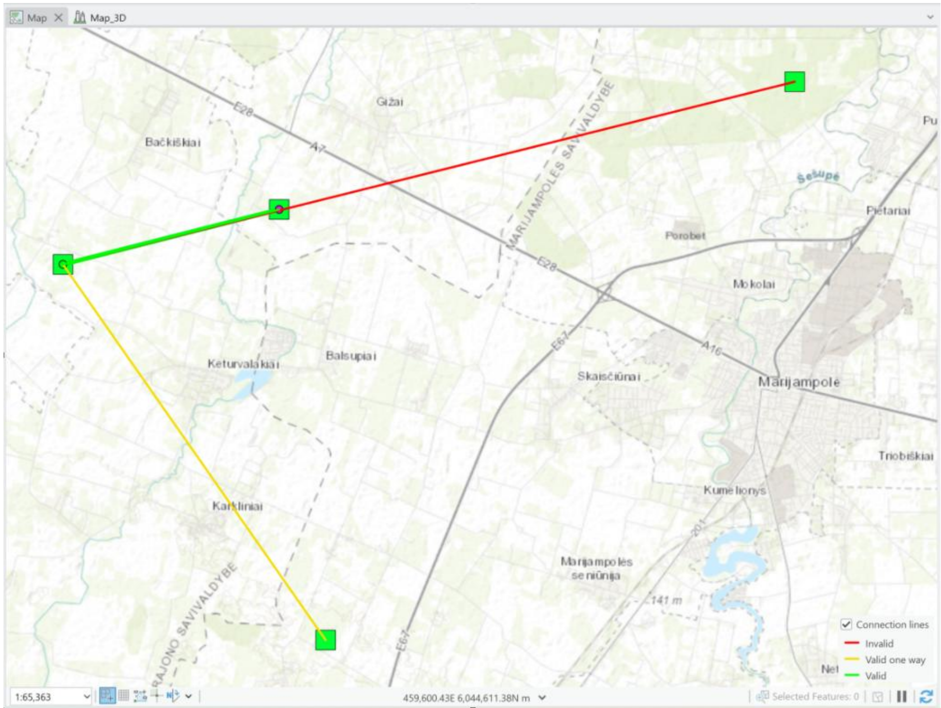
other.

- 🔴 **Red**: No connectivity – RSL falls below the sensitivity threshold in both directions.

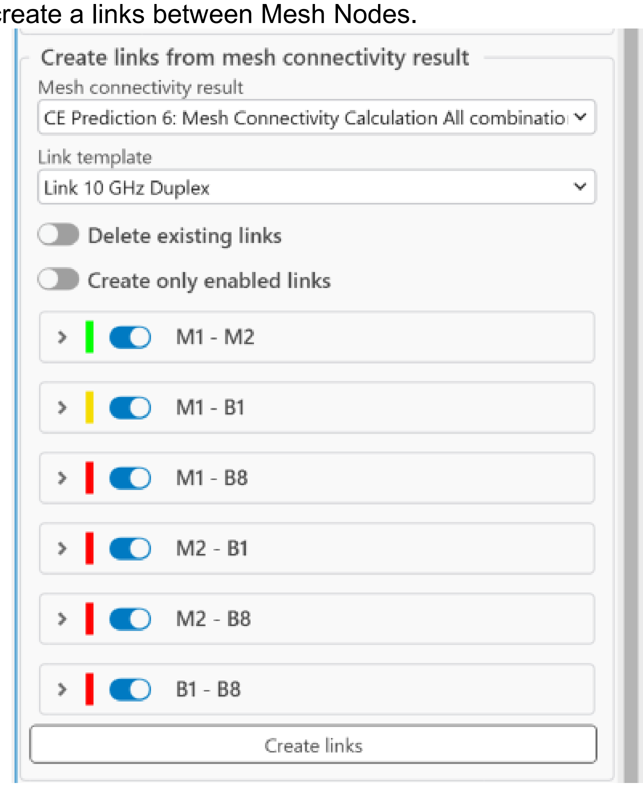
Result. All cominications
It shows all possible combinations of Mesh network connections.



Result. Connection lines
Shows Mesh connections where the tool will create Link object.



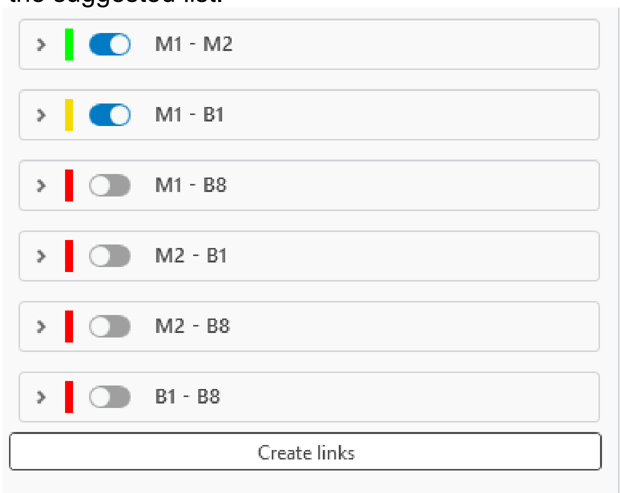
To create a Links between Mesh Nodes, define Mesh connectivity results. Once result is selected, it will add possible options to create a links between Mesh Nodes.



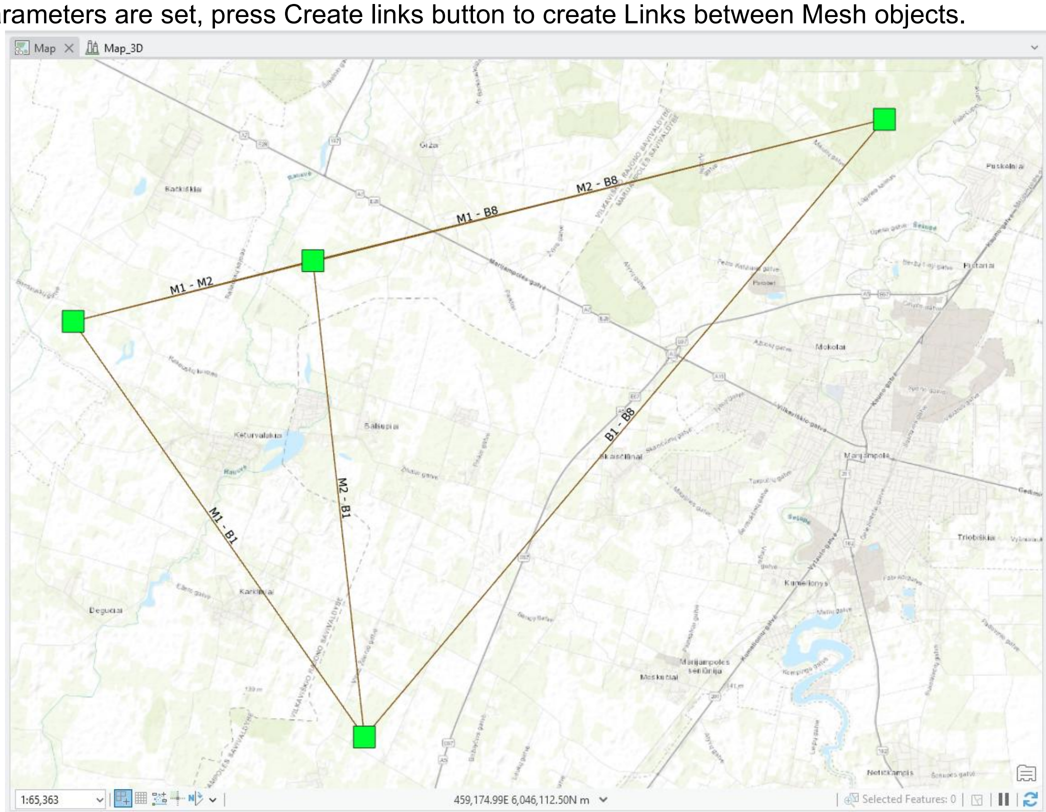
Delete existing links
If selected, existing links that have been created from mesh connectivity results will be deleted.

Create only enabled links
If checked, the links will be created only from the checked connectivity lines in the suggested list.

Enable/Disable Links from the suggested list.



Once parameters are set, press Create links button to create Links between Mesh objects.



10.2 Quick Mesh Connectivity

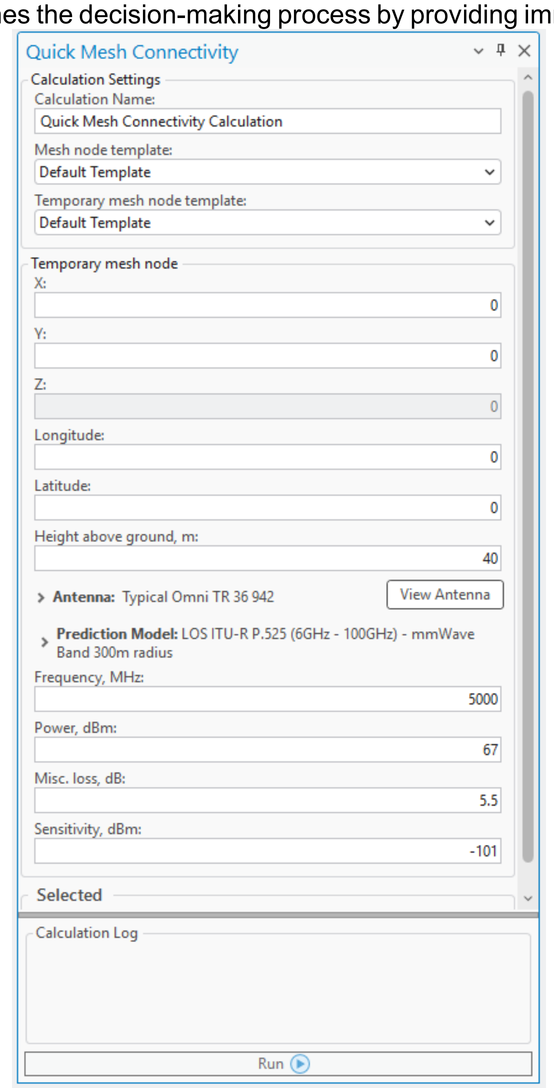
Click the 

button to open Quick Mesh Connectivity dialogue.

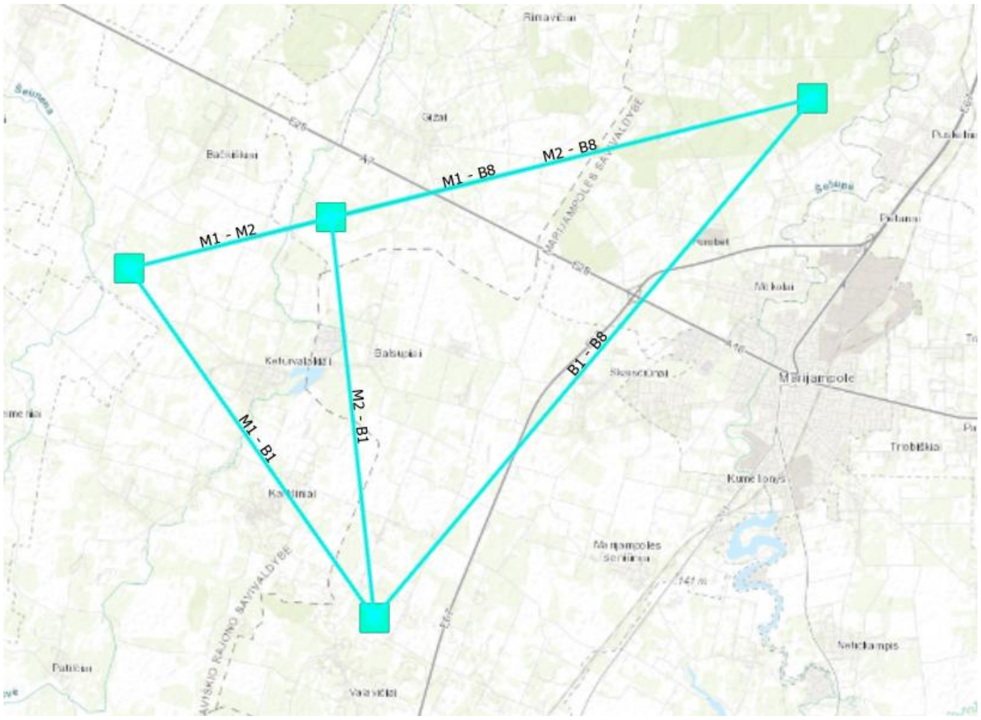
To enhance the efficiency of early-stage planning and dynamic mesh layout, the **Quick Mesh Connectivity** tool offers a fast and intuitive way to assess connectivity potential.

This lightweight yet powerful feature enables users to **instantly evaluate whether a Mesh Node can connect to an existing mesh network** based on either its current position or a proposed location.

Whether you are planning the deployment of a single node or simulating a mobile mesh scenario, **Quick Mesh Connectivity** streamlines the decision-making process by providing immediate, actionable feedback.

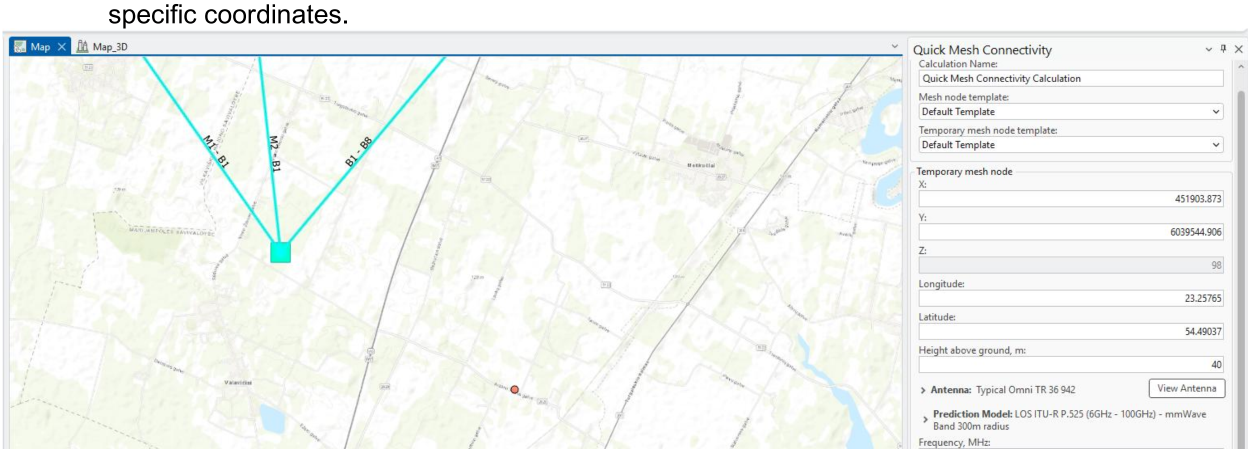


1. **Select Existing Mesh Nodes**
   On the map interface, select the Mesh Nodes that should be included in the connectivity calculation.



2. **Open the Quick Mesh Connectivity Tool**
   Launch the tool from the menu or toolbar.

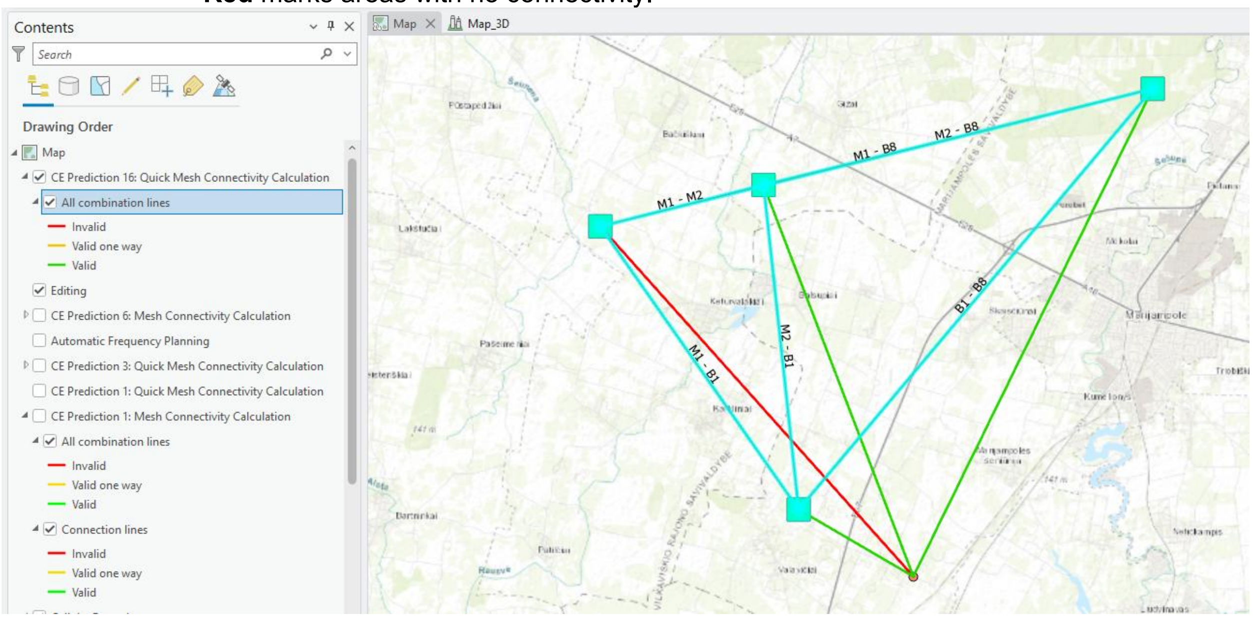
3. **Define the Proposed Node Location**
   Choose the location of the new Mesh Node either by clicking directly on the map or by entering specific coordinates.



4. **Configure Node Parameters**
   Input the required parameters for the proposed Mesh Node (e.g., transmission power, antenna type, height).

5. **Run the Calculation**
   Click the **Run** button to initiate the connectivity analysis.

6. **View the Results**
   Once the analysis is complete, the results will appear in the **Contents** panel and on the map:
   - **Green** indicates areas with successful two-way connectivity.
   - **Yellow** shows one-way connectivity.
   - **Red** marks areas with no connectivity.



Calculation Name
The name of calculations which will be used and loaded to Contents.

Mesh node template
The **Mesh Node Object Template** is automatically utilized whenever a Mesh object lacks the required parameters needed to complete connectivity calculations. This ensures that all essential data is available for accurate simulation and evaluation, even if the original Mesh object is incomplete.

Temporary mesh node template
The **Temporary Mesh Node Object Template** is automatically utilized whenever a newly proposed Mesh object lacks the required parameters needed to complete connectivity calculations. This ensures that all essential data is available for accurate simulation and evaluation, even if the original Mesh object is incomplete.

Latitude
Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

Longitude
Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

X
Coordinate in the projected coordinate system.

Y
Coordinate in the projected coordinate system.

Z

3D dimensions representing an object's height above sea level, used for visualizing objects in a 3D scene.

Height Above Ground
Mesh node object's height above the terrain.

Antenna
Antenna which will be used for Mesh node.

Prediction model
Prediction model used to calculate a path loss between Mesh Nodes.

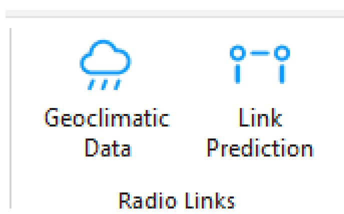
Frequency
Frequency in MHz for proposed Mesh Node.

Power
Tx power in dBm for proposed Mesh Node.

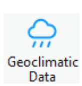
Misc. Loss
Total losses in dB for proposed Mesh Node.

Sensitivity
Receiving Signal Level threshold in dBm at Mesh Node.

11. Radio Links

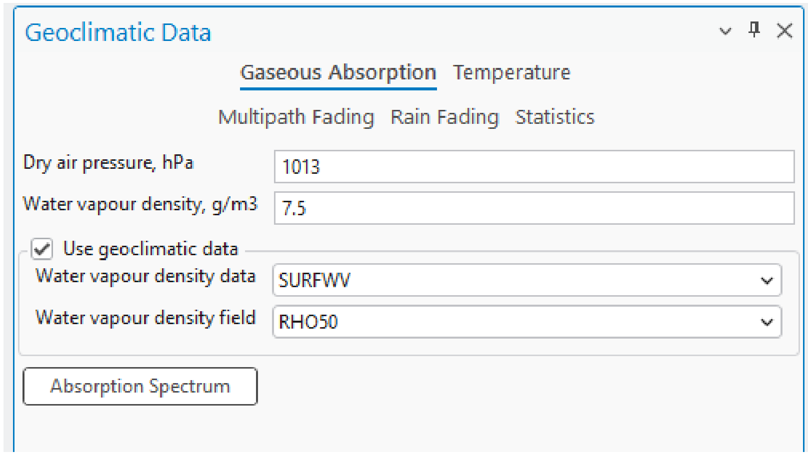


11.1 Geoclimatic Data

Click the 

button to open the **Geoclimatic Data** dialogue.

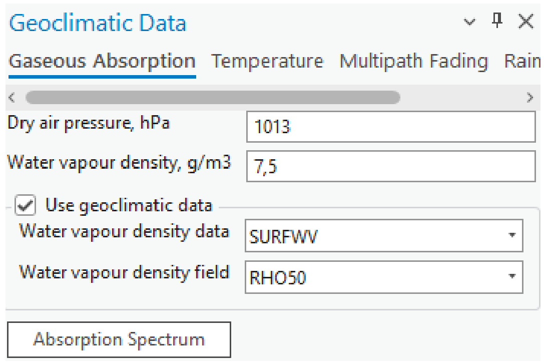
Geoclimatic Data is a tool that lets you adjust the geoclimatic settings that will be used in certain calculations (e.g. Link Prediction).



Save Changes
Save changes to the settings.

11.1.1 Gaseous Absorption

Gaseous absorption pages define values for dry air pressure and water vapour density. These values can be obtained from predefined geoclimatic data maps.



Water vapour density data according to ITU-R P.836-3. It is used in gaseous absorption evaluation.

Dry Air Pressure
The atmospheric pressure contributed by air that contains no water vapor, typically measured in hectopascals (hPa).

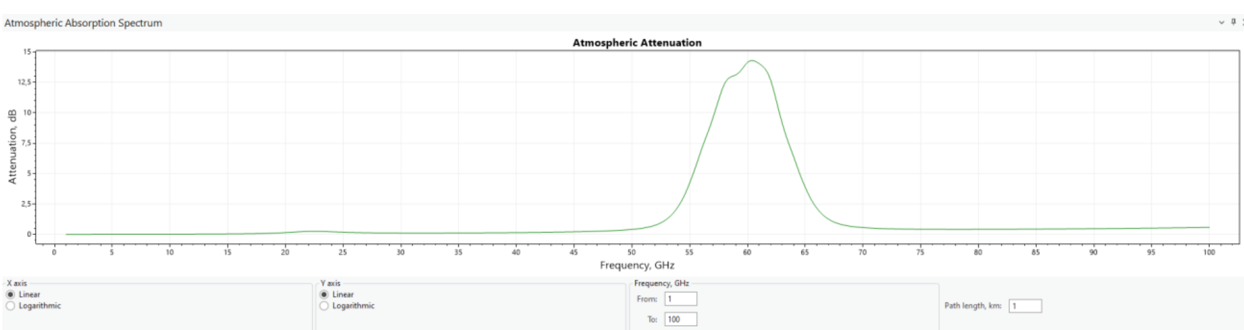
Water Vapour Density
The mass of water vapor present in a unit volume of air, typically measured in grams per cubic meter (g/m³).

Water Vapour Density Data
A dropdown menu that allows the user to select the source or type of data used to determine water vapor density.

Water Vapour Density Data
A dropdown menu to select the specific field or parameter from the chosen data source that provides the water vapor density information.

Use Geoclimatic Data
Enable/Disable this Geoclimatic data

Absorption Spectrum
A button that lets you see the visual representation of the absorption spectrum.



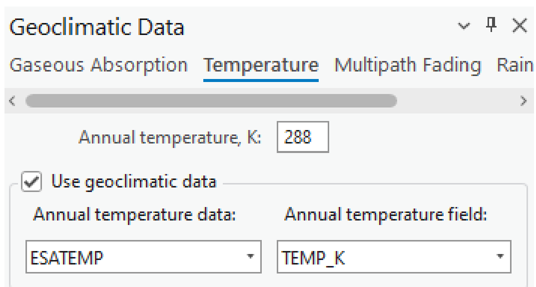
Linear/Logarithmic
Specifies how the axes values are calculated and displayed on the plot. Logarithmic is calculated as $\log_{10}$.

Frequency From/To
The frequency range of the absorption spectrum.

Path Length
The length of the spectrum.

11.1.2 Temperature

Annual mean surface temperature at 2m above the surface of the Earth according to ITU-R P.1510. The data is used to evaluate the thermal noise of a receiver.



Annual Temperature
Annual Temperature expressed in Kelvins.

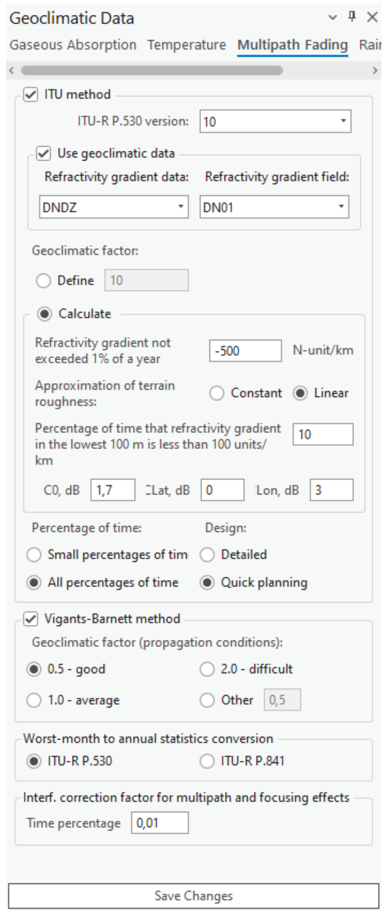
Annual Temperature Data
A dropdown menu allowing the user to select the dataset from which the annual temperature data should be sourced.

Annual Temperature Field
A dropdown menu to select the specific data field within the chosen dataset that contains the annual temperature information.

Use Geoclimatic Data
Enable/Disable this Geoclimatic data.

11.1.3 Multipath Fading

The Multipath Fading page defines refractivity data and calculation parameters for ITU and Vigants-Barnett methods.



Worst month-to-annual statistics conversion can be performed according to ITU-R P. 530 or ITU-R P. 841 methods.

Refractivity gradient data is based on ITU-R P.453-9. Refractivity gradient data is used for multipath fading analysis.

ITU Method
Enable/Disable the ITU Method

ITU-R P.530 version
A checkbox indicating the use of the International Telecommunication Union's method for calculations, with a dropdown to select the ITU-R P.530 version.

Use Geoclimatic Data
Enable/Disable this Geoclimatic data.

Refractivity Gradient Data / Field
Refractivity gradient data: DNDZ
Refractivity gradient field: DN01; DN10; DN50; DN90; DN99, KF01...KF99

- **DNx** - refractivity gradient not exceeded for x% of the average year in the lowest 65 m of atmosphere. DN01 is the parameter referred to as dN1 in ITU-R P.530.
- **KFx** - Effective Earth radius not exceeded for x% of the average year in the lowest 65 m of atmosphere. KFx is provided only for map preview and cannot be used in the Geoclimatic Data settings dialog.

Geoclimatic Factor (Define)
A field to input or calculate a factor used in propagation modeling, affecting the prediction of multipath fading.

Geoclimactic Factor (Calculate)
Calculate the geoclimactic factor used in propagation modeling, affecting the prediction of multipath fading.

- **Approximation of terrain roughness** - Options to select how terrain roughness is modeled in the calculations - as a constant value or as a linear approximation.
- **Percentage of time that refractivity gradient in the lowest 100 m is less than 100 units/km** - An input field for the percentage of time when the refractivity gradient at low altitudes meets a specific threshold.
- **Co, dB / Lat, dB / Lon, dB** - Fields to input correction values in decibels for specific geographic coordinates, possibly related to the orientation or position of the transmitter/receiver.

Percentage of Time / Design
Radio buttons to choose the granularity of the time percentage for which the calculations are relevant, and the level of detail required for planning.

Use Viggants-Barnett Method
Enabled/Disabled Viggants-Barnett Method.

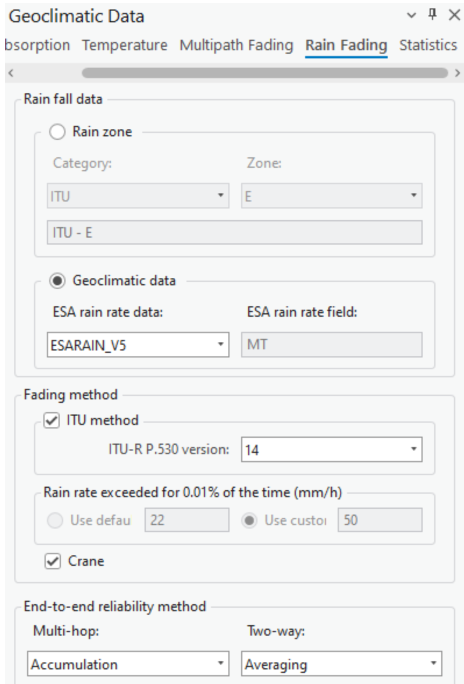
Geoclimatic Factor (Propagation Conditions)
A checkbox to select this specific method for calculations, with radio buttons to choose a geoclimatic factor indicating propagation conditions.

Worst-month-to-annual statistics conversion
A section to select the ITU recommendation (either ITU-R P.530 or ITU-R P.841) for converting worst-month statistics to annual statistics.

Interf. correction factor for multipath and focusing effects
Input for the percentage of time to apply an interference correction factor to account for multipath and focusing effects.

11.1.4 Rain Fading

The rain fading page defines rain regions (ITU and Crane) for rain rate statistics and calculation methods (ITU and Crane). The rain rate exceedance parameter for 0.01% of the time can be set manually or automatically according to the rain zone.



Rain rate data based on ITU-R P.837-4. This data is used to evaluate rain fading.

Rain Zone
Elections to categorize the geographic location by ITU rain zone standards, which are used to estimate rain attenuation in different regions.

Geoclimatic data - ESA rain rate data / ESA rain rate
Options to select the dataset (such as "ESARAIN_V5") and the specific field within that dataset (like "MT") for rain rate data, used for calculating rain attenuation.

Fading method - ITU method / ITU-R P.530 version
A checkbox to select the ITU method for rain fading calculations and a dropdown to choose the version of the ITU-R P.530 recommendation being applied.

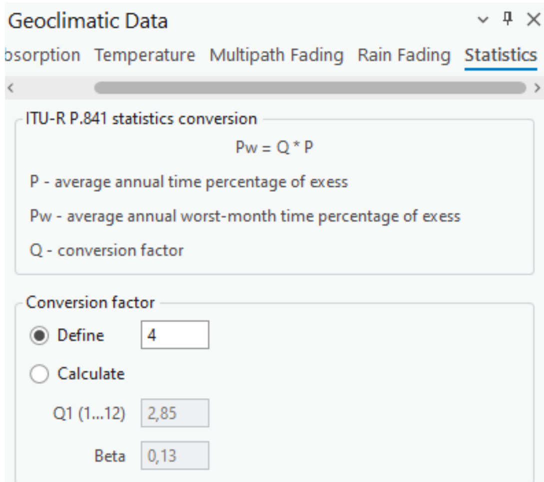
Rain rate exceeded for 0.01% of the time (mm/h)
Options to use either default or custom values for the rain rate that is exceeded for 0.01% of the time, which is a measure used to predict rain fading in telecommunications.

Crane
This is a model selection option related to the Crane model for calculating rain attenuation.

End-to-end reliability method - Multi-hop / Two-way
Selection options for the method of calculating the reliability of a signal in a communication path that may involve multiple hops or two-way transmission.

11.1.5 Statistics

These settings are used to handle statistical conversions for telecommunication planning, ensuring that systems are designed to cope with the worst-case scenarios based on historical data and predictive models.

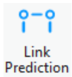


Conversion factor - Define / Calculate
Options to either define a fixed conversion factor or to calculate it based on additional inputs.

Q1 (1...12)
An input field likely used to define a conversion factor or another parameter for a specific month or set of months.

Beta
An input field for the beta parameter, which may be part of the statistical model or conversion formula within the ITU-R P.841 recommendation.

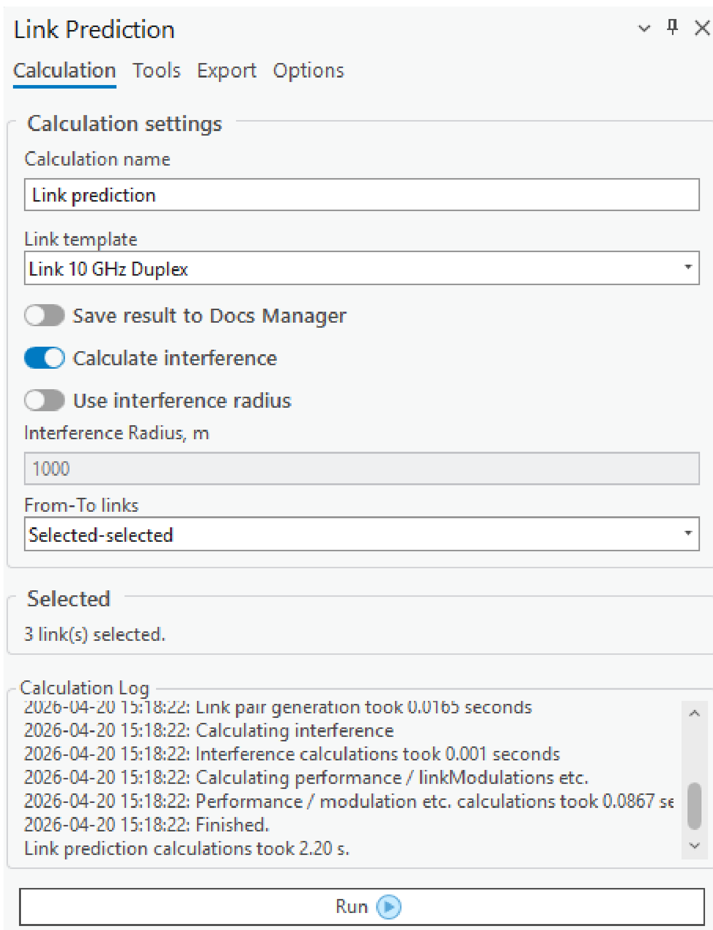
11.2 Link Prediction

Click the  button to open the **Link Prediction** dialogue.

Link Predictions is a tool specific to *CE for ArcGIS Pro* RLP license. The tool enables you to calculate Link predictions.

11.2.1 Calculation

The **Calculation** tab is found on the Link Prediction dockpane. Here you can see the selected number of carriers as well as some other parameters.

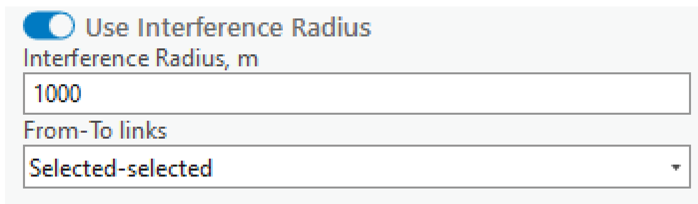


Calculation Name
Link Prediction identification.

Link Template
The template will fill all empty or not specified fields with default values that are not necessary for predictions.

Calculate Interference
If checked will calculate the interference of multiple links. If enabled, also allows to specify a certain radius from either link endpoint. Points outside this range will not be included in the calculation, speeding up the calculation process. If the radius is not specified, all links will be used in the calculations.

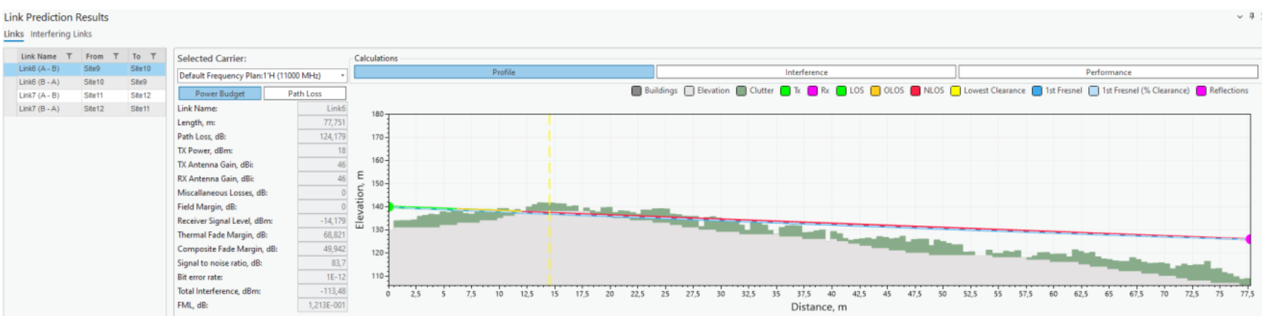
The “From-To links” dropdown settings allows the selection of links used for interference calculations. The available options are “Selected-selected” (default option), “Selected-all”, and “All-all”. When the “All-all” option is selected, it is not necessary to manually select all links on the map to perform the prediction calculations.



11.2.1.1 Links

After calculations, a new dockpane will open. In the **Links** tab. You will be able to view **Power Budget**, **Path Loss**, **Interference,** and **Performance** calculation results.

The Link Profile behaves in virtually the same way as a regular profile (you cannot adjust Link Profile data). Read more in *Profile*.



This section of the results contains **Power Budget** and **Path Loss** results. Select different carriers to view the results for each one of them.

Power Budget - the calculation of the balance between transmitted power, power losses in the system, and receiver sensitivity in a communication system.

Path Loss - the reduction in signal strength as it travels from the transmitter to the receiver.

Selected Carrier:

Default Frequency Plan:1'H (11000 MHz)

Power Budget | Path Loss

Link Name:	Link6
Length, m:	77,751
Path Loss, dB:	124,179
TX Power, dBm:	18
TX Antenna Gain, dBi:	46
RX Antenna Gain, dBi:	46
Miscallaneous Losses, dB:	0
Field Margin, dB:	0
Receiver Signal Level, dBm:	-14,179
Thermal Fade Margin, dB:	68,821
Composite Fade Margin, dB:	49,942
Signal to noise ratio, dB:	83,7
Bit error rate:	1E-12
Total Interference, dBm:	-113,48
FML, dB:	1,213E-001

Selected Carrier:

Default Frequency Plan:1'H (11000 MHz)

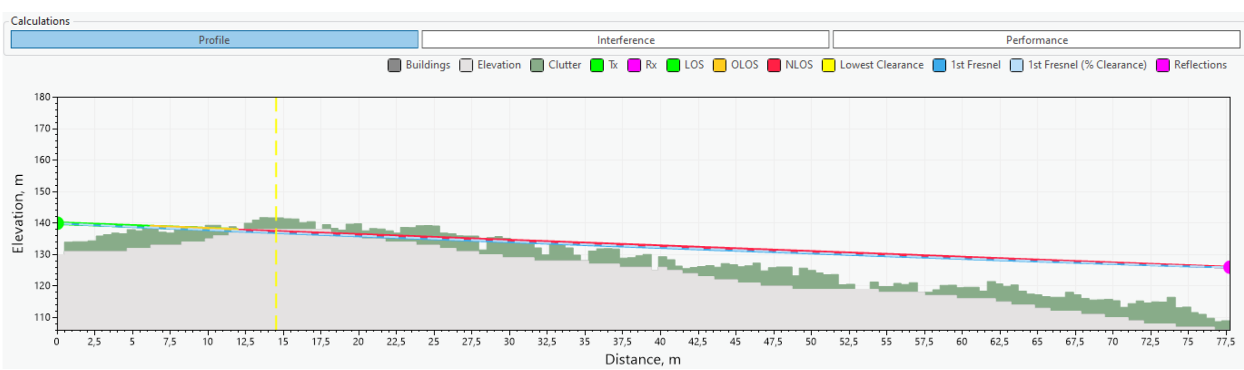
Power Budget | Path Loss

Link Name:	Link6
Length, m:	77,751
Link Azimuth, deg:	74
Link tilt, deg:	10,21
Clearance, m:	-4,34
Clearance Distance, m:	14,51
Distance to OLOS, m:	11,92
Distance to NLOS, m:	6,22
Path loss, dB:	124,179
Fade occurence factor, %	2,78E-009
ITU climatic factor	0,002
ITU C0 factor	1,7
V.-Barnett climatic factor	0,5
Refractivity grad., N-units/km	-351,15
Refractivity gradient, %	10

This section of the results contains the **Profile Plot**, **Interference,** and **Performance** results. Select different carriers to view the results for each one of them.

Interference - the undesired impact of one signal on another, leading to potential signal degradation.

Performance - assessments of key metrics like data rate, error rates, and signal-to-noise ratio, crucial for evaluating and optimizing a communication system's efficiency.



Calculations

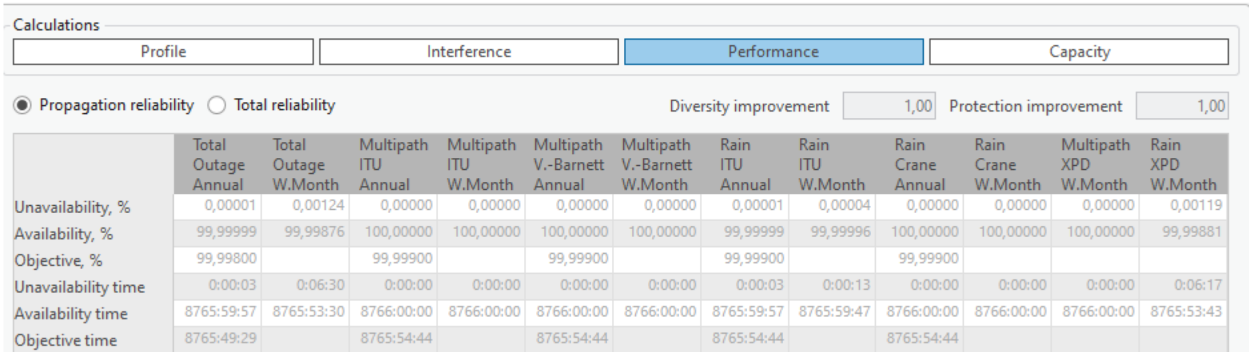
Profile | Interference | Performance

Interference from (interfering links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
Link7	Site11	Default Frequency Plan:1'H	56,8	87,915	-113,48	99,301	0
Link7	Site11	Default Frequency Plan:2'H	56,8	87,993	-213,558	199,379	0
Link7	Site12	Default Frequency Plan:1H	42,11	128,939	-260,458	246,279	0
Link7	Site12	Default Frequency Plan:2H	42,11	129,069	-260,588	246,409	0

Interference to (interfered links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
Link6	Site12	Default Frequency Plan:1'H	56,321	93,933	-119,235	96,421	0
Link6	Site12	Default Frequency Plan:2'H	56,321	93,933	-219,235	196,303	0
Link6	Site11	Default Frequency Plan:1H	41,863	85,264	-217,588	196,159	0
Link6	Site11	Default Frequency Plan:2H	41,863	85,264	-217,588	196,029	0



Calculations

Profile | Interference | Performance | Capacity

Propagation reliability ○ Total reliability — Diversity improvement 1,00 — Protection improvement 1,00

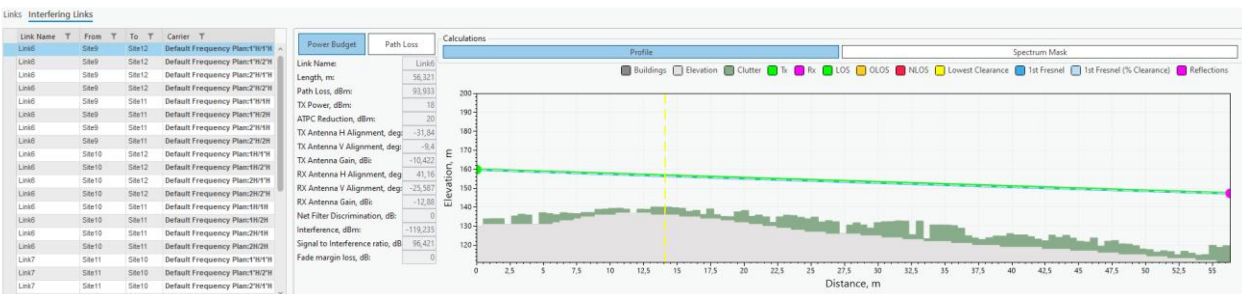
	Total Outage Annual	Total Outage W.Month	Multipath ITU Annual	Multipath ITU W.Month	Multipath V.-Barnett Annual	Multipath V.-Barnett W.Month	Rain ITU Annual	Rain ITU W.Month	Rain Crane Annual	Rain Crane W.Month	Multipath XPD W.Month	Rain XPD W.Month
Unavailability, %	0,00001	0,00124	0,00000	0,00000	0,00000	0,00000	0,00001	0,00004	0,00000	0,00000	0,00000	0,00119
Availability, %	99,99999	99,99876	100,00000	100,00000	100,00000	100,00000	99,99999	99,99996	100,00000	100,00000	100,00000	99,99881
Objective, %	99,99800		99,99900		99,99900		99,99900		99,99900			
Unavailability time	0:00:03	0:06:30	0:00:00	0:00:00	0:00:00	0:00:00	0:00:03	0:00:13	0:00:00	0:00:00	0:00:00	0:06:17
Availability time	8765:59:57	8765:53:30	8766:00:00	8766:00:00	8766:00:00	8766:00:00	8765:59:57	8765:59:47	8766:00:00	8766:00:00	8766:00:00	8765:53:43
Objective time	8765:49:29		8765:54:44		8765:54:44		8765:54:44		8765:54:44			

Select different links to see these results for them.

11.2.1.2 Interfering Links

Navigate to the **Interfering Links** tab on the Link Prediction result dockpane.

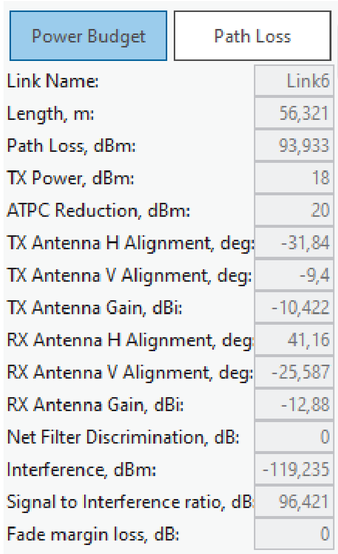
In this tab, you will be able to view interference **Power Budget**, **Path Loss**, **Profile,** and **Spectrum Mask** results in an unordered fashion.

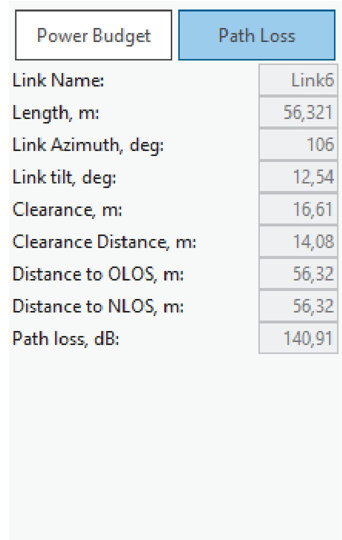


In this section, you can view **Power Budget** and **Path Loss** results. Change the selected site pair to see different results for each one of them

Power Budget - the calculation of the balance between transmitted power, power losses in the system, and receiver sensitivity in a communication system.

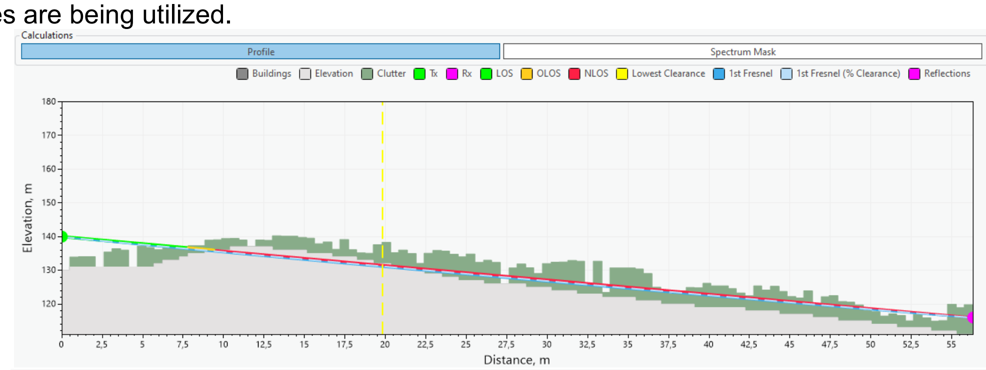
Path Loss - the reduction in signal strength as it travels from the transmitter to the receiver.

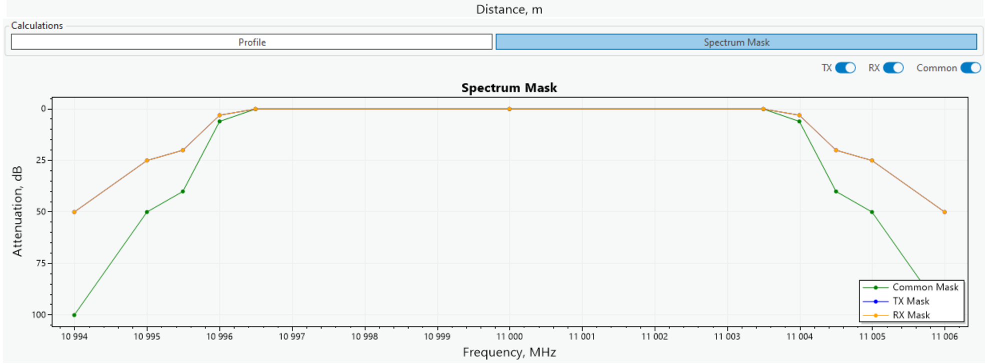




In this section, you can view **Profile Plot** and **Spectrum Mask** results. Change the selected site pair to see different results for each one of them

Spectrum Mask - the method used to assess the usage efficiency of a frequency spectrum in a telecommunications system, involving the measurement of how much and how effectively different frequencies are being utilized.





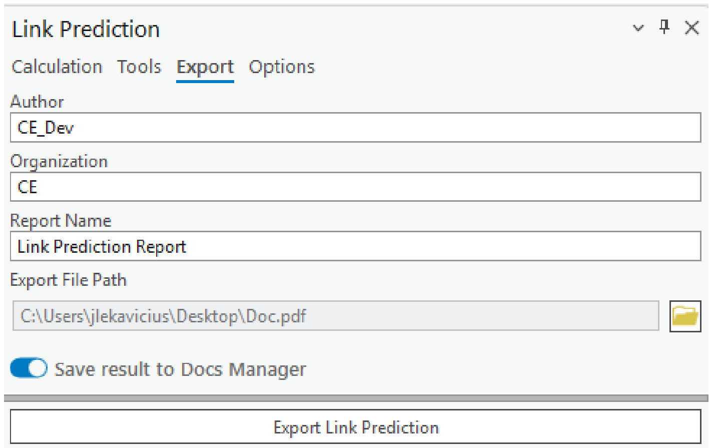
Tx, Rx, Common
Remove/Add certain parameters from the plot

11.2.2 Reflections

Reflections are calculated in the same way as a regular Profile. Read more in **Tools.**

11.2.3 Export (Link Prediction Report)

The calculation results can be automatically transferred into a **Link Prediction Report**. This report will show profile, prediction parameter and results, performance and propagation reliability. The report can be exported in PDF format. The Link Prediction Report can also be saved to Docs Manager by selecting **Save result to Docs Manager**.



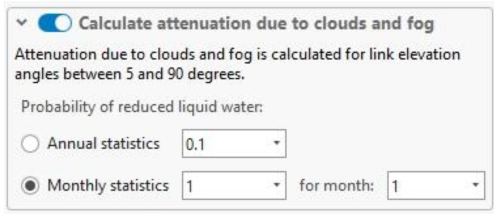
11.2.4 Options

The **Options** tab is found on the Link Prediction dockpane. Here you will see additional parameters that can be changed in accordance with your prediction needs.

Include Field Margin in Power Budget
If checked, it will take into account the specified field margin for Power Budget calculations.

Field Margin
Extra signal strength beyond the minimum required, providing a buffer for reliable communication against unforeseen signal losses or variations.

Thermal Fade Margin Prediction
The threshold below which thermal fade will not be calculated.

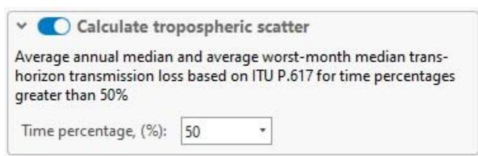


Calculate attenuation due to clouds and fog
Enabled/Disable this option.

Annual Statistics
The annual chance of reduced water in %.

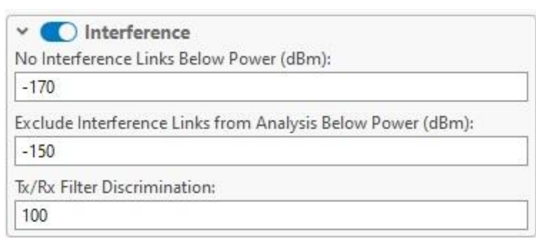
Monthly Statistics
Monthly chance of reduced water in %.

For Month
Specify which month has the monthly statistic.



Calculate Tropospheric Scatter
Enable/Disable this option.

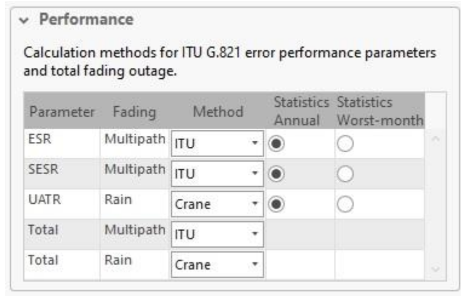
Time Percentage
Time Percentage in %.



No Interference Link Below Power
The power threshold below which no interference for links will be calculated.

Exclude Interference Links from Analysis Below Power
The power threshold below which no links with lesser interference will be included in the analysis

Tx/Rx Filter Discrimination
The ability of filters in a duplex system to effectively separate and prevent interference between Tx and Rx frequencies. No dBm greater than this will be accounted for in the calculation.



Parameter	Fading	Method	Statistics Annual	Statistics Worst-month
ESR	Multipath	ITU	◉	○
SESR	Multipath	ITU	◉	○
UATR	Rain	Crane	◉	○
Total	Multipath	ITU		
Total	Rain	Crane		

Parameter
The specific telecommunication performance metric being evaluated.

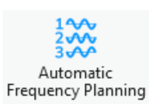
Fading
The type of signal disruption being accounted for in the metric.

Method
The standard used to calculate the performance metric.

Statistics Annual
A selection indicating if the metric is calculated based on yearly data.

Statistics Worst-month
A selection indicating if the metric is calculated based on the data from the month with the poorest performance.

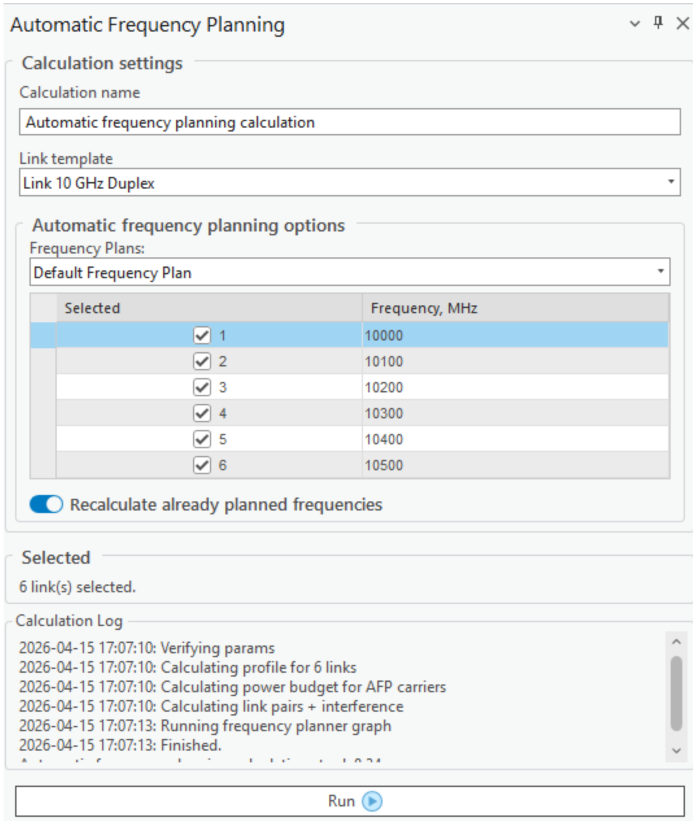
11.3 Automatic Frequency Planning

Click the 

button to open the Automatic Frequency Planning tool.

The tool suggests the best channel for each selected link based on the chosen frequency plan and channel list. It accounts for inter-link interference, evaluates the signal-to-interference ratio (SIR) across available carriers, and aims to maximize link performance while minimizing co-channel interference. The tool accounts for duplex link pairing, node-layer priority, and site location constraints (e.g., neighboring links and links sharing the same site). If automatic frequency planning is done with links created from Mesh Nodes, the tool avoids polarization conflicts by switching between horizontal and vertical polarizations for neighboring links sharing the same channel.

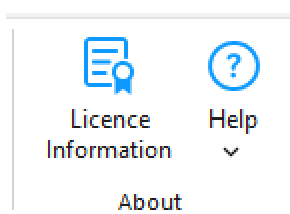
The planning results include the link names, carrier IDs, assigned frequencies, best available modulation, polarization, SIR score (in dB), and identifying color. The results are visualized on the map, as well as on the Automatic Frequency Planning Results table. The recommended carriers can be assigned for the links and later used in Link Prediction calculations.



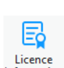
Recalculate already planned frequencies
Where applicable, the tool respects existing channel assignments. If frequency recalculation is explicitly enabled, i.e., the Recalculate already planned frequencies option is checked, the frequencies are reassigned for all links regardless of whether they have channel assignments.

Run
Run the automatic frequency planning tool.

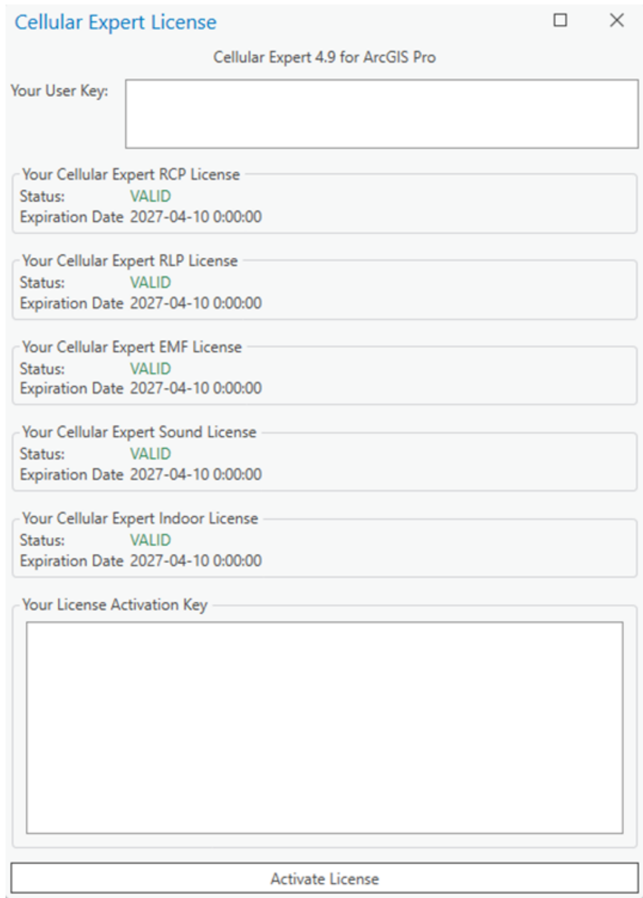
12. About



12.1 License information

Click the  button to open the License Information dialogue.

License information is a useful resource to see your current version of Cellular Expert for ArcGIS Pro, user key, currently active licenses, and their expiration dates. The license information window is also used when a user is enabling the *CE for ArcGIS Pro* extension on their computer. For more information about license activation see *Activation*.

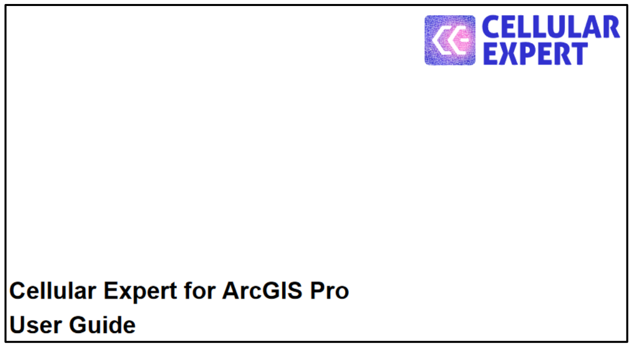


12.2 Help

12.2.1 Documentation

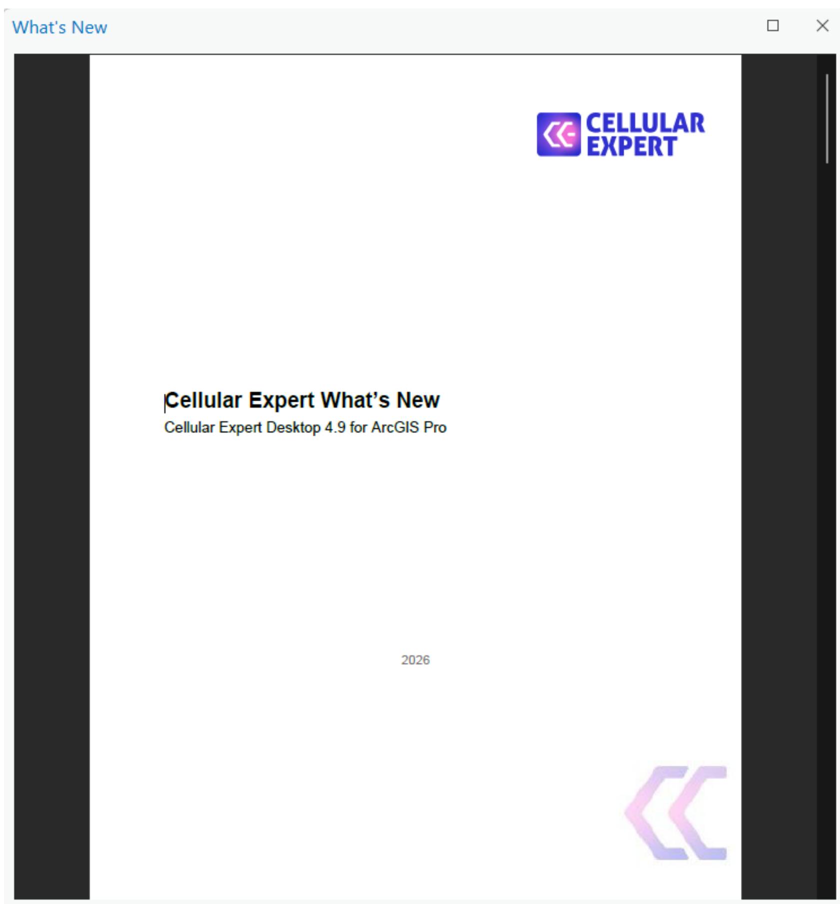
Click the Help button in the **Help** dropdown list to open the **Documentation** of *CE for ArcGIS Pro*.

Here you will find extensive documentation of the add-on. Also, this should be the first place you check when trying to figure out a problem.



12.2.2 What's New

Click the What's New button in the **Help** dropdown list to open the **What's New** document. This document is updated for each new release of Cellular Expert for ArcGIS Pro, and here you will find the changelog for the current installed version. This document serves as the introduction of added new features, enhancements, bug fixes, and other changes.



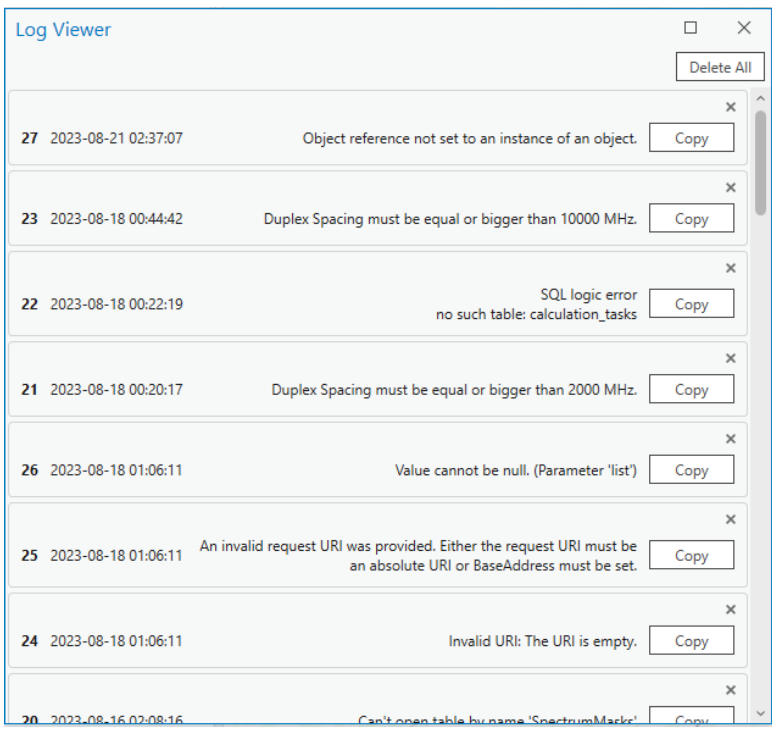
12.2.3 Technical Support

Click the Technical Support button in the **Help** dropdown list to open the **Technical Support** page in your default Internet browser. In the Cellular Expert Support page, you can browse the Knowledge base for instructions and solutions related to Cellular Expert for ArcGIS Pro, such as how to prepare geodata for a project. Additionally, you can log in or sign up to submit a new support ticket.

12.2.4 Error log

Click the Error Log button in the **Help** dropdown list to open the **Error Log** dialogue.

Here you will find information about errors that have occurred during certain processes of the *CE for ArcGIS Pro* extension's lifetime. These logs are crucial to improving the overall quality of the user experience. Thus, when an error happens and you decide to contact a Cellular Expert, you will be asked to send these logs so that the problems you encounter can be patched as soon as possible.



Copy
Copy the necessary error information into your clipboard and send it to Cellular Expert.

Delete All
If the error log ever gets too crowded, you can delete all errors from it.

13. Technical Support

For any issues, e-mail support (support@cellular-expert.com) and we will register your ticket.

When you contact us, please describe your issue precisely and completely. Please mention which steps you already tried, when the problem appears, etc. Attach a screenshot of the Cellular Expert log files.

Note: all provided information remains confidential and is used only for solving your issue. Other users cannot see your ticket and it is not published in any other way.

Thank you for using our software.
Cellular Expert Team

www.cellular-expert.com